The most entangled states of four and seven qubits:

Supplement

Antony Lewis
Department of Physics & Astronomy, University of Sussex, Brighton BN1 9QH, UK

Abstract

This supplement gives full proofs and certificate tables for the intermediate-order Rényi bounds, stability estimates, and the seven-qubit purity and entropy optima stated in the paper [1]. It also records invariant-dimension counts needed for certificate verification and further results used in the paper. Like the paper, it was prepared by AI agents under human supervision.

Contents

S1 About this supplement . . . 1
S2 Determining symmetric invariant identities . . . 2
S3 Proof of the general envelope . . . 3
S4 A direct proof of the spin-flip gradient identity . . . 6
S5 A weighted-variance proof of the sextic zero set . . . 8
S6 The Rényi bounds: full proofs and further results . . . 9
S7 Proofs of the stability theorems . . . 38
S8 Orbit stability: further proofs and results . . . 46
S9 A common replica hierarchy . . . 62
S10 The complex conference family . . . 67
S11 The five-copy level at seven qubits . . . 71
S12 The sharp mean entropy at seven qubits . . . 87
S13 Focused covariants of the four-qubit sextic . . . 105
S14 The complete cubic-SOS obstruction . . . 108
S15 Exact minima of the balanced purity . . . 109
S16 An exact eighteen-qubit balanced-purity minimum . . . 111
S17 Sharp purity–time-reversal tradeoffs . . . 112
S18 Thirteen qubits: purity interval and an entropy witness . . . 115
S19 Lean formalisation . . . 119
S20 Verification checks and certificate status . . . 123

S1 About this supplement

References without an S prefix, such as Theorem 3 or Appendix I, refer to the paper; numbers with the prefix S refer to this supplement. Each paper statement proved here is restated in full, under its paper number, immediately before the proof. The check labels (QG, SHD, MS, . . .) are described in Section S20; each identifies a verification check whose script and captured output are in the companion repository `cmbant/qubit-entropy` (the folder `verifiers/`; `verifiers/MAP.md` lists the script for each label). As in the paper, every stated result is proved, analytically or with the checks named in its proof; numerical observations are labelled as such.

For the four-qubit von Neumann maximum, the paper already contains the complete analytic proof. This supplement first supplies invariant-dimension counts (Section S2), the spin-flip gradient

identity (Section S4), and an alternative proof of the sextic zero-set classification (Section S5). Readers following the extensions should then turn to Section S6 for intermediate Rényi orders or to Sections S7 and S8 for orbit stability.

For seven qubits, Figure S1 separates the two proof chains. The replica hierarchy and complex conference family lead to the five-copy purity bound and its equality orbit; the entropy proof adds a spectral majorant and a restricted competitor region before identifying a different equality orbit (Sections S9–S12). Later sections develop further sextic covariants, purity–time-reversal inequalities, balanced-purity minima through eighteen qubits and an interval for the thirteen-qubit minimum, with an entropy witness at that size. Sections S19 and S20 record the Lean formalisation and the verification checks.

Seven-qubit notation. For a pure state of n qubits, $\pi_k(\psi) = \binom{n}{k}^{-1} \sum_{|S|=k} \operatorname{tr} \rho_S^2$ is the mean k-qubit purity, and at $n = 7$, E_3 is the mean von Neumann entropy (in bits) of the 35 three-qubit marginals. The icosahedral family ψ_x, $0 \le x \le 1/2$, is that of Theorem 41 and equation (14) of the paper; in the amplitude patterns U, V of Section S12.1, $\psi_x = \alpha U + \beta V$ with $\alpha = \sqrt{(1-2x)/30}$ and $\beta = \sqrt{x}$. Its member $\Psi_{\text{ico}} = \psi_{37/524}$ minimises π_3, and ψ_{x_*}, $x_* = 0.0698\ldots$, maximises E_3. The twirled m-copy moment of Appendix G of the paper, written X_m there, is written X_m^+ here.

S2 Determining symmetric invariant identities

This section supports the exact polynomial certificates, in particular the quadratic variance proof in Section S6.1. The elementary entropy proof of the paper does not use these counts. Statements here are proved by Schur–Weyl duality and finite-group character averages.

For N sites of local dimension r, let V_d be the permutation-symmetric local-unitary invariants of bidegree (d, d). Put $A_d = \bigoplus_{\lambda \vdash d,\ \ell(\lambda) \le r} V_\lambda \otimes V_\lambda^*$, with its left–right $S_d \times S_d$ action. Polarisation identifies invariant polynomials with operators on the symmetric copy subspace. Extend them by zero to all copies and apply Schur–Weyl duality at each site: the commutant is $A_d^{\otimes N}$. The left and right diagonal copy symmetrisers impose $S_d \times S_d$ invariance; physical-site symmetry then gives

$$V_d \cong \left(\operatorname{Sym}^N A_d\right)^{S_d \times S_d}. \tag{ID1}$$

Consequently

$$\dim V_d = \frac{1}{N!(d!)^2} \sum_{\sigma \in S_N} \sum_{g,h \in S_d} \prod_{C \text{ cycle of } \sigma} \left(\sum_{\lambda \vdash d,\ \ell(\lambda) \le r} \chi_\lambda(g^{|C|}) \chi_\lambda(h^{|C|}) \right). \tag{ID2}$$

Indeed, the trace of a tensor permutation is the product of the traces of powers over its cycles; averaging projects onto the fixed vectors. For bidegree (d, d), $SU(r)$ and $U(r)$ invariance agree since the scalar action cancels. Conjugation interchanges the two copy actions. Since the irreducible representations of S_d are real, it acts on each $V_\lambda \otimes V_\lambda^*$ as the swap, and the trace of (g, h) followed by the swap is $\chi_\lambda(gh)$. Inserting it in the average, a cycle of length L contributes $\chi_\lambda((gh)^{L/2})^2$ for even L and $\chi_\lambda((gh)^L)$ for odd L, so

$$\begin{aligned} \dim V_d^{\pm} &= \frac{1}{2}\Big(\dim V_d \pm \frac{1}{N!(d!)^2} \sum_{\sigma \in S_N} \sum_{g,h \in S_d} \prod_{C} \sum_{\lambda \vdash d,\ \ell(\lambda) \le r} \tau_\lambda(gh, |C|)\Big), \\ \tau_\lambda(k, L) &= \begin{cases} \chi_\lambda(k^{L/2})^2, & L \text{ even}, \\ \chi_\lambda(k^L), & L \text{ odd}. \end{cases} \end{aligned} \tag{ID3}$$

for the conjugation-even (+) and odd (−) invariants.

For four qubits the exact character checks SD-1, SD-2 and SD-4 give:

A dimension-D space admits D determining evaluations, but a proposed set must have a nonsingular evaluation matrix. Dimension alone never certifies particular points. The quadratic

d	All invariants	Qubit-symmetric	Conjugation-even	Conjugation-odd
3	20	5	5	0
4	98	18	16	2
5	293	37	30	7
6	1128	115	85	30
7	3409	273	183	90

Table S1: Dimensions of four-qubit local-unitary invariant spaces of bidegree (d, d). The last two columns split the qubit-symmetric space by complex conjugation. The value 85 is used in the quadratic variance certificate; these counts are not inputs to the elementary entropy proof.

variance certificates QVA and QG use the degree-six even count 85 and two independent zero evaluations, with a nonsingular 83-point raw-Gram matrix supplied by QG on their common kernel. The sextic entropy proof of Appendix A uses direct swap contractions and needs no determining-point argument.

S3 Proof of the general envelope

Notation is that of Appendix A of the paper: $N = \|z\|^2$, W_1, the site residuals $V_i - 8U_i$, the pair vectors $(R_i^{(i)} - R_j^{(j)})z$ and the sextic identity (S); X_i and c_{ij} are defined below.

Theorem 15 (general envelope and complete equality; exact certificate, FI, FG). *Every nonzero four-qubit vector satisfies, with $s = z^T \sigma_y^{\otimes 4} z$,*

$$F - \tfrac{3}{4}NW_1 - N|s|^2 + |s|^3 \;\geq\; \frac{7|s|^2}{8(N+|s|)}\, W_1. \tag{GE}$$

For unit states, with $\tau = |s|^2$,

$$\boxed{F \;\geq\; \Big(\frac{3}{4} + \frac{7\tau}{8(1+\sqrt{\tau})}\Big) W_1 + \tau - \tau^{3/2}.}$$

Equality in $F \geq \frac{3}{4}W_1 + \tau - \tau^{3/2}$ holds exactly on fully product states and on the critical equality family of Proposition 13, including its HS endpoint at $\tau = 0$ and Bell-pair-product endpoint at $\tau = 1$.

The optimality of the coefficient $\frac{7}{8}$ is shown in the paper.

Write $H = F - \frac{3}{4}NW_1 \geq 0$ (by (S)) and use the magic coordinates $z = \sum_{jk} Z_{jk} m_j \otimes m_k$ of Appendix B of the paper. There the spin flip $\tilde{z} = \epsilon^{\otimes 4}\bar{z}$ is $Z \mapsto \bar{Z}$ (because $\sigma_y^{\otimes 2}\bar{m}_k = -m_k$), and $s = \operatorname{tr} Z^T Z$ by (τ_G). Call z *real* if $\tilde{z} = z$, that is Z real. The flip is an antiunitary involution. It commutes with local $SU(2)$ transformations and qubit permutations and anticommutes with every $\sigma_a^{(i)}$, so it fixes all marginal spectra and every $\|\mathbf{r}_i\|$, and $H(\tilde{z}) = H(z)$. A real state has $\mathbf{r}_i = 0$, since $\langle a, \sigma a\rangle = -\overline{\langle a, \sigma a\rangle}$ for $\tilde{a} = a$.

Spin-flip split. After a global phase making $s \geq 0$, every unit z is

$$z_\kappa = \sqrt{\tfrac{1+\kappa}{2}}\, a + i\sqrt{\tfrac{1-\kappa}{2}}\, b, \qquad \kappa = s = \sqrt{\tau},$$

with a, b real and orthonormal. Indeed, with $Z = X + iY$, $s = \|X\|^2 - \|Y\|^2 + 2i\langle X, Y\rangle$ is real and nonnegative exactly when $\langle X, Y\rangle = 0$ and $\|X\|^2 - \|Y\|^2 = \kappa$. The path $t \mapsto z_t$, $0 \leq t \leq \kappa$, joins the flip-orthogonal midpoint $\phi = z_0 = (a + ib)/\sqrt{2}$ to z through unit states with $s(z_t) = t$.

The flip derivative. By (S), $H(x) = \|\mathcal{O}(x, x, \tilde{x})\|^2$ for every x, where $\mathcal{O}(x, y, t)$ is the vector $\big(\sqrt{3/8}\,(V_i - 8U_i)\big)_i \oplus \big(\sqrt{3/2}\, c_{ij}\big)_{i<j}$ computed from the three-copy tensor $x \otimes y \otimes t$ in place of w, symmetrised in x, y; in $c_{ij} = (R_i^{(i)} - R_j^{(j)})y$ the marginal is $R_i[a, b] = \sum x_{a\ldots}(\epsilon^{\otimes 4} t)_{b\ldots}$, which is R_i when $x = y = z$ and $t = \tilde{z}$. Write $X_i = (V_i - 8U_i)(z)$ for the site residual of z, a vector on six qubits (three slots at site i, one at each other site), and $\tilde{X}_i = \epsilon^{\otimes 6}\bar{X}_i$ for its spin flip.

Lemma S1 (spin-flip derivative)**.** *For every four-qubit vector z,*

$$\frac{d}{dt}\,H(z+t\tilde{z})\Big|_{t=0} = \frac{9}{4}\,\mathrm{Re}\sum_i \langle \tilde{X}_i, X_i\rangle + \mathrm{Re}\,s\left(4N^2 - 4|s|^2 + \frac{7}{2}W_1\right). \tag{FG}$$

Proof. For real t the flip of $z+t\tilde{z}$ is $\tilde{z}+tz$, since the flip is an antiunitary involution. As $\mathcal{O}$ is symmetric and bilinear in its first two arguments,

$$\frac{d}{dt}H(z+t\tilde{z})\Big|_{t=0} = 2\,\mathrm{Re}(\langle A, C\rangle + 2\langle D, A\rangle), \qquad A=\mathcal{O}(z,z,\tilde{z}),\ C=\mathcal{O}(z,z,z),\ D=\mathcal{O}(z,\tilde{z},\tilde{z}).$$

We show

$$\langle A, C\rangle + 2\langle D, A\rangle = \frac{9}{8}\sum_i X_i^T\epsilon^{\otimes 6}X_i + s\Big(\frac{5}{4}N^2 + \frac{11}{8}W_1 + \frac{3}{4}P - \frac{19}{8}|s|^2\Big). \tag{FGI}$$

Section S4 of the supplement derives this identity directly, without point evaluations, from the full symmetry of $V_i - 6U_i$ and the angular momentum of the copy labels. Here we give a shorter check, by determining its values in a finite-dimensional invariant space. Both sides are polynomials of bidegree $(4,2)$ in $(z,\bar{z})$, and both are $SU(2)^4$-invariant: L, M and the slot symmetrisers intertwine $g\otimes g\otimes g$ with g (respectively $g\otimes g\otimes g$) for $g\in SU(2)$, the flip commutes with local unitaries, and $g^T\epsilon g = \epsilon$. Replace $\bar{z}$ by an independent vector ζ; this is legitimate because all coefficients are real, so the conjugate of a vector built from $(z,\bar{z})$ is the same construction applied to $(\bar{z},z)$. The identity is then one between polynomials of bidegree $(4,2)$ on pairs (z,ζ) invariant under $(z,\zeta)\mapsto(gz,\bar{g}\zeta)$, $g\in SU(2)^4$. Since $\bar{g}$ is equivalent to g, the Weyl character formula shows that these invariants form a space of dimension 19 [FI-1]. Nineteen invariants that contract each site's six legs in pairs (ϵ between two z-legs or two ζ-legs, δ between a z-leg and a ζ-leg) take values at 24 integer points (z,ζ) whose 24×19 matrix has exact rank 19 [FI-2]. So an invariant of this bidegree that vanishes at these points is zero. Both sides of (FGI) agree exactly at the 24 points [FI-4], with $\|A\|^2 = H$ there as a check on the normalisation [FI-3]; hence (FGI) holds identically. Replacing $\frac{9}{8}$ by 1, or dropping the multiple of s, fails at every point tested [FI-5].

Now $\langle\tilde{X}, X\rangle = X^T(\epsilon^{\otimes 6})^T X = X^T\epsilon^{\otimes 6}X$ (six factors $\epsilon^T = -\epsilon$), and the purity identity (Lemma 2) in homogeneous form is $P = N^2 + \frac{1}{2}(W_1 + |s|^2)$. Taking twice the real part of (FGI) gives (FG) [FG-7]. □

Lemma S2 (zeros of the sharp polarisation remainder)**.** *For a unit four-qubit state, $H = F - \frac{3}{4}W_1$ vanishes exactly on fully product states, the two HS orbits, and the three Bell-pair-product orbits.*

Proof. By (S), $H=0$ forces $(R_i^{(i)} - R_j^{(j)})z = 0$ for every pair. If one Bloch vector $\mathbf{r}_i$ is zero, then $(\mathbf{r}_j\cdot\boldsymbol{\sigma}_j)z = 0$ for every j. Since $(\mathbf{r}_j\cdot\boldsymbol{\sigma}_j)^2 = \|\mathbf{r}_j\|^2 I$, all Bloch vectors vanish. The state is critical, $H=F$, and Proposition 14 gives HS or Bell products.

Otherwise diagonalise the four one-body marginals locally and write $R_i = (I + t_i Z)/2$, $t_i > 0$. Each supported computational string obeys $t_i(-1)^{x_i} = t_j(-1)^{x_j}$ for all i,j. Thus all t_i agree and every supported string has all four bits equal: $z = a|0000\rangle + b|1111\rangle$. Direct evaluation gives $H = 18|ab|^2 = (9/2)\tau$, so $H=0$ forces a product state [PZ-1, PZ-2]. Conversely the listed states have $H=0$, by (S) and Proposition 14. □

Proof of Theorem 15. Both sides of (GE) are local-unitary invariant and homogeneous of degree six, so take $N=1$ and $s=\kappa\geq 0$. If $\kappa=1$, equality in $|\langle z,\tilde{z}\rangle|\leq\|z\|^2$ makes z real up to phase, so $W_1=0$ and (GE) is $H\geq 0$. Let $0\leq\kappa<1$ and use the spin-flip split, with $h(t)=H(z_t)$ and $W_0 = W_1(\phi)$. Since a and b have zero Bloch vectors, the Bloch vectors of z_t are $\sqrt{1-t^2}$ times those of ϕ, so $W_1(z_t) = (1-t^2)W_0$. Differentiating, $2(1-t^2)\frac{d}{dt}z_t = \tilde{z}_t - tz_t$; as H is homogeneous of degree six, Lemma S1 at z_t gives

$$2(1-t^2)h'(t) = \frac{9}{4}\,\mathrm{Re}\sum_i\langle\tilde{X}_i, X_i\rangle + 4t(1-t^2) + \frac{7}{2}t(1-t^2)W_0 - 6t\,h(t).$$

The spin flip is antiunitary, so $\mathrm{Re}\langle \tilde{X}_i, X_i\rangle = \frac{1}{2}\|X_i + \tilde{X}_i\|^2 - \|X_i\|^2$, and by (S) $h = \frac{3}{8}\sum_i \|X_i\|^2 + \frac{3}{2}\sum_{i<j}\|c_{ij}\|^2$ with $c_{ij} = c_{ij}(z_t)$, where $c_{ij}(x) = (R_i^{(i)} - R_j^{(j)})x$ with the marginals of x. Hence the calibrated remainder

$$\mathcal{M}(t) = \frac{h(t)}{(1-t)^3} - \Big(1 + \frac{7}{8}W_0\Big)\frac{t^2}{(1-t)^2}$$

satisfies the exact transport law [FG-7, FG-8]

$$\mathcal{M}'(t) = \frac{9}{16(1-t)^4(1+t)}\Big[\sum_i \|X_i + \tilde{X}_i\|^2 + 8\sum_{i<j}\|c_{ij}\|^2\Big] \;\geq\; 0, \tag{SC}$$

all vectors evaluated at z_t. So $\mathcal{M}(\kappa) \geq \mathcal{M}(0) = H(\phi)$, that is,

$$H(z) \;\geq\; (1-\kappa)^3 H(\phi) + \kappa^2(1-\kappa)\Big(1 + \frac{7}{8}W_0\Big).$$

Drop $H(\phi) \geq 0$ and substitute $W_0 = W_1/(1-\kappa^2)$: $H \geq \kappa^2 - \kappa^3 + \frac{7\kappa^2}{8(1+\kappa)}W_1$, which is (GE).

Equality. For $0 < \tau < 1$ the added term is positive unless $W_1 = 0$, so an equality state is critical, and Proposition 13 gives exactly its equality family. At $\tau = 1$ the state is real up to phase, hence critical, and equality is $F = 0$, which gives the Bell-pair products (Proposition 14). At $\tau = 0$ equality is $H = 0$; by Lemma S2 this gives exactly the product states and the HS orbits, since Bell-pair products have $\tau = 1$. Conversely these states attain the bound, by Proposition 13 and product-state sharpness ($F = 3 = \frac{3}{4}W_1$, $\tau = 0$). □

The transport law (SC) integrates exactly. The result is a sum of squares for the defect in (GE), which shows which terms vanish on the equality states.

Proposition S3 (Exact spin-flip calibration)**.** *In the notation of the proof of Theorem 15, let* $\kappa = \sqrt{\tau} < 1$ *and put*

$$A_i = X_i(a), \qquad E_i = X_i(\phi) + \tilde{X}_i(\phi), \qquad c_{ij} = c_{ij}(\phi), \qquad C_0 = \sum_{i<j}\|c_{ij}\|^2.$$

Then the defect in the stronger envelope (GE) is exactly

$$\begin{aligned} &H(z_\kappa) - \kappa^2(1-\kappa)\Big(1 + \frac{7}{8}W_0\Big) \\ &\quad= (1-\kappa)^3 H(\phi) + \frac{9\kappa}{32}\sum_i \|\kappa A_i + \sqrt{2}(1-\kappa)E_i\|^2 + \frac{3\kappa^3}{32}\sum_i \|A_i\|^2 \\ &\qquad + \frac{9}{2}\kappa(1-\kappa)^2 C_0 + \frac{9}{8}\kappa^2(1-\kappa)\sum_{i<j}\|c_{ij} - \tilde{c}_{ij}\|^2. \end{aligned} \tag{SC-SOS}$$

In particular, since $\frac{3}{8}\sum_i \|A_i\|^2 = F(a)$,

$$H(z_\kappa) \geq \kappa^2(1-\kappa)\Big(1 + \frac{7}{8}W_0\Big) + (1-\kappa)^3 H(\phi) + \frac{\kappa^3}{4}F(a).$$

Proof. Integrate (SC) from 0 to κ and multiply by $(1-\kappa)^3$; the initial value is $\mathcal{M}(0) = H(\phi)$. To integrate exactly, set $p = \sqrt{(1+t)/2}$ and $q = \sqrt{(1-t)/2}$. All coefficients of $\mathcal{O}$ are real and its maps commute with spin flip. Cubic expansion in the flip-fixed vectors a, b therefore gives

$$X_i(z_t) + \tilde{X}_i(z_t) = p\big[(1+t)A_i + (1-t)B_i\big], \qquad B_i = \sqrt{2}E_i - A_i.$$

Only the terms containing zero or two occurrences of ib survive. With $r = (1+t)/(1-t)$ the site contribution to $\mathcal{M}'(t)dt$ is $\frac{9}{64}\sum_i \|rA_i + B_i\|^2 dr$. For $R = (1+\kappa)/(1-\kappa)$, completing the square gives

$$\int_1^R \|rA + B\|^2 dr = (R-1)\|B + \tfrac{R+1}{2}A\|^2 + \frac{(R-1)^3}{12}\|A\|^2.$$

Multiplication by $(1-\kappa)^3$ gives the two site terms of (SC-SOS).

The traceless marginal differences scale by $\sqrt{1-t^2}$ and anticommute with spin flip. Writing $z_t = \alpha\phi + \beta\tilde{\phi}$, where $\alpha^2+\beta^2=1$ and $2\alpha\beta = t$, gives

$$c_{ij}(z_t) = \sqrt{1-t^2}(\alpha c_{ij} - \beta\tilde{c}_{ij}), \qquad \sum \|c_{ij}(z_t)\|^2 = (1-t^2)(C_0 - tD_0),$$

where $D_0 = \mathrm{Re}\sum\langle c_{ij}, \tilde{c}_{ij}\rangle$. Its integrated contribution is $\frac{9}{4}\kappa(1-\kappa)[(2-\kappa)C_0 - \kappa D_0]$. Use $C_0 - D_0 = \frac{1}{2}\sum\|c_{ij}-\tilde{c}_{ij}\|^2$ to obtain the last two terms. The integrations are checked in FG-8; this check involves only scalar rational identities, not state evaluations. □

S4 A direct proof of the spin-flip gradient identity

Section S3 proves the identity (FGI) behind Lemma S1 by evaluating both sides at 24 points of a 19-dimensional space of invariants. This section derives it without a normal form or point evaluations. The key is the fully symmetric cubic part of the residual, whose norm is determined by the angular momentum of the copy labels. The notation (L, M, S_i, U_i, V_i and the contractions (C), (U), (Pr)) is that of Appendix A of the paper; X_i and the pair vectors c_{ij} are those of Section S3.

The cubic and its polarisation defect. Symmetrise U_i in its first two inputs and put $\mathcal{T}_i = V_i - 6U_i$. Its ordinary-site operator is

$$\prod(L+M) - 3(\prod L + \prod M).$$

It is invariant under $(L,M)\mapsto(M,L)$ and $(L,M)\mapsto(-L,M-L)$, the two generating input transpositions (the latter is Schouten). Thus $\mathcal{T}_i$ is fully symmetric, while U_i has zero fully symmetric part. Write

$$C_i(z) = \mathcal{T}_i(z,z,z), \qquad Y_i = \mathcal{T}_i(z,z,\tilde{z}), \qquad X_i = Y_i - 2U_i, \qquad K_c = \sum_{i<j}\|c_{ij}\|^2.$$

The key identity is a polarisation defect of this one cubic:

$$\boxed{\sum_i \|C_i(z)\|^2 - \sum_i \|Y_i\|^2 = \frac{32}{3}N(N^2-|s|^2) - 4NW_1 - \frac{8}{3}K_c.} \qquad \text{(CP)}$$

We derive it by angular momentum of the copy labels, then polarise once. This gives the flip derivative directly, without a state normal form.

Why the cubic has this norm. At a site, three copies decompose into the symmetric spin-3/2 space and spin 1/2 with the two-dimensional standard representation of S_3. Let P_3 select three standard sites and one symmetric site inside the global symmetric cube. Then

$$\sum_i \|C_i(z)\|^2 = 54\langle z^{\otimes 3}, P_3 z^{\otimes 3}\rangle.$$

Indeed the cubic annihilates the symmetric component at its three ordinary sites and selects the unique invariant in three standard factors. In an orthonormal standard basis, the local maps L, M have rows $\ell = (\sqrt{3/2}, -1/\sqrt{2})$ and $m = (\sqrt{3/2}, 1/\sqrt{2})$. The row $(\ell+m)^{\otimes 3} - 3(\ell^{\otimes 3} + m^{\otimes 3})$ is $\frac{3\sqrt{6}}{2}(1,0,0,-1,0,-1,-1,0)$, with squared norm 54 [FG-3]. This fixes the normalisation on the whole spin module. The sector also has a direct test interpretation: a state product across a balanced cut has matching S_3 labels at the two sites of each side, hence an even number of standard sites and zero P_3 projection.

Here is the copy-label calculation relating P_3 to the physical moments. Let K count standard sites. On those K factors write $\mathbf{J} = \frac{1}{2}\sum\boldsymbol{\sigma}$ and $Y = \sum\sigma_y$; these Pauli matrices act on copy multiplicities, not physical qubits. A three-cycle is a rotation by $2\pi/3$ in the standard plane. Global copy invariance therefore forces Y to be a multiple of 3. For $K \le 4$, the only nonzero possibility is $Y = \pm 3$ at $K = 3$:

$$Y^2 = 9P_3, \qquad K^2 - 4K + 2\mathbf{J}^2 = \tfrac{1}{2}Y^2 + 12P_+.$$

For the second identity, two standard factors have their invariant in spin 1, three in spin 3/2, and four in spin 2 or spin 0; P_+ denotes the spin-2 invariant line in the last case. There is no invariant with one standard factor or in the spin-1 summands of four standard factors. These statements follow by restricting the usual spins to the triangle's rotation and reflection; at $K = 3$ only the symmetric cube contains the rotation charges ± 3, and at $K = 4$ spin 1 has no invariant.

For completeness the associated moment operators are

$$\begin{aligned} NW_1 &\longleftrightarrow 4I - 2K, & N|s|^2 &\longleftrightarrow 6P_+, \\ K_c &\longleftrightarrow 2K - K^2 + \mathbf{J}^2 - \tfrac{1}{4}Y^2, & F &\longleftrightarrow 3I - \tfrac{5}{2}K(K+1) + \tfrac{17}{2}\mathbf{J}^2 + \tfrac{1}{8}Y^2. \end{aligned} \tag{CM}$$

All operators are compressed to the global symmetric cube. The first is the one-body purity identity. The second follows by summing the three four-singlet projectors: their compression is $9P_+/8$, while each has expectation $N|s|^2/16$ (a reflection-antisymmetric fourth tensor power has invariant projection norm $(2 + 4/16)/6 = 3/8$). To derive the other two, set $\Pi = \sum_{i<j} \mathrm{tr}[R_{ij}(R_i \otimes R_j)]$. Its operator is the average of distinct transpositions at sites i, j. The three standard reflections have vanishing sum and angles $2\pi/3$, so

$$\Pi \longleftrightarrow \tfrac{1}{2}(4-K)(3-K) - \tfrac{1}{2}\mathbf{J}^2 + \tfrac{1}{8}Y^2 + \tfrac{1}{4}K.$$

Insert this in (Pr) for K_c. For F, the symmetrised swap expression is $(17/6)A - 3B - 12I$, where A sums equal transpositions over site pairs and B sums equal three-cycles and their inverses. Using $\sum \text{reflections} = 3S$ and $c + c^{-1} = 3S - I$, $c - c^{-1} = \pm i\sqrt{3}\sigma_y$ for the slot permutations at one site, with S the projector onto its symmetric component and σ_y acting on its two-dimensional standard component, gives explicitly

$$A = \tfrac{3}{2}(4-K)(3-K) + 3\mathbf{J}^2 - \tfrac{3}{4}Y^2 - \tfrac{3}{2}K, \qquad B = 12 - \tfrac{21}{2}K + \tfrac{9}{4}K^2 - \tfrac{3}{4}Y^2.$$

Their combination is the displayed operator for F. Thus (CM) uses only the standard-representation Casimirs [FG-2]. The globally invariant sectors and the values of these operators on them are:

K	j	Y^2	projector	NW_1	$N\lvert s\rvert^2$	K_c	F	$54P_3$
0	0	0		4	0	0	3	0
2	1	0		0	0	2	5	0
3	$\frac{3}{2}$	9	P_3	-2	0	$-\frac{3}{2}$	6	54
4	2	0	P_+	-4	6	-2	4	0
4	0	0		-4	0	-8	-47	0

Here K counts standard sites, j is the copy-label spin of those K factors, P_3 is the $K = 3$ line and P_+ the spin-2 line at $K = 4$; N^3 acts as 1. For each choice of the K standard sites, the $K = 4$, $j = 0$ sector has multiplicity two; the other listed sectors have multiplicity one. The displayed operators are scalar on both copies of the spin-0 sector. No invariant occurs with $K = 1$ or with $j = 1$ at $K = 4$ [FG-2]. Substitution of the two selection rules gives

$$54P_3 \longleftrightarrow \frac{8}{3}F + \frac{32}{3}N(N^2 - |s|^2) - \frac{14}{3}NW_1 - 12K_c.$$

On the other hand, the already proved contractions (C), (U), (Pr) give

$$\sum \|Y_i\|^2 = \frac{8}{3}F - \frac{2}{3}NW_1 - \frac{28}{3}K_c, \quad \sum \|U_i\|^2 = \frac{1}{3}NW_1, \quad \mathrm{Re}\sum \langle U_i, Y_i\rangle = \frac{2}{3}NW_1 - \frac{4}{3}K_c. \tag{CY}$$

Subtract the first of (CY) to obtain (CP) [FG-2, FG-4]. No new contraction table is required.

A vanishing quartic covariant. Use the bilinear output pairing $B(v, w) = v^T \epsilon^{\otimes 6} w$. For every i, $B(C_i(z), U_i(z, z, w)) = 0$. Indeed the local formula $U_i = -S_i(z \otimes R_i \epsilon^T)$ makes this pairing a contraction of R_i with a holomorphic quartic spin-1 covariant at site i, singlet at the other sites. Such a covariant is zero: in four copy labels, the singlets have multiplicity representation [22],

on which the Klein four group acts trivially. Spin 1 has multiplicity [31], the standard traceless representation on four letters, which has no Klein-four fixed vector. Hence their tensor product has no globally symmetric component [FG-3]. This is the only vanishing statement needed below.

One polarisation gives the transport identity. Now regard $w = \tilde{z}$ as independent of z and differentiate in the w direction z, denoting this derivative by $\mathcal{D}$. With the four-qubit pairing $B_4(x,y) = x^T\epsilon^{\otimes 4}y$, write $\bar{s} = B_4(w,w)$, $n = B_4(z,w)$; physically these are $\bar{s}, N$. All Hermitian expressions above extend polynomially by replacing their conjugate tensors by their spin flips with z, w exchanged. Put

$$u = \sum B(U_i,U_i), \quad q = \sum B(Y_i,U_i), \quad p = \sum c_{ij}^T\epsilon^{\otimes 4}c_{ij},$$
$$y = \sum B(Y_i,Y_i), \quad d = \sum B(C_i(z),\mathcal{T}_i(z,w,w)).$$

Since the fully symmetric parts of U and c vanish, $\mathcal{D}\sum\|U_i\|^2 = -u$ and $\mathcal{D}K_c = -p$. The vanishing quartic covariant, differentiated in z along w, gives $\sum B(C_i(z),U_i(w,w,z)) = 3q$. Consequently $\mathcal{D}\,\mathrm{Re}\sum\langle U_i,Y_i\rangle = 2q$. Using $\mathcal{D}n = s$, $\mathcal{D}W_1 = 0$ and (CY) yields

$$u = -\tfrac{1}{3}sW_1, \qquad q = \tfrac{1}{3}sW_1 + \tfrac{2}{3}p.$$

The derivative of the left side of (CP) is $2(d-y)$. Its right side gives

$$d - y = \tfrac{16}{3}s(n^2 - s\bar{s}) - 2sW_1 + \tfrac{4}{3}p.$$

Finally $X = Y - 2U$ and the same cubic polarisation give

$$\Delta := \mathcal{D}H = \tfrac{3}{8}(d + 2y - 8q - 4u) - \tfrac{3}{2}p,$$
$$b := \sum B(X_i,X_i) = y - 4q + 4u.$$

The cancellation is now transparent:

$$\Delta - \tfrac{9}{8}b = \tfrac{3}{8}(d-y) + \tfrac{3}{2}q - 6u - \tfrac{3}{2}p = s(2n^2 - 2s\bar{s} + \tfrac{7}{4}W_1).$$

The purity identity $P = n^2 + (W_1 + s\bar{s})/2$ turns this precisely into (FGI) [FG-4, FG-9]. Thus the spin-flip transport law follows from cubic symmetry, the copy-label Casimirs, and one absent quartic representation, rather than normal-form block evaluations.

S5 A weighted-variance proof of the sextic zero set

The paper derives the zero set of F (Proposition 14) from the critical envelope. This section gives an independent proof: on the critical normal form, F is an explicit sum of twelve squared moduli, whose simultaneous vanishing is solved directly. The notation is that of Appendices A and B of the paper.

Second proof of Proposition 14. The identity (S) makes every nonzero zero critical. Use the elementary normal form and write the state in its G_{abcd} coordinates $v = (a,b,c,d)$. These differ only by fixed coordinate signs from the magic-basis diagonal coordinates. Its three cut spectra are the squared moduli of v, H_+v, H_-v, where

$$H_+ = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad H_- = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{pmatrix}.$$

This follows by the same two-block Hadamard diagonalisation for each cut. Put $p_i = |v_i|^2$, $x_i = v_i^2$, $s = \sum_i x_i$, and $B = \sum_i p_i x_i$. The critical-slice identity is

$$\boxed{F(G_{abcd}) = \sum_i \sum_{\substack{j<k \\ j,k\neq i}} \left|\bar{v}_i(v_j^2 - v_k^2) + 2v_i(p_j - p_k)\right|^2.} \tag{Z}$$

Here is a direct derivation. The rows of $2H_+$ and $2H_-$ together are all four-sign vectors modulo an overall sign. Expanding the sixth power and cancelling terms with an odd sign multiplicity gives

$$T = \frac{3}{4}N^3 + \frac{9}{8}N|s|^2 - \frac{9}{4}N\sum_i p_i^2 + 3\sum_i p_i^3 - \frac{3}{2}\,\mathrm{Re}(\bar{s}B).$$

Also $P = N^2 + |s|^2/2$: this is the purity identity (Lemma 2) in homogeneous form for a critical state, where $P = N^2 + \tau/2$ and $\tau = |s|^2$ by (τ_G) (the signs relating G_{abcd} to the magic-basis coordinates do not affect the squares x_i). Substitution gives

$$F = 27N\sum_i p_i^2 - 36\sum_i p_i^3 - 4N^3 - 5N|s|^2 + 18\,\mathrm{Re}(\bar{s}B).$$

Expanding the right side of (Z) gives this same expression: for each i use $\sum_{j<k}|y_j - y_k|^2 = 3\sum_j |y_j|^2 - |\sum_j y_j|^2$ on the three remaining indices, and its bilinear version for the cross terms. Thus (Z) is an elementary weighted-variance identity.

We now classify its zeros; rescale to $N = 1$. For $p_i > 0$ let $u_i = x_i/p_i$, so $|u_i| = 1$. Vanishing of each summand says

$$x_j - x_k = -2u_i(p_j - p_k), \qquad i, j, k \text{ distinct}, \quad p_i > 0. \tag{ZE}$$

If between two and three coordinates are nonzero, choose nonzero i, j and zero k. Then $|x_j| = 2p_j$, impossible. A single nonzero coordinate is a Bell-pair product across $12|34$.

It remains to take all $p_i > 0$. If they are equal, (ZE) makes all x_i equal too. Thus, up to phase, $v = (\pm 1, \pm 1, \pm 1, \pm 1)/2$. One of H_+v, H_-v has just one nonzero coordinate: the sign parity selects which. These are Bell-pair products across the other two cuts.

Otherwise choose $p_1 \neq p_2$. Equation (ZE) for the pair $1, 2$ with $i = 3, 4$ gives $u_3 = u_4 =: u$. If $p_3 \neq p_4$, the same reasoning for the pair $3, 4$ gives $u_1 = u_2 =: w$. Now (ZE) for the pair $3, 4$ with $i = 1$ reads $u(p_3 - p_4) = -2w(p_3 - p_4)$, that is $u = -2w$, and for the pair $1, 2$ with $i = 3$ it reads $w(p_1 - p_2) = -2u(p_1 - p_2)$, that is $w = -2u$; but $|u| = |w| = 1$, which is impossible. Hence $p_3 = p_4 =: t$ and $x_3 = x_4 = ut$. Relabel $1, 2$ so $p_1 \neq t$. Applying (ZE) to the pair $1, 3$, with $i = 2, 4$, gives $u_2 = u$. The equations for the pairs $1, 2$ and $1, 3$ now give

$$x_1 = u(3p_2 - 2p_1) = u(3t - 2p_1).$$

Thus $p_2 = t$, and $|x_1| = p_1$ forces $p_1 = 3t$ and $u_1 = -u$. Normalisation yields, up to position and common phase,

$$v = (\sqrt{3}, \pm i, \pm i, \pm i)/\sqrt{6}. \tag{HS}$$

These sign families are exactly the two M_4 LU orbits, by the explicit normal-form identification in the proof of Proposition 4; the fixed coordinate signs relating G_{abcd} to $G(v)$ preserve the displayed set of sign choices. Conversely, all the listed vectors make every term of (Z) vanish. □

S6 The Rényi bounds: full proofs and further results

The intermediate-order Rényi proof works with a *relaxation* of the quantum problem. Take three arbitrary four-point probability vectors $\lambda^{(j)}$, $j = 2, 3, 4$, and set $P = \sum_{j,i}(\lambda_i^{(j)})^2$ and $T = \sum_{j,i}(\lambda_i^{(j)})^3$. Impose the necessary constraints $P \geq 1$, $F = 17P - 12T - 12 \geq 0$, and the quadratic purity-variance bound (QK). We do not assume these inequalities characterise physically realisable cut spectra. An upper bound for this larger set proves the quantum bound; identifying the equality states requires the separate rigidity arguments of the paper. The bound (QK) is deduced below from the sharper six-copy certificate (QV).

The remaining optimisation has two anchor orders. Determinant monotonicity reduces individual spectra to repeated-root charts; the supporting bounds then cover an HS region, a strict intermediate band, a capped region and a narrow cluster region. Single crossing transfers the HS anchor to lower orders, and monotonicity of Rényi entropy transfers the cluster anchor to higher orders. The final subsections give independent estimates and explain why local filtering cannot replace the unrestricted compatibility argument.

Proofs of results of the paper

S6.1 Proof of Theorem 27 and Corollary 28

Theorem 27 (Sharp quadratic purity variance; exact-rational certificate, QVA). *Every unit pure four-qubit state satisfies*

$$\boxed{F^2 + 3P(P-1) \geq 6V_P, \qquad V_P = \sum_{j=2}^{4} (p_j - P/3)^2.} \tag{QV}$$

Equality holds along the whole cluster-to-Bell family (QF), so the coefficient 3 *cannot be decreased.*

Write $p_j = \operatorname{tr} \rho_{1j}^2 \in [1/4, 1]$ and $U = \sum_j p_j^2$. Maximising the convex function U at fixed P over the box $[1/4, 1]^3$, filling one coordinate at a time towards an endpoint, gives the *purity box*

$$U \leq \begin{cases} (P-1/2)^2 + 1/8, & 1 \leq P \leq 3/2, \\ 1 + (P-5/4)^2 + 1/16, & 3/2 \leq P \leq 9/4, \\ 2 + (P-2)^2, & 9/4 \leq P \leq 3. \end{cases} \tag{JB}$$

The family (QF) has $P = 1 + r^2/2$, $F = (1-r^2)/2$ and $V_P = (1+2r^2)^2/24$, which gives equality and, for $r > 0$, $(6V_P - F^2)/(P(P-1)) = 3$ exactly (QV-3). Proposition S22 covers critical states, including $P = 1$ by Lemma 2. When $F \geq 3/2 - P$, the purity box (JB) gives the inequality directly by substituting into $F^2 + 5P^2 - 3P - 6U$ (QV-2). Thus only noncritical states with $1 < P < 3/2$ and $F < 3/2 - P$ need the certificate, which treats all states at once.

Proof. For $z \in \mathbb{C}^{16}$ put $N = \|z\|^2$, use unnormalised marginals in P, T, U, and set

$$\mathfrak{F} = 17NP - 12T - 12N^3, \qquad \mathfrak{D} = \mathfrak{F}^2 + 5N^2P^2 - 3N^4P - 6N^2U.$$

Then $\mathfrak{D}(z) = N^6(F^2 + 3P(P-1) - 6V_P)(z/\sqrt{N})$, so it suffices to prove $\mathfrak{D} \geq 0$. It is a local-unitary invariant of bidegree $(6,6)$, symmetric under qubit permutations and complex conjugation, and it vanishes at HS and at the cluster state, which are equality cases. We exhibit it as a sum of squared norms.

Put $\widetilde{z} = \epsilon^{\otimes 4} \bar{z}$, where $\epsilon = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Since $\epsilon \bar{U} = U\epsilon$ for $U \in SU(2)$, apply the six-copy spin intertwiners to $z^{\otimes a} \otimes \widetilde{z}^{\otimes(6-a)}$, $a \in \{3, 4\}$, and stack all 24 qubit permutations. For a spin tuple $p = (j_1, \ldots, j_4)$, one twice-spin $j_i \in \{0, 2, 4, 6\}$ of the six copies at each qubit, call the resulting map $W_{a,p}$ and equip its magnetic rows with the positive weights

$$w_p = \bigotimes_{j \in p} \left(\frac{\operatorname{lcm}_r \binom{j}{r}}{\binom{j}{k}} \right)_{k=0}^{j},$$

repeated over the permutations. These are the invariant spin inner products: the spin-$j/2$ lowering basis has weights $1/\binom{j}{k}$. The data in the verifier map specifies integer multiplicity vectors v_{ki}, integer weights q_{ki} and an integer denominator D. Put $\Phi_{ki} = W_{a_k, p_k} v_{ki}$. Intertwining makes their squared norms local-unitary invariant; summing over permutations makes them qubit-symmetric, and the real intertwiners and weights make them conjugation-even.

There are three finite checks: the highest-weight identities and vanishing of each Φ_{ki} at HS and cluster (QVA-1); positivity of D and every q_{ki} (QVA-2); and the identity

$$2D\, \mathfrak{D}(z) = \sum_{k,i} q_{ki} \|\Phi_{ki}(z)\|_{w_{p_k}}^2 \tag{S6.1}$$

at the 83 determining states (QVA-3). Both sides lie in the space $\mathcal{V}$ of qubit-symmetric, conjugation-even local-unitary invariants of bidegree $(6,6)$ vanishing at HS and cluster. The

ambient space has dimension 85 (Section S2, SD-4); HS and cluster evaluations are independent, since the normalised values of (N^6, N^3T) are $(1, 5/12)$ and $(1, 3/8)$. Thus $\dim \mathcal{V} = 83$. QG supplies 83 Gram entries in $\mathcal{V}$, built from five-copy covariants multiplied by N or coupled to one further copy (QG-1), whose evaluation matrix at these states has determinant 627608 modulo 1000003 (QG-3). Evaluation is therefore injective on $\mathcal{V}$, and (S6.1) holds identically. Every summand is nonnegative, so $\mathfrak{D} \geq 0$. □

Corollary 28 (Quadratic purity variance). *Every unit pure four-qubit state satisfies*

$$\boxed{F^2 + \frac{9}{2}(P-1) \geq 6V_P.} \tag{QK}$$

Equality holds exactly for HS cut spectra, cluster cut spectra, or products of two Bell pairs, up to ordering of cuts and eigenvalues. The coefficient $9/2$ *cannot be decreased, and the coefficient* 6 *cannot be increased while retaining* $9/2$*.*

Proof. Write $\mathcal{K} = F^2 + \frac{9}{2}(P-1) - 6V_P$ and $\mathcal{V}_Q = F^2 + 3P(P-1) - 6V_P$, so that $\mathcal{K} - \mathcal{V}_Q = 3(P-1)(3/2-P)$. By Lemma 2, $P \geq 1$. For $1 \leq P \leq 3/2$, Theorem 27 gives $\mathcal{K} \geq 3(P-1)(3/2-P) \geq 0$, strictly for $1 < P < 3/2$. For $P \geq 3/2$ the purity box (JB) gives $\mathcal{K} \geq F^2 + 4(P - \frac{3}{2})(\frac{27}{8} - P)$ on $[3/2, 9/4]$ and $\mathcal{K} \geq F^2 + (P - \frac{9}{4})(\frac{39}{2} - 4P) + \frac{27}{8} > 0$ on $[9/4, 3]$ (QV-2), so $\mathcal{K} \geq 0$, strictly for $P > 3/2$. Equality therefore forces $P = 1$ or $P = 3/2$.

At $P = 3/2$ the box gives $\mathcal{K} \geq F^2$, so equality forces $F = 0$ and $U = 9/8$; the purities are a permutation of $(1, 1/4, 1/4)$. The marginal of purity one makes the state a product across that cut. On either crossed cut its marginal is a product of two one-qubit states, and purity $1/4$ forces both to be maximally mixed, so both factors are Bell pairs. At $P = 1$, Proposition S21 gives $\mathcal{K} = F^2 - 6V_P \geq s^4(1-s)(2-s)$ with $0 \leq s \leq 1$ (QV-1). Equality requires $s \in \{0, 1\}$ and equality in (RF), which gives exactly HS or cluster spectra. All three types attain equality. Bell products force the coefficient $9/2$, and cluster spectra force the coefficient 6. □

The Lean formalisation proves (S6.1) for all z directly, without the dimension count (Section S19). An older direct certificate for (QK), from eighty squared covariant norms in sixteen modules, is kept as a regression (QG-2, QG-4).

S6.2 Proof of Lemma 29

At fixed trace, second moment and third moment, the determinant is the only remaining elementary symmetric function of a four-point spectrum. The lemma shows that increasing this determinant improves Rényi entropy for $1 < q < 2$, reducing each spectral optimisation to a repeated-root boundary.

Lemma 29 (Determinant monotonicity at fixed second and third moments). *For* $1 < q < 2$*, let* λ, λ' *be four-point spectra with the same second moment* p *and third moment* t*. If* $\prod_i \lambda_i < \prod_i \lambda'_i$*, then* $H_q(\lambda) < H_q(\lambda')$*. Consequently, at fixed* p *and* t*, the maximum of* H_q *is attained at a spectrum of one of the two types*

$$(a, b, c, c) \quad \text{or} \quad (a, a, b, c), \qquad a \geq b \geq c \geq 0, \tag{DM}$$

including their degenerations.

Proof. Newton's identities fix $e_1 = 1$, $e_2 = (1-p)/2$ and $e_3 = (1 - 3p + 2t)/6$ (UR-3). Hence, for $s > 0$ and $d = \prod_i \lambda_i$,

$$\prod_i (\lambda_i + s) = M(s) + d, \qquad \sum_i \prod_{j \neq i} (\lambda_j + s) = M'(s), \qquad M(s) = s^4 + s^3 + e_2 s^2 + e_3 s,$$

where M depends only on (p,t) and both quantities are positive. Since $x^2/(x+s) = x - s + s^2/(x+s)$,

$$\sum_i \frac{\lambda_i^2}{\lambda_i + s} = 1 - 4s + \frac{s^2 M'(s)}{M(s) + d},$$

which is strictly decreasing in d for every $s > 0$. For $1 < q < 2$ and $x \geq 0$, the substitution $s = xu$ and $\int_0^\infty u^{q-2}(1+u)^{-1}\,du = \pi/\sin(\pi(q-1))$ give

$$x^q = \frac{\sin(\pi(q-1))}{\pi} \int_0^\infty s^{q-2}\,\frac{x^2}{x+s}\,ds.$$

Each of these integrals converges, so summing over the four eigenvalues gives, for $d < d'$,

$$\sum_i \lambda_i^q - \sum_i \lambda_i'^q = \frac{\sin(\pi(q-1))}{\pi} \int_0^\infty s^q M'(s)\Big(\frac{1}{M(s)+d} - \frac{1}{M(s)+d'}\Big)ds > 0,$$

and $H_q = (1-q)^{-1}\log_2 \sum_i \lambda_i^q$ gives $H_q(\lambda) < H_q(\lambda')$.

For (DM), the ordered spectra $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4 \geq 0$ with trace one and the given p, t form a compact set, so the determinant attains its maximum on it, and by the first part a spectrum of maximal determinant maximises H_q. Suppose it has $\lambda_1 > \lambda_2$ and $\lambda_3 > \lambda_4$. Let $g(z) = \prod_i (z - \lambda_i)$, $m_1 = (\lambda_1 + \lambda_2)/2$ and $m_2 = (\lambda_3 + \lambda_4)/2$. Then $g(m_1) < 0$, $g(m_2) < 0$ and $g(\lambda_2) = g(\lambda_4) = 0$, so for $0 < \delta < \min\{-g(m_1), -g(m_2)\}$ the quartic $g + \delta$ changes sign on each of (λ_4, m_2), (m_2, λ_2), (λ_2, m_1) and (m_1, ∞). Its four roots are therefore distinct and exceed $\lambda_4 \geq 0$. They have the same e_1, e_2, e_3, hence the same p, t, and product $d + \delta$, contradicting maximality. So $\lambda_1 = \lambda_2$ or $\lambda_3 = \lambda_4$, which is (DM). □

S6.3 The classical relaxation: multiplier table

The proof of Theorem 36 of the paper uses the multiplier table below and the supporting lemmas, at the orders $q_s = 1587/1000$ and $q_u = 8/5$ of that theorem. The next subsection combines them; the supporting inequalities follow afterwards.

Multiplier table. The table specifies the supporting function Λ_θ of Lemma 34, with $\theta = (\alpha, \beta, \gamma)$ multiplying the centred cut purity $p - \frac{1}{3}$, the moment slack $f - 4$ and minus the centred squared purity, $-(p^2 - \frac{1}{9})$, respectively. Here $\varepsilon = P - 1$, $F = 17P - 12T - 12$, and h_q is the mean Rényi entropy of M_4. Only two rational multiplier rows are needed: L for the strict band and HT for the cluster sliver, both at $q_s = 1587/1000$. Divide the integer entries by 10000. The HS supporting inequality at $q_u = 8/5$ has its own explicit multiplier (Lemma S6); the cubic bounds use none. The partition in Table S3 specifies where each is used. The L estimate is proved on the slightly larger rectangle $[0, 1/9] \times [3/20, 11/25]$ in (ε, F); the HT estimate uses $\varepsilon \leq 1/100$ and distinguishes one cut of purity above $47/100$ from two of purity at most $29/100$.

Band	10000α	10000β	10000γ
L	39700	1500	15900
HT	35425	1872	11235

Table S2: Exact supporting multipliers at the lower anchor q_s; divide by 10000. L supplies a strict common bound, while HT uses separate bounds for the distinguished cut and the two near-uniform cuts.

S6.4 Proof of Theorem 36

Theorem 36 (Two Rényi anchors). *Let $q_s = 1587/1000$ and $q_u = 8/5$. Every triple satisfying $P \geq 1$, $F \geq 0$ and (QK) obeys one of the following.*

1. *If* $F \geq 3/20$ *or* $20\varepsilon + 2F \geq 1$, *then* $\mathcal{E}_{q_s} \leq 5/3$, *with equality only at a permutation of* (c_1, c_2, c_2).

2. *Otherwise* $\mathcal{E}_{q_u} \leq h_{q_u}$, *with equality only when all three spectra are HS spectra.*

Proof. The following partition is exhaustive and disjoint. In the last four rows, $20\varepsilon + 2F < 1$; in the last two, $F > 11/25$.

Region (first applicable row)	Order	Bound used
$20\varepsilon + 2F \geq 1$	q_s	Cubic envelope (BC)
$F < 3/20$	q_u	HS support (HD)
$3/20 \leq F \leq 11/25$	q_s	Strict band L (LT, SHD, LHD)
$\max_j p_j \leq 47/100$	q_s	Capped cubic bound (CG)
$\max_j p_j > 47/100$	q_s	Cluster sliver (NU, SL)

Table S3: The anchor partition. Rows are applied in order: each triple is handled by the first row whose region contains it, so every row excludes the regions above it. HD gives HS equality; BC gives cluster equality; the other three regions are strict. SHD and CG both use the monotonicity of A_q in Lemma S8, proved by Cauchy's formula; SHD adds the quartic checks SHD-3.

Cubic envelope. Corollary 33, with $q_s > q_b$, handles the first row and classifies its equality triples. Henceforth $\varepsilon < 1/20$, $F < 1/2$, and

$$p_j \leq P - 1/2 < 11/20 < 9/16 < 49/75.$$

These are the purity caps required by HD and L; they are consequences of the region, not hypotheses on physical states.

HS zone. For $F < 3/20$, Lemma S6 gives $\Lambda_\theta \leq h_{q_u}$, strictly off HS, and $\ell_\theta \geq 0$ on $[0, 1/9] \times [0, 3/20]$, which contains (ε, F). The dual lemma therefore gives $\mathcal{E}_{q_u} \leq h_{q_u}$, with equality only at three HS spectra.

Strict band. For $3/20 \leq F \leq 11/25$, Lemma S7 gives $\Lambda_\theta < W - 10^{-5}$, where $3W = 5 + \min \ell_\theta$ on the L rectangle. Thus

$$3\mathcal{E}_{q_s} \leq \sum_j \Lambda_\theta - \ell_\theta < 5 + \min \ell_\theta - \ell_\theta - 3 \cdot 10^{-5} < 5.$$

Near the cluster triple. The remaining region has $F > 11/25$ and $\varepsilon < (1 - 2F)/20 < 3/500 < 1/100$. If every cut has $p_j \leq 47/100$, Corollary S13 gives the strict ceiling. Otherwise label a cut with $p_2 > 47/100$. It is unique, since two such cuts would give $P > 47/50 + 1/4 > 101/100$. The other two have $p_j \leq P - p_2 - 1/4 < 29/100$. For this last case only, use determinant replacement (Corollary 30) to put each cut on an ordered double-root chart, preserving every p_j, t_j and the constraints.

Cluster sliver. Set $X = (p_3 - 1/4) + (p_4 - 1/4) = 1/2 + \varepsilon - p_2 \leq 1/25$. Writing $x_j = p_j - 1/4$, Lemma 35 for cuts $3, 4$ gives

$$F \in I = [F_c - \kappa_a X, F_c + \kappa_a X], \qquad F_c = -1 + 8\varepsilon + 9p_2 - 12t_2, \qquad \kappa_a = 4\sqrt{3}/5,$$

since the error is at most $4\sqrt{3}(x_3^{3/2} + x_4^{3/2}) \leq 4\sqrt{3}X^{3/2} \leq \kappa_a X$. The actual F lies in $I \cap (11/25, 1/2)$, exactly the window needed by SL. Use the HT row $\theta_s = (3.5425, 0.1872, 1.1235)$ and put

$$\Lambda_i^\star = \Lambda_{\theta_s}(c_i), \qquad m = \min_{f \in \partial I} \ell_{\theta_s}(\varepsilon, f), \qquad \ell_0 = \ell_{\theta_s}(0, 1/2).$$

NU bounds each of the two near-uniform cuts by $\Lambda_2^\star$. SL bounds the distinguished cut by $\Lambda_1^\star + m - \ell_0$. Concavity of ℓ_{θ_s} in F gives $\ell_{\theta_s}(\varepsilon, F) \geq m$; hence Lemma 34 and the contact identity (DC) give

$$3\mathcal{E}_{q_s} \leq \Lambda_1^\star + 2\Lambda_2^\star - \ell_0 + m - \ell_{\theta_s}(\varepsilon, F) \leq \Lambda_1^\star + 2\Lambda_2^\star - \ell_0 = 5.$$

Here NU and SL are Lemmas S16 and S17.

Equality. HD forces three HS spectra. The strict band and capped region are strict. In the sliver, equality in SL forces the distinguished cut to be c_1 and $\varepsilon = 0$. Then $X = 0$, so both other cuts are c_2, giving $F = 1/2$, excluded in this region. Thus the replaced triple, and hence the original triple, has a strict bound there. Only the HS equality of HD and the cluster equality of the cubic envelope remain. □

S6.5 A monotone path from the repeated-largest chart

Lemma S4 (Low third-moment tilt; analytic)**.** *Let $3/2 \le q \le 8/5$ and $0 \le \omega \le 9/5$. Among spectra (a, a, b, c) with $a \ge b \ge c \ge 0$ of a fixed purity, $H_q - \omega t$ is maximised only by two equal pairs. Consequently, whenever $0 \le \beta \le 3/20$, the maximum at fixed purity of the supporting function Λ_θ of Lemma 34 is attained on the single pattern (a, b, c, c).*

Proof. The path. Write a repeated-largest spectrum of purity p as

$$(1/4 + u,\ 1/4 + u,\ 1/4 - u + d,\ 1/4 - u - d), \qquad x = p - 1/4 = 4u^2 + 2d^2, \quad 0 \le d \le 2u,$$

the last condition being the ordering. Increase u to $u_* = \sqrt{x}/2$ at fixed x, with $d = \sqrt{(x - 4u^2)/2}$. Then $d' = -2u/d$ and $d \le 2u$, so the middle entry $1/4 - u + d$ decreases, stays at most $1/4 + u$, and the smallest entry has derivative $2u/d - 1 \ge 0$: it never falls below its initial value $c \ge 0$. The endpoint is two equal pairs. Along the path

$$t = \frac{1}{16} + \frac{3}{4}x + 12u^3 - 3xu, \qquad \frac{dt}{du} = 36u^2 - 3x = 6(4u^2 - d^2) \ge 0,$$

strictly in the interior unless the path is constant.

The slope. Put $r = q - 1 \in [1/2, 3/5]$, $L = \ln 2$ and $Z = \sum_i \lambda_i^q$, so that $H_q = \log_2 Z/(1 - q)$. The path preserves trace and purity and moves the three distinct entries a, b, c; interpolating x^r quadratically at them gives $dZ = \frac{q}{3}[a, b, c]x^r\, dt$, hence

$$\frac{dH_q}{dt} = \eta(a, b, c) := -\frac{q[a, b, c]x^r}{3rLZ}.$$

For distinct $a, b, c \ge 0$ with mean m and $0 < r < 1$,

$$-[a, b, c]x^r \ge \frac{r(1 - r)}{2}\, m^{r-2}. \tag{DD}$$

Indeed, let T be the cubic Taylor polynomial of x^r at m. Then $(x^r - T)'' = -r(1-r)\{x^{r-2} - m^{r-2} - (r-2)m^{r-3}(x-m)\} \le 0$ by the tangent inequality for the convex function x^{r-2}, so $x^r - T$ is concave on $[0, \infty)$ and $[a, b, c](x^r - T) \le 0$. As $[a, b, c](x-m)^2 = 1$ and $[a, b, c](x-m)^3 = a+b+c-3m = 0$, $[a, b, c]T = -\frac{1}{2}r(1 - r)m^{r-2}$. Here $m = (1 - a)/3$. With $Z \le a^r \sum_i \lambda_i = a^r$ this gives

$$\eta(a, b, c) \ge \frac{q(1 - r)3^{2-r}}{6L\, a^r(1 - a)^{2-r}}, \qquad a^r(1 - a)^{2-r} = [a(1 - a)]^r(1 - a)^{2-2r} \le \frac{9}{16}\left(\frac{4}{9}\right)^r \le \frac{3}{8},$$

using $1/4 \le a \le 1/2$ and $r \ge 1/2$. With $q(1 - r) = q(2 - q) \ge 16/25$, $L < 7/10$ and $3^{2-r} \ge 3^{7/5} > 9/2$ (as $3^7 > (9/2)^5$),

$$\eta(a, b, c) > \frac{64}{35} > \frac{9}{5} \ge \omega. \tag{LT}$$

Thus $d(H_q - \omega t)/du = (\eta - \omega)\, dt/du > 0$ on every nonconstant path, and continuity covers zero or coalescing entries. This proves the first claim.

For the second, Lemma 29 reduces a fixed-purity maximum of Λ_θ to the patterns (a, b, c, c) and (a, a, b, c). At fixed purity the variable part of Λ_θ is $H_q - 12\beta t$ with $12\beta \le 9/5$, and two equal pairs already belong to the first pattern. □

S6.6 An analytic HS supporting inequality

At the upper anchor $q_u = 8/5$, choosing $\beta = 1/12$ makes the fixed-purity optimisation a bound on $H_{8/5} - t$. We first show that its maximiser has one large eigenvalue, then prove the resulting one-variable supporting inequality. The outer dual is nonnegative for $F \le 3/20$, which determines the split between the two anchor regions. Throughout this subsection put $q = 8/5$, $r = q-1 = 3/5$ and $L = \ln 2$.

Lemma S5 (A third-moment tilt at fixed purity; proved)**.** *Among four-point probability vectors of a fixed purity $p \le 9/16$, $H_{8/5}(\lambda) - \sum_i \lambda_i^3$ is maximised only by the one-large spectrum*

$$\lambda(s) = ((1+3s)/4, (1-s)/4, (1-s)/4, (1-s)/4), \qquad p = (1+3s^2)/4,$$

up to permutation (uniform when $p = 1/4$).

Proof. Let $Z = \sum_i \lambda_i^q$ and let a, b, c be three distinct entries. Their mean is at most $1/3$, so the divided-difference bound (DD) gives $-[a,b,c]x^r \ge \frac{1}{2} r(1-r)3^{2-r}$. Also $Z = \sum_i \lambda_i \lambda_i^r \le p^r$ by concavity of x^r. Consequently

$$\eta(a,b,c) := -\frac{q[a,b,c]x^r}{3rLZ} \ge \frac{8}{75L}\frac{3^{7/5}}{(9/16)^{3/5}} > \frac{16}{105}(3\cdot 2^{12})^{1/5} > 1. \tag{HTS}$$

The last comparisons use $L < 7/10$ and $105^5 = 12762815625 < 3 \cdot 2^{32} = 12884901888$.

Lemma S4, with $\omega = 1$, reduces the maximum to the pattern (a, b, c, c). Write it as

$$(1/4 + u + d, 1/4 + u - d, 1/4 - u, 1/4 - u), \qquad x = p - 1/4 = 4u^2 + 2d^2, \quad 0 \le d \le 2u.$$

Decrease u to $u_0 = \sqrt{x/12}$ at fixed x, setting $d = \sqrt{(x - 4u^2)/2}$. Ordering and positivity persist, and $b = c$ at the endpoint: it is the one-large spectrum. Along this path

$$t = \frac{1}{16} + \frac{3}{4}x + 3xu - 12u^3, \qquad \frac{dt}{du} = 3x - 36u^2 \le 0.$$

As in the preceding chart reduction, $dH_q/dt = \eta(a,b,c) > 1$. Thus decreasing u strictly increases $H_q - t$ unless the spectrum is already one-large. Continuity handles zero or coalescing entries. At equality the strict paths force every determinant-maximal replacement to be one-large. That replacement preserves p, t, so equality in the sharp third-moment bound (Lemma 35) forces the original spectrum to be the same. The uniform case $p = 1/4$ is immediate. □

Lemma S6 (HS supporting inequality; proved)**.** *At $q = q_u = 8/5$ put $u = 3^{3/5}$ and*

$$k = \frac{16}{3L}\frac{u-1}{u+1}, \qquad \beta = \frac{1}{12}, \qquad \gamma = \frac{10}{3}, \qquad \alpha = \frac{3}{2}k + \frac{65}{36}.$$

Every four-point probability vector with $p \le 9/16$ satisfies $\Lambda_{(\alpha,\beta,\gamma)}(\lambda) \le h_{8/5}$, with equality only at HS. For $0 \le \varepsilon \le 1/9$ and $0 \le F \le 3/20$,

$$\ell_{(\alpha,\beta,\gamma)}(\varepsilon, F) \ge \frac{149}{324}\varepsilon + F\left(\frac{1}{12} - \frac{5}{9}F\right) \ge 0.$$

Proof. At fixed p, the only variable part of Λ is $H_q - t$. Lemma S5 reduces its maximum to $\lambda(s)$. Write $f(p) = H_q(\lambda(s))$, $p_* = 1/3$, and

$$A(s) = \frac{9}{2}(s - \tfrac{1}{3})^2(s + \tfrac{1}{6}), \qquad W(p) = f(p) - \beta A(s) - \gamma(p - p_*)^2.$$

Direct moment evaluation gives

$$\Lambda(\lambda(s)) = W(p) + \frac{3}{2}k(p - p_*), \qquad W(p_*) = h_q, \qquad W'(p_*) = -\frac{3}{2}k, \qquad \frac{d^2A}{dp^2} = \frac{6}{s}.$$

We show that W is first convex and then concave, allowing either interval to be empty. The needed scalar calculation is explicit below. Put

$$x = \left(\frac{1+3s}{1-s}\right)^{1/5}, \quad J = \frac{Lf''(p)}{2\gamma + 6\beta/s}, \quad h_4(x) = 1 + x + x^2 + x^3 + x^4.$$

Since $p \le 9/16$, $s < 2/3$, so $1 < x < 2$ and $0 < x^5 - 1 < 8$. Differentiating

$$L\frac{dH_q(\lambda(s))}{ds} = -\frac{2(x^5+3)(x^3-1)}{x^8+3}$$

gives the identity

$$\frac{dJ}{dx} = -\frac{8(x^5+3)^3 Q(x)}{5x^3(43x^5-31)^2(x^8+3)^3 h_4(x)^3}. \tag{HCJ}$$

Here, with $y = x - 1$ and $R = x^5 - 1$, $Q(x) = \sum_{j=0}^{7} R^j P_j(y)$, where the five columns below are the coefficients of $1, y, y^2, y^3, y^4$ in P_j.

j	1	y	y^2	y^3	y^4
0	225600	457600	471040	240000	48640
1	582160	1108000	1118304	557584	110864
2	690608	1195520	1018512	418768	68128
3	756896	1304520	1151162	522393	96703
4	516296	613195	462027	220948	46208
5	181031	120245	477	−23717	−6797
6	−259	2795	1978	774	129
7	43	0	0	0	0

This displayed identity proves the sign without a parameter certificate: $P_0, \dots, P_4, P_7$ have positive coefficients, and, for $0 \le y < 1$,

$$P_5(y) \ge 181031 - 23717 - 6797 = 150517, \qquad P_6(y) \ge -259.$$

Thus $P_5 + RP_6 \ge 150517 - 8 \cdot 259 > 0$. Hence $Q > 0$ and J strictly decreases. As $W'' = (2\gamma + 6\beta/s)(J/L - 1)$, the asserted curvature pattern follows.

To turn this curvature pattern into a supporting line, subtract the tangent at p_* and write $D(p) = W(p) - W(p_*) - W'(p_*)(p - p_*)$. Then $D(p_*) = D'(p_*) = 0$, and D' first increases and then decreases. If $D(1/4) < 0$, the contact p_* must lie in the strictly concave portion: otherwise $D' < 0$ throughout its left side, which would give $D(1/4) > 0$. In the concave portion, the tangent strictly dominates to the right of p_*; to its left, D has no maximum except at $1/4$ or p_*. Thus $D < 0$ away from p_*. In the present case the left-endpoint gap is strictly positive, so $D(1/4) < 0$:

$$\Delta = -D(1/4) = h_q + \frac{k}{8} - 2 + \frac{\beta}{12} + \frac{\gamma}{144} = h_q + \frac{k}{8} - 2 + \frac{13}{432} > \frac{1}{5000}. \tag{HUC}$$

For completeness, this small constant has an elementary rational proof. Let $u_0 = 19331/10000$, $v = 1/(2u_0 + 1)$ and

$$L_- = 2\sum_{j=0}^{5} \frac{1}{(2j+1)3^{2j+1}}, \qquad B_+ = 2\sum_{j=0}^{4} \frac{v^{2j+1}}{2j+1} + \frac{2v^{11}}{11(1-v^2)}.$$

The formula $h_q = 8/3 - (5/(3L))\ln(1 + 1/u)$ and the atanh series give $L > L_-$ and $\ln(1 + 1/u_0) < B_+$. Since $u_0^5 < 27$, $u > u_0$; the function $-5\ln(1 + 1/u) + 2(u-1)/(u+1)$ increases with $u > 0$. The purely rational comparison

$$\frac{301}{144}L_- - 5B_+ + \frac{2(u_0 - 1)}{u_0 + 1} > \frac{21}{50000}$$

therefore gives $3L\Delta > 21/50000 > 3L/5000$, using $L < 7/10$. This proves (HUC). The tangent bound is strict away from p_* by the strict curvature calculation. Thus $\Lambda \le h_q$, with equality only at HS.

Finally $u > 19/10$ and $L < 7/10$ give $k > 480/203 > 7/3$. Substitution into the outer dual yields

$$\ell = \left(\alpha - \gamma(\tfrac{17}{12} + \tfrac{\varepsilon}{3})\right)\varepsilon + F\left(\tfrac{1}{12} - \tfrac{5}{9}F\right) \ge \frac{149}{324}\varepsilon + F\left(\tfrac{1}{12} - \tfrac{5}{9}F\right).$$

The last term is nonnegative for $F \le 3/20$, as required. All global maximisation and sign arguments above are analytic; HD checks the displayed identities and rational comparisons. □

S6.7 The strict supporting band

The bounded-head and capped-cubic proofs below use the same near-uniform moment inequality. Readers following dependencies rather than the case partition can first consult Lemma S14 and the moment-rectangle argument in Section S6.11, then return to these two applications.

Lemma S7 (Strict dual band; analytic with exact polynomial sign certificates)**.** *For band L of Table S2, set $W = 5/3 + \min \ell_\theta/3$, with the minimum over its rectangle's corners. At $q_s = 1587/1000$, every four-point spectrum with $p \le 49/75$ satisfies $\Lambda_\theta < W - 10^{-5}$.*

Proof. At fixed purity the variable part is $H_q - (9/5)t$. Lemma S4 reduces the maximum to (a, b, c, c). Lemma S9 handles $a \le 1/2$ and Lemma S11 handles $1/2 \le a \le 4/5$. These cover the domain: $p \ge a^2 + (1-a)^2/3$ and its value $49/75$ at $a = 4/5$ force $a \le 4/5$. The corner minimum of the outer dual is $1837/125000$, so $W - 10^{-5} = 2507333/1500000$ as required. □

Lemma S8 (Binary quartic correction; analytic)**.** *Let $1 < q < 2$. Put $v = (1-2u)^2$,*

$$m_q(v) = \frac{(1+\sqrt{v})^q + (1-\sqrt{v})^q}{2}, \qquad D_q(v) = \frac{\ln m_q(v)}{(q-1)\ln 2} = 1 - H_q(u, 1-u).$$

Then $D_q'''(v) > 0$ on $0 < v < 1$, and

$$A_q(u) = \frac{4u(1-u) - H_q(u, 1-u)}{u(1-u)(1-2u)^2}$$

is strictly increasing on $0 < u < 1/2$. Consequently, a quartic bound $H_q(u, 1-u) \le 4u(1-u) - au(1-u)(1-2u)^2$ at $u = u_0$ holds throughout $u_0 \le u \le 1/2$.

Proof. Write $a = \tan^2(\frac{\pi}{2q})$; since $\pi/4 < \pi/(2q) < \pi/2$, $a > 1$. Put $f = \ln m_q$. The proof is Cauchy's formula for $\ln m_q$ on the slit plane, after removing the one zero of m_q.

Analyticity. Let $g(w) = \frac{1}{2}((1+w)^q + (1-w)^q)$ with principal powers. It is holomorphic on $\mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty))$ and even, so $m_q(v) = g(\sqrt{v})$ does not depend on the sign of the root and is holomorphic on $\Omega = \mathbb{C} \setminus [1, \infty)$; at $v = 0$ it is the convergent series $\sum_k \binom{q}{2k} v^k$. It is real on the real axis and $m_q(\bar{v}) = \overline{m_q(v)}$. For $\operatorname{Im} v \ge 0$ take $w = \sqrt{v}$ in the closed first quadrant. Then $\arg(1-w) = \arg(w-1) - \pi$, so

$$m_q(v) = \tfrac{1}{2}\left[(1+w)^q + e^{-i\pi q}(w-1)^q\right], \qquad \arg(w-1) \in [0, \pi]. \tag{S6.2}$$

The right side is continuous on the closed first quadrant, so m_q extends continuously to the closed upper half-plane, and to the closed lower half-plane by conjugation.

Zeros. Let $m_q(v) = 0$ and $\operatorname{Im} v \ge 0$, $w = \sqrt{v}$. Then $|1+w|^q = |1-w|^q$, so $\operatorname{Re} w = 0$ and $v = -y^2$ is real, $w = iy$, $y \ge 0$. Now $(1 \pm iy)^q = (1+y^2)^{q/2} e^{\pm iq \arctan y}$, so $m_q(-y^2) = (1+y^2)^{q/2} \cos(q \arctan y)$. As $0 < q \arctan y < q\pi/2 < \pi$ for $y > 0$, this vanishes only at $q \arctan y = \pi/2$, that is $v = -a$. (For $v \in [0, 1)$, $m_q > 0$.) The zero is simple: $g'(w) = \frac{q}{2}((1+w)^{q-1} - (1-w)^{q-1})$, and at $w = iy_0$, $y_0 = \sqrt{a}$,

$$g'(iy_0) = iq(1+y_0^2)^{(q-1)/2} \sin((q-1)\arctan y_0) \ne 0,$$

since $0 < (q-1)\arctan y_0 < \pi/2$; and $m_q'(v) = g'(w)/(2w)$. Hence $h(v) = m_q(v)/(1+v/a)$ is holomorphic and zero-free on Ω, and continuous and zero-free on the closed upper half-plane away from the cut; on the cut see the next step.

Boundary values on the cut. For $t > 1$, (S6.2) gives $2m_q(t+i0) = (1+\sqrt{t})^q + e^{-i\pi q}(\sqrt{t}-1)^q$ with $\operatorname{Im} 2m_q(t+i0) = -\sin(\pi q)(\sqrt{t}-1)^q > 0$, because $\pi < \pi q < 2\pi$. So $m_q(t+i0) \neq 0$ and

$$\theta_q(t) = \arg\bigl(2m_q(t+i0)\bigr) \in (0,\pi)$$

is continuous on $[1,\infty)$, with $\theta_q(t) \to 0$ as $t \downarrow 1$, since $m_q(1) = 2^{q-1} > 0$.

The logarithm. Ω is simply connected and h is zero-free on it with $h > 0$ on $(-\infty, 1)$ (there m_q and $1 + v/a$ change sign together at $-a$). So $F = \ln h$ with $F(0) = 0$ is holomorphic on Ω, real on $(-\infty,1)$, and $F(\bar v) = \overline{F(v)}$. The closed upper half-plane is simply connected and h is continuous and zero-free on it, so F extends continuously to it. On the cut, $\operatorname{Im} F(t+i0)$ is continuous and equals $\theta_q(t)$ modulo 2π (as $1 + t/a > 0$). Both vanish at $t = 1$, so $\operatorname{Im} F(t+i0) = \theta_q(t)$ for all $t \geq 1$ and $F(t+i0) - F(t-i0) = 2i\theta_q(t)$. In particular F is bounded near $v = 1$.

Growth. For $|w| > 1$ in the first quadrant, $(1+w)^q = w^q(1+1/w)^q$ and $(w-1)^q = w^q(1-1/w)^q$ (the arguments add without wrapping), so by (S6.2)

$$m_q(v) = w^q\bigl(c + \varepsilon(1/w)\bigr), \qquad c = \tfrac{1}{2}\bigl(1 + e^{-i\pi q}\bigr) \neq 0,$$

with ε holomorphic on $|z| < 1$ and $\varepsilon(0) = 0$; $c \neq 0$ because q is not an odd integer. Hence $F(v) - \bigl(\frac{q}{2}\operatorname{Log} v - \operatorname{Log}(1+v/a) + \operatorname{Log} c\bigr)$ has exponential tending to 1 as $|v| \to \infty$ in the closed half-plane, which is connected, so it tends to a constant in $2\pi i\mathbb{Z}$. Thus $|F(v)| \leq C \ln|v|$ for large $|v|$, in both half-planes.

The integral formula. Fix $0 < v < 1$. By Cauchy's formula for F''' on the region $\{|\zeta| < R\}$ with the strip $\{\operatorname{Re}\zeta \geq 1,\ |\operatorname{Im}\zeta| \leq \epsilon\}$ removed,

$$F'''(v) = \frac{3!}{2\pi i}\oint \frac{F(\zeta)}{(\zeta - v)^4}\,d\zeta.$$

Let $\epsilon \downarrow 0$ and then $R \to \infty$. F is bounded near 1, so the short segment near the tip $\zeta = 1$ contributes $O(\epsilon)$; the arc contributes $O(R^{-3}\ln R)$; and by the jump of F across the cut the two edges contribute $2i\int_1^R \theta_q(t)(t-v)^{-4}\,dt$. Therefore

$$F'''(v) = \frac{6}{\pi}\int_1^\infty \frac{\theta_q(t)}{(t-v)^4}\,dt > 0.$$

Since $m_q = (1+v/a)e^F$ and $(\ln(1+v/a))''' = 2/(a+v)^3$,

$$f'''(v) = (\ln m_q)'''(v) = \frac{2}{(a+v)^3} + \frac{6}{\pi}\int_1^\infty \frac{\theta_q(t)}{(t-v)^4}\,dt > 0 \qquad (-a < v < 1),$$

which is $D_q''' > 0$ on $0 < v < 1$.

Monotonicity of A_q. $D_q(0) = 0$ and $D_q(1) = 1$, so

$$\frac{A_q(u)}{4} = \frac{D_q(v) - v}{v(1-v)} = -[0,v,1]D_q.$$

For $0 < v < v' < 1$, $[0,v,1]D_q - [0,v',1]D_q = (v-v')[0,v,v',1]D_q$, and by the mean value theorem for divided differences (Rolle's theorem; D_q is continuous on $[0,1]$) the last factor is $D_q'''(\xi)/6 > 0$. So A_q decreases in $v = (1-2u)^2$: it is strictly increasing in u. □

S6.8 The bounded-head chart

Lemma S9 (Supporting bound on the bounded-head chart; analytic; exact rational checks)**.** *At $q = 1587/1000$ and the multiplier $\theta = (397/100, 3/20, 159/100)$, every ordered spectrum (a, b, c, c) with $a \le 1/2$ satisfies*

$$\Lambda_\theta \le \frac{8023}{4800} < \frac{5}{3} + \frac{1837}{375000} - 10^{-5}.$$

So on this chart the band-L dual bound needs only rational arithmetic: rational enclosures of eight node values (SHD-1) for the binary quartic correction of Lemma S8 and eight one-variable quartic sign checks (SHD-3).

Proof. Put $r = q - 1$, $L = \ln 2$, $h(u) = H_q(u, 1-u)$, $d = u(1-u)$, and write

$$\lambda = \Big(\frac{1-u+\delta}{2}, \frac{1-u-\delta}{2}, \frac{u}{2}, \frac{u}{2}\Big), \qquad v = \delta^2,$$

a head pair of total weight $1-u$ and a uniform tail pair of weight u. The head cap and ordering give $0 \le u \le 1/2$ and $0 \le \delta \le \min\{u, 1-2u\}$. We treat $u \le 1/4$ by a head-deficit estimate and $u \ge 1/4$ by the near-uniform moment bound.

Small tail, $0 \le u \le 1/4$. For the conditional vectors of the two pairs, Jensen's inequality for the logarithm (as in the proof of (CG3)) gives

$$H_q(\lambda) \le h(u) + (1-w)\big(1 - D_q(z)\big) + w = 1 + h(u) - (1-w)D_q(z), \qquad w = \frac{u^q}{(1-u)^q + u^q},$$

where $z = v/(1-u)^2 \le u^2/(1-u)^2 \le 1/9$ is the squared imbalance of the head pair and D_q is the function of Lemma S8. That lemma shows that $(D_q(z) - z)/(z(1-z)) = A_q/4$ decreases in $z \in (0,1)$. Put $\kappa = \frac{81}{8}\big(D_q(1/9) - \frac{1}{9}\big)$, its value at $z = 1/9$. Here $D_q(1/9) = 1 - H_q(1/3, 2/3) \ge 249/2000 > 1/9$, since $H_q(1/3, 2/3) < 1751/2000$ (power sums and the atanh series, as for the nodes below), so $\kappa > 0$. For $0 < z \le 1/9$,

$$D_q(z) \ge z + \kappa z(1-z) \ge z\big(1 + \tfrac{8}{9}\kappa\big) = 9D_q(1/9)\, z \ge \frac{2241}{2000} z,$$

since $9 \cdot 249/2000 = 2241/2000$.

Next $1 - w = (1-u)^2 k_q(u)$ with

$$k_q(u) = \frac{(1-u)^{q-2}}{(1-u)^q + u^q} = \frac{(1-u)^{-2}}{1 + (u/(1-u))^q},$$

which increases with q because $u \le 1-u$; hence $k_q \ge k_{3/2}$. With $t = \sqrt{u/(1-u)}$,

$$\frac{k_{3/2}(u)}{1+2u} = \Big[1 + \Big(\frac{t(1-t)}{1+t^2}\Big)^3\Big]^{-1} \ge \frac{64}{65},$$

since $0 \le t(1-t) \le 1/4$ and $1 + t^2 \ge 1$, so the cube is at most $1/64$. Combining, $(1-w)D_q(z) \ge \frac{2241}{2000} \cdot \frac{64}{65}(1+2u)v = \frac{8964}{8125}(1+2u)v$, so

$$H_q(\lambda) \le 1 + h(u) - \frac{11}{10}(1+2u)v. \tag{SHD}$$

The binary entropy is controlled by the quartic correction of Lemma S8: $h(u) = 4d - A_q(u)\, d(1-4d)$ with $1 - 4d = (1-2u)^2 \ge 0$, and A_q is increasing. Take the nodes and values

u_i	0	$\frac{1}{100}$	$\frac{3}{100}$	$\frac{1}{20}$	$\frac{2}{25}$	$\frac{13}{100}$	$\frac{39}{200}$	$\frac{47}{200}$
a_i	$\frac{99}{1000}$	$\frac{217}{1000}$	$\frac{289}{1000}$	$\frac{83}{250}$	$\frac{377}{1000}$	$\frac{43}{100}$	$\frac{12}{25}$	$\frac{101}{200}$

and $u_8 = 1/4$. They satisfy $A_q(u_i) \ge a_i$: at $u_0 = 0$ in the sense $A_q(0^+) = 4 - q/(rL) > 99/1000$, and at the other seven points by interval enclosures of the power sum $u^q + (1-u)^q$ and of the

logarithm (atanh series) [exact certificate; SHD-1]. Since A_q is increasing, $A_q \ge a_i$ on $[u_i, u_{i+1}]$, hence $h(u) \le 4d - a_i d(1-4d)$ there.

Substituting (SHD) in the definition of Λ_θ gives

$$\Lambda_\theta \le \frac{1999}{1200} + h(u) - \frac{179}{50}d - \frac{159}{100}d^2 + \left(\frac{3}{200} + \frac{37}{50}u - \frac{159}{100}u^2\right)v - \frac{159}{400}v^2.$$

The coefficient of v is positive on $[0, 1/4]$. Using $v \le u^2$, dropping the last term and inserting the bound for h leaves, on $[u_i, u_{i+1}]$, the quartic

$$P_i(u) = \frac{1999}{1200} + \frac{21}{50}d - a_i\, d(1-4d) - \frac{159}{100}d^2 + \left(\frac{3}{200} + \frac{37}{50}u - \frac{159}{100}u^2\right)u^2.$$

It remains to show $P_i \le 8023/4800$ on $[u_i, u_{i+1}]$. Each $8023/4800 - P_i$ is a quartic in u; written in the Bernstein basis of its interval, all five coefficients are positive [exact-rational certificate; SHD-3]. The smallest coefficient is $1/6400$, at $u = 1/4$.

Large tail, $1/4 \le u \le 1/2$. Here we show that Λ_θ does not exceed its uniform value. The coordinate bounds give $p \le 11/32$, by maximising a convex quadratic separately on $[1/4, 1/3]$ and $[1/3, 1/2]$. Put $x = p - 1/4$, $y = \sum(\lambda_i - 1/4)^3$. Here $y \ge 0$, so $v = y/x$ lies in $[0, 9/50]$, and $x \le X = 3/32$. With $b = 12\beta = 9/5$ and $c = \alpha + 8\beta - \gamma/2 = 35/8$, the identity (UD), valid for every multiplier, gives $\Lambda_\theta((1/4)^4) - \Lambda_\theta(\lambda) = 2 - H_q - cx + by + \gamma x^2$. We apply Lemma S15 with the SHD column of Table S4. On the stated rectangle, $17/10 < V < 19/10$, $-3/50 < W < 3/8$, $1 \le N < 6/5$ and $|B + 2Cv + XD_5| < 19/10$, with A, B, C, D_5 those of Lemma S14. The bound $V^2 - 2W > 8/3$ from the capped-cubic proof gives $Q = 8/3$, and $8/3 - 2X(19/10)(3/50) > (7/4)(6/5)^2$ gives $m = 7/4 > 2\rho\gamma$; $C > 7/5$ gives $\Phi_{vv}(X, v) \ge (2C - X(19/10)^2)/(\rho(6/5)^2) > 4 = \kappa$. At the single point $v_* = 27/200$,

$$\Phi(X, v_*) > 1/800, \qquad |\Phi_v(X, v_*)| < 1/100,$$

using the eight-term alternating lower sum for $\ln(1 + XV + X^2W)$ and $693147/10^6 < L < 693148/10^6$. The lemma gives a dual difference greater than $(1/800 - 1/80000)x > 0$. Thus $\Lambda_\theta \le \Lambda_\theta((1/4)^4) = 8023/4800$. The uniform case $x = 0$ is immediate.

The small-tail and large-tail cases together prove the stated bound. □

S6.9 The large-head chart by two Hermite bounds

Throughout this subsection $q = 1587/1000$, $r = q - 1$, $L = \ln 2$ and $\theta = (397/100, 3/20, 159/100)$. Put

$$f(a, z) = \Lambda_\theta\left(a, \frac{(1-a)(1+2z)}{3}, \frac{(1-a)(1-z)}{3}, \frac{(1-a)(1-z)}{3}\right),$$

$$B = \frac{1+2z^2}{3}, \qquad C = \frac{1+6z^2+2z^3}{9}, \qquad K = \frac{(1+2z)^q + 2(1-z)^q}{3^q}.$$

Here $1/2 \le a \le 4/5$, $0 \le z \le 1$, and the purity and third moment are $p = a^2 + B(1-a)^2$, $t = a^3 + C(1-a)^3$. The objective is $f < T = 2507333/1500000$. The first lemma supplies the shape condition $f_{aaa} \le 0$. The second then bounds f by a quadratic in a, replaces $K(z)$ by one cubic minorant, and proves positivity of that quadratic's gap by completing the square. Thus the two-dimensional chart reduces to univariate polynomial signs.

Lemma S10 (Largest-entry curvature; exact polynomial sign certificates)**.** *On this whole chart,* $f_{aaa} \le 0$.

Proof. Write $h(a, K) = -\ln(a^q + K(1-a)^q)/(rL)$. The remaining part of f is a polynomial in a whose only fourth derivative comes from $-\gamma p^2$, so $f_{aaaa} = h_{aaaa} - 24\gamma(1+B)^2$.

The shape of h_{aaaa} *in* K. Set $x = 2a - 1 \in [0, 3/5]$, $\rho = (1-a)/a = (1-x)/(1+x) \in [1/4, 1]$, $w = K\rho^q$ and

$$y = \frac{a^q - K(1-a)^q}{a^q + K(1-a)^q} = \frac{1-w}{1+w}.$$

Apart from the positive factor $16q/(rL(1-x^2)^4)$, h_{aaaa} is

$$J(y) = 6q^3y^4 - 24q^2xy^3 + (-8q^3 + 36qx^2 + 8q)y^2 \\ + (24q^2x - 24x^3 - 24x)y + 2q^3 - 36qx^2 - 8q + 6x^4 + 36x^2 + 6.$$

On $0 \le x \le 3/5$,

$$J'(y)\big|_{y=x} = 8x(q-1)\{3(q-1)^2x^2 - 2q^2 + q + 3\} \le 0, \qquad J'''(y) = 144q^2(qy - x) \ge 0 \quad (y \ge x).$$

So on $[x, 1]$ the derivative J' is convex and starts nonpositive. Hence J decreases and then increases, and has no interior maximum on any subinterval:

$$J(y) \le \max\{J(y_\ell), J(y_h)\} \qquad (x \le y_\ell \le y \le y_h \le 1).$$

Envelopes for y. Write $K = \lambda_1^q + 2\lambda_2^q$ with $\lambda = (\frac{1+2z}{3}, \frac{1-z}{3}, \frac{1-z}{3})$. Convexity of s^q gives $K \ge 3^{-r} > 1/2$, since $3^r < 2$. Concavity of s^r with weights λ_i gives $K = \sum \lambda_i\lambda_i^r \le B^r \le 1$. If $B \le 1/2$, then $K \le 2^{-r} < 2/3$, because $2^q > 3$. For $0 < \rho \le 1$ we have $\rho^2 \le \rho^q \le \rho^{3/2} \le \rho(1+\rho)/2$ and $\rho^q \le \rho$. Hence $w \ge \rho^2/2$, $w \le \rho(1+\rho)/2 = (1-x)/(1+x)^2$ always, and $w \le 2\rho/3$ when $B \le 1/2$. Since $y = (1-w)/(1+w)$ decreases in w, we get $y_\ell \le y \le y_h$ with

$$y_h = \frac{(1+x)^2 - (1-x)^2/2}{(1+x)^2 + (1-x)^2/2}, \quad y_A = \frac{(1+x) - \frac{2}{3}(1-x)}{(1+x) + \frac{2}{3}(1-x)}, \quad y_B = \frac{(1+x)^2 - (1-x)}{(1+x)^2 + (1-x)},$$

and $y_\ell = y_A$ if $B \le 1/2$, $y_\ell = y_B$ otherwise. Moreover $y_A - x$ and $y_B - x$ have the numerators $\frac{1}{3}(1-x^2)$ and $(1-x)(x+x^2)$ over the positive denominators, so $x \le y_\ell$, and $y_h \le 1$.

Three polynomial inequalities. Let $L_0 = 0.693147 < L$. For $0 \le x \le 3/5$,

$$J(x, y_A) \le \tfrac{1696}{25}\,\frac{rL_0(1-x^2)^4}{16q}, \qquad J(x, y_h) \le \tfrac{1696}{25}\,\frac{rL_0(1-x^2)^4}{16q},$$
$$J(x, y_B) \le \tfrac{4293}{50}\,\frac{rL_0(1-x^2)^4}{16q}.$$

Clearing the denominators (D_A^4 with D_A linear, D_B^4 and D_h^4 with quadratic denominators) turns each into a polynomial inequality of degree 12, 16 and 16 in x. Each is certified by the Bernstein expansion on $[0, 3/5]$, all of whose coefficients are positive (LHD-1). Thus $h_{aaaa} < 1696/25 = 24\gamma(16/9)$ if $B \le 1/2$, and $h_{aaaa} < 4293/50 = 24\gamma(9/4)$ otherwise. Since $B \ge 1/3$, respectively $B \ge 1/2$, this is $h_{aaaa} < 24\gamma(1+B)^2$, that is, $f_{aaaa} < 0$.

The point $a = 1/2$. There, with $v = (1-K)/(1+K)$,

$$f_{aaa} = \frac{16q(q+1)v - 16q^3v^3/r}{L} - \frac{54}{5}(1-C) - \frac{477}{25}(1-B^2).$$

The first term increases with $v \le 1/3$. Since $K^{(4)} > 0$ (because $q(q-1)(q-2)(q-3) > 0$), cubic Hermite interpolation at $0, 1$ gives, with $k = 5247/10000 < K(0) = 3^{-r}$,

$$K \ge K_*(z) = k + (3 - 3k - 2q/3)z^2 + (2q/3 - 2 + 2k)z^3.$$

Replacing K by K_* and L by L_0, and clearing $r(1+K_*)^3$, reduces $-f_{aaa}(1/2, z) \ge 0$ to $(1-z)\sum_{j=0}^{12} c_j z^j \ge 0$, where every c_j is positive (LHD-2). Lowering $K(0)$ to k preserves the minorant because its coefficient is $1 - 3z^2 + 2z^3 = (1-z)^2(1+2z) \ge 0$. Together with $f_{aaaa} < 0$ this proves the lemma. □

Lemma S11 (The whole large-head chart; exact polynomial sign certificates)**.** *Every spectrum on the above chart satisfies $\Lambda_\theta < T = 2507333/1500000$.*

Proof. Put $a_0 = 127/200$, $S = 1 - a_0 = 73/200$, $d = a_0 - 1/2 = 27/200$ and $\xi = (a - a_0)/d \in [-1, 11/9]$. By Lemma S10, the quadratic Hermite interpolant in a at the single node $1/2$ and the double node a_0 is a global upper bound, since its remainder has the sign of $f_{aaa}(\eta, z)(a - 1/2)(a - a_0)^2 \le 0$:

$$f(a,z) \le Q = f_0(1-\xi^2) + f_l\xi^2 + dD(\xi + \xi^2), \quad f_0 = f(a_0,z),\ f_l = f(\tfrac12, z),\ D = f_a(a_0,z).$$

Write $f = h(a, K) + P$, where P is a polynomial in (a, B, C) and does not depend on K. Then only the three values $h(a_0, K)$, $h(\frac12, K)$, $h_a(a_0, K)$ in Q depend on K.

Q decreases in K on $[.52, 1]$. Let $R = (a_0/S)^q \in [12/5, 5/2]$, $c = rL$ and $\lambda = dq/(a_0 S) = 42849/46355$. Differentiating,

$$h_K(a_0,K) = \frac{-1}{c(R+K)}, \quad h_K(\tfrac12,K) = \frac{-1}{c(1+K)}, \quad h_{aK}(a_0,K) = \frac{qR}{c\,a_0 S(R+K)^2},$$

and therefore

$$-c(R+K)^2(1+K)\,Q_K = (1+K)\,(R + K - \lambda R\xi) + \xi^2\{(R+K)(R-1) - \lambda R(1+K)\}.$$

For $-1 \le \xi \le 11/9$ both terms are nonnegative, the first strictly positive. The first is linear in ξ, positive at both ends: $\frac{11}{9}\lambda < 1.13$ and $K > .13R$. The brace increases in R and decreases in K, and is $\frac{17}{5}\cdot\frac{7}{5} - \frac{24}{5}\lambda > 0$ at $R = 12/5$, $K = 1$ (LHD-3).

A cubic minorant of K. Because $K^{(4)} > 0$, cubic Hermite interpolation at the double nodes $3/20, 17/20$ gives

$$K(z) \ge \kappa(z) = .5238033 + .0142244z + .4204281z^2 + .0384307z^3, \qquad .52 < \kappa < 1. \tag{LH1}$$

Each coefficient of κ is a rational lower rounding of the interpolant's coefficient (LHD-3). Since $.52 < \kappa \le K \le 1$, we have $Q(K) \le Q(\kappa)$.

Rational bounds for the logarithms. Set

$$\rho = 2.40792594, \quad u = 2.45774282, \quad H_0 = 1.10489073, \quad H_l = 1.32818582.$$

They approximate R, $1/(rL)$, $h(a_0, 3/4)$, $h(\frac12, 3/4)$ respectively with error less than $2\cdot10^{-8}$. For $v \in [.52, 1]$ define

$$\widehat{h}_0 = H_0 - 2u\frac{v - 3/4}{2\rho + v + 3/4}, \quad \widehat{h}_l = H_l - 2u\frac{v-3/4}{v+11/4}, \quad \widehat{h}_a = -\frac{2qu}{1-(27/100)^2}\Big(\frac{\rho - v}{\rho + v} - \frac{27}{100}\Big).$$

The identity $\ln((1+y)/(1-y)) = 2y + e_y$ has $|e_y| \le 2|y|^3/(3(1-y^2))$. Here $h(a_0, v) - h(a_0, \frac34) = -\frac{1}{rL}\ln\frac{R+v}{R+3/4}$ and $h(\frac12, v) - h(\frac12, \frac34) = -\frac{1}{rL}\ln\frac{1+v}{7/4}$. The logarithms equal $\ln\frac{1+y}{1-y}$ with $y = (v - \frac34)/(2R + v + \frac34)$, respectively $y = (v - \frac34)/(v + \frac{11}{4})$, and $|y| < .0383$, respectively $|y| < .0704$. Rounding the constants and the remainder give

$$|h(a_0,v) - \widehat{h}_0| < e_0 = 10^{-4}, \quad |h(\tfrac12,v) - \widehat{h}_l| < e_l = 6\cdot10^{-4}, \quad |h_a(a_0,v) - \widehat{h}_a| < e_a = 10^{-6}. \tag{LH2}$$

These include the rounding errors; LHD-3 checks the scalar arithmetic. On $[-1, 11/9]$ we have $|1 - \xi^2| \le 1$ and $|d(\xi + \xi^2)| \le 11/30 < 1$, so with $P_0 = P(a_0)$, $P_l = P(\frac12)$ and $P_a = \partial_a P(a_0)$,

$$Q(\kappa) \le F = (\widehat{h}_0 + P_0)(1-\xi^2) + (\widehat{h}_l + P_l)\xi^2 + d(\widehat{h}_a + P_a)(\xi + \xi^2) + e_0 + e_a + e_l\xi^2, \qquad v = \kappa(z).$$

The completed square. Put $E = (v + 2\rho + 3/4)(v + 11/4)(v + \rho) > 0$. Then $E\,(T - F) = c_0 + c_1\xi + c_2\xi^2$, where after substituting $v = \kappa(z)$ and the polynomials B, C the c_i are polynomials in z. On $[0, 1]$,

$$c_2 \ge \tfrac{1}{10}, \qquad 4c_0c_2 - c_1^2 \ge \tfrac{3}{100},$$

because all Bernstein coefficients of c_2 (degree 13) and of $4c_0c_2 - c_1^2$ (degree 26) are at least these values (LHD-4). Thus the quadratic in ξ is positive for every real ξ, so $f \le Q \le F < T$. □

S6.10 A strengthened cubic bound below purity 47/100

We now fix $q = 1587/1000$, $r = q - 1$, $L = \ln 2$ and $x = p - 1/4$.

Lemma S12 (Capped cubic slack; analytic with explicit scalar arithmetic)**.** *Every four-point probability vector with $p \le 47/100$ satisfies*

$$H_q(\lambda) \le \frac{10}{3} - 6p + \frac{8}{3}t - \frac{11}{10}(p - 1/4)(1/2 - p). \tag{CG}$$

Equality holds only at the uniform-four spectrum.

Proof. In centred moments $y = \sum_i (\lambda_i - 1/4)^3$, the assertion is

$$2 - H_q \ge \frac{171}{40}x - \frac{8}{3}y - \frac{11}{10}x^2.$$

We first prove it for $p \le 21/50$, then use binary grouping for the rest.

The moment argument, $x \le 17/100$. The claim follows from (MR) of Lemma S15 with $b = 8/3$, $c = 171/40$, $\gamma = 11/10$ and $\mu > 0$; we apply that lemma with the CG column of Table S4. Here $N = 1 + xV + x^2W$, $V = A + Bv + Cv^2$, $W = C/4 + D_5 v$, $v = y/x$, and A, B, C, D_5 are those of Lemma S14; the rectangle lemma does not require its smaller purity domain. The third-moment bound gives $|v| < 6/25$. On $0 \le x \le X = 17/100$, $|v| \le 6/25$, direct rational bounds give

$$5/3 < V < 9/4, \quad -21/100 < W < 1, \quad 1 \le N < 3/2, \qquad V^2 - 2W \ge \frac{1706}{625} - \frac{9}{10}|v| + 5v^2 > \frac{8}{3},$$

and $|B + 2Cv + XD_5| < 213/100$. Then $(8/3 - 2X(9/4)(21/100))/(3/2)^2 > 1 = m$, so $(\ln N)_{xx} < -1$, and $m > (11/5)\rho = 2\rho\gamma$; $C > 144/100$ gives $\Phi_{vv}(X, v) \ge (2C - X(213/100)^2)/(\rho(3/2)^2) > 2 = \kappa$. At $v_* = 0$ put $z = XA + X^2C/4$, use the eight-term lower sum $\ln(1 + z) \ge \sum_{j=1}^{8} (-1)^{j+1} z^j/j$, and use $693147/10^6 < L < 693148/10^6$: then $\Phi(X, 0) > 1/250$ and $|\Phi_v(X, 0)| < 1/20$. The lemma gives $\mu \ge 1/250 - (1/20)^2/4 > 0$. This proves the claim strictly when $x > 0$; $x = 0$ is the uniform spectrum.

Two binary bounds from endpoint values. Write $h(u) = H_q(u, 1 - u)$ and $d_0 = u(1 - u)$. The monotonicity of A_q in Lemma S8 (proved there by Cauchy's formula) gives

$$h(u) \le 4d_0 - \frac{11}{40}d_0(1 - 4d_0) \quad (3/100 \le u \le 1/2), \qquad h(u) \le 4d_0 - \frac{3}{8}d_0(1 - 4d_0) \quad (1/12 \le u \le 1/2). \tag{BQ}$$

Only the two lower endpoints require checking. Explicit sufficient bounds are

u_0	$u_0^q >$	$(1 - u_0)^q >$	$h(u_0) <$
3/100	19/5000	1191/1250	1093/10000
1/12	193/10000	871/1000	5713/20000

The power bounds follow by raising to the thousandth power. The power sums are then at least $S = 4783/5000$ and $S = 8903/10000$, and the entropy caps follow from $-\ln(1 - z) \le z + z^2/2 + z^3/(3(1 - z))$ with $z = 1 - S$; the two quartics exceed them (CG-2).

It follows that $4u(1 - u) - h(u) > (11/10)d(1/4 - d)$ whenever $3/100 \le u \le 4/15$ and $3/100 \le d \le \min\{2/25, d_0\}$. For $u \le 1/12$, use the first quartic and the fact that $d(1/4 - d)$ increases on $[0, 1/8]$. For $u \ge 1/12$, use the second quartic: the concave function $d_0(1 - 4d_0)$ has its minimum at an endpoint of $11/144 \le d_0 \le 44/225$, and

$$\frac{3}{8}\min\left\{\frac{11}{144}\frac{25}{36}, \frac{44}{225}\frac{49}{225}\right\} > \frac{187}{12500} \ge \frac{11}{10}d(1/4 - d).$$

The two pairs, $21/50 \le p \le 47/100$. Determinant monotonicity preserves p, t and increases H_q, so it suffices to use the two ordered charts in (SV). Put $v = \delta^2$, $d = 1/2 - p = u - u^2 - v/2 \in [3/100, 2/25]$. Ordering gives $3/100 < u < 4/15$ on S and $3/100 < u < 3/32$ on B. For S use

$p \le 1 - 3u + 3u^2$; for B use $p \le 1/2 - u + 3u^2/2$, $u \le 1/2$, and convexity of this quadratic on $[3/32, 1/2]$. Also on S,

$$\frac{v}{(1-u)^2} \le \frac{2(u - u^2 - 3/100)}{(1-u)^2} < \frac{2}{3}.$$

For two conditional probability vectors, Jensen's inequality for the logarithm gives

$$H_q((1-u)a, ub) \le h(u) + (1-w)H_q(a) + wH_q(b), \qquad w = \frac{u^q}{(1-u)^q + u^q}.$$

One conditional vector in each chart is uniform-two. We claim

$$H_q(\lambda_S) \le 1 + h(u) - (1+2u)v, \qquad H_q(\lambda_B) \le 1 + h(u) - (3-2u)v. \tag{CG3}$$

For S, the conditional squared imbalance $z = v/(1-u)^2$ is less than $2/3$. Its smaller probability is therefore greater than $1/12$; the second bound in (BQ) gives a deficit $z + (3/32)z(1-z) \ge (33/32)z$. Its weight times z is $v/[(1-u)^2(1+\rho^q)]$ with $\rho = u/(1-u) < 1$, and $\rho^q \le \rho^{3/2} \le \rho(\rho + 1/4)$ because $\sqrt{\rho} \le \rho + 1/4$. Hence $(1-u)^2(1+\rho^q) \le (1-u)^2 + u^2 + u(1-u)/4$, and the S inequality follows from

$$32(1+2u)\Big((1-u)^2 + u^2 + \frac{u(1-u)}{4}\Big) = 32 + 8u(1 - 7u + 14u^2) \le 33.$$

Indeed $1 - 7u + 14u^2 = 14(u - 1/4)^2 + 1/8$ is at most 1 for $u \le 1/8$ and at most $11/32$ for $1/8 \le u \le 4/15$, so $8u(1 - 7u + 14u^2) \le \max\{1, 11/15\}$.

For B the binary cubic bound gives deficit at least z at order q. Here $M = (1-u)^q + u^q \le 1 - u$, because $q \ge 3/2$, $\sqrt{u} < 1/3$, and $(1-u)^{3/2} \le (1-u)(1-u/2)$. Thus $u^{q-2}/M \ge u^{-2/5}/(1-u) \ge 3 - 2u$ on $u \le 3/32$, as $u^2((1-u)(3-2u))^5 \le 1$ there: its logarithmic derivative has the sign of $24u^2 - 35u + 6 > 0$, and at $u = 3/32$ the inequality is $(3/32)^2(1305/512)^5 < 1$. This proves (CG3).

In these coordinates the cubic bound is $1 + 4u - 4u^2 - (1+2u)v$ on S and $1 + 4u - 4u^2 - (3-2u)v$ on B, and the correction is $(11/10)d(1/4 - d)$. By (CG3) the slack is at least $4u(1-u) - h(u)$, which exceeds the correction by the binary estimate above, since $d \le u - u^2 = d_0$. This completes the proof, with strictness. □

Corollary S13 (The capped cluster region; analytic)**.** *For three four-point spectra with $1 \le P \le 101/100$, $F \ge 11/25$ and every $p_j \le 47/100$, one has $\mathcal{E}_{q_s} < 5/3$. No quadratic variance bound is required.*

Proof. Put $\varepsilon = P - 1$. Convexity gives $U \le (47/100)^2 + (7/25 + \varepsilon)^2 + 1/16$. The sum of the corrections in (CG) is therefore at least $(11/10)(33/2500 + 19\varepsilon/100 - \varepsilon^2)$. The cubic envelope gives the uncorrected bound $3\mathcal{E}_{q_s} \le 5 + (1 - 20\varepsilon - 2F)/9$. Subtracting the correction leaves at most $5 - 89/75000 < 5$: the maximum is at $F = 11/25$, $\varepsilon = 0$. □

S6.11 An analytic moment bound near the uniform spectrum

The following argument bounds the near-uniform cuts in the cluster sliver. It applies to all four-point spectra, without a chart reduction or physical compatibility assumption. Its rectangle form, Lemma S15, is shared with the bounded-head chart and the capped region.

Lemma S14 (Near-uniform moment minorant; analytic)**.** *Let λ be a four-point probability vector. Put $w_i = \lambda_i - 1/4$, $x = \sum w_i^2 \le 1/25$, and $y = \sum w_i^3$. For $q = q_s = 1587/1000$, write $r = q - 1$ and*

$$A = 2qr, \quad B = \frac{8}{3}qr(q-2), \quad C = \frac{8}{3}qr(q-2)(q-3), \quad D_5 = \frac{2}{3}(q-4)C.$$

If $x > 0$, set $v = y/x$, $V = A + Bv + Cv^2$, $W = C/4 + D_5 v$. The notation D_5 distinguishes this Taylor coefficient from the global third-moment sum T. Then $|v| < 3/25$, $V, W > 0$, and

$$2 - H_q(\lambda) \ge \frac{\ln(1 + xV + x^2W)}{r \ln 2}. \tag{NU}$$

Proof. The sixth derivative of z^q is positive on $z > 0$. Taylor's formula through degree five at $z = 1/4$, summed over the four entries, therefore gives

$$4^{q-1}\sum_i \lambda_i^q \geq 1 + Ax + By + C\sum_i w_i^4 + D_5 xy.$$

Here the trace-zero Newton identity is $\sum w_i^5 = 5xy/6$. The estimate extends to zero entries by continuity. The elementary square identity

$$\sum_i \left(w_i^2 - \frac{x}{4} - \frac{y}{x}w_i\right)^2 = \sum_i w_i^4 - \frac{x^2}{4} - \frac{y^2}{x} \geq 0$$

proves the required moment lower bound, since $C > 0$. Lemma 35 gives $|y| \leq x^{3/2}/\sqrt{3}$, hence $|v| \leq 1/\sqrt{75} < 3/25$. The logarithm is increasing, and its division by $r\ln 2$ gives (NU). For completeness, the rational bounds at q_s

$$A \geq \tfrac{9}{5}, \quad |B| \leq \tfrac{21}{20}, \quad C \geq \tfrac{7}{5}, \quad -\tfrac{12}{5} \leq D_5 < 0$$

give $V \geq \frac{837}{500}$ and $W \geq \frac{7}{20} - \frac{12}{5}\cdot\frac{3}{25} = \frac{31}{500} > 0$ for $|v| < 3/25$. This proves the asserted positivity without relying on a later case split. □

The bound (NU) is used three times in the same way: on a rectangle in (x, v) it is concave enough in x to reduce to the far edge $x = X$, and convex enough in v there to reduce to one point.

Lemma S15 (Moment rectangle; analytic)**.** *Keep $q = q_s$ and the notation of Lemma S14, put $L = \ln 2$, $N(x,v) = 1 + xV(v) + x^2W(v)$ and $\rho = 21/50 = (3/5)(7/10)$, so that $rL < \rho$. For constants b, c and $\gamma \geq 0$ let*

$$\Phi(x,v) = \frac{\ln N(x,v)}{rLx} + bv - c + \gamma x \qquad (x > 0).$$

Let $X > 0$ and $v_* \in \mathcal{I} = [v_{\min}, v_{\max}]$, and suppose that on $[0, X] \times \mathcal{I}$

$$1 \leq N \leq N_+, \quad V \leq V_+, \quad W \geq -W_-, \quad V^2 - 2W \geq Q, \quad |B + 2Cv + XD_5| \leq S, \tag{MR1}$$

with $W_- \geq 0$, and that the constants m, κ, f_0, η satisfy

$$Q - 2XV_+W_- \geq mN_+^2, \quad m > 2\rho\gamma, \quad 2C - XS^2 \geq \kappa\rho N_+^2, \quad \kappa > 0, \tag{MR2}$$

$$\Phi(X, v_*) > f_0, \qquad |\Phi_v(X, v_*)| < \eta. \tag{MR3}$$

Then Φ is strictly decreasing in x and $\Phi > \mu := f_0 - \eta^2/(2\kappa)$ on $(0, X] \times \mathcal{I}$. Consequently every four-point probability vector with $0 < x \leq X$ and $v = y/x \in \mathcal{I}$ satisfies

$$2 - H_q(\lambda) - cx + by + \gamma x^2 > \mu x. \tag{MR}$$

Proof. Fix $v \in \mathcal{I}$ and put $\ell(x) = \ln N$. Then

$$\ell''(x) = \frac{2W - V^2 - 2xW(V + xW)}{N^2}.$$

Since $N \geq 1$, $V + xW \geq 0$ for $x > 0$. The last term is therefore nonpositive when $W \geq 0$; when $W < 0$ it is at most $2XW_-V_+$, as $V + xW \leq V$. The numerator is at most $-Q + 2XV_+W_- \leq -mN_+^2 < 0$, so $\ell'' \leq -m$. Put $\mathcal{F}(x) = rLx\Phi = \ell(x) - rLx(c - bv - \gamma x)$. Then $\mathcal{F}(0) = 0$ and $\mathcal{F}'' = \ell'' + 2rL\gamma \leq -m + 2\rho\gamma < 0$, so $\mathcal{F}(x)/x$, and with it Φ, is strictly decreasing in x on $(0, X]$.

At $x = X$, since $1 \leq N \leq N_+$, $C > 0$ and $2C - XS^2 > 0$,

$$\Phi_{vv} = \frac{2CN - X(B + 2Cv + XD_5)^2}{rLN^2} \geq \frac{2C - XS^2}{\rho N_+^2} \geq \kappa.$$

Completing the tangent square at v_* gives $\Phi(X,v) \geq \Phi(X,v_*) - \Phi_v(X,v_*)^2/(2\kappa) > \mu$, and so $\Phi(x,v) \geq \Phi(X,v) > \mu$.

The Taylor and projection inequality in the proof of Lemma S14 holds for every four-point vector; $x \leq 1/25$ entered only the bound on $|v|$. As $N \geq 1 > 0$, it gives $2 - H_q(\lambda) \geq \ln N/(rL) = x\Phi + cx - by - \gamma x^2$, which is (MR). □

In each application (MR3) is checked at one point by rational arithmetic. With $z = XV(v_*) + X^2W(v_*) \in (0,1)$ and an even k, $\ln(1+z) \geq \sum_{j=1}^{k}(-1)^{j+1}z^j/j$; if this sum is positive, dividing it by rL_+X bounds $\ln N/(rLX)$ below whenever $L < L_+$. The slope $\Phi_v(X,v_*) = (B + 2Cv_* + XD_5)/(rLN(X,v_*)) + b$ is monotone in L, so it is bounded by substituting the two rational endpoints of an enclosure of L. Table S4 lists the three instances; each application derives (MR1) from its own coefficient bounds.

	NU	SHD (large tail)	CG (moment part)
target	$\Lambda_\theta(c_2) - \Lambda_\theta(\lambda) \geq dx$	$\Lambda_\theta(\lambda) \leq \Lambda_\theta((1/4)^4)$	(CG), $p \leq 21/50$
X; $\mathcal{I}$; v_*	$1/25$; $[-3/25, 3/25]$; $1/20$	$3/32$; $[0, 9/50]$; $27/200$	$17/100$; $[-6/25, 6/25]$; 0
b; c; γ	12β; $\alpha + 8\beta - \gamma/2$; γ (HT)	$9/5$; $35/8$; $159/100$	$8/3$; $171/40$; $11/10$
N_+; V_+; W_-	$13/12$; $41/20$; 0	$6/5$; $19/10$; $3/50$	$3/2$; $9/4$; $21/100$
Q; m	$(837/500)^2 - 4/3$; $6/5$	$8/3$; $7/4$	$8/3$; 1
S; κ	$753/500$; 5	$19/10$; 4	$213/100$; 2
f_0; η	$21/2500$; $1/50$	$1/800$; $1/100$	$1/250$; $1/20$
k; (L_-, L_+)	4; $(6931/10^4, 1733/2500)$	8; $(693147, 693148)/10^6$	8; $(693147, 693148)/10^6$
μ	$> d = 1/125$	$= 1/800 - 1/80000$	$= 1/250 - 1/1600$

Table S4: The three instances of Lemma S15, all at q_s with $\rho = 21/50$. For NU and SHD, (α, β, γ) is the multiplier of the lemma and the target is the dual difference $2 - H_q - cx + by + \gamma x^2$; for CG the target is the centred form of (CG). The hypotheses are checked by NU-3, SHD-4 and CG-1 through one shared function.

S6.12 An analytic uniform-cut dual bound

Lemma S16 (Analytic uniform-cut dual bound; NU)**.** *At the order $q_s = 1587/1000$, with the HT multiplier row, every four-point spectrum with $p \leq 29/100$ satisfies*

$$\Lambda_\theta(c_2) - \Lambda_\theta(\lambda) \geq d\,(p - 1/4), \qquad c_2 = (1/4)^4. \tag{ND}$$

HT labels the cluster-sliver multiplier row. The order and multipliers are exactly those of the row HT of Table S2, and $d = 1/125$. Equality occurs only at c_2.

The proof uses two further constants, checked by NU-3: $f_0 = 21/2500$ and $\eta = 1/50$.

Proof. Put $b = 12\beta$, $c = \alpha + 8\beta - \gamma/2$ and $x = p - 1/4$. The uniform moment identities give

$$\Lambda_\theta(c_2) - \Lambda_\theta(\lambda) = 2 - H_q(\lambda) - cx + by + \gamma x^2. \tag{UD}$$

By Lemma S14, $x \leq X = 1/25$ and $|v| < D = 3/25$ when $x > 0$. We apply Lemma S15 with the NU column of Table S4. At $q = q_s$ the coefficients satisfy

$$\frac{9}{5} \leq A \leq \frac{19}{10}, \quad -\frac{21}{20} \leq B < 0, \quad \frac{7}{5} \leq C \leq \frac{3}{2}, \quad -\frac{12}{5} \leq D_5 < 0.$$

These are rational comparisons at the single order q_s (NU-2). Consequently

$$\frac{837}{500} \leq V \leq \frac{41}{20}, \quad 0 < W < \frac{2}{3}, \quad 1 \leq N < \frac{13}{12}, \quad |B + 2Cv + XD_5| \leq \frac{21}{20} + 3D + \frac{12}{5}X = \frac{753}{500},$$

and $V^2-2W \geq Q = (837/500)^2-4/3$. This is (MR1) with $W_- = 0$. For (MR2), $Q > (6/5)(13/12)^2$, $2\rho\gamma \leq (21/25)(6/5) < 6/5$ as $0 \leq \gamma \leq 6/5$, and $C \geq 7/5$ gives $14/5 - (1/25)(753/500)^2 > 5\rho(13/12)^2$. For (MR3) take $k = 4$ and

$$\frac{6931}{10000} < L < \frac{1733}{2500}.$$

The latter follows by summing ten terms of $L = 2\sum_{k\geq 0}((2k+1)3^{2k+1})^{-1}$ and bounding the geometric tail. Direct rational substitution of the multiplier row gives $\Phi(X, v_*) > f_0$ and $|\Phi_v(X, v_*)| < \eta$ at $v_* = 1/20$. Since $\mu = f_0 - \eta^2/10 > d$, (MR) and (UD) prove (ND). At $x = 0$ the spectrum is c_2; otherwise $d > 0$ makes it strict. □

S6.13 An analytic bound on the distinguished cluster cut

We now treat the distinguished cut in the cluster sliver. Elementary entropy estimates reduce the required bound to finitely many polynomial signs, with no interval covering or derivative grid. Write $q_s = 1587/1000$ and let θ_s be the HT row of Table S2. Set

$$\kappa = 4\sqrt{3}/5, \quad X = \tfrac{1}{2} + \varepsilon - p, \quad F_c = -1 + 8\varepsilon + 9p - 12t.$$

Lemma S17 (Analytic cluster sliver; SL)**.** *At the order $q_s = 1587/1000$, on either ordered chart, suppose*

$$p \geq 47/100, \quad X \geq 0, \quad 0 \leq \varepsilon \leq 1/100, \quad [F_c - \kappa X, F_c + \kappa X] \cap [11/25, 1/2] \neq \varnothing.$$

For either sign, put

$$G_\pm = \Lambda_{\theta_s}(c_1) - \Lambda_{\theta_s}(\lambda; q_s) + \ell_{\theta_s}(\varepsilon, F_c \pm \kappa X) - \ell_{\theta_s}(0, 1/2).$$

Then $G_\pm \geq 0$, with equality only at $\lambda = c_1$, $\varepsilon = 0$.

Proof. The proof first bounds the two chart coordinates, then replaces the entropy by an explicit minorant of its deficit. The resulting four polynomial bounds are reduced to one-variable endpoint signs by removing ε and using concavity in v.

Coordinates and domains. Put $v = \delta^2$. In the two charts use

$$\begin{aligned} \lambda_S &= ((1-u+\delta)/2, (1-u-\delta)/2, u/2, u/2), \\ \lambda_B &= ((1-u)/2, (1-u)/2, (u+\delta)/2, (u-\delta)/2), \\ p &= \tfrac{1}{2} - u + u^2 + v/2, \\ t_S &= \tfrac{1}{4} - \tfrac{3}{4}u + \tfrac{3}{4}u^2 + \tfrac{3}{4}(1-u)v, \qquad t_B = \tfrac{1}{4} - \tfrac{3}{4}u + \tfrac{3}{4}u^2 + \tfrac{3}{4}uv. \end{aligned} \tag{SV}$$

In both charts $0 \leq u < 1/16$, $0 \leq v < 1/20$. Here are explicit domain reductions. Ordering in S gives $u \leq 1/2$, $v \leq (1-2u)^2$ and $p \leq 1 - 3u + 3u^2$; hence $u < 1/4$. Since $X \leq 1/25$ and $\kappa < 7/5$, the window condition gives $(9/2)(1-2u)v \leq 49/250$. Successively this gives $v < 9/100$, $u < 1/10$, $v < 11/200$, $u < 1/16$ and $v < 1/20$, using $u - u^2 \leq 3/100 + v/2$ each time (verifier-checked, SL-3). In B, ordering gives $u \leq 1/2$, $v \leq u^2$ and $p \leq 1/2 - u + 3u^2/2$; thus already $u < 1/30$.

Entropy estimates. Set $(K, c_s) = (7801/2000, 5/2)$. On $0 \leq u \leq 1/16$, the binary entropy obeys

$$H_{q_s}(u, 1-u) \leq Ku - c_s u^{8/5} + \frac{12}{5}u^2. \tag{SE}$$

Indeed $(1-u)^q \geq 1 - qu + q(q-1)u^2/2$. If $z = qu - q(q-1)u^2/2 - u^q$, then $0 \leq z \leq qu$ and $\ln(1-z) \geq -z - z^2/(2(1-z))$ give the sharper bound

$$H_q(u, 1-u) \leq \frac{q}{(q-1)\ln 2}u - \frac{u^q}{(q-1)\ln 2} + \frac{q^2/(2(1-q/16)) - q(q-1)/2}{(q-1)\ln 2}u^2.$$

The last coefficient is less than $12/5$. The linear coefficient is less than K, while $16^{8/5-q_s} > 51/50$ exceeds $c_s(q_s-1)\ln 2$. These are rational comparisons, integer-power comparisons and the atanh bounds $693147/10^6 < \ln 2 < 693148/10^6$.

Separating the head and tail pairs in the power sum gives

$$H_{q_s}(\lambda) \le 1 + Ku - c_s u^{8/5} + \frac{12}{5}u^2 - \begin{cases} \frac{113}{100}v + \frac{3}{2}uv - \frac{3}{8}v^2, & S, \\ 3v, & B. \end{cases} \tag{SH}$$

To see the head estimate, let $M = (1-u)^q + u^q$. The even Taylor expansion of the head power gives an increase of at least $(q(q-1)/2)(1-u)^{q-2}v$ over M. The factor

$$\frac{(1-u)^{q-2}}{M} = \frac{(1-u)^{-2}}{1+(u/(1-u))^q} \ge 1 + \frac{3}{2}u$$

follows from $(u/(1-u))^q \le 7u/25$ and $(1-u)^{-2} \ge 1+2u$. The coefficient $\frac{q(q-1)}{2}(1-u)^{q-2}/M$ of v in the resulting relative increase is at most $\frac{q(q-1)}{2}(1-u)^{-2} \le \frac{q(q-1)}{2}\left(\frac{16}{15}\right)^2 < 27/50$. Applying $\ln(1+z) \ge z - z^2/2$ proves the S estimate, since $q/(2\ln 2) > 113/100$ and $(27/50)^2/(2(q-1)\ln 2) < 3/8$. In B the same argument acts on the tail: its factor is $u^{q-2}/M \ge 3$, and the logarithm increment is at most $1/100$ when $v \le u^2$. The resulting coefficient exceeds 3. All estimates extend to $u = 0$ by continuity.

Polynomial lower bounds. Inserting (SH) in $G_\pm$ gives the following lower bound, throughout $u \le 1/16$, $v \le 1/20$, $\varepsilon \le 1/100$:

$$P(u,v) + \varepsilon(A + Bv), \qquad P = au + bv + cu^2 + duv + fv^2 + gu^2v + c_s u^{8/5}. \tag{SP}$$

The coefficients are exact rationals. In each row label, the first letter is the chart (S or B) and the second is the sign in $G_\pm$:

	a	b	c	d	f	g	A	B
$S-$	2/125	7/500	−571/100	−33/10	−281/100	6	185/100	8
$S+$	2/125	7/500	−6	69/50	−103/20	0	147/100	16
$B-$	2/125	19/100	−6	3	−5	0	1/2	0
$B+$	2/125	19/100	−571/100	−3/2	−5/2	0	1/2	0

(ST)

For clarity, this table involves polynomial algebra only. Expand the formula for G using (SV), enclose κ between 13856/10000 and 13857/10000, and bound negative terms of higher degree by $u \le 1/16$, $v \le 1/20$, $\varepsilon \le 1/100$. At $\varepsilon = 0$, charge them to u^2, uv or v^2; in the coefficient of ε, retain positive terms proportional to v. The difference from every row in (ST) then has nonnegative coefficients (verifier-checked, SL-2). No transcendental sign is left.

Removing ε (verifier-checked, SL-3). In S the window and $X \ge 0$ imply

$$v \le \min\{2(u-u^2) + 2\varepsilon,\ V(u) + 21\varepsilon/10\}, \qquad V(u) = \frac{3/50 + (7/5)(u-u^2)}{26/5 - 9u}.$$

Put $v_0 = \min\{v, 2(u-u^2), V(u)\}$. Then $v - v_0 \le 21\varepsilon/10$. From the two S rows, $P_v \ge -C_0 + 2fv$, where $C_0 = \max\{0, -b + \max(0,-d)/16\}$. Direct substitution gives

$$A - \tfrac{21}{10}C_0 + \tfrac{1}{20}\min\{0, B + \tfrac{21}{5}f\} > 9/10.$$

Thus $G_\pm \ge P(u,v_0) + 9\varepsilon/10$. Hence it remains to prove $P \ge 0$ with $\varepsilon = 0$, $v \le \min\{2(u-u^2), V(u)\}$ and $u \le 1/16$. In chart B simply discard the positive $\varepsilon/2$ term; there $u < 1/30$ and $v \le u^2$.

One-variable signs. The function P is concave in v, so its minimum is at an endpoint. At $v = 0$ the claim follows from $u^{8/5} \ge 3u^2$ on $0 \le u \le 1/16$. In chart B, $b + du + fv \ge 3/40$, so this also proves the $v = u^2$ case.

For the S endpoint $v = 2(u - u^2)$, that endpoint is the smaller one only when $u < 1/140$. Dividing P by $u > 0$ and dropping positive terms gives

$$P/u \geq a + 2b + c_s u^{3/5} - Du, \quad D = 2b - c - 2\min\{d, (139/140)d\} - 4f.$$

This is a concave function of u. It is positive at both 0 and $1/140$; for the latter check use $(1/140)^{3/5} > 51/1000$.

Finally $v = V(u)$ is the smaller endpoint only for $u > 1/150$. Use the following quadratic minorant of $u^{8/5}$: for $Z = 23/40$, $U = Z^5$, $z = 2/5$, $u_0 = z^5$, put

$$Q_{Z,z}(u) = z^8 + \frac{8}{5}z^3(u - u_0) + \frac{Z^8 - z^8 - (8/5)z^3(U - u_0)}{(U - u_0)^2}(u - u_0)^2. \tag{SQ}$$

Since the third derivative of $u^{8/5}$ is negative, Hermite's remainder gives $u^{8/5} \geq Q_{Z,z}(u)$ for $0 \leq u \leq U$. After replacing $u^{8/5}$ by this minorant, multiply $P(u, V(u))$ by $(26/5 - 9u)^2$. The resulting polynomial R has degree at most five. For $L = 1/150$, $U = Z^5$, write

$$R(L + (U - L)x) = \sum_{i=0}^{n} B_i \binom{n}{i} x^i (1 - x)^{n-i}.$$

Every B_i is positive. Explicit integer lower bounds on $10^6 B_i$ are

	n	coefficient lower bounds					
$S-$	5	793	27467	39030	38325	27813	9606
$S+$	4	238	51693	86240	106860	116069	

(SB)

These are two polynomial expansions on single intervals, with no subdivision. They prove the last endpoint signs. Together with the strict $v = 0$ and ε bounds, this proves the asserted equality case. □

Further results

The proof of Corollary 9 in the paper uses none of the results of Sections S6.14–S6.21. They give an independent near-one defect estimate with its quantitative form, the sharp purity-variance profile on the critical locus with an analytic proof of (QV) on critical states (used for the equality case of (QK) at $P = 1$), and a shorter proof of the envelope for states with a rank-two cut.

S6.14 An independent near-one defect estimate

Use H_q, h_q and $\mathcal{E}_q$ from Section 3, and set $m_q = 2^{-q} + 3\,6^{-q}$. Entropies are in bits throughout this section; the proof below uses $H_q^{\mathrm{nat}} = H_q \ln 2$ when working in nats. The Tsallis result alone does not give an upper Rényi bound when $q > 1$. The anchor-order argument proves the full HS–cluster envelope of Corollary 9 for all $q > 1$. Here we give an independent near-one proof using the nonnegative Hermite interpolation error to control the logarithm's curvature; it also yields the quantitative defect estimate (RE).

Theorem S18 (Rényi interval above one)**.** *For every pure four-qubit state and every* $1 < q \leq 5/4$,

$$\boxed{\mathcal{E}_q(\psi) \leq \frac{\log_2 m_q}{1 - q}.}$$

Equality holds exactly on the two M_4, $\overline{M_4}$ *LU orbits.*

The argument rests on a scalar slack estimate. Set $p = 1/6$, $s = 1/2$, and

$$f_q(x) = \frac{x^q - x}{1 - q}, \qquad f_1(x) = -x \ln x, \qquad f_q(0) = 0.$$

Let ℓ_q be the secant of f_q through p, s, and $\widehat{h}_q$ its cubic Hermite interpolant there, as in Corollary 7.

Lemma S19 (Scalar slack control). *For $1 \le q \le 5/4$ and $0 \le x \le 1$,*

$$\big(f_q(x) - \ell_q(x)\big)^2 \le \frac{141}{100}\big(\widehat{h}_q(x) - f_q(x)\big). \tag{RA}$$

Consequently, for any four-component probability vector λ, put $S_q = \sum_i f_q(\lambda_i)$, $S_q^* = f_q(s)+3f_q(p)$, and $D_q = \sum_i(\widehat{h}_q - f_q)(\lambda_i)$. Then

$$(S_q - S_q^*)^2 \le \frac{141}{25} D_q. \tag{RB}$$

The proof (Section S6.15; check RA-1) puts $G = \widehat{h}_q - f_q - \frac{100}{141}(f_q - \ell_q)^2$, which vanishes with its derivative at p and s, and shows $G^{(4)} > 0$ on $(0,1]$ from an elementary bound $\Phi_q < 141/100$ on the factor its fourth derivative contains; the Hermite remainder then gives $G \ge 0$, and Cauchy–Schwarz gives the summed bound.

Proof of Theorem S18. Put $r = q - 1$, $L = 4^{-r}$, $M = \sum_i \lambda_i^q = 1 - rS_q$, and $m = m_q = 1 - rS_q^*$. Convexity of x^q gives $M \ge L$ and $m \ge L$. For $u > 0$, integration of $(t-1)/t$ gives

$$u - 1 - \ln u \le \frac{(u-1)^2}{2\min(u,1)}.$$

Apply this with $u = M/m$; since $\min(u,1) \ge L/m$, it follows that

$$H_q^{\mathrm{nat}}(\lambda) - H_q^{\mathrm{nat}}(s,p,p,p) \le \frac{S_q - S_q^*}{m} + \frac{r(S_q - S_q^*)^2}{2mL}.$$

By (RB) and $S_q = \sum_i \widehat{h}_q(\lambda_i) - D_q$, this is at most

$$\frac{\sum_i \widehat{h}_q(\lambda_i) - S_q^*}{m} - \frac{1 - 141r/(50L)}{m} D_q. \tag{RD}$$

For $r \le 1/4$, the last coefficient is positive, because

$$\frac{141r}{50L} \le \frac{141}{200}\sqrt{2} < 1 \qquad \left(\left(\tfrac{141}{200}\sqrt{2}\right)^2 = \tfrac{19881}{20000}\right).$$

Write a_q, b_q for the cubic and quadratic coefficients of $\widehat{h}_q$ and $\alpha_q = -b_q - 17a_q/12$. Corollary 7 proves $a_q, \alpha_q > 0$. Sum (RD) over the three cuts and use $T = (17P - 12 - F)/12$. With $\mathcal{D}_q = \sum_j D_q(\rho_{1j})$ and $\kappa = 1 - 141\sqrt{2}/200 > 0$, the result is the defect estimate

$$3m_q \ln 2\big(\mathcal{E}_q(M_4) - \mathcal{E}_q(\psi)\big) \ge \alpha_q(P-1) + \frac{a_q}{12}F + \kappa\mathcal{D}_q \ge 0. \tag{RE}$$

Equality forces $P = 1, F = 0$, so Proposition 14 gives the two HS orbits; both attain the bound. □

S6.15 Proof of Lemma S19

Notation is that of Section S6.14: $p = 1/6$, $s = 1/2$, $f_q(x) = (x^q - x)/(1-q)$, ℓ_q the secant of f_q through p, s, and $\widehat{h}_q$ its cubic Hermite interpolant there.

The proof rests on an elementary bound for the function Φ_q that appears in the fourth derivative.

Lemma S20 (An elementary fourth-derivative bound; proved, RA-1). *Write $\ell_q(x) = \mu_q x + \nu_q$ and, for $1 < q \le 5/4$, put*

$$A(q) = \frac{2(q+1)}{3-q}\left(\frac{1}{q-1} - \mu_q\right), \qquad D(q) = \frac{4(2q-1)(3-2q)}{(q-1)(3-q)(2-q)},$$

and $\Phi_q(x) = 2\nu_q + A(q)x - D(q)x^q$. Define Φ_1 by continuity. Then $\Phi_q(x) < 141/100$ for $1 \le q \le 5/4$ and $0 \le x \le 1$.

Proof. Set $r = q - 1$, $p = 1/6$, $s = 1/2$, and $\delta = r(2-r)(1-r)$. The function Φ_q is concave in x. Its derivative at p is $p^r N_p/\delta$, where

$$N_p = (2+r)(1-r)(3 \cdot 3^r - 1) - 4(1+r)(1-4r^2).$$

Since $\ln 3 > 1$, $3^r \geq 1 + r$, and substitution gives $N_p \geq 11r^2 + 13r^3 \geq 0$. Thus its maximum on $[0,1]$ is attained in $[p,1]$.

We show that $\Phi_q(x)$ increases with q on this latter interval. Three endpoint derivatives suffice. Define

$$\begin{aligned}
N_0 &= 6(1-r)3^r - 6 + 5r + 7r^2, \\
N_1 &= (2+r)(1-r)(3 \cdot 2^{-r} - 6^{-r}) - 4(1+r)(1-4r^2), \\
J_0 &= (N_0' - (\ln 6)N_0)\delta - N_0\delta', \\
J_p &= (N_p' - (\ln 6)N_p)\delta - N_p\delta', \qquad J_1 = N_1'\delta - N_1\delta'.
\end{aligned}$$

Primes here mean r derivatives. Direct substitution into Φ_q gives

$$\partial_q \Phi_q(p) = \frac{p^r J_0}{3\delta^2}, \qquad \partial_q \partial_x \Phi_q(p) = \frac{p^r J_p}{\delta^2}, \qquad \partial_q \partial_x \Phi_q(1) = \frac{J_1}{\delta^2}.$$

All three J are positive for $0 < r \leq 1/4$. Here are coarse bounds that permit this assertion to be checked with small rational numbers. Write $J = \sum_{k=2}^{6} c_k r^k + R$; the constant and linear coefficients are zero. The five coefficients have the following lower bounds, and $|R| \leq Mr^7$:

	c_2	c_3	c_4	c_5	c_6	M
J_0	4	−6	15	−14	0	3
J_p	27	−2	−33	52	−29	3
J_1	4	86	−75	2	0	14

For completeness, these bounds require only $69/100 < \ln 2 < 7/10$ and $109/100 < \ln 3 < 11/10$. Both follow from the first five terms of $\ln u = 2\sum_{j\geq 0} z^{2j+1}/(2j+1)$, $z = (u-1)/(u+1)$, with tail at most $2z^{11}/[11(1-z^2)]$. Expand the displayed N and J through degree six and substitute these two logarithm intervals to get the five columns. To check the remainder, each J is a polynomial of degree at most six plus terms $Q(r)e^{\alpha r}$ with $\deg Q \leq 5$. For $Q(r) = \sum_j Q_j r^j$, its tail after degree six has absolute value at most

$$r^7 e^{|\alpha|/4} \sum_j \frac{|Q_j|\,|\alpha|^{7-j}}{(7-j)!}.$$

Use $3^{1/4} < 4/3$, $6^{1/4} < 8/5$ and $2^{1/4} < 6/5$ for the three possible exponential factors. The resulting sums are respectively less than $3, 3, 14$, giving the last column. In particular the table yields $J_0 > 2r^2$, $J_p > 24r^2$ and $J_1 > 3r^2$ on $0 < r \leq 1/4$; for the last row use $86r - 75r^2 \geq 0$.

Now let $W(x) = \partial_q \partial_x \Phi_q(x)$ and $k(q) = q(q-1)D(q) > 0$. Differentiating $\partial_x^2 \Phi_q = -k(q)x^{q-2}$ gives

$$W'(x) = -k(q)x^{q-2}(\partial_q \ln k(q) + \ln x).$$

The bracket increases with x, so W has no interior minimum. Its endpoint values at $p, 1$ are positive by the table. Thus $W \geq 0$, and $\partial_q \Phi_q(x) \geq \partial_q \Phi_q(p) > 0$ on $[p,1]$. It remains only to bound $\Phi_{5/4}$.

Put $a = 6^{-1/4}$, $b = 2^{-1/4}$, and $t = x^{1/4}$. Then

$$\Phi_{5/4}(x) = 2(b-a) + \frac{36}{7}(3b-a)t^4 - \frac{64}{7}t^5 \leq 2(b-a) + \frac{36 \cdot 9^4}{35 \cdot 20^4}(3b-a)^5,$$

by maximising the last two terms over $t \geq 0$; their maximum is at $t = 9(3b-a)/20$. Integer fourth powers give $a > 319/500$ and $b < 841/1000$. Consequently

$$\Phi_{5/4}(x) < \frac{203}{500} + \frac{36 \cdot 9^4}{35 \cdot 20^4}\left(\frac{377}{200}\right)^5 < \frac{141}{100}.$$

Concavity in x and monotonicity in q give the assertion everywhere; the endpoint $q = 1$ follows by continuity. □

Proof of Lemma S19. Let $g = f_q - \ell_q$ and $G = \widehat{h}_q - f_q - (100/141)g^2$. Both G and G' vanish at p, s. Since $f_q^{(4)} = -q(2-q)(3-q)x^{q-4} < 0$, differentiation gives

$$G^{(4)}(x) = -f_q^{(4)}(x)\left(1 - \frac{100}{141}\Phi_q(x)\right) > 0 \qquad (x > 0).$$

The cubic Hermite remainder, with repeated nodes p, p, s, s, gives $G(x) = G^{(4)}(\xi)(x-p)^2(x-s)^2/24 \geq 0$ for $x > 0$; continuity gives the result at zero. No derivative at zero is needed. Finally $\sum_i \ell_q(\lambda_i) = S_q^*$, since ℓ_q is affine and $\sum_i \lambda_i = s + 3p = 1$. Cauchy–Schwarz gives the summed bound. □

S6.16 Critical-locus variance refinements

The universal bound (QK) suffices for the anchor argument. We now give its sharper critical-locus refinements: the fixed-variance profile on $P = 1$, used to classify equality in (QK), and the analytic proof of the sharp bound (QV) on every critical state. Put $p_j = \operatorname{tr} \rho_{1j}^2$ and $U = \sum_j p_j^2$.

Proposition S21 (Sharp purity-variance tradeoff at minimal purity)**.** *Suppose a pure four-qubit state has $P = 1$. Set*

$$V = \sum_{j=2}^{4} (p_j - 1/3)^2, \qquad s = (24V)^{1/4}.$$

Then $0 \leq s \leq 1$ and the sharp bound at each fixed V is

$$\boxed{F \geq \frac{3}{2}s^2 - s^3.} \tag{RF}$$

For $a = (1+2s)/3$, $b = (1-s)/3$, equality holds exactly when, up to ordering of cuts and eigenvalues, the three spectra are

$$\begin{aligned} &\left(\frac{1}{2}, \frac{a}{2}, \frac{b}{2}, \frac{b}{2}\right), \\ &\left(\frac{1+b+2\sqrt{ab}}{4}, \frac{1+b-2\sqrt{ab}}{4}, \frac{1-b}{4}, \frac{1-b}{4}\right) \quad \textit{on each of the other two cuts.} \end{aligned} \tag{RF-eq}$$

These spectra are attained for every $s \in [0,1]$. In particular, $T + U \leq 3/4$, equivalently $F \geq 12V$; equality in this weaker bound occurs only at the HS spectra ($s = 0$) or cluster spectra ($s = 1$).

[Proved under the hypothesis $P = 1$.]

Proof. By Lemma 2, the state is critical with $\tau = 0$. As in the proof of Proposition 13, its Bloch correlation matrix is then $\Theta = \mathbf{m}\mathbf{n}^T$ with unit $\mathbf{m}, \mathbf{n} \in \mathbb{R}^3$ (Lemma S36(2) at $\kappa = 0$), and the cut $c \in \mathbb{Z}/3$ has the four eigenvalues $(1 + \sigma \cdot g^{(c)})/4$, $g_a^{(c)} = m_{a+c} n_a$, as σ runs over the sign triples with $\sigma_1\sigma_2\sigma_3 = -1$ (Lemma S33). Put $A_a = m_a^2$, $B_a = n_a^2$ and $k = -m_1 m_2 m_3 n_1 n_2 n_3$. The three vectors $g^{(c)}$ have squared norms

$$u_c = \sum_a A_{a+c} B_a, \qquad \sum_c u_c = 1,$$

and the same coordinate product $-k$. The sign sums $\sum_\sigma \sigma \cdot g = 0$, $\sum_\sigma (\sigma \cdot g)^2 = 4|g|^2$ and $\sum_\sigma (\sigma \cdot g)^3 = -24 g_1 g_2 g_3$ give, for the corresponding cut,

$$p_j = \frac{1 + u_c}{4}, \qquad \operatorname{tr} \rho_{1j}^3 = \frac{1 + 3u_c + 6k}{16}.$$

Set $X = 1 - 3e_2(A)$ and $Y = 1 - 3e_2(B)$, where $e_2(A) = \sum_{i<j} A_i A_j$. Then $0 \le X, Y \le 1$, and expanding the three cyclic sums gives

$$\sum_c u_c^2 = \frac{1 + 2XY}{3}.$$

We sharpen the elementary Schur estimate by keeping X and Y. For any nonnegative triple A of sum one and $X = 1 - 3e_2(A)$,

$$27e_3(A) \le h(X), \qquad h(X) = 1 - 3X + 2X^{3/2} = (1 - \sqrt{X})^2(1 + 2\sqrt{X}). \tag{RP}$$

This is the cubic discriminant identity

$$4X^3 - \big(27e_3(A) - 1 + 3X\big)^2 = 27 \prod_{i<j} (A_i - A_j)^2 \ge 0.$$

Taking the upper square-root bound proves (RP). Equality requires two equal coordinates, and the upper sign selects $((1+2r)/3, (1-r)/3, (1-r)/3)$, $r = \sqrt{X}$, up to permutation. This includes the uniform triple and coordinate-vector endpoints.

The transfer from X, Y to their geometric mean is algebraic as well. Set $g(t) = (1-t)^2(1+2t)$, $\xi = X^{1/4}$, $\zeta = Y^{1/4}$, $q = \xi\zeta$, $d = (\xi - \zeta)^2$, and $w = (1 - \xi^2)(1 - \zeta^2)$. Direct factorisation gives

$$\begin{aligned} h(\sqrt{XY})^2 - h(X)h(Y) &= g(\xi\zeta)^2 - g(\xi^2)g(\zeta^2) \\ &= d\{w[6q(q+2) + 2d] + d(1+2q)^2\} \ge 0. \end{aligned} \tag{S6.3}$$

Every factor is nonnegative for $0 \le X, Y \le 1$, and the difference is strictly positive when $X \ne Y$. Consequently

$$27k \le 27\sqrt{e_3(A)e_3(B)} \le \sqrt{h(X)h(Y)} \le h(\sqrt{XY}).$$

The moment formulas above give

$$V = \frac{XY}{24}, \qquad F = \frac{1}{2} - \frac{27}{2}k.$$

Thus $s = (XY)^{1/4}$ and $F \ge (1 - h(s^2))/2 = 3s^2/2 - s^3$. For $0 < s < 1$, equality requires $X = Y = s^2$, $k > 0$, and both A, B to be permutations of (a, b, b). The shift c that aligns their distinguished coordinates gives $|g_a^{(c)}| = (a, b, b)$ up to order; the other two shifts give $(\sqrt{ab}, \sqrt{ab}, b)$. As every $g^{(c)}$ has negative coordinate product, all sign choices give the same spectral multisets, and the eigenvalue formula proves (RF-eq). At $s = 0$, equality forces $k = 1/27$ and hence both triples uniform; at $s = 1$ both are coordinate vectors. These give the stated endpoint spectra.

For attainability, take $\mathbf{m} = -\mathbf{n} = (\sqrt{a}, \sqrt{b}, \sqrt{b})$, so $A = B = (a, b, b)$ and $k = ab^2 = e_3(A)$. The product auxiliary vector with Bloch vectors $\mathbf{m}, \mathbf{n}$ has $\Theta = \mathbf{m}\mathbf{n}^T$ and $\tau = 0$, and it is $\sum_k v_k m_k$ for a critical $G(v)$, as at the end of the proof of Proposition 13. Finally,

$$\frac{3}{2}s^2 - s^3 - 12V = \frac{s^2}{2}(1 - s)(3 + s) \ge 0,$$

with equality exactly at $s = 0, 1$. This proves the weaker bound and its exhaustive equality statement as a corollary. □

S6.17 A joint-cut proof on the critical locus

Proposition S22 (Sharp quadratic variance on critical states)**.** *Every critical unit four-qubit state satisfies*

$$F^2 + 3P(P - 1) \ge 6V_P. \tag{CQ}$$

Equality holds at HS spectra and along the full cluster-to-Bell family

$$\big((1 + \kappa)/2, (1 - \kappa)/2, 0, 0\big), \qquad (1/4)^4, \qquad (1/4)^4, \qquad 0 \le \kappa \le 1,$$

up to ordering of cuts and eigenvalues, and only there. In particular, the defect in (QK) is at least $3\tau(1-\tau)/4$ on the critical locus. The proof uses no variance certificate. At $P = 1$ the sharper profile (RF) of Proposition S21 holds.

Proof. Use the auxiliary correlation matrix of Proposition 13:

$$\Theta = U\,\mathrm{diag}(\kappa, -\kappa, 1)V^T, \qquad U, V \in SO(3), \qquad \tau = \kappa^2, \qquad P = 1 + \kappa^2/2.$$

Write $r = Ue_3$, $s = Ve_3$, $A_i = r_i^2$, $B_i = s_i^2$, and $X = 1 - 3e_2(A)$, $Y = 1 - 3e_2(B)$. Thus $0 \le X, Y \le 1$. We retain these two directional parameters in both the permanent and purity estimates.

The permanent bound with directions retained. Put $z = \sqrt{XY}$ and

$$b_\kappa(x) = \kappa^2 + \frac{2}{3}\kappa(1-\kappa)^2(1-x) + \frac{2}{9}(1-\kappa)^3(1 - 3x + 2x^{3/2}).$$

The polar-decomposition Cauchy–Schwarz bound from Proposition 13, now using (RP) instead of discarding X, Y, gives

$$|\operatorname{perm}\Theta| \le \sqrt{b_\kappa(X)b_\kappa(Y)} \le b_\kappa(z).$$

For completeness, the last inequality is geometric concavity. Write $b_\kappa(x) = a - bx + cx^{3/2}$, where

$$a = (4\kappa^3 + 3\kappa^2 + 2)/9, \quad b = 2(1-\kappa)^2/3, \quad c = 4(1-\kappa)^3/9.$$

The sign of $d^2 \log b_\kappa(e^v)/dv^2$ is the negative of $4ab - 9act + bct^3$, $t = \sqrt{x}$. This polynomial decreases on $[0, 1]$, since $b \le 3a$, and its value at one is $4\kappa^2(1-\kappa)^2(4\kappa^2 + \kappa + 1)/9 \ge 0$. The cases with zero arguments follow by continuity. Consequently the exact formula for F gives

$$F \ge \kappa^2(1-\kappa) + \frac{3}{2}(1-\kappa)^2 z - (1-\kappa)^3 z^{3/2} \ge f_\kappa(z),$$

where, using $\sqrt{z} \le (1+z)/2$,

$$f_\kappa(z) = (1-\kappa)\kappa^2 + \frac{1}{2}(1-\kappa)^2(1+2\kappa)z + \frac{1}{2}(1-\kappa)^3 z(1-z) \ge 0.$$

The three purities as one Fourier coefficient. Let $\omega = e^{2\pi i/3}$, $D = \mathrm{diag}(1, \omega, \omega^2)$, and

$$u_c = \sum_{j=0}^{2} \Theta_{j+c,j}^2, \qquad Z = \sum_c \omega^c u_c = \mathrm{tr}(D\Theta\overline{D}\Theta^T).$$

Indices are cyclic. The cut formula $p_c = (1 + u_c)/4$ implies $6V_P = |Z|^2/4$. The symmetric unitary matrices $M = U^T D U$ and $N = V^T \overline{D} V$ have $|M_{33}|^2 = X$, $|N_{33}|^2 = Y$. Their last-row off-diagonal squared norms are $1-X, 1-Y$; their leading 2×2 blocks have squared Frobenius norms $1+X, 1+Y$. Expanding Z in these blocks and applying Cauchy–Schwarz gives

$$|Z| \le z + 2\kappa\sqrt{(1-X)(1-Y)} + \kappa^2\sqrt{(1+X)(1+Y)} \le B_\kappa(z) := (1-\kappa)^2 z + 2\kappa + \kappa^2.$$

To see the last step, fix z and put $S = X + Y \ge 2z$. The last two terms become $2\kappa\sqrt{1 - S + z^2} + \kappa^2\sqrt{1 + S + z^2}$, which decreases with S for $0 \le \kappa \le 1$; evaluate at $S = 2z$.

An elementary scalar comparison. For $0 \le \kappa, z \le 1$,

$$4f_\kappa(z)^2 + 9\kappa^2 - B_\kappa(z)^2 - 3\kappa^2(1-\kappa^2) = (1-\kappa)^2(1-z)R_\kappa(z),$$

where, putting $d = 1 - \kappa$,

$$R_\kappa(z) = -d^4 z^3 + 3d^3(1+\kappa)z^2 - 4\kappa d^2 z + 2\kappa^2(1 + 2\kappa^2).$$

Since $z^3 \le z^2$,

$$R_\kappa(z) \ge 2d^3(1+2\kappa)z^2 - 4\kappa d^2 z + 2\kappa^2(1+2\kappa^2) \ge 0.$$

The final quadratic is nonnegative: its discriminant is $-16\kappa^3 d^3(3+2\kappa+4\kappa^2) \le 0$ and its leading and constant coefficients are nonnegative (the endpoints follow directly). Combining the estimates proves

$$4F^2 + 9\tau - 24V_P \ge 3\tau(1-\tau),$$

which is exactly (CQ).

For $0 < \kappa < 1$ the scalar comparison is strict unless $z = 1$. Then $X = Y = 1$, and equality in the Fourier triangle bound forces all nonzero entries of Θ onto one cyclic diagonal. Their magnitudes are $1, \kappa, \kappa$, with negative product since $\det\Theta = -\kappa^2$. The cut eigenvalue formula gives precisely the displayed family. At $\kappa = 0$, equality requires $z = 0$ or $z = 1$; the latter is its cluster endpoint, while the former requires $F = 0$ and hence HS by the critical envelope equality case. At $\kappa = 1$, equality requires $F = 0$ and $|Z| = 3$, forcing the Bell endpoint. Conversely the family spectra are attained by the correlation matrices $\mathrm{diag}(\kappa, -\kappa, 1)$, and direct substitution gives equality. □

S6.18 Independent arguments on restricted domains

The proof of Corollary 9 in the paper needs no eigenvalue cap and no rank condition. The following result gives a shorter, independent proof of the envelope for states with a rank-two cut, at every order (for $q \ge 2$ see Section F.1).

Proposition S23 (Sharp fixed-spectrum rank-two cut envelope)**.** *Suppose one of the three balanced cuts of a pure four-qubit state has Schmidt rank at most two and nonzero eigenvalues $a, 1-a$, where $1/2 \le a \le 1$. Then for every $q > 0$,*

$$\boxed{\mathcal{E}_q(\psi) \le \frac{4}{3} + \frac{1}{3}H_q(a, 1-a) \le \frac{5}{3}.} \tag{Rk2}$$

For each fixed a, the first bound is sharp; equality holds precisely when the other two balanced cut spectra are both $(1/4)^4$. The attained cluster-to-Bell family (QF) realises equality for every $a \in [1/2, 1]$ and every order $q > 0$. In particular the global HS–cluster Rényi envelope holds on this entire rank-two boundary without any purity or other cap hypothesis.

Proof. A density matrix on a four-dimensional cut has Rényi entropy at most $\log_2 4 = 2$, with equality exactly for the uniform-four spectrum. The stipulated cut has entropy $H_q(a, 1-a)$, so bounding each of the other two cut entropies by 2 proves the first inequality and its equality classification. A binary probability vector has entropy at most 1, proving the second inequality. The physical states in (QF) have the stipulated binary spectrum and both crossed cuts uniform-four for all $a = (1+r)/2 \in [1/2, 1]$; they therefore attain the bound at every $q > 0$. Since the HS entropy h_q is at least $5/3$ up to q_*, and the cluster ceiling is $5/3$ thereafter, the final assertion follows. □

S6.19 A cap-free purity-controlled Rényi bound

The defect estimate of Theorem S18 also gives a ceiling at every order $q \ge 5/4$ that improves linearly with the purity sum. It is not used in the proof of Corollary 9.

Corollary S24 (Cap-free purity-controlled Rényi bound; exact-rational check, RA-5)**.** *For every unit pure four-qubit state and every $q \ge 5/4$,*

$$\boxed{\mathcal{E}_q(\psi) \le h_{5/4} - \frac{13}{20}(P-1), \qquad h_{5/4} = \frac{\log_2 m_{5/4}}{1-5/4}.} \tag{RH}$$

In particular, $P \ge 10/9$ implies $\mathcal{E}_q(\psi) < 5/3$ for $q \ge 5/4$. The bound (RH) holds for every triple of four-point spectra with $P \ge 1$ and $F = 17P - 12T - 12 \ge 0$, since the proof uses only these and the per-cut estimate (RD).

Proof. At $q_0 = 5/4$, the defect estimate (RE) and $F, \mathcal{D}_{q_0} \geq 0$ give

$$\mathcal{E}_{q_0}(\psi) \leq h_{q_0} - c(P-1), \qquad c = \frac{\alpha_{q_0}}{3m_{q_0} \ln 2}.$$

The exact-rational enclosures RA-5 establish

$$20\alpha_{q_0} - 39m_{q_0} \ln 2 > 0, \qquad 4 \ln m_{q_0} + \frac{313}{180} \ln 2 > 0.$$

Equivalently $c > 13/20$ and $h_{q_0} < 313/180$. Rényi entropy is nonincreasing in its order, so $\mathcal{E}_q(\psi) \leq \mathcal{E}_{q_0}(\psi)$ for $q \geq q_0$. Since $P \geq 1$, the first assertion follows. If $P \geq 10/9$,

$$\mathcal{E}_q(\psi) < \frac{313}{180} - \frac{13}{20}\left(\frac{10}{9} - 1\right) = \frac{5}{3},$$

as asserted. □

S6.20 The unrestricted joint purity bound

Theorem S25 (Joint purity compatibility; consequence of QK)**.** *Every pure four-qubit state satisfies*

$$\boxed{T + \sum_{j=2}^{4} p_j^2 \leq 3P - \frac{9}{4}.} \tag{JV}$$

Equality holds exactly for HS cut spectra, cluster cut spectra, or products of two Bell pairs, up to ordering of the cuts and eigenvalues. In the HS case the states belong to the two HS LU orbits.

Proof. This is an elementary consequence of the stronger quadratic variance bound (QK), together with $F \geq 0$ and $p_j \in [1/4, 1]$. Set

$$U = \sum_j p_j^2, \qquad G = 3P - \frac{9}{4} - T - U, \qquad Q = F^2 + \frac{9}{2}(P-1) - 6(U - P^2/3) \geq 0.$$

Since $T = (17P - 12 - F)/12$, direct algebra gives the useful identity

$$12G = F(1-2F) + 2(P-1)(3-2P) + 2Q. \tag{JI}$$

It proves $G \geq 0$ whenever $1 \leq P \leq 3/2$ and $0 \leq F \leq 1/2$.

For the remaining cases, use the purity box (JB) of Section S6.1, the maximum of the convex function U at fixed P over $[1/4, 1]^3$. Using $12G = 19P - 15 + F - 12U$, the first piece, for $1 \leq P \leq 3/2$ and $F \geq 1/2$, yields

$$12G \geq F - \frac{1}{2} + (P-1)(19 - 12P) \geq 0.$$

For $3/2 \leq P \leq 9/4$ the second piece yields

$$12G \geq F + \frac{1}{2}(2P-3)(31 - 12P) \geq 0,$$

and for $9/4 \leq P \leq 3$ the last piece yields

$$12G \geq F - 12P^2 + 67P - 87 \geq F + 3 > 0.$$

The last inequality follows from concavity of the quadratic and its values $3, 6$ at the two endpoints.

Equality in the first case forces $Q = 0$, so the equality classification of Corollary 28 gives HS, cluster or Bell-pair spectra. In the $F \geq 1/2$ case equality requires $P = 1$, $F = 1/2$, $U = 3/8$, again giving $Q = 0$ and the cluster spectra. In the second box range equality requires $P = 3/2$, $F = 0$ and $U = 9/8$; its purity triple is $(1, 1/4, 1/4)$, and the state is a product of two Bell pairs. The final range is strict. All three listed types attain equality. □

S6.21 What local filtering can and cannot reduce

Lemma S26 (One-site filtering majorises a locally maximally mixed pair)**.** *Let ρ be a two-qubit density matrix with both one-qubit marginals equal to $I/2$. Let K be any nonzero filter on one of its qubits, and normalise $\rho_K = (K \otimes I)\rho(K^\dagger \otimes I)$ by its trace (with the factors interchanged if the filter acts on the second qubit). Then*

$$\operatorname{spec}(\rho) \prec \operatorname{spec}(\rho_K). \tag{FM}$$

In particular, filtering a single qubit of any pure critical four-qubit state cannot increase the Rényi entropy of any *of the three balanced cuts, at any $q > 0$.*

Proof. The positive factor in the polar decomposition of K suffices, since the unitary factor only conjugates ρ_K. By a positive rescaling assume $\operatorname{tr} K^\dagger K = 2$; the given single-qubit marginal makes the filtered state already normalised. Then $K^\dagger K = I + s\,\mathbf{n}\cdot\boldsymbol{\sigma}$, $0 \le s \le 1$, where $\mathbf{n}$ is a real unit vector. Put $N = (\mathbf{n}\cdot\boldsymbol{\sigma}) \otimes I$.

Let $\Theta = (\sigma_y \otimes \sigma_y)\mathcal{K}$ be the two-qubit spin-flip antiunitary, where $\mathcal{K}$ is complex conjugation in the computational basis. Expanding ρ in Pauli matrices shows that $\Theta\rho\Theta^{-1} = \rho$, because the two one-site Bloch vectors vanish and spin flip changes the sign of each individual Pauli factor. Also $\Theta N \Theta^{-1} = -N$. The matrices

$$B_\pm = \rho^{1/2}(I \pm sN)\rho^{1/2}$$

are positive, have the same spectrum by conjugation with Θ, and satisfy $(B_+ + B_-)/2 = \rho$. The nonzero eigenvalues of B_+ coincide with those of $(K \otimes I)\rho(K^\dagger \otimes I)$, because these matrices have the form $X^\dagger X$ and $XX^\dagger$. Convexity of every Ky Fan sum under averaging therefore gives $\operatorname{spec}(\rho) \prec \operatorname{spec}(B_+) = \operatorname{spec}(\rho_K)$.

For a critical four-qubit state, each qubit marginal is $I/2$. On each balanced cut choose the two-qubit side containing the filtered site. Its reduced state has both one-qubit marginals $I/2$; the two sides of a pure bipartition have the same nonzero spectrum. Apply the result above and Schur concavity of Rényi entropy. □

Two filters need not preserve this ordering. The single-site conclusion is *not* an orbitwise monotonicity theorem. For an exact counterexample let

$$z = (2+2i)(|0011\rangle + |1100\rangle) + |0101\rangle - |0110\rangle - |1001\rangle + |1010\rangle.$$

It has squared norm 20 and all four one-qubit marginals maximally mixed. Apply $\operatorname{diag}(2,1)$ at *both* sites 2 and 3, and normalise the resulting unnormalised vector z' of squared norm 89. The three cut spectra of $z/\sqrt{20}$ are, respectively,

$$(2/5, 2/5, 1/5, 0), \qquad (13/20, 1/4, 1/20, 1/20), \qquad (13/20, 1/4, 1/20, 1/20),$$

whereas those of $z'/\sqrt{89}$ are

$$(32, 32, 25, 0)/89, \quad (65, 16, 4, 4)/89, \quad (52, 20, 16, 1)/89.$$

Consequently, their products of cut purities are

$$\prod_j p_j(z/\sqrt{20}) = \frac{21609}{250000} \;>\; \frac{40544579889}{496981290961} = \prod_j p_j(z'/\sqrt{89}). \tag{FN}$$

This proves that the *mean* Rényi-2 entropy increases under two local filters. The same is true at $q = 3/2$, inside the intermediate-order interval: if $M(\lambda) = \sum_i \lambda_i^{3/2}$, the original moments obey $M_1 > 59/100$ and $M_2 = M_3 > 67/100$, while the filtered moments obey $M_1' < 581/1000$,

$M_2' < 720/1000$, and $M_3' < 631/1000$. Each strict bound follows by squaring rational bounds on the square roots of the displayed rational eigenvalues. Thus

$$\prod_j M_j > \frac{264851}{10^6} > \frac{3299499}{12500000} > \prod_j M_j',$$

so $\mathcal{E}_{3/2}(z'/\sqrt{89}) > \mathcal{E}_{3/2}(z/\sqrt{20})$. Both states obey the global HS–cluster envelope of Corollary 9. The valid single-filter lemma therefore cannot be iterated to replace all noncritical states by critical representatives.

S7 Proofs of the stability theorems

This section proves the global and local stability results of Appendix D of the paper. We first write the surrogate Q intrinsically in the magic matrix and prove the sharp global overlap bound. We then compute the transverse quadratic forms and obtain explicit local radii. The two sharp critical-locus theorems of Appendix D.1 are proved in Section S8.

Theorem 19 (Explicit global LU-orbit stability; from Theorem 18). *For every unit pure four-qubit state,*

$$d(\psi)^2 \le \frac{12}{6+\sqrt{6}}\, Q(\psi), \qquad \boxed{\delta(\psi) \ge k\,\frac{6+\sqrt{6}}{12}\, d(\psi)^2}, \qquad k\,\frac{6+\sqrt{6}}{12} > \frac{1}{20}.$$

For every critical state the optimal bound is $\delta \ge C_* d^2$, with $C_* \approx 0.19058$ (Theorem 21). On an explicit neighbourhood of the two orbits the coefficient improves:

$$d(\psi) \le 1/20 \quad \Longrightarrow \quad \delta(\psi) \ge \frac{3}{10} d(\psi)^2.$$

Its sharp limiting coefficient is $c_* = (2-\ln 3)/(2\ln 2)$, in the precise sense of (S7.12). The power two is optimal. No optimal global coefficient is claimed; the equality states of Theorem 21 show that it is at most C_*.

The global bounds are proved at the end of Section S7.2, the explicit radius and the limiting coefficient in Section S7.4; the critical coefficient C_* is Theorem 21, proved in Section S8.

S7.1 The intrinsic moment identity

Here is a convenient arbitrary-state identity. For a state with magic matrix Z (Appendix B of the paper) put $s = \operatorname{tr}(Z^T Z)$, $\mathcal{N} = ZZ^T Z - sZ/2$, and $C = \sum_j e_3(\rho_{1j})$, where e_3 is the third elementary symmetric polynomial of the eigenvalues. Then

$$C = \frac{3}{2}\|\operatorname{adj} Z\|_F^2 + \frac{1}{2}\|\mathcal{N}\|_F^2, \qquad Q = 5 - 36C \quad (\|Z\|_F = 1). \tag{S7.1}$$

For a direct derivation let D_{12}, D_{13}, D_{14} be the three flattening determinants in the computational basis, and set $J = \operatorname{tr}((Z^T Z)^2)/4 - s^2/8$. Expanding the four-by-four determinants in the stated magic basis gives $D_{12} = \det Z$, $D_{13} = (\det Z + J)/2$, $D_{14} = (-\det Z + J)/2$. If M_{1j} is the flattening matrix with $\rho_{1j} = M_{1j}M_{1j}^\dagger$, then $e_3(\rho_{1j})$ is the sum of the 3×3 principal minors of $M_{1j}M_{1j}^\dagger$, which by Cauchy–Binet is $\|\operatorname{adj} M_{1j}\|_F^2$; the holomorphic gradient of $\det M_{1j}$ is the transposed adjugate, and the flattening and the magic change of basis are unitary reshapings, so $e_3(\rho_{1j}) = \|\nabla D_{1j}\|^2$. Here $\nabla D_{12} = \nabla \det Z$ has norm $\|\operatorname{adj} Z\|_F$, and $\nabla J = \mathcal{N}$. By the parallelogram law $\|\nabla D_{13}\|^2 + \|\nabla D_{14}\|^2 = \frac{1}{2}(\|\operatorname{adj} Z\|_F^2 + \|\mathcal{N}\|_F^2)$, and adding $\|\nabla D_{12}\|^2$ proves the first identity in (S7.1). The second is Newton's identity $6e_3(\rho) = 1 - 3\operatorname{tr}\rho^2 + 2\operatorname{tr}\rho^3$ for $\operatorname{tr}\rho = 1$, summed over $j = 2,3,4$: $36C = 18 - 18P + 12T$, while $Q = 18P - 12T - 13$ for unit states.

Section S8.1 also gives a constructive distance certificate from three chosen Bell-pair products.

S7.2 Closest-orbit geometry and the sharp overlap bound

We prove the global overlap comparison by a rational polynomial certificate on a normal slice. The first step shows that this slice contains a closest representative of every state; the second identifies the constraints that make the certificate nonnegative there.

Theorem 18 (Sharp HS overlap stability; exact rational certificate, GO-1–GO-3). *For every unit pure four-qubit state,*

$$Q(\psi) \geq 1 - \Phi(\psi) \geq \frac{6 + \sqrt{6(1 + \sqrt{\tau(\psi)})}}{12}\, d(\psi)^2 \geq \frac{6 + \sqrt{6}}{12}\, d(\psi)^2, \qquad \frac{6 + \sqrt{6}}{12} > \frac{7}{10}.$$

Equality in $Q \geq 1 - \Phi$ holds on $\mathcal{M}$ ($Q = 0$, $\Phi = 1$) and on the three Bell-pair-product LU orbits ($Q = \Phi = 1/2$), so the coefficient one is optimal.

Coverage by the normal slice. Take a closest HS representative and transform it to $Z_* = \mathrm{diag}(\sqrt{3}, i, i, i)/\sqrt{6}$ (here and in the rest of the supplement, magic-matrix indices run over $0, \ldots, 3$, index i labelling the magic vector m_{i+1} of Appendix B of the paper); choose phase so $h = \langle Z_*, Z\rangle \geq 0$ is real. Up to global phase, the union $\mathcal{M}$ is the $O(4) \times O(4)$ orbit of Z_*: the proof of Proposition 4 of the paper puts every HS state in the $SO(4) \times SO(4)$ orbit of Z_* or of $\overline{Z_*}$, and $\mathrm{diag}(1, -1, -1, -1)$ on one side maps Z_* to $\overline{Z_*}$. The arbitrary-state identity (S7.1) gives

$$\begin{aligned} Q(Z) &= 27N\,\mathrm{tr}[(ZZ^\dagger)^2] - 18\,\mathrm{tr}[(ZZ^\dagger)^3] - 18\|ZZ^TZ - sZ/2\|_F^2 - 4N^3, \\ s &= \mathrm{tr}(Z^TZ), \\ W_1(Z) &= \|ZZ^\dagger - \overline{ZZ^\dagger}\|_F^2 + \|Z^\dagger Z - \overline{Z^\dagger Z}\|_F^2. \end{aligned} \tag{S7.2}$$

The W_1 formula follows because the six one-qubit Pauli matrices on either magic pair form an orthogonal basis of imaginary skew matrices, each with Frobenius norm squared four (BF-2). These formulas show directly that N, Q, W_1 and s are invariant under both real orthogonal factors. Thus the transformation preserves every quantity needed below.

A closest representative maximises the real height $\sqrt{6}\,\mathrm{Re}\langle Z_*, Z\rangle = 3X_{00} + \sum_j Y_{jj}$ over phase and both orthogonal factors. Write $Z = \sqrt{3}X + iY$ with X, Y real. The first-order conditions for the six left and six right generators and for the phase have rank ten (simultaneous lower rotations fix Z_*), and are exactly

$$Y_{00} = \sum_j X_{jj}, \qquad X_{j0} = Y_{0j}/3, \qquad Y_{j0} = 3X_{0j}, \qquad Y_{jk} = Y_{kj} \tag{S7.3}$$

($1 \leq j < k \leq 3$ in the last relation). Simultaneous lower rotations preserve (S7.3) and conjugate the symmetric matrix $\mathrm{Sym}(X_{jk}) + 2(Y_{jk})$, so one of them makes it diagonal: three further linear conditions (NSL-1–NSL-3). Consequently every closest-orbit problem has a representative $Z = \sqrt{3}X + iY$ with exactly the following 19 real coordinates:

$$\begin{aligned} X_{00} &= t, & Y_{00} &= u_1 + u_2 + u_3, \\ X_{jj} &= u_j, & Y_{jj} &= t + a_j, \\ X_{0j} &= p_j, & Y_{0j} &= q_j, \\ X_{j0} &= q_j/3, & Y_{j0} &= 3p_j, \\ X_{jk} &= e_{jk} + b_{jk}, & X_{kj} &= e_{jk} - b_{jk}, \\ Y_{jk} &= Y_{kj} = -e_{jk}/2 \qquad (1 \leq j < k \leq 3). \end{aligned}$$

Here and below $1 \leq j \leq 3$ for cross-block and diagonal entries. Set

$$h_0 = 6t + a_1 + a_2 + a_3 = \sqrt{6}h, \quad N = 3\|X\|_F^2 + \|Y\|_F^2, \quad e = N - h_0^2/6.$$

For a normalised state, $e = 1 - \Phi$. This coverage does not assume criticality or distinct singular values.

Positive constraints at a closest point. For any unit HS ray η put

$$g_\eta = h^2 - |\langle \eta, Z\rangle|^2 \geq 0.$$

Write $Y_3 = (Y_{jk})_{j,k=1}^3$.

Fixing either orthogonal factor and reflecting the other across an arbitrary real hyperplane gives the stronger one-factor Procrustes constraints

$$K_0 = \begin{pmatrix} 3t & q^T \\ q & Y_3 \end{pmatrix} \succeq 0, \qquad K_1 = \begin{pmatrix} 3t & 3p^T \\ 3p & Y_3 \end{pmatrix} \succeq 0.$$

Indeed these are $\sqrt{6}\,\mathrm{Re}(ZZ_*^\dagger)$ and $\sqrt{6}\,\mathrm{Re}(Z_*^\dagger Z)$, respectively, and both are symmetric on the normal slice. A reflection $I - 2vv^T$ changes the real height by $-2v^T K_i v$, which cannot be positive at a closest point. Reflections are admissible because both HS orbits are included. Their principal blocks also give $t \geq 0$ and $Y_3 \succeq 0$. These constraints retain the correlation between each cross block and the positive lower block.

Let G_{ij} be the real skew matrix with entries $(i,j) = 1$, $(j,i) = -1$. Use the following ten infinitesimal actions on Z: left $Z \mapsto G_{0j}Z$ and right $Z \mapsto -ZG_{0j}$, left G_{jk} for $1 \leq j < k \leq 3$, and phase multiplication by $i\sqrt{3}$. Denote them by T_ℓ. The height Hessian satisfies

$$H_{\ell m}(Z) = -\sqrt{6}\,\mathrm{Re}\left\langle Z, \frac{T_\ell T_m + T_m T_\ell}{2} Z_* \right\rangle, \qquad H(Z) \succeq 0. \tag{S7.4}$$

Indeed the chosen representative maximises real height along each group curve. Every entry of H is rational linear in the 19 coordinates.

A final family tests a continuous set of lower-block axes. Put $S = \mathrm{Sym}(X_{jk})_{j,k=1}^3$. For $\sigma_0, \sigma \in \{\pm 1\}$ and $(c,s) \in \{(1,0), (1/2,1/2), (-1/2,1/2), (-1,0), (-1/2,-1/2), (1/2,-1/2)\}$, define

$$r_0 = \sigma_0 Y_{00} + \sigma\,\mathrm{tr}\,Y_3, \qquad i_0 = -\sigma_0 X_{00} - \sigma\,\mathrm{tr}\,S,$$
$$L_{\sigma_0,\sigma,c,s} = (h_0 - cr_0 - 3si_0)I - c(3S - \sigma Y_3) - 3s(Y_3 + \sigma S).$$

Then $L_{\sigma_0,\sigma,c,s} \succeq 0$. To prove this, for each real unit $v \in \mathbb{R}^3$ use the HS matrix with top component $\sigma_0 i/\sqrt{6}$ and lower block $vv^T/\sqrt{2} + \sigma i(I - vv^T)/\sqrt{6}$. The closest-height inequality at phase ϕ, where $\cos\phi = c$ and $\sin\phi = \sqrt{3}s$, is exactly $v^T L v \geq 0$.

The fidelity floor in Lemma S27 also gives a positive quadratic matrix. Put

$$s_R = 3\|X\|_F^2 - \|Y\|_F^2, \quad s_I = 2\sum_{ij} X_{ij}Y_{ij}, \quad \ell = h_0^2 - N, \qquad B_0 = \begin{pmatrix} \ell + s_R & s_I \\ s_I & (\ell - s_R)/3 \end{pmatrix} \succeq 0.$$

Indeed $s = s_R + i\sqrt{3}s_I$ and the homogeneous fidelity floor gives $\ell \geq |s|$. Both diagonal entries of B_0 are nonnegative and $3\det B_0 = \ell^2 - |s|^2 \geq 0$.

The positive module. Below, $n = N = 3\|X\|_F^2 + \|Y\|_F^2$ denotes the homogeneous norm squared, so every block has total degree six. Let $\mathcal{A}$ average over simultaneous lower signed permutations. They preserve Z_*, the normal slice and all closest-point inequalities. The target polynomial is invariant under this averaging. For a positive matrix B and a positive rational Gram matrix G, write

$$\mathcal{T}_G(B; v) = \sum_{a,b} G_{ab} v_a^T B v_b \geq 0.$$

A factorisation of G makes this a sum of nonnegative quadratic forms; no exact factorisation is required in the certificate checker. The weighted matrices used below are

$$\mathcal{C}_0 = \Big\{1, \frac{g_\eta}{6}, \frac{h_0 t}{6}, \frac{B_0}{36}, \frac{nh_0 H}{216}, \frac{nh_0 Y_3}{36}, \frac{h_0 H}{216}, \frac{h_0 Y_3}{36}, \frac{h_0 K_0}{216}, \frac{h_0 K_1}{216}\Big\}. \tag{S7.5}$$

Here η ranges over the supplied HS rays. All members are positive at a closest point. The supplied HS rays are checked independently by $N = 1, W_1 = \tau = Q = 0$, so the established equality classification proves that every one is HS.

Lemma S27 (An HS fidelity floor from real singular values)**.** *For every unit pure four-qubit state,*

$$\Phi \geq \frac{1+\sqrt{\tau}}{6} \geq \frac{1}{6}, \qquad d^2 \leq 2\left(1 - \sqrt{\frac{1+\sqrt{\tau}}{6}}\right) \leq 2(1 - 1/\sqrt{6}).$$

The fidelity inequality is attained by the GHZ state, for which $\tau = 1$ and $\Phi = 1/3$.

Proof. Choose phase so that $s = \operatorname{tr}(Z^T Z)$ is real and nonnegative. Then $\|\operatorname{Re} Z\|_F^2 = (1 + |s|)/2$. Apply a real singular-value decomposition to $\operatorname{Re} Z$, writing its nonnegative diagonal entries as $m \geq x \geq y \geq z$. The transformed imaginary part can be arbitrary. Two HS representatives, with large real component $1/\sqrt{2}$ in the m position and the other three components all $+i/\sqrt{6}$ or all $-i/\sqrt{6}$, give

$$\Phi \geq \tfrac{1}{2}m^2 + \tfrac{1}{6}(x+y+z)^2 \geq \tfrac{1}{3}(m^2+x^2+y^2+z^2).$$

The first inequality follows by averaging their squared overlaps; the imaginary diagonal entries contribute further nonnegative squares. For the second, six times its difference is

$$m^2 + (x+y+z)^2 - 2(x^2+y^2+z^2) = (m^2-x^2) + y(2x-y) + z(2x-z) + 2yz \geq 0.$$

Both real orthogonal transformations preserve the phase-inclusive union of the two HS orbits, so these comparisons are admissible. Finally $|s| = \sqrt{\tau}$, since $\tau = |\operatorname{tr} Z^T Z|^2$ in the magic basis. For GHZ the magic diagonal is $(1,0,0,-1)/\sqrt{2}$, signed-permutation equivalent to $(1,1,0,0)/\sqrt{2}$. Formula (S8.2) gives maximum squared overlap $1/3$ over the 32 HS representatives (RP-3), proving the equality claim. □

Proof of Theorem 18. Place a closest HS representative at $Z_* = \operatorname{diag}(\sqrt{3}, i, i, i)/\sqrt{6}$ so that the state has a representative $Z = \sqrt{3}X + iY$ on the 19-real-coordinate stationary normal slice (S7.3), with $h_0 = 6t + a_1 + a_2 + a_3 = \sqrt{6}h \geq 0$, $n = 3\|X\|_F^2 + \|Y\|_F^2$, and $e = n - h_0^2/6$. For a unit state, $n = 1$ and $e = 1 - \Phi(\psi) \geq 0$ by Cauchy–Schwarz. The homogeneous sextic target $(Q - n^2 e)/36$ is a polynomial in the 19 slice coordinates.

Enlarge the positive module $\mathcal{C}_0$ of (S7.5) to

$$\mathcal{C}_Q = \mathcal{C}_0 \cup \left\{ \frac{e}{6},\ \frac{h_0 L_{\sigma_0,\sigma,c,s}}{216} \right\}, \tag{S7.6}$$

where η ranges over 50 supplied unit HS rays, checked independently in GO-1 by $n = 1$ and $Q = W_1 = \tau = 0$. Only the four lower-frame choices

$$(\sigma_0, \sigma, c, s) \in \{(-1,-1,-1/2,1/2),\ (-1,1,1/2,1/2),\\ (1,-1,-1,0),\ (1,1,1,0)\}$$

are needed in this module. The additional scalar constraint $e/6 \geq 0$ is multiplied by quadratic polynomial vectors in the slice coordinates, and the lower-frame matrices $h_0 L_{\sigma_0,\sigma,c,s}/216 \succeq 0$ are paired with quadratic matrix multipliers; on the $B_0/36$ and lower-frame blocks where the simultaneous signed lower-permutation representation splits into conjugate two-dimensional S_3 characters, both paired character components are retained in the polynomial-vector bases p_B.

The certificate supplies 277 blocks $B \in \mathcal{C}_Q$ with explicit rational polynomial-vector bases p_B and positive rational Gram matrices G_B such that, with the two explicit nonnegative terms below, the polynomial identity

$$\frac{Q - n^2 e}{36} = R_{\mathrm{sq}} + R_{\mathrm{ax}} + \mathcal{A} \sum_{B \in \mathcal{C}_Q} \mathcal{T}_{G_B}(B; p_B) \tag{S7.7}$$

holds identically in $\mathbb{Q}[x_1,\dots,x_{19}]$ across all 3,277 degree-≤ 6 signed-permutation monomial orbits (including the six dependent core equations as well as the 3,271 independent ones). GO-2 checks every Gram matrix $G_B \succ 0$ by the compact integer test below, expands every block, applies the signed Reynolds average $\mathcal{A}$, and verifies that all 3,277 orbit coefficients match $(Q - n^2 e)/36$ identically over $\mathbb{Q}$. Evaluating (S7.7) at a closest representative of a unit state gives $Q(\psi) \ge e = 1 - \Phi(\psi)$.

Analytic replacement of the linear lower-frame blocks. Put $\gamma = 10^{-7}$, $\omega = 794424/10^{12}$ and define

$$\begin{aligned} J &= 5\left[\sum_j (p_j^2 + q_j^2) + \sum_{j<k}(e_{jk}^2 + b_{jk}^2)\right] + 2\sum_{j<k}\left[(u_j - u_k)^2 + (a_j - a_k)^2\right], \\ z &= (b_{23}, -b_{13}, b_{12})^T, \qquad L_* = L_{-1,-1,-1/2,1/2}, \\ R_{\mathrm{sq}} &= \frac{\gamma}{27} n h_0^2 J, \qquad R_{\mathrm{ax}} = \frac{\omega}{216} n h_0 z^T L_* z. \end{aligned}$$

The first term is manifestly nonnegative, and the second is nonnegative because $n, h_0 \ge 0$ and $L_* \succeq 0$. Together they replace all 96 linear lower-frame Gram blocks, so none of those blocks is supplied numerically.

The reduction follows from linearity in L. For each of four common polynomial-vector bases, the 24 lower-frame matrices have the same Gram contribution γI, and

$$\sum_{\sigma_0,\sigma,\,(c,s)} L_{\sigma_0,\sigma,c,s} = 24 h_0 I,$$

where the sum is over both independent signs and the six phase pairs used in the lower-frame construction. The signed-permutation average of the sum of squared components of those four bases is $J/3$: coefficients $(2,2,1)$ on each of the p, q, e, b triples average to $5/3$, and coefficients $(1,1,0)$ on each of the u, a difference triples average to $2/3$. Their contribution is $(\gamma n h_0/216)(24 h_0)(J/3) = R_{\mathrm{sq}}$. The only additional Gram contribution is $\omega v v^T$, with $v = (0,1,0,-1,0,1)$ in the six-vector basis whose selected components are b_{23}, b_{13}, b_{12}. Its contribution is R_{ax}. Both remainders are invariant under the signed lower permutations: z is the axial vector of the real antisymmetric lower block, so it transforms by $(\det O)O$, while L_* transforms by conjugation. GO-2 independently checks their invariance and the complete reduced polynomial identity. The lower-frame conditions used by the finite certificate are consequently reduced from 24 to 4.

Compact integer Gram data. The remaining Gram matrices and their positivity witnesses are stored compactly. Write $g = 10^{12}$, c for the supplied correction denominator, and $s = 2^{46}$. For each block of size m, the data consist of a rectangular integer matrix A, symmetric integer matrices Δ, C (the latter sparse), and the reconstruction rule

$$H_{ij} = \left\lfloor \frac{g(AA^T)_{ij}}{s^2} + \frac{1}{2} \right\rfloor, \qquad G_{ij} = \frac{H_{ij} + \Delta_{ij} + C_{ij}/c}{g}.$$

Put $D_{ij} = c\Delta_{ij} + C_{ij}$. The only positivity test is

$$2\left(D_{ii} - \sum_{j\ne i} |D_{ij}|\right) > mc \qquad \text{for every } i. \tag{S7.8}$$

Indeed each rounding error $H_{ij} - g(AA^T)_{ij}/s^2$ has absolute value at most $1/2$. Thus (S7.8) makes $G - AA^T/s^2$ strictly diagonally dominant with positive diagonal. Its positive definiteness follows, for example, by applying $2|v_i v_j| \le v_i^2 + v_j^2$ to its quadratic form; adding $AA^T/s^2 \succeq 0$ proves $G \succ 0$. No square roots, eigenvalues, divisions in an ordered field, or matrix factorisation are needed in this finite check: rounding is integer quotient and the final tests are integer inequalities. The 277 numerical blocks use 2,222 factor columns and 178,221 factor entries; 11 need no factor. The exact polynomial identity uses 50 HS comparisons and the two explicit remainders (GO-1–GO-2).

Integer identity check. All basis coefficients have denominators dividing 6 and all constraint coefficients become integral after multiplication by the supplied map denominator, so GO-2 compares the identity at a common integer scale, keeping the integer Gram parts $H+\Delta$ and the sparse corrections C in separate integer accumulators; the correction denominator c enters only the final coefficient comparisons.

Finally, since $d(\psi)^2 = 2(1-\sqrt{\Phi(\psi)})$, we have

$$1-\Phi(\psi) = \frac{1+\sqrt{\Phi(\psi)}}{2}\, d(\psi)^2.$$

Lemma S27 gives $\Phi(\psi) \geq (1+\sqrt{\tau(\psi)})/6 \geq 1/6$, which yields the two middle inequalities in (4). Because $(2/5)^2 = 4/25 < 1/6$, we have $1/\sqrt{6} > 2/5$ and $(6+\sqrt{6})/12 = (1+1/\sqrt{6})/2 > (1+2/5)/2 = 7/10$ (GO-3). Attainment of $Q = 1-\Phi$ at the HS orbits and the Bell-pair-product orbits is established in Theorem 20. □

Global part of Theorem 19. Theorem 18 gives $d^2 \leq 12Q/(6+\sqrt{6})$, and (3) gives $\delta \geq kQ \geq k\frac{6+\sqrt{6}}{12}d^2$. With $k > 16/225$ and $\sqrt{6} > 2449/1000$, $k(6+\sqrt{6})/12 > \frac{16}{225}\cdot\frac{8449}{12000} > \frac{1}{20}$ (GO-4). □

S7.3 The transverse quadratic form and an explicit local radius

Use the magic representative $Z_* = \mathrm{diag}(a, ib, ib, ib)$, $a = 1/\sqrt{2}$, $b = 1/\sqrt{6}$, which belongs to one of the HS orbits. For a perturbation write

$$H = \begin{pmatrix} c & u^T \\ v & D \end{pmatrix}, \quad S = (D+D^T)/2, \quad A = (D-D^T)/2.$$

The coefficient $Q_2(H)$ of t^2 in $Q(Z_* + tH)$ is

$$\begin{aligned} Q_2(H) = {} & 6(|c|^2 + \|S\|_F^2 - |ac - ib\,\mathrm{tr}\,S|^2) + 20\|\mathrm{Re}\,A\|_F^2 \\ & + 30\sum_{j=1}^{3}[(a\,\mathrm{Im}\,u_j + b\,\mathrm{Re}\,v_j)^2 + (b\,\mathrm{Re}\,u_j + a\,\mathrm{Im}\,v_j)^2]. \end{aligned} \tag{S7.9}$$

This is an explicit orthogonal decomposition. On the complex symmetric block together with c, it is six times the norm squared perpendicular to the complex line through Z_*. The real antisymmetric block has coefficient twenty; the imaginary antisymmetric block is tangent to the orbit. For each cross-block pair the two displayed linear forms are orthogonal and have norm squared $a^2 + b^2 = 2/3$, giving coefficient twenty on the two normal directions. The other two directions are the left and right real-rotation tangents. Thus the real quadratic-form operator has eigenvalues $0, 6, 20$ with multiplicities $11, 12, 9$. Its kernel is exactly the radial direction and the tangent space of $\mathcal{M}$ (including phase). The actual Hessian eigenvalues are twice these quadratic coefficients.

For clarity, the formula can be derived with only quadratic matrix expansions. On each cut let C and B be the flattenings of Z_* and H in the computational basis, and put $R = CC^\dagger$, $L = CB^\dagger + BC^\dagger$, $V = BB^\dagger$. Then

$$P_2 = \sum_{\text{cuts}} \mathrm{tr}(L^2 + 2RV), \quad T_2 = 3\sum_{\text{cuts}} \mathrm{tr}(R^2V + RL^2), \quad Q_2 = 18P_2 - 12T_2 - 21\|H\|_F^2 - 3n_1^2,$$

where $n_1 = 2\,\mathrm{Re}\langle Z_*, H\rangle$; substitution gives (S7.9). This also specifies every matrix entry of the quadratic form without a numerical eigendecomposition.

To bound all higher terms, represent $Q(z)$ as a matrix element on $z^{\otimes 3}$. Each $N\,\mathrm{tr}\,R_j^2$ is a two-copy swap matrix element with an identity third copy, and each $\mathrm{tr}\,R_j^3$ is a three-cycle matrix element. Taking Hermitian parts if necessary gives an operator of norm at most $18\cdot 3 + 12\cdot 3 + 13 = 103$. Consequently, for $d = \|H\| \leq r$,

$$|Q(Z_* + H) - Q_2(H)| \leq 103 d^3(20 + 15r + 6r^2 + r^3). \tag{S7.10}$$

The constant and linear terms vanish because Q is nonnegative and zero at Z_*. The binomial coefficients count the six tensor slots.

Take a closest $\chi \in \mathcal{M}$ to ψ and $H = \psi - \chi$. Stationarity makes H real-orthogonal to the orbit tangent, which contains the phase direction $i\chi$, so $\langle \chi, H\rangle$ is real, while $\mathrm{Re}\langle \chi, H\rangle = -d^2/2$. Therefore $Q_2(H) \geq 6(d^2 - d^4/4)$. The same bound applies at either orbit by LU invariance and conjugation. Inserting $r = 2^{-10}$ in (S7.10) proves $Q \geq 3d^2$ for $d \leq 2^{-10}$, used for the local spectral-majorant transfer of Corollary 22 of the paper.

S7.4 The sharp entropy Hessian and a larger local neighbourhood

Let

$$c_* = \frac{2 - \ln 3}{2 \ln 2}.$$

The transverse quadratic coefficients of the entropy gap are

coefficient	c_*	$1/\ln 2$	$(2 + 3\ln 3)/(3\ln 2)$
real multiplicity	10	2	9

(S7.11)

(the Hessian eigenvalues are twice these coefficients). The ten orbit tangents are its kernel on the unit sphere. In particular

$$\lim_{r \downarrow 0} \inf_{0 < d(\psi) \leq r} \frac{\delta(\psi)}{d(\psi)^2} = c_*. \tag{S7.12}$$

The Hessian determines the limiting coefficient. To obtain a bound on a whole neighbourhood, we also control the higher-order terms through the following radius estimate.

Lemma S28 (Explicit local radius)**.** *For $0 < r \leq 1/20$ put*

$$\epsilon_r = 2\sqrt{6}\, r + 6r^2, \qquad c(r) = \frac{1 - r^2/4}{3 \ln 2} \left[(3 - \epsilon_r) \left(1 - \sqrt{3}\, r - \frac{r^2}{2} \right)^2 - \frac{3}{2} \ln 3 \right].$$

Then

$$d(\psi) \leq r \quad \Longrightarrow \quad \delta(\psi) \geq c(r) d(\psi)^2, \qquad c(1/20) > 3/10. \tag{S7.13}$$

The bracket is positive and decreasing on this interval, and $c(r) \to c_*$.

Proof. Fix a closest $\chi \in \mathcal{M}$, and write $\psi = h\chi + s\nu$, where $h = 1 - d^2/2$, $s^2 = d^2 - d^4/4$, $\|\nu\| = 1$. The vector ν is complex-orthogonal to χ and real-orthogonal to its orbit tangents. On cut j write C_j, B_j for the flattenings of χ, ν, and set

$$\sigma_j = C_j C_j^\dagger = I/6 + \Pi_j/3, \quad L_j = C_j B_j^\dagger + B_j C_j^\dagger, \quad V_j = B_j B_j^\dagger, \quad K = \sum_j \Pi_j \otimes I - \frac{3}{2} I.$$

The sum is over the three cuts, with Π_j acting on the corresponding pair containing qubit one. Direct substitution at Z_* gives $K\chi = 0$. The relative entropy identity, with natural logarithms, is exact:

$$3 \ln 2\, \delta(\psi) = \ln 3\, \langle \psi, K\psi\rangle + \sum_j D(\rho_j \| \sigma_j). \tag{S7.14}$$

Let $\mathcal{L}_j$ be the derivative of matrix logarithm at σ_j, and put $U(\nu) = \sum_j \mathrm{tr}(L_j \mathcal{L}_j(L_j))$. The quadratic entropy term is most simply diagonalised before estimating U. Put $\Pi = \Pi_j$, $\Pi^\perp = I - \Pi$ on each cut and define the rational forms

$$\mathfrak{a} = \sum_j (\mathrm{tr}[(\Pi L_j \Pi)^2] + 3\,\mathrm{tr}[(\Pi^\perp L_j \Pi^\perp)^2]), \qquad \mathfrak{b} = \langle \nu, K\nu\rangle + 3 \sum_j \|\Pi L_j \Pi^\perp\|_F^2.$$

The divided differences of ln at $1/2, 1/6$ give

$$U = 2\big[\mathfrak{a} + \ln 3\, (\mathfrak{b} - \langle \nu, K\nu\rangle)\big].$$

Thus the coefficient of s^2 in (S7.14) is $(\mathfrak{a} + \ln 3\, \mathfrak{b})/(3 \ln 2)$: the K terms cancel.

Here are explicit coordinates for the full real normal space at Z_*:

$$H = \begin{pmatrix} i\sqrt{3}\, z & (b\xi + ia\eta)^T \\ b\eta + ia\xi & zI_3 + E + A \end{pmatrix}, \qquad a = 1/\sqrt{2}, \quad b = 1/\sqrt{6}, \tag{S7.15}$$

where $z \in \mathbb{C}$, E is complex symmetric and traceless, A is real antisymmetric, and $\xi, \eta \in \mathbb{R}^3$. Indeed complex orthogonality to Z_* imposes $H_{00} = i\sqrt{3}\, z$; orthogonality to lower rotation tangents makes the antisymmetric part real; the two cross-block tangents impose the displayed relations. These mutually orthogonal blocks have real dimensions $10, 2, 9$ and norm squares

$$N_0 = \|E\|_F^2, \qquad N_1 = 6|z|^2, \qquad N_2 = \|A\|_F^2 + \tfrac{2}{3}(|\xi|^2 + |\eta|^2).$$

Substitution in the cut formulas gives

$$\mathfrak{a} = 3N_0 + 3N_1 + 2N_2, \qquad \mathfrak{b} = -\tfrac{3}{2}N_0 + 3N_2. \tag{S7.16}$$

This proves the three coefficients and multiplicities in (S7.11) without spectral projectors. For reproducibility, use the two unit traceless diagonal matrices and three unit symmetric off-diagonal matrices in E, and their imaginary multiples; use $z = 1/\sqrt{6}, i/\sqrt{6}$; use the three unit antisymmetric matrices in A; finally use each coordinate axis for ξ, η, scaled by $\sqrt{3/2}$. EH-1 substitutes this orthonormal basis in all cut forms and checks every coefficient of (S7.16).

Direct substitution gives $K = -3/2$ on the E block and $K = 0$ on the z block. Hermiticity makes their orthogonal complement invariant. On that complement $K \le 3/2$, since it is the sum of three projectors minus $3I/2$. Consequently $U = 6$ per unit norm on the first two blocks ($\mathfrak{b} - \langle \nu, K\nu \rangle = 0$ there), and $U \ge 4 + 3\ln 3 > 6$ per unit norm on the last block. By (S7.16), $\mathfrak{a}$ and $\mathfrak{b}$ have no cross terms between the three blocks, and neither has $\langle \nu, K\nu \rangle$, since the first two blocks are eigenspaces of K; so $U \ge 6\|\nu\|^2$ on all normals, the least value being on the E and z blocks. The radius estimate below needs only the bound $K \ge -3/2$, which holds on the whole space by the definition of K.

For the radius bound use the norm induced by $\mathcal{L}_j$. Since $\|\mathcal{L}_j\| \le 6$, $\operatorname{tr} \sigma_j^2 = 1/3$ and $\operatorname{tr}(V_j \sigma_j) \ge 1/6$,

$$\sum_j \|V_j - \sigma_j\|_{\mathcal{L}_j}^2 \le 6 \sum_j \operatorname{tr}[(V_j - \sigma_j)^2] \le 18.$$

Also $\rho_j \le (1 + \epsilon_r)\sigma_j$: write the cut matrix as $C_j + E_j$ with $\|E_j\| \le d$. The least singular value of C_j is $1/\sqrt{6}$, so $\|C_j^{-1} E_j\| \le \sqrt{6}\, d$, and $\rho_j = C_j(I + C_j^{-1}E_j)(I + C_j^{-1}E_j)^\dagger C_j^\dagger \le (1 + \sqrt{6}\, r)^2 \sigma_j = (1 + \epsilon_r)\sigma_j$. The resolvent formula for the logarithmic derivative implies $\|X\|_{\mathcal{L}_A}^2 \ge a^{-1}\|X\|_{\mathcal{L}_\sigma}^2$ when $A \le a\sigma$. To see the order directly, inverse resolvents are ordered, their tensor squares are ordered, and integration with a rescaled variable gives the factor a^{-1}. Twice integrating the entropy Hessian along the segment therefore gives

$$D(\rho_j \| \sigma_j) \ge \left(\int_0^1 \frac{1 - v}{1 + v\epsilon_r}\, dv \right) \|\rho_j - \sigma_j\|_{\mathcal{L}_j}^2 \ge \left(\frac{1}{2} - \frac{\epsilon_r}{6} \right) \|\rho_j - \sigma_j\|_{\mathcal{L}_j}^2.$$

Now $\rho_j - \sigma_j = hsL_j + s^2(V_j - \sigma_j)$. The reverse triangle inequality in the sum of the three induced norms gives, since $h\sqrt{6} \ge s\sqrt{18}$ for $r \le 1/20$,

$$\sum_j \|\rho_j - \sigma_j\|_{\mathcal{L}_j}^2 \ge s^2 (h\sqrt{6} - s\sqrt{18})^2.$$

Insert this into (S7.14), use $\langle \psi, K\psi \rangle = s^2 \langle \nu, K\nu \rangle \ge -3s^2/2$, and substitute $h \ge 1 - r^2/2$, $s \le r$ and $s^2 \ge d^2(1 - r^2/4)$; since $(h\sqrt{6} - s\sqrt{18})^2 = 6(h - \sqrt{3}\, s)^2$ and $6(\frac{1}{2} - \frac{\epsilon_r}{6}) = 3 - \epsilon_r$, this proves (S7.13). The rational bounds $\ln 2 < 7/10$, $\ln 3 < 1099/1000$, $\sqrt{6} < 49/20$ and $\sqrt{3} < 1733/1000$ certify $c(1/20) > 3/10$ (EH-2).

Finally the unit normal $H = \operatorname{diag}(0, 1, -1, 0)/\sqrt{2}$ belongs to the ten-dimensional least-coefficient space. Along $(Z_* + tH)/\|Z_* + tH\|$, the distance to $\mathcal{M}$ is $|t| + O(t^2)$ and the gap is $c_* t^2 + O(t^3)$. This proves (S7.12), and also optimality of the power two. □

S8 Orbit stability: further proofs and results

We supplement the stability results of Appendix D of the paper and Section S7 with a constructive distance certificate, the proofs of the two sharp critical-locus theorems, a one-parameter refinement on $P = 1$, and the proof of the sharp entropy coefficient on the critical locus. Throughout, $\mathcal{M}$ is the union of the two HS local-unitary orbits, including global phase, and

$$d(\psi) = \min_{\chi\in\mathcal{M}} \|\psi - \chi\|, \qquad \Phi(\psi) = \max_{\chi\in\mathcal{M}} |\langle\psi, \chi\rangle|^2, \qquad d^2 = 2(1 - \sqrt{\Phi}).$$

For unit states, $Q = F + (P - 1)$ is the sextic moment defect and $\delta = E(M_4) - E$ is the entropy deficit, as in the paper.

S8.1 A direct certificate from three Bell-pair products

Let $\tilde{w} = \epsilon^{\otimes 4}\bar{w}$ be the spin flip, as in the paper; in the magic matrix coordinates it is ordinary conjugation (Section S3). Hence a unit state w with $\langle\tilde{w}, w\rangle = 0$ has the form $(x + iy)/\sqrt{2}$ for orthonormal real vectors x, y in the magic coordinates; put $V = \mathrm{span}_{\mathbb{R}}(x, y)$. A *Bell-pair product* is a product of two maximally entangled Bell states with arbitrary local unitaries on the four qubits, with its pairing specified; it is real up to phase in these coordinates.

Proposition S29 (Three-Bell-frame certificate)**.** *Let w be unit with $\langle\tilde{w}, w\rangle = 0$. Choose one unit Bell-pair product b_j in each of the three pairings, and put*

$$\varepsilon_j = 1 - 2|\langle b_j, w\rangle|^2, \qquad E_B = \varepsilon_1 + \varepsilon_2 + \varepsilon_3.$$

Then $\varepsilon_j \geq 0$ and

$$\boxed{d(w)^2 \leq \frac{9}{2} E_B.}$$

The vectors b_j need not maximise their overlaps. The proof constructs at most six candidate HS states, two for each unordered pair of pairings.

Proof. Choose phases so the b_j are real. Their squared distances to V are exactly ε_j. Bell-pair products in distinct pairings have $|\langle b_i, b_j\rangle| \leq 1/2$: in the magic matrix coordinates for the first pairing they are a real rank-one matrix and $O/2$, $O \in O(4)$. The proper and improper components describe the two other pairings. This description follows by expanding the standard Bell products and using $SU(2) \otimes SU(2) \to SO(4)$.

Let G be the 3×3 real Gram matrix of the b_j, and let λ_3 be its smallest eigenvalue. The rank-two projection on V gives $\lambda_3 \leq E_B$. Since $\mathrm{tr}\, G = 3$,

$$\mathrm{tr}\, G^2 \geq \frac{(3 - \lambda_3)^2}{2} + \lambda_3^2 \geq \frac{9}{2} - 3E_B.$$

Consequently, for $\Delta_{ij} = 1/4 - |\langle b_i, b_j\rangle|^2$,

$$\Delta_{ij} \geq 0, \qquad \sum_{i<j} \Delta_{ij} \leq \frac{3}{2} E_B. \tag{S8.1}$$

Fix a pair (i, j) and use real rotations to make $b_i = E_{00}$ and $b_j = O/2$. Adjust its real sign so $c = O_{00}/2 = |\langle b_i, b_j\rangle| \in [0, 1/2]$. The lower 3×3 block of O has singular values $1, 1, 2c$: its product with its transpose is $I - zz^T$, with $\|z\|^2 = 1 - 4c^2$. Choose its polar factor $R \in O(3)$, including a factor on any nullspace, and put $O' = \mathrm{diag}(1, R)$, $b'_j = O'/2$. The real plane $S = \mathrm{span}(b_i, b'_j)$ is spanned by E_{00} and $\mathrm{diag}(0, R)/\sqrt{3}$. Its two normalised complex-null rays are precisely HS states, since their magic matrices are $\mathrm{diag}(1/\sqrt{2}, \pm iR/\sqrt{6})$.

Put $S_0 = \mathrm{span}(b_i, b_j)$. Its Gram matrix has smallest eigenvalue $1 - c$. Consequently its leakage from V satisfies

$$\ell_{ij} := \mathrm{tr}(P_{S_0}(I - P_V)) \leq \frac{\varepsilon_i + \varepsilon_j}{1 - c} \leq 2(\varepsilon_i + \varepsilon_j).$$

The planes S_0, S share b_i. Their other principal-angle cosine is

$$r_c = \frac{1+c}{\sqrt{3(1-c^2)}} = \sqrt{\frac{1+c}{3(1-c)}}.$$

Indeed the polar-aligned lower block of O has trace $2+2c$. The distance squared between the nearer normalised null rays of these planes is $1-r_c$. Moreover

$$g(c) := 1 - r_c \le \frac{7}{4}(1/4 - c^2) = \frac{7}{4}\Delta_{ij}.$$

To check the last inequality, square its nonnegative right-hand comparison with r_c. After multiplication by $3(1-c)$ the difference is $(1-2c)[13 + c(525 - 462c) + 588c^3(1-2c)]/256 \ge 0$.

For any two real two-planes with principal-angle cosines s_1, s_2, choose orthonormal bases in which their overlap matrix has singular values s_1, s_2. Reversing an orientation replaces a null ray $(x+iy)/\sqrt{2}$ by its conjugate. Thus the largest overlap between their normalised null rays is $(s_1+s_2)/2$, and their minimum squared distance (after a phase choice) is $2 - s_1 - s_2 \le 2 - s_1^2 - s_2^2$, the projection leakage. The triangle inequality therefore gives $d(w) \le \sqrt{\ell_{ij}} + \sqrt{g(c)}$. Average its square over the three unordered pairs, use (S8.1), and apply Cauchy–Schwarz to the cross term:

$$d(w)^2 \le \left(\sqrt{\frac{4}{3}} + \sqrt{\frac{7}{8}}\right)^2 E_B \le \frac{9}{2} E_B,$$

since $(\sqrt{4/3} + \sqrt{7/8})^2 < 9/2$. The last inequality is strict when $E_B > 0$. □

For a general unit state choose phase and real orthonormal x, y so $\psi = \sqrt{(1+t)/2}\,x + i\sqrt{(1-t)/2}\,y$, where $t = |\langle\tilde{\psi}, \psi\rangle|$, and set $w = (x+iy)/\sqrt{2}$. At $t = 1$, choose any real unit y orthogonal to x. The same constructive certificate gives

$$d(\psi) \le \sqrt{2 - \sqrt{1+t} - \sqrt{1-t}} + \sqrt{9E_B(w)/2}.$$

This individual-state certificate requires only chosen Bell products and linear algebra; the entropy-only bound is proved in Section S7.2.

S8.2 The critical locus: signed Birkhoff reduction and real-density model

A state is critical when all four one-qubit marginals are $I/2$. On this locus, both the overlap comparison and its Hilbert-space distance version have optimal constants.

Theorem 20 (Sharp critical-state orbit stability; from Theorem 18, CO-1–CO-3). *For every critical unit pure four-qubit state, with Φ, d, Q, k as in Appendix D,*

$$\begin{aligned} Q(\psi) &\ge \max\{1 - \Phi(\psi),\ 3/2 - 2\Phi(\psi)\}, \\ Q(\psi) &\ge \frac{2+\sqrt{2}}{4} d(\psi)^2, \\ \delta(\psi) &\ge k\big(\max\{1-\Phi, 3/2 - 2\Phi\} + 5\tau/2\big) \ge k\,\frac{2+\sqrt{2}}{4}\,d(\psi)^2. \end{aligned}$$

The coefficient one in $Q \ge 1 - \Phi$, and the coefficient $(2+\sqrt{2})/4$ in (6), are both optimal: Bell-pair products attain them. The comparison $Q \ge 1 - \Phi$ holds for all states (Theorem 18). If the critical state also has $\tau = 0$, equivalently if $P = 1$, then

$$\Phi \ge \frac{2+\sqrt{3}}{6}, \qquad Q \ge \frac{3}{4-\sqrt{3}}(1-\Phi) \ge \frac{3+\sqrt{3}}{4} d^2.$$

These coefficients and the fidelity floor are optimal, attained by cluster states.

Section S8.3 below proves a one-parameter sharpening of (8) in the Bloch coordinates of (S8.4). The optimal entropy coefficient on the whole critical locus is larger: $\delta \geq C_* d^2$ with $C_* = 0.19057894701\ldots$ (9) (Theorem 21), optimal by its equality states; $k(2+\sqrt{2})/4$ in (7) is the weaker coefficient of the moment route.

We first remove the continuous orbit optimisation exactly. For complex diagonal matrices D_v, D_w of any common size,

$$\max_{A,B\in O(m)} |\operatorname{tr}(D_v^\dagger A D_w B^T)| = \max_{\pi,\,\varepsilon_i=\pm 1} \left|\sum_i \bar{v}_i \varepsilon_i w_{\pi(i)}\right|. \tag{S8.2}$$

Indeed $|A_{ij}B_{ij}|$ is doubly substochastic, by Cauchy–Schwarz on each row and column. Complete it to a doubly stochastic matrix; for any common phase the real overlap is bounded by the largest assignment sum of the absolute real coefficient entries. Birkhoff's permutation decomposition gives this bound, and signed permutation matrices attain it. Maximising the phase proves (S8.2).

The magic normal form makes a critical state D_v. Here and below the coordinates of $v \in \mathbb{C}^4$, and the rows and columns of 4×4 matrices, are indexed $0,\ldots,3$: v_i is the coefficient of $m_{i+1}\otimes m_{i+1}$ in $G(v)$ of Appendix B of the paper, so $D_v = \operatorname{diag}(v)$. Up to global phase, the union $\mathcal{M}$ is the $O(4)\times O(4)$ orbit of Z_*: a common orientation reversal in both factors fixes Z_* up to proper rotations, whereas a single reversal gives its conjugate up to proper rotations. Thus its overlap with D_v is exactly the maximum over the following 32 diagonal representatives:

$$\mathcal{H} = \Big\{\eta :\ \eta_i = 1/\sqrt{2},\quad \eta_j = \varepsilon_j i/\sqrt{6}\ (j\neq i),\quad i = 0,1,2,3\Big\}.$$

Its exact spin-flip-dependent covering profile is proved in Lemma S30 below.

The real cut transforms are $H_0 = I$ and

$$H_1 = \frac{1}{2}\begin{pmatrix} 1 & -1 & 1 & -1\\ -1 & 1 & 1 & -1\\ 1 & 1 & 1 & 1\\ -1 & -1 & 1 & 1\end{pmatrix}, \qquad H_2 = \frac{1}{2}\begin{pmatrix} 1 & -1 & -1 & -1\\ -1 & 1 & -1 & -1\\ 1 & 1 & -1 & 1\\ -1 & -1 & -1 & 1\end{pmatrix}.$$

With $\Delta = \operatorname{diag}(1,-1,1,-1)$, so that Δv are the coordinates (a,b,c,d) of G_{abcd}, these are the transforms $H_\pm$ of Section S5 conjugated by Δ: $H_1 = \Delta H_+ \Delta$, and H_2 is $\Delta H_- \Delta$ with its rows permuted and re-signed, which only permutes the squared moduli $|e_i^T H_2 v|^2$ of the cut spectrum.

Proof of Theorem 20. The comparison $Q(\psi) \geq 1-\Phi(\psi)$ holds for all states by Theorem 18. When $\Phi(\psi)\geq \frac{1}{2}$, $1-\Phi(\psi)\geq \frac{3}{2}-2\Phi(\psi)$, so (5) holds. Assume henceforth that $\Phi(v)\leq \frac{1}{2}$ for a unit critical normal vector $v\in\mathbb{C}^4$.

The three real orthogonal cut transforms $H_0 = I, H_1, H_2$ displayed above preserve $N = \|v\|^2 = 1$, $\tau = |\sum_i v_i^2|^2$ and $Q(v)$, and permute the 32 representatives $\mathcal{H}$ up to the sixth-root-of-unity phases $(\pm 1 \pm i\sqrt{3})/2$: for any $\eta\in\mathcal{H}$, one coordinate is $1/\sqrt{2}$ and three are $\pm i/\sqrt{6}$, so under H_1 or H_2 the row whose relative signs on those three coordinates match η has inner product $\frac{\pm 1\pm i\sqrt{3}}{2\sqrt{2}}$ and the other three rows have $\frac{\pm 1\pm i\sqrt{3}}{2}(\mp i/\sqrt{6})$ (CO-1). Thus H_0, H_1, H_2, coordinate permutations, global phase and coordinate sign flips preserve both $Q(v)$ and $\Phi(v)$. Replace v by $H_m v$ $(m\in\{0,1,2\})$ and permute coordinates so that $p_0 := |v_0|^2 = \max_{m,i}|e_i^T H_m v|^2$ is the largest of all twelve cut probabilities. Choose global phase so that $v_0 = r_0 > 0$, and flip signs $v_j \mapsto \pm v_j$ $(1\leq j\leq 3)$ so that $y_j := \operatorname{Im} v_j \geq 0$, writing $v_j = x_j + iy_j$ and $p_j = x_j^2 + y_j^2 \leq p_0 = r_0^2$. Put $X_2 := \sum_{j=1}^3 x_j^2$, $Y_2 := \sum_{j=1}^3 y_j^2 = 1 - p_0 - X_2$, $S_x := \sum_{j=1}^3 |x_j|$ and $S_y := \sum_{j=1}^3 y_j$.

Testing $\Phi(v)$ against $Z_* = (1/\sqrt{2}, i/\sqrt{6}, i/\sqrt{6}, i/\sqrt{6})^T \in \mathcal{H}$ gives

$$\Phi(v) \geq \Phi_{\mathrm{im}}(v) := \left(\frac{r_0}{\sqrt{2}} + \frac{S_y}{\sqrt{6}}\right)^2 = \frac{1}{2} - \frac{1}{2}X_2 + \frac{1}{3}\Big(\sqrt{3}\, r_0 S_y - Y_2 + \sum_{1\leq k<\ell\leq 3} y_k y_\ell\Big).$$

Since $\Phi_{\mathrm{im}}(v) \le \Phi(v) \le \frac{1}{2}$, we have $S_y \le \sqrt{3}(1-r_0)$, hence $Y_2 \le S_y^2 \le 3(1-r_0)^2$ and $X_2 = 1 - r_0^2 - Y_2 \ge 2(1-r_0)(2r_0-1)$. Moreover, since $0 \le y_j \le \sqrt{p_j} \le r_0$ and $\sqrt{3} < \frac{26}{15} < \frac{7}{4}$ ($26^2 = 676 > 675$), we also have $Y_2 \le r_0 S_y \le \frac{26}{15} r_0(1-r_0) \le \frac{7}{4} r_0(1-r_0)$ and $X_2 \ge \frac{1}{15}(1-r_0)(15-11r_0)$.

Up to sign, the eight rows of H_1 and H_2 are $\frac{1}{2}(1, \varepsilon_1, \varepsilon_2, \varepsilon_3)$ for $\varepsilon \in \{\pm 1\}^3$, so the eight cut probabilities in $H_1 v$ and $H_2 v$ are

$$\lambda_\varepsilon = \frac{1}{4}\left[\left(r_0 + \sum_{j=1}^{3} \varepsilon_j x_j\right)^2 + \left(\sum_{j=1}^{3} \varepsilon_j y_j\right)^2\right] = \frac{1}{4}\left(1 + 2\sum_{j=1}^{3} \varepsilon_j c_{0j} + 2 \sum_{1 \le k < \ell \le 3} \varepsilon_k \varepsilon_\ell c_{k\ell}\right) \le p_0,$$

where $c_{ij} := \mathrm{Re}(\bar{v}_i v_j) = x_i x_j + y_i y_j$ ($c_{0j} = r_0 x_j$). Choosing $\varepsilon_j = \mathrm{sgn}(x_j)$ gives $(r_0 + S_x)^2 \le 4p_0$, so $S_x \le r_0$. For each $j \in \{1,2,3\}$, averaging $\lambda_\varepsilon \le p_0$ over the four sign vectors with $\varepsilon_j = \mathrm{sgn}(x_j)$ cancels the other terms and yields $\frac{1}{4}(1 + 2r_0|x_j|) \le r_0^2$, so $|x_j| \le \frac{4r_0^2 - 1}{2r_0}$ and $X_2 \le (\max_j |x_j|) S_x \le 2r_0^2 - \frac{1}{2}$. Combining this with $Y_2 \le \frac{7}{4} r_0(1-r_0)$ gives

$$1 = r_0^2 + X_2 + Y_2 \le 3r_0^2 - \frac{1}{2} + \frac{7}{4} r_0(1-r_0), \qquad 0 \le \frac{1}{4}(5r_0 - 3)(r_0 + 2),$$

hence $r_0 \ge \frac{3}{5}$. Furthermore, for $\varepsilon_j = \mathrm{sgn}(x_j)$ we have $(\sum_{j=1}^3 \varepsilon_j y_j)^2 \ge Y_2 - 2\sum_{k<\ell} y_k y_\ell = 2Y_2 - S_y^2 \ge 2(1 - p_0 - X_2) - 3(1-r_0)^2$, so

$$(r_0 + S_x)^2 \le 4p_0 - 2(1 - p_0 - X_2) + 3(1-r_0)^2 = (3r_0 - 1)^2 + 2X_2 \le \left(3r_0 - 1 + \frac{X_2}{3r_0 - 1}\right)^2,$$

and therefore $S_x \le \min(r_0,\, 2r_0 - 1 + \frac{X_2}{3r_0 - 1})$. For $s_2 := \sum_{1 \le k < \ell \le 3} |x_k||x_\ell| = \frac{1}{2}(S_x^2 - X_2)$, if $X_2 \le (1-r_0)(3r_0-1)$ then

$$2s_2 \le \left(2r_0 - 1 + \frac{X_2}{3r_0 - 1}\right)^2 - X_2 = (2r_0 - 1)^2 - \frac{X_2((1-r_0)(3r_0-1) - X_2)}{(3r_0-1)^2} \le (2r_0 - 1)^2,$$

while if $X_2 \ge (1-r_0)(3r_0-1)$ then $2s_2 \le r_0^2 - X_2 \le (2r_0-1)^2$. Thus $2s_2 \le (2r_0-1)^2$ and $q_4 := \sum_{1 \le k < \ell \le 3} x_k^2 x_\ell^2 \le s_2^2 \le \frac{1}{4}(2r_0 - 1)^4$ for all $X_2 \ge 0$.

On the critical locus, expanding $Q(v)$ in terms of $p_i = |v_i|^2$ and $c_{ij} = \mathrm{Re}(\bar{v}_i v_j)$ gives the exact identity (CO-1)

$$Q(v) = \frac{1}{2}\Big(\sum_{i=0}^{3} p_i\Big)^3 + 18 \sum_{j=1}^{3} D_j (c_{0j}^2 - c_{k\ell}^2), \tag{S8.3}$$

where for each $j \in \{1,2,3\}$ and $\{k,\ell\} = \{1,2,3\} \setminus \{j\}$ we set $D_j := p_0 + p_j - p_k - p_\ell = 2(p_0 + p_j) - 1 = 1 - 2(p_k + p_\ell) \le D_{\max} := \min(1, 4p_0 - 1)$. Using $c_{0j}^2 = p_0 x_j^2$, $c_{k\ell}^2 \le 2(x_k^2 x_\ell^2 + y_k^2 y_\ell^2)$ when $D_j \ge 0$, and $c_{k\ell}^2 \ge 0$ when $D_j < 0$, we obtain with $D_j^+ := \max(D_j, 0)$ the decoupled lower bound

$$Q(v) + 2\Phi(v) - \frac{3}{2} \ge Q(v) + 2\Phi_{\mathrm{im}}(v) - \frac{3}{2} \ge \mathcal{Y}(r_0, y) + \mathcal{X}(p; x^2),$$

where

$$\mathcal{Y}(r_0, y) := \frac{2}{3}\Big(\sqrt{3}\, r_0 S_y - Y_2 + \sum_{1 \le k < \ell \le 3} y_k y_\ell\Big) - 36 \sum_{j=1}^{3} D_j^+ y_k^2 y_\ell^2,$$

$$\mathcal{X}(p; x^2) := \sum_{j=1}^{3} (18 p_0 D_j - 1) x_j^2 - 36 \sum_{j=1}^{3} D_j^+ x_k^2 x_\ell^2.$$

We show that both $\mathcal{Y}(r_0, y)$ and $\mathcal{X}(p; x^2)$ are nonnegative (CO-2). For $\mathcal{Y}(r_0, y)$, the four-term AM–GM inequality on $(2r_0 + 1) + 3(2r_0 - 1) + 4(1 - r_0) + 4(1 - r_0) = 6$ gives $48(4r_0^2 - 1)(1-r_0)^2 \le$

$(3/2)^4 = 81/16$, so $9D_{\max}S_y^2 \le 27(4r_0^2-1)(1-r_0)^2 \le 729/256 < 3$. Since $Y_2 = S_y^2 - 2\sum_{k<\ell} y_k y_\ell$ and $\sum_{k<\ell} y_k^2 y_\ell^2 \le \frac{1}{4}S_y^2 \sum_{k<\ell} y_k y_\ell$ (as $y_k y_\ell \le \frac{1}{4}(y_k+y_\ell)^2 \le \frac{1}{4}S_y^2$), we have

$$\mathcal{Y}(r_0,y) \ge \frac{2}{3}S_y(\sqrt{3}\,r_0 - S_y) + (2 - 9D_{\max}S_y^2) \sum_{1\le k<\ell\le 3} y_k y_\ell \ge \frac{2}{3}S_y(\sqrt{3}\,r_0 - S_y) - \sum_{1\le k<\ell\le 3} y_k y_\ell.$$

Using $\sum_{k<\ell} y_k y_\ell \le \frac{1}{3}S_y^2$ and $S_y \le \sqrt{3}(1-r_0)$ gives $\mathcal{Y}(r_0,y) \ge \frac{1}{3}S_y(2\sqrt{3}\,r_0 - 3S_y) \ge \frac{\sqrt{3}}{3}S_y(5r_0-3) \ge 0$.

For $\mathcal{X}(p;x^2)$, substituting $D_j = (2p_0-1) + 2x_j^2 + 2y_j^2 \ge (2p_0-1) + 2x_j^2$ and $\sum_{j=1}^3 x_j^4 = X_2^2 - 2q_4$ into the first sum yields

$$\mathcal{X}(p;x^2) \ge (18p_0(2p_0-1)-1)X_2 + 36p_0(X_2^2 - 2q_4) - 36\sum_{j=1}^{3} D_j^+ x_k^2 x_\ell^2.$$

If $X_2 = 0$, then $\mathcal{X} = 0$; assume $X_2 > 0$. On $r_0 \in [\frac{19}{25}, 1]$, we have $p_0 \ge \frac{361}{625} > \frac{1}{2}$, so $D_j = 1 - 2(p_k + p_\ell) \ge 0$ and $D_j \le 1 - 2(x_k^2 + x_\ell^2)$. Hence $\sum_{j=1}^3 D_j^+ x_k^2 x_\ell^2 \le q_4 - 2(X_2 q_4 - 3x_1^2 x_2^2 x_3^2) \le (1 - \frac{4}{3}X_2)q_4$ by $3x_1^2x_2^2x_3^2 \le \frac{1}{3}X_2 q_4$, and since $2p_0 + 1 - \frac{4}{3}X_2 > 0$ and $q_4 \le \frac{1}{3}X_2^2$,

$$\begin{aligned}\frac{\mathcal{X}(p;x^2)}{X_2} &\ge 18p_0(2p_0-1) - 1 - 12(1-p_0)X_2 + 16X_2^2 \\ &= \left(4X_2 - \frac{3}{2}(1-p_0)\right)^2 + \frac{135p_0^2 - 54p_0 - 13}{4} \ge \frac{16573}{78125} > 0.\end{aligned}$$

On $r_0 \in [\frac{3}{5}, \frac{19}{25}]$, using $D_j^+ \le D_{\max}$ and $q_4 \le \frac{1}{4}(2r_0-1)^4$ gives

$$\frac{\mathcal{X}(p;x^2)}{X_2} \ge 18p_0(2p_0-1) - 1 + 36p_0X_2 - \frac{9(2p_0 + D_{\max})(2r_0-1)^4}{X_2},$$

which is strictly increasing in $X_2 > 0$. On $[\frac{3}{5}, \frac{16}{25}]$, setting $D_{\max} = 4p_0 - 1$ and $X_2 \ge X_{2,1}(r_0) := \frac{1}{15}(1-r_0)(15-11r_0)$ and multiplying by $15X_{2,1}(r_0) > 0$ gives the degree-six polynomial $P_1(r_0) = 15((36r_0^4 - 18r_0^2 - 1)X_{2,1}(r_0) + 36r_0^2 X_{2,1}(r_0)^2 - 9(6r_0^2-1)(2r_0-1)^4)$, which is strictly concave on $[\frac{3}{5}, 1]$ ($P_1''(\frac{3}{5}+u) = -368208u^4 - \frac{2057376}{5}u^3 - \frac{2990232}{25}u^2 - \frac{1591704}{125}u - \frac{226078}{625} < 0$ for all $u \ge 0$) with $P_1(\frac{3}{5}) = \frac{3693}{78125} > 0$ and $P_1(\frac{16}{25}) = \frac{216609912}{1220703125} > 0$. On $[\frac{16}{25}, \frac{19}{25}]$, setting $D_{\max} = 1$ and $X_2 \ge X_{2,2}(r_0) := 2(1-r_0)(2r_0-1)$ and multiplying by $\frac{X_{2,2}(r_0)}{2r_0-1} = 2(1-r_0) > 0$ gives the degree-five polynomial $P_2(r_0) = 72r_0^5 - 432r_0^4 + 432r_0^3 - 54r_0^2 - 52r_0 + 7$, which is strictly concave on $[\frac{3}{5}, 1]$ ($P_2''(\frac{3}{5}+u) = -36(8u^2(9-5u) + \frac{288}{5}u + 3) < 0$ for $u \in [0, \frac{2}{5}]$) with $P_2(\frac{16}{25}) = \frac{988047}{9765625} > 0$ and $P_2(\frac{19}{25}) = \frac{570453}{9765625} > 0$. Thus $\mathcal{X}(p;x^2) \ge 0$ in all cases, proving (5).

Let $x = \sqrt{\Phi}$ and $c = (2+\sqrt{2})/4$. The 32 overlaps $q_\eta = |\langle \eta, v\rangle|^2$ sum to 8, since the frame operator is $8I$ (CO-1), so $\Phi = \max_\eta q_\eta \ge 1/4$ and $x \ge 1/2$. For $x \ge 1/\sqrt{2}$, the first bound gives $1 - x^2 \ge 2c(1-x)$. For $1/2 \le x \le 1/\sqrt{2}$, the second gives

$$3/2 - 2x^2 - 2c(1-x) = 2(1/\sqrt{2} - x)(x - (2-\sqrt{2})/4) \ge 0.$$

Since $d^2 = 2(1-x)$, this proves (6). Equation (3) gives (7), with $\mathcal{D} = Q + 5\tau/2$ on critical states and the bound just proved as its last step. Bell products have $Q = \Phi = 1/2$, hence $d^2 = 2 - \sqrt{2}$; both constants are sharp. The null-critical refinement is proved in the next paragraph. □

Two auxiliary Bloch spheres on the null-critical locus. Let $M = (m_1, m_2, m_3, m_4)$ be the unitary 4×4 matrix whose columns are the magic basis vectors of Appendix B of the paper, written in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, so that $Mv = \sum_k v_k m_{k+1}$ for $v \in \mathbb{C}^4$. For the critical normal form D_v, regard Mv as an auxiliary two-qubit vector. Its coefficient matrix has determinant $\sum_j v_j^2/2$. Thus $\tau = 0$ is equivalent to $Mv = a \otimes b$ for unit auxiliary qubits a, b. These are coordinates on

the critical normal form, not physical tensor factors of the original four-qubit state. Let $r, s \in S^2$ be their Bloch vectors. Direct Bell probabilities give

$$Q = \frac{1}{2} + \frac{27}{2} r_1 r_2 r_3 s_1 s_2 s_3, \qquad \Phi = \frac{1}{4} \max_{\substack{\varepsilon,\eta\in\{\pm1\}^3 \\ \prod_j \varepsilon_j\eta_j=-1}} \left(1 + \frac{\varepsilon\cdot r}{\sqrt{3}}\right)\left(1 + \frac{\eta\cdot s}{\sqrt{3}}\right). \tag{S8.4}$$

Indeed, the four probabilities $|v_j|^2$ are $(1 + \sum_\alpha \chi_{j\alpha} r_\alpha s_\alpha)/4$, where

$$(\chi_{j\alpha}) = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & -1 & -1 \\ -1 & 1 & 1 \end{pmatrix}.$$

Their second and third power sums in (S7.2), with $Z = D_v$ and $\operatorname{tr}(Z^T Z) = 0$, give the first identity. The 32 HS representatives become exactly the auxiliary product rays whose Bloch vectors are cube vertices $\varepsilon/\sqrt{3}, \eta/\sqrt{3}$ of opposite parity. The product-state overlap formula and (S8.2) give the second identity. These finite identities are checked independently by NC-1–NC-3.

Here is an elementary proof of the geometric claims in (8); NC-4 checks its scalar identities and constants. Write $A = 1 - 1/\sqrt{3}$, $B = 1 + 1/\sqrt{3}$, and $f_0 = B^2/4 = (2 + \sqrt{3})/6$. First we prove $\Phi \geq f_0$. If the signs of the coordinates of r, s have opposite parity, align both cube vertices with those signs; $\|r\|_1, \|s\|_1 \geq 1$ gives the claim. A zero coordinate allows either parity. In the remaining case put $L = \|r\|_1$, $m = \min_j |r_j|$ and define $f_r = 1 + L/\sqrt{3}$, $g_r = 1 + (L - 2m)/\sqrt{3}$, with analogous quantities for s. Flipping the sign at a smallest coordinate gives

$$4\Phi \geq \max(f_r g_s, g_r f_s) \geq \sqrt{f_r g_r f_s g_s}.$$

For the ordered coordinate moduli $x \geq y \geq z = m$ of either unit vector,

$$f_r g_r - B^2 = \frac{2}{\sqrt{3}}(x + y - 1) + \frac{2}{3}(xy - z^2) \geq 0.$$

Here $xy \geq z^2$ and $(x + y)^2 = 1 - z^2 + 2xy \geq 1$. This proves the fidelity floor in all cases.

If $r_1 r_2 r_3 s_1 s_2 s_3 \geq 0$, then $Q \geq 1/2$, and with $\Phi \geq f_0$ this gives $Q \geq (1 - \Phi)/(2(1 - f_0)) = \frac{3}{4-\sqrt{3}}(1-\Phi)$. Otherwise put $\mu = 3\sqrt{3}|r_1 r_2 r_3|$ and $\nu = 3\sqrt{3}|s_1 s_2 s_3|$, so $0 \leq \mu, \nu \leq 1$ and $2Q = 1 - \mu\nu$. For a nonnegative unit triple x, y, z,

$$x + y + z - 1 = \frac{2(xy + yz + zx)}{x + y + z + 1} \geq (9 - 3\sqrt{3})xyz.$$

For positive coordinates this follows from $1/x + 1/y + 1/z \geq 3\sqrt{3}$ and $x + y + z \leq \sqrt{3}$; zero coordinates follow by continuity. Thus $\|r\|_1 \geq 1 + (\sqrt{3} - 1)\mu$ and similarly for s. Aligning opposite-parity signs in (S8.4) gives $4\Phi \geq \Lambda := (B + A\mu)(B + A\nu)$. The identity

$$\Lambda - B^2 - (4 - B^2)\mu\nu = AB\big(\mu(1 - \nu) + \nu(1 - \mu)\big) \geq 0$$

gives $1 - \Phi \leq (1 - f_0)(1 - \mu\nu) = 2(1 - f_0)Q$, the infidelity bound. In both cases the distance bound follows pointwise: $1 - \Phi = \frac{1+\sqrt{\Phi}}{2}\, d^2 \geq \frac{1+\sqrt{f_0}}{2}\, d^2$, and $\frac{3}{4-\sqrt{3}} \cdot \frac{1+\sqrt{f_0}}{2} = \frac{3+\sqrt{3}}{4}$ because $\sqrt{f_0} = B/2$. The critical cluster representative $v = (1, i, 0, 0)/\sqrt{2}$ has $Q = 1/2$ and $\sqrt{\Phi} = B/2$, attaining all three geometric bounds.

Lemma S30 (Real-density geometry and sharp critical covering profile; exact certificate, RP, FP)**.** *Write a unit critical normal vector as $v = x + iy$, with $x, y \in \mathbb{R}^4$, and put $A = xx^T + yy^T$. Then A ranges over exactly the real positive semidefinite matrices of trace one and rank at most two, and*

$$\lambda_{ji} = e_i^T H_j A H_j^T e_i, \qquad \tau = 2\operatorname{tr} A^2 - 1. \tag{S8.5}$$

Here H_0, H_1, H_2 are the three real orthogonal cut transforms displayed above. Their bases are mutually unbiased. For their twelve unit vectors $h_{ji} = H_j^T e_i$ and rank-one projectors $B_{ji} = h_{ji}h_{ji}^T$,

$$A = \sum_{j,i} \lambda_{ji} B_{ji} - \tfrac{1}{2} I, \qquad \sum_{j,i} \lambda_{ji}^2 = \operatorname{tr} A^2 + \tfrac{1}{2}. \tag{S8.6}$$

For each coordinate axis e_k and each of the four lines $u = (0, \pm 1, \pm 1, \pm 1)/\sqrt{3}$ in its orthogonal coordinate three-space, put $P = e_k e_k^T + uu^T$ and $A_P = P/2$. There are sixteen such planes. With $F_{\mathrm{B}}(A, B) = (\operatorname{tr}\sqrt{\sqrt{B}A\sqrt{B}})^2$,

$$\Phi(v) = \max_P F_{\mathrm{B}}(A, A_P), \qquad F_{\mathrm{B}}(A, A_P) = \frac{1}{2}\left[\operatorname{tr}(PA) + 2\sqrt{\det((e_k, u)^T A (e_k, u))}\right]. \tag{S8.7}$$

Thus $d(v)$ is exactly the minimum Bures distance from A to these sixteen real density matrices. The entire critical locus obeys the sharp bound

$$\Phi \geq f_{\mathrm{cov}}(\tau) := \frac{1}{3} + \frac{\sqrt{1-\tau}}{2\sqrt{3}}, \qquad \max_{\substack{\psi \text{ critical} \\ \tau(\psi) = t}} d(\psi)^2 = 2 - 2\sqrt{f_{\mathrm{cov}}(t)} \quad (0 \leq t \leq 1). \tag{S8.8}$$

The bound is attained for every t by the cluster-to-GHZ family below. In particular every critical state has $d^2 \leq 2 - 2/\sqrt{3}$, with equality for GHZ. For any one such plane, put $\lambda_{ji}^P = \operatorname{tr}(B_{ji}A_P)$ and $D_P = \sum_{j,i} \lambda_{ji} \ln(\lambda_{ji}/\lambda_{ji}^P)$. The entropy gap has the exact form

$$3 \ln 2\, \delta = D_P - \tfrac{3}{2} \ln 3\, J_P, \qquad J_P = 1 - \operatorname{tr}(PA) = \|A - A_P\|_F^2 - \tau/2. \tag{S8.9}$$

Proof. The three physical cut vectors are $H_j v$, so their squared moduli are the stated real measurements. Conversely, for any real A of the specified rank, its spectral decomposition gives $v = \sqrt{a_1}e_1 + i\sqrt{a_2}e_2$; rank one is included. The identity $|v^T v|^2 = 2 \operatorname{tr} A^2 - 1$ proves the spin-flip claim. The twelve real lines are the coordinate lines and the half-sign lines of the regular 24-cell. Their tight frame identity, valid for every real symmetric matrix C, is

$$\sum_{j,i} \operatorname{tr}(B_{ji}C) B_{ji} = C + \tfrac{1}{2} \operatorname{tr}(C) I.$$

Taking its trace pairing with C proves both tomography formulas (RP-1).

The two HS rays associated with P are $(e_k \pm iu)/\sqrt{2}$. Their squared overlaps with v have sum $\operatorname{tr}(PA)$ and half their difference has squared value $\det((e_k, u)^T A (e_k, u))$. Maximising over this pair therefore gives (S8.7), by the two-dimensional formula $(\operatorname{tr}\sqrt{C})^2 = \operatorname{tr} C + 2\sqrt{\det C}$. The signed-Birkhoff reduction already proved shows that maximising these sixteen pairwise maxima is the full physical HS fidelity.

For $q_\eta = |\langle \eta, v \rangle|^2$, the finite HS frame satisfies

$$\sum_{\eta \in \mathcal{H}} q_\eta = 8, \qquad \sum_{\eta \in \mathcal{H}} q_\eta^2 = \tfrac{8}{9}(4 - \tau).$$

Both follow by expanding and cancelling odd sign powers (RP-3). The stronger sharp profile (S8.8) follows from the area-weighted calibration proved below; the moment identity alone does not give that profile.

Each reference distribution has one entry $1/2$ in each cut and nine entries $1/6$ in total. The sum of the three projectors selecting its large entries is $3P/2$ (RP-3). Expanding D_P then gives (S8.9). Finally $\operatorname{tr} A_P^2 = 1/2$ and $2 \operatorname{tr}(AA_P) = \operatorname{tr}(AP)$ give the leakage formula. All matrix and compression identities here are independently checked by RP; the covering calibration uses FP. □

Area-weighted covering calibration. We prove the profile in (S8.8) using an explicit probability measure on the sixteen HS planes. Its only polynomial positivity input is the following finite exact certificate.

Lemma S31 (Four-point cubic inequality; exact rational certificate)**.** *If $X, Y \in \mathbb{R}^4$ satisfy $\sum_i X_i = \sum_i Y_i = 0$ and $\sum_i X_i^2 = \sum_i Y_i^2 = 4$, then*

$$\sum_{i,j=1}^{4} |X_i - Y_j|^3 \geq 64. \tag{S8.10}$$

Equality holds at $X = Y = (-1, -1, 1, 1)$.

Proof. Put $X = Er$, $Y = Es$, where

$$E = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}, \qquad E^T E = 4I,$$

so that $\|r\| = \|s\| = 1$. Every zero-sum four-vector is represented. Sort X and Y separately and interleave their entries in increasing order. There are $\binom{8}{4} = 70$ words with four letters of each kind. Swapping X, Y and simultaneously negating both vectors reduce these to exactly twenty-three classes (FP-1). Ties may use either order. For each word let $z_1, \dots, z_8$ be its ordered entries and $g_i = z_{i+1} - z_i \geq 0$. The absolute values then give a cubic polynomial $p = \sum_{ij} |X_i - Y_j|^3/16 - 4$. For each of the twenty-three words the supplied rational data give the identity

$$p = B_0^T G_0 B_0 + \sum_{i=1}^{7} g_i B_i^T G_i B_i + (\|r\|^2 - 1)\ell_r + (\|s\|^2 - 1)\ell_s. \tag{S8.11}$$

The ℓ are quadratic polynomials and all eight G_i are positive definite. FP-2 rebuilds all 210 coefficients through degree four in each class and proves positivity of all 184 Grams using integer arithmetic. Thus the right side is nonnegative on each ordering cone and the two spheres, proving the full inequality. The bases and exact matrix representation are specified in Section S20. The stated equality follows by substitution. □

Let $U = (U_1, U_2)$ be a real orthonormal 4×2 frame with plane projector $\Pi = UU^T$. For each HS plane P use its orthonormal frame $V_P = (e_k, u)$ and put $c_P = |\det(U^T V_P)|$. The sixteen planes obey

$$\sum_P c_P^2 = \frac{8}{3}, \qquad \sum_P c_P^2 P = \frac{8}{9}(I + \Pi). \tag{S8.12}$$

Indeed, the bivector $\Delta = U_1 U_2^T - U_2 U_1^T$ satisfies $\Delta\Delta^T = \Delta^T\Delta = \Pi$, and $\det(U^T V_P) = e_k^T \Delta u$. Fix k, put $g = \Delta^T e_k$ (so $g_k = 0$, $|g|^2 = \Pi_{kk}$) and sum over the four lines u for this k, using $u_j = \pm 1/\sqrt{3}$ $(j \neq k)$: $\sum_u (g \cdot u)^2 = \frac{4}{3}|g|^2$ and $\sum_u (g \cdot u)^2 uu^T = \frac{4}{9}|g|^2(I - e_k e_k^T) + \frac{8}{9}(gg^T - \mathrm{diag}(g_j^2))$. Summing over k with $\sum_k gg^T = \Pi$ and $\sum_k \mathrm{diag}(g_j^2) = \mathrm{diag}(\Pi)$ gives both identities. FP-3 also checks them without assuming orthonormality first, using the homogeneous squared area and its associated matrix. Therefore the explicit weights $w_P = 3c_P^2/8$ satisfy

$$\sum_P w_P = 1, \qquad \sum_P w_P U^T P U = \frac{2}{3} I_2, \qquad \sum_P w_P c_P \geq \frac{1}{\sqrt{3}}. \tag{S8.13}$$

To prove the last inequality, let M be the magic-basis matrix defined before (S8.4). Form the null critical vector $z = (U_1 + iU_2)/\sqrt{2}$. Its auxiliary product representation $Mz = a \otimes b$ has unit Bloch vectors r, s, as in (S8.4). For the two HS rays $\eta_P = (e_k + iu)/\sqrt{2}$ and $\bar{\eta}_P$, expansion gives

$$c_P = \big| |\langle \eta_P, z \rangle|^2 - |\langle \bar{\eta}_P, z \rangle|^2 \big|.$$

In the cube-pair description of the HS rays given after (S8.4), conjugation reverses both cube vertices. Indeed $M = \epsilon^{\otimes 2}\bar{M}$, where $\epsilon = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ (FP-3), and the qubit spin flip reverses the Bloch vector. Choose the four even-parity cube vertices e_i, the rows of E. The sixteen conjugate pairs are represented once by $(\varepsilon, \eta) = (e_i, -e_j)$, and hence

$$c_P = \frac{|e_i \cdot r - e_j \cdot s|}{2\sqrt{3}}.$$

The four numbers $X_i = e_i \cdot r$, and likewise $Y_j = e_j \cdot s$, have sum zero and sum of squares four. Lemma S31 therefore gives

$$\sum_P c_P^3 = \frac{\sum_{ij} |X_i - Y_j|^3}{24\sqrt{3}} \geq \frac{8}{3\sqrt{3}},$$

which proves the last calibration inequality.

Now write the real density matrix of any critical state as $A = U\Lambda U^T$, with $\operatorname{tr}\Lambda = 1$ and $\det\Lambda = (1-\tau)/4$. Rank one is included by completing its support to any plane. The exact Bures formula gives

$$\begin{aligned} \Phi &\geq \sum_P w_P F_{\mathrm{B}}(A, P/2) \\ &= \frac{1}{2}\operatorname{tr}\left(\Lambda \sum_P w_P U^T P U\right) + \sqrt{\det\Lambda}\sum_P w_P c_P \\ &\geq \frac{1}{3} + \frac{\sqrt{1-\tau}}{2\sqrt{3}}. \end{aligned}$$

This proves the profile for the entire critical locus.

Finally, for $a, b \geq 0$, $a + b > 0$, the critical normal family

$$v = \frac{(a+ib, -a+ib, 0, 0)}{\sqrt{2(a^2+b^2)}}, \qquad \tau = \frac{(a^2-b^2)^2}{(a^2+b^2)^2}, \qquad \Phi = \frac{1}{3} + \frac{ab}{\sqrt{3}(a^2+b^2)} \tag{S8.14}$$

attains equality. FP-4 checks every one of the 32 ray overlaps against this maximum. The parameters cover $0 \leq \tau \leq 1$, proving pointwise optimality and the conditional covering radius. The endpoint $a = b$ is a cluster state; $b = 0$ is GHZ up to LU.

This model expresses both the entropy and the optimised orbit distance exactly. In particular, a bound on the classical relative entropy D_P that controls the explicit plane leakage J_P can improve the entropy constant without first improving a moment surrogate. Its domain is a real rank-two density matrix; replacing it by all real density matrices would admit $I/4$, whose entropy exceeds the critical-state maximum.

S8.3 A one-parameter family of sharp geometric inequalities on the slice $P = 1$

Theorem S32 (Sharper geometric family on $P = 1$; MS)**.** *Let ψ be a critical unit pure four-qubit state with $\tau = 0$, equivalently $P = 1$, and let Φ, d, Q be as in Theorem 20. Use the two auxiliary Bloch vectors of (S8.4) to put $X = 3\sum r_j^4 - 1$ and $Y = 3\sum s_j^4 - 1$. Then*

$$\begin{gathered} \Phi \geq \frac{2+\sqrt{3}}{6}, \qquad 0 \leq \theta \leq \frac{1}{24}, \\ Q + \theta XY \geq (1+8\theta)\frac{3}{4-\sqrt{3}}(1-\Phi) \geq (1+8\theta)\frac{3+\sqrt{3}}{4}d^2. \end{gathered} \tag{S8.15}$$

For every θ, both geometric coefficients and the fidelity floor are optimal, attained by cluster states.

The case $\theta = 0$ is the inequality (8) of the paper. The endpoint $\theta = \frac{1}{24}$ is proved by an elementary multiaffine vertex check in the opposite-parity sector and a univariate polynomial minorant in the same-parity sector.

Sharp mixed-moment geometry. With the auxiliary Bloch vectors r, s of (S8.4), put $X = 3\sum r_j^4 - 1$ and $Y = 3\sum s_j^4 - 1$. We prove the endpoint $\theta = 1/24$ in (S8.15) by elementary polynomial inequalities in both parity sectors. Interpolation with the already proved $\theta = 0$ bound, which is the inequality (8) of the paper, then covers $0 \le \theta \le 1/24$.

First, every nonnegative unit triple a, b, c obeys the strengthened norm inequality

$$L := a+b+c \ge 1 + (\sqrt{3}-1)u + \tfrac{1}{4}(2 - X - 2u), \quad u = 3\sqrt{3}abc, \quad X = 3(a^4+b^4+c^4) - 1. \tag{S8.16}$$

To see this, put $B = 9 - 9\sqrt{3}/2$ and use $a^4 + b^4 + c^4 = 1 - (L^2-1)^2/2 + 4Labc$. The desired residual is

$$L - 1 - \tfrac{3}{8}(L^2-1)^2 + (3L - B)abc.$$

For $1 \le L \le \sqrt{2}$, its first three terms equal $(L-1)[1 - 3(L-1)(L+1)^2/8] \ge 0$, since $(\sqrt{2}-1)(\sqrt{2}+1)^2 = 1 + \sqrt{2} < 8/3$; also $3L > B$. For $\sqrt{2} \le L \le \sqrt{3}$, Schur gives $abc \ge L(L^2-2)/9$. The resulting lower bound is

$$f(L) = -L^4/24 + (\sqrt{3}/2 - 1)L^3 + L^2/12 + (3 - \sqrt{3})L - 11/8.$$

It is concave on that interval, with $f(\sqrt{2}) = \sqrt{2} - 11/8 > 0$ and $f(\sqrt{3}) = 0$, proving the claim. All these finite identities are checked by MS-1.

Suppose first that $\prod_j r_j s_j \le 0$. Put $u = 3\sqrt{3}|r_1 r_2 r_3|$, $v = 3\sqrt{3}|s_1 s_2 s_3|$. For a unit triple r, put $t_i = r_i^2$ and $q_r = \sum_{i<j} t_i t_j$. The identity

$$q_r^2 - 3t_1t_2t_3 = \tfrac{1}{2}\left[(t_1t_2 - t_2t_3)^2 + (t_2t_3 - t_3t_1)^2 + (t_3t_1 - t_1t_2)^2\right] \ge 0$$

gives the elementary bound

$$0 \le u \le 3q_r = 1 - X/2, \qquad 0 \le v \le 1 - Y/2. \tag{S8.17}$$

Write $p = 1-u$, $q = 1-v$, $X = 2p(1-\alpha)$ and $Y = 2q(1-\beta)$, with $p, q, \alpha, \beta \in [0,1]$; a zero p or q is handled by continuity. Aligning the two cube signs is admissible in this parity sector, so (S8.16) gives

$$4\Phi \ge f_r f_s, \qquad f_r = 2 - (1 - 1/\sqrt{3})p + \frac{p\alpha}{2\sqrt{3}}, \quad f_s = 2 - (1 - 1/\sqrt{3})q + \frac{q\beta}{2\sqrt{3}}.$$

Set $a_{24} = 4/(4 - \sqrt{3})$. The polynomial

$$\frac{p + q - pq}{2} + \frac{pq(1-\alpha)(1-\beta)}{6} - a_{24}(1 - f_r f_s/4)$$

is affine in each of its four variables separately. At the sixteen cube vertices its distinct values are

$$0, \quad \frac{3+4\sqrt{3}}{26}, \quad \frac{-6+5\sqrt{3}}{78}, \quad \frac{2+7\sqrt{3}}{52}, \quad \frac{3+4\sqrt{3}}{78},$$

all nonnegative (MS-1). This proves $Q + XY/24 \ge a_{24}(1 - \Phi)$ in this sector.

For the remaining parity sector ($\prod_{j=1}^3 r_j s_j \ge 0$), sort the coordinate moduli of r as $a_r \ge b_r \ge c_r \ge 0$ and those of s as $a_s \ge b_s \ge c_s \ge 0$. Swap r and s if necessary so that $c_r \ge c_s$, and set

$$x = \sqrt{3}\, c_r \in [0,1], \quad y = \sqrt{3}\, c_s \in [0, x], \qquad \tilde{w}_r = 3a_r b_r, \quad \tilde{w}_s = 3a_s b_s.$$

For a sorted unit triple $a \ge b \ge c \ge 0$ with $x = \sqrt{3}\, c$ and $\tilde{w} = 3ab$, one has $a + b = \sqrt{1 - c^2 + 2ab} = \sqrt{1 + (2\tilde{w} - x^2)/3}$, $3\sqrt{3}\, abc = x\tilde{w}$, and $\tilde{w} \in [x\sqrt{3 - 2x^2}, (3 - x^2)/2]$, where the lower and upper bounds correspond to $b = c$ and $a = b$. Moreover (MS-2),

$$X = 2 - 2x^2\Big(1 - \frac{x^2}{3}\Big) - \frac{2}{3}\tilde{w}^2 \ge X_{\mathrm{lin}}(x, \tilde{w}) := 2 - 2x^2\Big(1 - \frac{x^2}{3}\Big) - \Big(1 - \frac{x^2}{3}\Big)\tilde{w} \ge 0,$$

since $X-X_{\mathrm{lin}} = \frac{2}{3}\tilde{w}(\frac{3-x^2}{2}-\tilde{w}) = 3ab(a-b)^2 \geq 0$ and $X_{\mathrm{lin}}(x,\tilde{w}) = \frac{1}{2}(1-x^2)^2+(1-\frac{x^2}{3})(\frac{3-x^2}{2}-\tilde{w}) \geq 0$. Using the branch $(1+(a_r+b_r+c_r)/\sqrt{3})(1+(a_s+b_s-c_s)/\sqrt{3})$ of 4Φ gives $4\Phi \geq f_r(x,\tilde{w}_r)\, g_s(y,\tilde{w}_s)$ with

$$f_r(x,\tilde{w}_r) = 1+\frac{1}{\sqrt{3}}\sqrt{1+\frac{2\tilde{w}_r-x^2}{3}}+\frac{x}{3}, \qquad g_s(y,\tilde{w}_s) = 1+\frac{1}{\sqrt{3}}\sqrt{1+\frac{2\tilde{w}_s-y^2}{3}}-\frac{y}{3}.$$

Hence $Q + XY/24 - a_{24}(1-\Phi)$ is bounded below by

$$\mathcal{F}(x,y,\tilde{w}_r,\tilde{w}_s) := \frac{1+xy\tilde{w}_r\tilde{w}_s}{2}+\frac{X_{\mathrm{lin}}(x,\tilde{w}_r)\,Y_{\mathrm{lin}}(y,\tilde{w}_s)}{24}+\frac{a_{24}}{4}f_r(x,\tilde{w}_r)\,g_s(y,\tilde{w}_s)-a_{24}.$$

Since $(3-2x^2)-\left(\frac{5-2x}{3}\right)^2 = \frac{2(1-x)(1+11x)}{9} \geq 0$, we have $\tilde{w}_r \geq x\sqrt{3-2x^2} \geq \tilde{w}_{\mathrm{low}}(x) := \frac{x(5-2x)}{3} \geq x$ and $\tilde{w}_s \geq \tilde{w}_{\mathrm{low}}(y) \geq y$. Consequently $2\tilde{w}_s - y^2 \geq \frac{y(10-7y)}{3} \geq 0$, so

$$Y_{\mathrm{lin}}(y,\tilde{w}_s) \leq 2-\frac{2}{3}y \leq \frac{2\sqrt{3}}{\sqrt{3}+1}\Big(1+\frac{1}{\sqrt{3}}-\frac{y}{3}\Big) \leq \frac{2\sqrt{3}}{\sqrt{3}+1}\,g_s(y,\tilde{w}_s),$$

and likewise $X_{\mathrm{lin}}(x,\tilde{w}_r) \leq 2 \leq \frac{2\sqrt{3}}{\sqrt{3}+1}\,f_r(x,\tilde{w}_r)$. Since $1+(2\tilde{w}_r-x^2)/3 \leq 2$, the partial derivative with respect to $\tilde{w}_r$ satisfies

$$\frac{\partial\mathcal{F}}{\partial\tilde{w}_r} \geq Y_{\mathrm{lin}}(y,\tilde{w}_s)\Big(\frac{\sqrt{3}+1}{18\sqrt{2}\,(4-\sqrt{3})}-\frac{1}{24}\Big) \geq 0,$$

and the same calculation gives $\partial\mathcal{F}/\partial\tilde{w}_s \geq 0$ (MS-2). Monotonicity in $(\tilde{w}_r,\tilde{w}_s)$ therefore reduces the inequality to $\tilde{w}_r = \tilde{w}_{\mathrm{low}}(x)$ and $\tilde{w}_s = \tilde{w}_{\mathrm{low}}(y)$.

At $\tilde{w} = \tilde{w}_{\mathrm{low}}(x)$ one has $x\tilde{w}_{\mathrm{low}}(x) = \frac{x^2(5-2x)}{3} \geq x^2$,

$$X_{\mathrm{lin}}(x,\tilde{w}_{\mathrm{low}}(x)) - \frac{4}{9}x^2(1-x)^2 = \frac{1}{9}(1-x)(18+3x-13x^2) =: X_b(x) \geq 0,$$

and $\sqrt{1+(2\tilde{w}_{\mathrm{low}}(x)-x^2)/3} = \frac{1}{3}\sqrt{9+10x-7x^2} \geq S_b(x) := 1+\frac{1}{2}x-\frac{7}{20}x^2$, since $(9+10x-7x^2)-9S_b(x)^2 = \frac{x(10-7x)(63x^2-90x+40)}{400} \geq 0$ on $[0,1]$ (MS-2). Putting $\hat{f}(x) = 1+S_b(x)/\sqrt{3}+x/3$ and $\hat{g}(y) = 1+S_b(y)/\sqrt{3}-y/3$, we obtain

$$Q+\frac{XY}{24}-a_{24}(1-\Phi) \geq P(x,y) := \frac{1+x^2y^2}{2}+\frac{X_b(x)X_b(y)}{24}+\frac{a_{24}}{4}\hat{f}(x)\hat{g}(y)-a_{24} \tag{S8.18}$$

for $0 \leq y \leq x \leq 1$. Write $B = 1+1/\sqrt{3}$. Three univariate polynomials on $[0,1]$ close the bound (MS-3):

1. $\hat{g}(y) - \frac{B}{2}X_b(y) = \frac{y}{2}q_2(y) \geq 0$, where $q_2(y) = (1+\frac{8\sqrt{3}}{9})+(\frac{16}{9}+\frac{97\sqrt{3}}{270})y-(\frac{13}{9}+\frac{13\sqrt{3}}{27})y^2$ is concave on $[0,1]$ with $q_2(0)>0$ and $q_2(1) = \frac{4}{3}+\frac{23\sqrt{3}}{30} > 0$.

2. Consequently, for $0 \leq y \leq x \leq 1$,

$$\frac{\partial P}{\partial x}(x,y) = xy^2+\frac{a_{24}}{4}\hat{f}'(x)\hat{g}(y)+\frac{X_b'(x)}{24}X_b(y) \geq \frac{X_b(y)}{4-\sqrt{3}}\,q_1(x) \geq 0,$$

   where $q_1(x) := (4-\sqrt{3})(\frac{a_{24}B}{8}\hat{f}'(x)+\frac{X_b'(x)}{24}) = (\frac{11}{20}-\frac{3}{5}x)^2+(\frac{5\sqrt{3}}{24}-\frac{1189}{3600})+(\frac{17\sqrt{3}}{540}-\frac{133}{2700})x+(\frac{163}{450}-\frac{13\sqrt{3}}{72})x^2 > 0$ on $[0,1]$.

3. Hence $P(x,y) \geq P(y,y) = yq_3(y) \geq 0$ on $[0,1]$, where

$$(4-\sqrt{3})\,q_3(y) = \Big(\frac{13}{25}-\frac{26}{25}y-\frac{1}{10}y^2\Big)^2+\Big(\frac{11\sqrt{3}}{18}-\frac{5896}{5625}\Big)+\Big(\frac{737}{7500}-\frac{19\sqrt{3}}{360}\Big)y$$

$$+\Big(\frac{173411}{202500}-\frac{79\sqrt{3}}{162}\Big)y^2+\Big(\frac{756757}{486000}-\frac{419\sqrt{3}}{972}\Big)y^3$$
$$-\Big(\frac{21043}{24300}-\frac{52\sqrt{3}}{243}\Big)y^4+\frac{169(4-\sqrt{3})}{1944}y^5>0,$$

since every coefficient through y^3 and at y^5 is strictly positive while the y^3 coefficient ($>4/5$) exceeds the y^4 coefficient ($<1/2$) on $0\le y\le 1$. Since $P(0,0)=0$, this proves $P(x,y)\ge 0$ on the entire triangle $0\le y\le x\le 1$.

Finally,

$$Q+\theta XY=(1-24\theta)Q+24\theta(Q+XY/24)\ge(1+8\theta)\frac{3}{4-\sqrt{3}}(1-\Phi).$$

The fidelity floor already proved converts this to the displayed distance bound. Clusters have $Q=1/2$, $X=Y=2$ and $\Phi=(2+\sqrt{3})/6$, attaining both coefficients for every θ in the stated interval (MS-4). Thus the full geometric family is sharp.

S8.4 The minimum-purity slice

Statements are proved; the finite verifications are interval-certified (outward-rounded ball arithmetic, every sign decision a certain comparison) or exact, and are listed in Section S20 (PM, PS). This section gives the coordinates, symmetries and equality geometry on the slice $P=1$ ($\tau=0$) that Section S8.5 uses for the whole critical locus.

Geometry of the equality states. At an equality point δ is stationary in every direction, because one Bloch vector is an axis and δ is even in the vanishing coordinate of the other, while four admissible cube pairs tie (PM-4) and, computationally, their d^2-gradients form a regular tetrahedron around 0 (explanatory; not used). The distance d^2 therefore has a conical corner and $\delta-C_*d^2$ has a conical minimum in the three transverse directions; the one smooth direction, which rotates the second vector within its coordinate plane, is the wall of Lemma S35, where the wall function G_w is stationary. This picture is explanatory; the proof uses the exact identities of Lemma S39.

Coordinates and inputs

Let r,s be the unit auxiliary Bloch vectors of (S8.4); indices are taken mod 3.

Lemma S33 (Cut spectra; PM-1)**.** *For $k\in\mathbb{Z}/3$ put $u_a^{(k)}=r_{a+k}s_a$ and let*

$$\mathcal{S}=\{(-,-,-),(-,+,+),(+,-,+),(+,+,-)\}.$$

The three physical cuts have the spectra $\{(1+\sigma\cdot u^{(k)})/4:\sigma\in\mathcal{S}\}$, $k=0,1,2$ in some order. Hence $E=\frac{1}{3}\sum_k g(u^{(k)})$, $g(u)=\sum_{\sigma\in\mathcal{S}}\eta((1+\sigma\cdot u)/4)$, $\eta(x)=-x\log_2 x$, $\delta=1+\frac{1}{2}\log_2 3-E$, and $\sum_k|u^{(k)}|^2=|r|^2|s|^2=1$.

Proof. Let $h_{ji}=H_j^Te_i$ be the unit vectors of Lemma S30. Since h_{ji} is real, Mh_{ji} is a Bell ray; write $C^{ji}_{\alpha\beta}=\langle Mh_{ji}|\sigma_\alpha\otimes\sigma_\beta|Mh_{ji}\rangle$ for its correlation matrix, which is real orthogonal with determinant -1. Its projector is $\frac{1}{4}(I+\sum_{\alpha\beta}C^{ji}_{\alpha\beta}\sigma_\alpha\otimes\sigma_\beta)$, so the probability $\lambda_{ji}=|h_{ji}^Tv|^2=|\langle Mh_{ji},Mv\rangle|^2$ is $(1+r^TC^{ji}s)/4$, as $Mv=a\otimes b$. *Cut* 0. From the definitions of the m_k, $(\sigma_\alpha\otimes\sigma_\alpha)m_k=s_k^\alpha m_k$ with $s^x=(+,+,-,-)$, $s^y=(-,+,-,+)$, $s^z=(+,-,-,+)$. So C^{0i} has diagonal (s_i^x,s_i^y,s_i^z), entries ±1, and being orthogonal it is diagonal; the four diagonals $(+,-,+),(+,+,-),(-,-,-),(-,+,+)$ are the elements of $\mathcal{S}$. *Cuts* 1, 2. Let $u=\frac{1}{2}(-I+i(\sigma_z-\sigma_x-\sigma_y))$, a unitary with $u\sigma_\beta u^\dagger=\sum_\alpha R_{\alpha\beta}\sigma_\alpha$, where $Re_1=e_2$, $Re_2=-e_3$, $Re_3=-e_1$. Direct multiplication gives $MH_2M^\dagger=u\otimes u^\dagger$ and $MH_1M^\dagger=(u\otimes u^\dagger)\,\mathrm{SWAP}$, where SWAP exchanges the two auxiliary qubits and transposes

correlation matrices. Since $Mh_{ji} = (MH_jM^\dagger)^\dagger m_i$, this gives $C^{2i} = R^T C^{0i} R^T$ and $C^{1i} = RC^{0i}R$. As $R_{\alpha\beta} \neq 0$ only for $\alpha = \beta + 1$, these are the signed permutations $C^{ji}_{\alpha\beta} = \sigma_\beta[\alpha = \beta + k_j]$ with $(k_j) = (0, 2, 1)$; the four column-sign triples σ of each cut are distinct and have product $\det C^{ji} = -1$, so they run over $\mathcal{S}$. Hence $r^T C^{ji} s = \sigma \cdot u^{(k_j)}$. PM-1 recomputes all twelve C^{ji} exactly from the physical magic basis. □

Lemma S34 (Symmetries; PM-2)**.** *δ and d^2 are invariant under: even sign changes of the six coordinates; cyclic shifts of r alone and of s alone; and the exchange $r \leftrightarrow s$. (The stated operations permute both the admissible cube pairs and the twelve probabilities. Fidelity alone is also invariant under arbitrary coordinate permutations of either vector.)*

The wall

Let $h(t) = H_b((1 + t)/2)$, with H_b the binary entropy defined in Appendix D.1. For $r = e_1$ and $s = (\cos\theta, 0, \sin\theta)$ (a point of the wall) write $\delta_w(\theta) = \frac{1}{2}\log_2 3 - \frac{1}{3}(1 + h(\cos\theta) + h(\sin\theta))$, $d_w^2(\theta) = 2 - \sqrt{B(1 + (\cos\theta + \sin\theta)/\sqrt{3})}$, $B = 1 + 1/\sqrt{3}$, and $G_w = \delta_w - C_* d_w^2$.

Lemma S35 (Wall)**.** *$G_w(\pi/4) = 0$ by (9), and $G_w(\theta) = G_w(\pi/2 - \theta)$; hence $G_w'(\pi/4) = 0$.*

Proof. The reflection $\theta \mapsto \pi/2 - \theta$ exchanges $\cos\theta$ and $\sin\theta$, so it exchanges $h(\cos\theta)$ and $h(\sin\theta)$ and fixes $\cos\theta + \sin\theta$; thus δ_w and d_w^2 are both invariant. G_w is smooth near $\pi/4$ (there $\cos\theta, \sin\theta \in (0, 1)$, where h is analytic), and an even function of $\theta - \pi/4$ has zero derivative at $\pi/4$. □

S8.5 The sharp entropy coefficient on the whole critical locus

Statements are proved; the finite verifications are interval-certified (outward-rounded ball arithmetic, every sign decision a certain comparison) or exact, and are listed in Section S20 (TC, EX, LCC, CCV). This section proves Theorem 21 for all critical states. Section S8.4 supplies the coordinates, symmetries, wall function and equality geometry on the slice $\tau = 0$ ($P = 1$) that are used here.

The constant C_* is defined in (9); the equality states attain it exactly (PM-4), and PS-1 encloses its decimal value.

Theorem 21 (Sharp entropy stability on the critical locus; interval-certified). *Every critical unit pure four-qubit state satisfies*

$$\delta(\psi) \;\geq\; C_*\, d(\psi)^2,$$

and equality with $d > 0$ holds exactly for states whose critical normal form has one of the following 144 Bloch-coordinate pairs, all with $P = 1$. In the auxiliary Bloch coordinates $(r, s) \in S^2 \times S^2$ of the null-critical normal form they are: one of r, s is a signed coordinate axis and the other is $(\pm e_b \pm e_c)/\sqrt{2}$, $b \neq c$ ($6 \cdot 12$ pairs, twice for the swap). These pairs attain the bound, so C_* is the optimal coefficient on the critical locus. Consequently every universal coefficient in $\delta \geq Cd^2$ is at most C_*, and the question left open is whether C_* is optimal over all states.

Strategy. On the critical locus the twelve cut probabilities are linear in a 3×3 correlation matrix T, and $\delta = \delta(T)$ is convex. The fidelity is the maximum of sixteen explicit plane functions. A critical state is determined by (r, s, w) with unit vectors r, s (the $P = 1$ coordinates) and a disc coordinate $w \in \mathbb{R}^2$, $|w|^2 = \tau$; the equality states sit at $w = 0$. Analytic bounds settle $\tau \geq 0.36$ (their crossing is at $\tau \approx 0.35879$, TC-6); for $\tau \leq 0.37$ a six-dimensional interval covering, together with a local cone lemma at the equality points, closes the proof.

Critical states as correlation matrices

Let M be the magic matrix, v a unit critical normal vector, $A = \mathrm{Re}(vv^\dagger)$, and $\psi = Mv$ the auxiliary two-qubit vector. Put $T_{\alpha\beta} = \langle\psi|\sigma_\alpha \otimes \sigma_\beta|\psi\rangle$, $t_{ji} = 4\lambda_{ji} - 1$ with λ_{ji} the cut probabilities of Lemma S30.

Lemma S36 (Correlation-matrix model; analytic, exact check TC-1)**.** *1. The twelve rays Mh_{ji} are Bell rays without local terms; their correlation matrices C^{ji} are signed permutation matrices (those of Lemma S33), and $t_{ji} = \langle C^{ji}, T\rangle$. Consequently, by (S8.6), $A = \frac{1}{4}I + \frac{1}{4}\sum_{j,i} t_{ji}B_{ji}$: A is an affine function of T.*

2. $T = R_a \,\mathrm{diag}(\kappa, -\kappa, 1)R_b^T$ with $R_a, R_b \in SO(3)$ and $\kappa = \sqrt{\tau}$. Writing $r = R_a e_3$, $s = R_b e_3$ and x_a, y_a (x_b, y_b) for the first two columns of R_a (R_b),

$$T = rs^T + \kappa(x_a x_b^T - y_a y_b^T).$$

Rotating the frames about r and s by θ_a, θ_b turns $\kappa(x_a x_b^T - y_a y_b^T)$ into $w_1(x_a x_b^T - y_a y_b^T) + w_2(x_a y_b^T + y_a x_b^T)$ with $w = \kappa(\cos(\theta_a + \theta_b), \sin(\theta_a + \theta_b))$.

3. For a plane $\Pi = VV^T$ of Lemma S30 the matrix $m_\Pi = V^T A V$ is affine in T and $\Phi = \max_\Pi \Phi_\Pi$, $\Phi_\Pi = \frac{1}{2}\,\mathrm{tr}\, m_\Pi + \sqrt{\det m_\Pi}$. Equivalently $\Phi = \max \frac{1}{4}(1 + \sqrt{1-\tau}\,(\varepsilon\cdot r + \eta\cdot s)/\sqrt{3} + \varepsilon^T T\eta/3)$ over the 32 cube pairs of (S8.4).

Proof. (1) The projector onto a maximally entangled ray is $\frac{1}{4}(I + \sum C_{\alpha\beta}\sigma_\alpha \otimes \sigma_\beta)$, with no local terms; the proof of Lemma S33 derives the twelve C^{ji}, and $t_{ji} = \langle C^{ji}, T\rangle$ because λ_{ji} is the expectation of that projector in ψ (PM-1 and TC-1 recompute them, TC-1 together with the identity $\lambda_{ji} = (\|v\|^2 + \langle C^{ji}, T\rangle)/4$ exactly for a random rational v). (2) A pure two-qubit vector is $(U\otimes V)(\cos\varphi|00\rangle + \sin\varphi|11\rangle)$ with $0 \le \varphi \le \pi/4$; its correlation matrix is $\mathrm{diag}(\sin 2\varphi, -\sin 2\varphi, 1)$ in the frames rotated by U, V, and the concurrence $\sin 2\varphi = |\sum_j v_j^2| = \sqrt{\tau}$ (the coefficient matrix of Mv has determinant $\sum_j v_j^2/2$). The local Bloch vectors are $\cos 2\varphi\, r$ and $\cos 2\varphi\, s$, of length $\sqrt{1-\tau}$. (3) m_Π is affine in A. The first formula is (S8.7). The second is the overlap of ψ with a product ray with Bloch vectors $\varepsilon/\sqrt{3}, \eta/\sqrt{3}$: $\langle\psi|\frac{1}{2}(1+\hat\varepsilon\cdot\sigma)\otimes\frac{1}{2}(1+\hat\eta\cdot\sigma)|\psi\rangle = \frac{1}{4}(1+\hat\varepsilon\cdot r_A + \hat\eta\cdot r_B + \hat\varepsilon^T T\hat\eta)$. TC-2 and TC-3 check the affine representation of A, the equality of the two maxima for 200 random states, and the singular values $(1, \kappa, \kappa)$, $\det T = -\kappa^2$. □

Thus $\tau = |w|^2$, and a critical state is determined (up to the discrete choices) by (r, s, w). At $\tau = 0$ one recovers the auxiliary Bloch vectors of (S8.4). In these variables

$$\delta = \mathrm{EMAX} - \tfrac{1}{3}\sum_{j,i} \mathrm{ent}\left(\tfrac{1+t_{ji}}{4}\right), \qquad d^2 = 2 - 2\sqrt{\max_\Pi \Phi_\Pi},$$
$$G := \delta - C_* d^2 = \max_\Pi G_\Pi, \qquad G_\Pi = \delta - C_*(2 - 2\sqrt{\Phi_\Pi}),$$

with $\mathrm{EMAX} = 1 + \frac{1}{2}\log_2 3$ and $\mathrm{ent}(x) = -x\log_2 x$. Each G_Π is analytic wherever all $\lambda_{ji} > 0$ and $\det m_\Pi > 0$, and δ is a convex function of T.

The chart

For $\zeta_1, \zeta_2, \zeta_1', \zeta_2' \in [0, 1]$ and a sign $\mathrm{sg} = \pm 1$ put

$$r = \tfrac{1}{N}(1, \zeta_1, \zeta_2),\ x_a = \tfrac{1}{n}(\zeta_1, -1, 0),\ y_a = r \times x_a,$$
$$s = \tfrac{1}{N'}(\mathrm{sg}, \zeta_1', \zeta_2'),\ x_b = \tfrac{1}{n'}(\zeta_1', -\mathrm{sg}, 0),\ y_b = s \times x_b,$$

$N = \sqrt{1 + \zeta_1^2 + \zeta_2^2}$, $n = \sqrt{1 + \zeta_1^2}$ (and primed), and for $w \in \mathbb{R}^2$, $|w| \le 1$,

$$T(y) = rs^T + w_1(x_a x_b^T - y_a y_b^T) + w_2(x_a y_b^T + y_a x_b^T), \qquad y = (\zeta_1, \zeta_2, \zeta_1', \zeta_2', w_1, w_2). \tag{S8.19}$$

Lemma S37 (Reduction to the chart; analytic). *δ and d are invariant under $T \mapsto S_a T S_b^T$, where $S_a = D_a\Pi_a$, $S_b = D_b\Pi_b$ with Π_a, Π_b cyclic permutation matrices and D_a, D_b diagonal sign matrices,* $\det D_a \det D_b = 1$ *(an even total number of sign changes), and under the transposition $T \mapsto T^T$. Every critical state is equivalent under such maps to one with $T = T(y)$, y in the chart $[0,1]^4 \times \{|w| \le 1\}$ and* $\mathrm{sg} = \pm 1$*, and then $\tau = |w|^2$. In the chart the transposition is $(\zeta_1, \zeta_2, \zeta_1', \zeta_2', w_1, w_2) \mapsto (\zeta_1', \zeta_2', \zeta_1, \zeta_2, w_1, \mathrm{sg}\, w_2)$, so it suffices to treat $\zeta_1 \le \zeta_1'$.*

Proof. The t_{ji} are permuted: this is a linear identity in T, and holds on the rank-one matrices rs^T by Lemma S34, which span all matrices. The set of 32 cube pairs is invariant (PM-2: the parity $\prod \varepsilon_j \eta_j$ is preserved because $\det D_a \det D_b = 1$), the local vectors transform with S_a, S_b up to a common sign which is absorbed by $(\varepsilon, \eta) \mapsto -(\varepsilon, \eta)$, and $\varepsilon^T T \eta$ is invariant; so Φ is invariant by Lemma S36(3). Given T, take r, s as in Lemma S36(2); cyclic shifts of r alone and of s alone make $|r_1|, |s_1|$ maximal, and flips of an even number of the six coordinate signs make $r_1, r_2, r_3, s_2, s_3 \ge 0$ (the sign of s_1 is then determined and is sg). The sign patterns of r and of s may have $\det D_a = \det D_b = -1$; then $D_a = -D_a'$, $D_b = -D_b'$ with $\det D_a' = \det D_b' = 1$ and $D_a T D_b^T = D_a' T D_b'^T$, i.e. the common sign flip of (r, s), which leaves rs^T fixed, is absorbed. (r, s) is determined by T up to this common sign when $\tau < 1$; at $\tau = 1$ the top singular value is degenerate and r, s are not unique, but $\tau = 1$ lies in the analytic range and the chart is used only for $\tau \le 0.37$. Then $r = (1, \zeta_1, \zeta_2)/N$, $s = (\mathrm{sg}, \zeta_1', \zeta_2')/N'$ with $\zeta_1, \zeta_2, \zeta_1', \zeta_2' \in [0,1]$, and $T - rs^T$ lies in the span of the two matrices of (S8.19) by Lemma S36(2), with $|w| = \kappa$. Transposition permutes the twelve t_{ji} (again a linear identity, on rank-one matrices Lemma S34 gives the swap $r \leftrightarrow s$) and exchanges the roles of the two cube vertices in Φ. Its action on the chart coordinates follows by direct substitution: $T(y)^T = DT(y')D$, with y' the image coordinates displayed in the lemma and $D = I$ for $\mathrm{sg} = 1$, $D = \mathrm{diag}(1, -1, -1)$ for $\mathrm{sg} = -1$, an admissible sign change (TC-4 also checks it numerically). □

Large τ

Lemma S38 (Analytic range; TC-6). *If $\tau \ge 0.36$ then $\delta \ge C_* d^2$, strictly.*

Proof. Put $k = (\ln 3 - 1)/(2 \ln 2)$. By (7) and (5), $\delta \ge k(1 - \Phi + 5\tau/2)$, and $\Phi \ge f_{\mathrm{cov}}(\tau)$ by (S8.8). With $x = \sqrt{\Phi}$ the function $k(1 - x^2) + 5k\tau/2 - 2C_*(1 - x)$ is concave in x, hence on $[\sqrt{f_{\mathrm{cov}}}, 1]$ it is at least its smaller endpoint value; at $x = 1$ it is $5k\tau/2 > 0$. At $x = \sqrt{f_{\mathrm{cov}}(\tau)}$, writing $\tau = 1 - u^2$, $f = f_{\mathrm{cov}} = \frac{1}{3} + \frac{u}{2\sqrt{3}}$, it is $\psi(u) = k(1 - f) + \frac{5}{2}k(1 - u^2) - 2C_*(1 - \sqrt{f})$, which is concave on $0 \le u \le 4/5$: $k(1 - f)$ is affine, $-\frac{5}{2}ku^2$ is concave, and $\sqrt{f}$ is the square root of a positive affine function, hence concave, with coefficient $2C_* > 0$. So ψ is positive on $[0, 4/5]$ once it is positive at the two endpoints, which TC-6 encloses in ball arithmetic ($\psi(0) > 0.064$, $\psi(4/5) > 1.7 \cdot 10^{-4}$). □

The local cone lemma

At $w = 0$ the equality points of Theorem 21 lie in the chart at the eight points $(\zeta_1, \zeta_2, \zeta_1', \zeta_2') \in \{(0,0,0,1), (0,0,1,0), (0,1,0,0), (1,0,0,0)\}$, one set for each sign sg. Let $z_i = |y_i - y_i^{\mathrm{eq}}|$ for the four chart coordinates and $z_{5,6} = w_{1,2}$; the box $\mathcal{B}_\rho$ is $|z|_\infty \le \rho$, $\rho = 1/256$.

Lemma S39 (Local cone; LCC). *At each equality point exactly four planes Π are tied: $G_\Pi = 0$ and the others are below -0.05. Their gradients (in z) sum to 0 and have rank three. In $\mathcal{B}_\rho$ one has $G > 0$ except at the point itself.*

Proof. By Taylor's formula with the Hessian enclosed over the box, $G_\Pi(t\omega) = t\, g_\Pi \cdot \omega + \frac{t^2}{2} \omega^T H_\Pi(\xi) \omega$ for ω on the unit sphere of $|\cdot|_\infty$ (in the orthant of the four chart coordinates) and $0 < t \le \rho$, with $H_\Pi(\xi)$ in an interval matrix $[H_\Pi]$, and $G \ge \sum_\Pi \mu_\Pi G_\Pi$ for weights $\mu_\Pi \ge 0$ summing to one on the tied planes. Write $\mu = \frac{1}{4} + \nu$. Because $\sum g_\Pi = 0$, the first-order term of the average depends on ν only. The tie and this identity are exact, and are proved symbolically (EX, no floating point or enclosures) at all eight points: exactly four planes have $\Phi_\Pi = \Phi_*$, where $4\Phi_* = (1 + 1/\sqrt{3})(1 + \sqrt{2/3})$

and all other Φ_Π are smaller; the twelve probabilities are $(2\pm\sqrt{2})/8$ and $\frac{1}{4}$, so $\delta = \frac{1}{2}\log_2 3 - \frac{1}{3}(1 + 2H_b(p))$ and $\delta = C_*(2 - 2\sqrt{\Phi_*})$, i.e. $G_\Pi = 0$ at the point for the four tied planes, which is the definition of C_* in (9); and $\sum_{\Pi \text{ tied}} \partial_k G_\Pi = 0$ for all six k, as an identity in $C_*, \ln p, \ln(1-p), \ln 2$. The reason is the sign symmetry $z \mapsto (s_1 z_1, s_2 z_2, s_1 s_2 z_3, z_4, s_2 z_5, s_1 z_6)$ at $(0,0,0,1)$, which EX also checks exactly: $T(\sigma y) = DT(y)D'$ with $D = \mathrm{diag}(1, s_1, s_2)$, $D' = \mathrm{diag}(1, s_1 s_2, 1)$ permutes the twelve t_{ji} inside each cut and the sixteen planes, so $G_\Pi \circ \sigma = G_{\pi(\Pi)}$ and only the z_4 component of $\sum_\Pi g_\Pi$ can survive; the z_4 component vanishes in EX (it is the derivative of the wall function G_w at $\pi/4$, zero by the reflection symmetry of Lemma S35). By the sign symmetry the tied gradients have the sign form $(s_1 a, s_2 a, -s_1 s_2 c, 0, -s_2 d, -s_1 d)$ of the remark below, so their rank is exactly three (the characters $s_1, s_2, s_1 s_2$ carry independent vectors, as the enclosures give $a, c \neq 0$); the bound -0.05 for the other planes and the Hessian enclosures are ball arithmetic (LCC). A piece Ω of the sphere is discharged if, for some ν, $\ell = \sum \nu_\Pi g_\Pi \cdot \omega \geq 0$ and $\ell + \rho q > 0$ (or $q > 0$) with $q = \frac{1}{2}\omega^T [H_\mu]\omega$, evaluated in ball arithmetic. This suffices: on Ω the average is at least $t\ell + t^2 q = t(\ell + tq)$, and $\ell + tq$ is positive for $0 < t \leq \rho$, being affine in t with $\ell \geq 0$ at $t = 0$ and $\ell + \rho q > 0$ at $t = \rho$ (or increasing if $q > 0$). LCC finds such weights on 9012 cells for the eight points. □

The covering

Proposition S40 (Interval covering; CCV)**.** *Let $\mathcal{R}$ be the part of the chart with $|w|^2 \leq 37/100$, both signs* sg*. Every box of the bisection tree is discharged by one of: (G) a certified positive lower bound of $\delta - C_* d^2$; (HS) $d^2 \leq 1/400$ (then $\delta \geq \frac{3}{10} d^2$, strictly greater than $C_* d^2$ when $d > 0$, by Theorem 19); (core) it lies in a box $\mathcal{B}_\rho$ of Lemma S39.*

Proof. The starting closed boxes cover $[0,1]^4 \times [-5/8, 5/8]^2$ in each sign sector, hence the required disc. A split replaces one box by two closed children whose union is the parent; their common boundary is retained. Boxes crossing the disc or ordering boundary are retained unless they are wholly outside the domain reduced by symmetry. Thus complete coverage means that every retained branch ends in one of the three certified alternatives in the statement. A failed or inconclusive sign test only triggers subdivision; it never certifies a box.

There is no need to choose a unique maximising fidelity plane. Since $G = \max_\Pi G_\Pi$, any convex combination G_μ is a lower bound for G, including on a tie. Near equality, the exact contacts and gradient cancellation of the local cone lemma replace a strict sign test at the apex.

On a box with centre c and half-widths h (in y) use the weights μ uniform on the best one to four planes. (i) δ is convex in the twelve numbers t_{ji}, hence $\delta(y) \geq \delta(c) + \sum_i D_i(t_i(y) - t_i(c))$ with $D_i = (1 + \ln \lambda_i(c))/(12 \ln 2)$, valid for every point of the box since every chart point is a state, so that $t_i(y) \geq -1$ (the tangent of a convex function holds on its whole domain); a ray whose probability at c is not certified positive is left out of the tangent and bounded by the maximum of $-x \log_2 x$ over its probability box ($-x \log_2 x$ increases on $[0, 1/e]$). (ii) Φ_Π is concave in the three entries m of m_Π, so $d_\Pi^2 = 2 - 2\sqrt{\Phi_\Pi}$ is convex there: $d_\Pi^2(m) \leq d_\Pi^2(m_c) + a_\Pi \cdot (m - m_c) + B$, where the Bregman remainder B is convex in $m - m_c$ and so at most its maximum at the corners of the box $|m - m_c| \leq r$; r is a bound for $|m(y) - m(c)|$. (iii) $t(y) - t(c)$ and $m(y) - m(c)$ are expanded in $y - c$ to second order, with the exact gradient at c and the interval second derivatives of the nine entries $T_{\alpha\beta}(y)$ over the box (closed forms, ball arithmetic). With $g_k = \partial_k G_\mu(c)$ this gives $G \geq G_\mu(c) - \sum_k |g_k| h_k - R$, where R is the sum of the second-order terms of (iii) inserted into the tangent of (i) and, with factor C_*, into the linear term of (ii), and of $C_* \max B$, the first-order terms being exact. Bisection of the widest side (the floating-point routine `critical_float.py` only orders the attempts and certifies nothing) discharges the chart region $\zeta_1 \leq \zeta_1'$, $|w|^2 \leq 37/100$ with 208953 boxes (G) in the sector sg $= +1$, and 246044 (G) and 436 (HS) for sg $= -1$, besides 42 boxes inside the local cone boxes in each sector; the ordering comparison $\zeta_1 > \zeta_1'$ is exact on the dyadic endpoints, while the outside-disc screen has a conservative positive margin exceeding its floating-point roundoff, so it discards only boxes strictly outside $|w|^2 \leq 37/100$. The orbit point of the chart ($d = 0$, for sg $= -1$) is covered by (HS). □

Proof of Theorem 21 for critical states. If $\tau \geq 0.36$, apply Lemma S38. Otherwise Lemma S37

reduces to the chart with $|w|^2 \le 37/100$, which Proposition S40 and Lemma S39 cover: $G > 0$, or $d > 0$ and $\delta \ge \frac{3}{10}d^2 > C_* d^2$, or $d = 0$, except at the equality points y^{eq} where $G = 0$ by (9). The equality points have $w = 0$ and are the 144 points of Theorem 21. □

Remark S41 (Structure at the equality points; numerical picture). In the coordinates $z = (\zeta_1, \zeta_2, \zeta_1', 1 - \zeta_2', w_1, w_2)$ at the equality point with $(\zeta_1, \zeta_2, \zeta_1', \zeta_2') = (0, 0, 0, 1)$, the four tied plane functions have the gradients $g = (s_1 a,\ s_2 a,\ -s_1 s_2 c,\ 0,\ -s_2 d,\ -s_1 d)$, $s_1, s_2 = \pm 1$, with $a = 0.059039$, $c = 0.036251$, $d = 0.018765$ (ball enclosures, LCC). They sum to zero and have rank three, with singular values $0.1239, 0.1239, 0.0725$: G has a conical minimum in three directions and is smooth in the other three. The average of the four tied Hessians is positive definite on the three-dimensional kernel of the gradients, with eigenvalues 0.0725 (the wall direction θ) and $0.7285, 0.7285$ (two combinations of the chart coordinates with the w-plane). In particular the dependence on w is first order, in the shape of a square $(\pm 1, \pm 1)$, so that the loss of fidelity under moving off $\tau = 0$ is compensated at first order by the cone and costs nothing along the smooth directions. Numerically (not a proof; numerical evidence, NV-1) the minimum of $\delta - C_* d^2$ over critical states with fixed τ is about $0.31\,\tau$ for every $\tau \in (0, 1]$ (about 0.30τ at $\tau = 0.4$ and 0.5), attained near the equality points for small τ and, at $\tau = 1$, at a real state with $d^2 = 2 - \sqrt{2}$.

Remark S42 (Non-critical states; numerical evidence only). The theorem is for critical states; whether C_* is optimal over all states is open. Evidence (not a proof): at the equality state, random positive local filters of size $\varepsilon \le 0.2$ (the twelve non-critical directions of the 18-dimensional LU quotient; 12 directions per size) give $\delta - C_* d^2 \ge 3\varepsilon^2$ (NV-2); on states obtained by filtering critical states within 0.4 of the equality state (normal-form coordinates) with filters of size 0.05 to 0.4 the minimum of δ/d^2 is 0.27. These sampled tests found no value below C_*. Reduction of a non-critical state to its critical representative does not preserve δ (by Section S6.21, one filter on one qubit of a critical state cannot raise any balanced-cut entropy, but filters on two qubits can raise the mean Rényi entropy of order 3/2), and the general coefficient $k(6 + \sqrt{6})/12 \approx 0.05$ is a factor 3.8 below C_*, so a proof needs a quantitative treatment of the filter directions: near the critical locus a Taylor bound in the filter variables over this six-parameter chart (a covering of the same size), far from it a stronger distance–entropy comparison.

S9 A common replica hierarchy

Section S9.2 bounds the mean three-qubit purity from below using four copies of the state. This section places that bound in a hierarchy of semidefinite relaxations indexed by the number m of copies. It is a bosonic-extension hierarchy with PPT constraints of the Doherty–Parrilo–Spedalieri type [3], applied to the pure-state moment problem rather than to separability; what is new here are the twirled reductions, the sharp certificates and the orbit recovery. Level $m = 2$ is exactly the linear programme of Rains (Proposition S43); the spin-flip blocks give all higher levels, which increase to the true minimum (Theorem S44); and levels $m = 3, 4$ are the three- and four-copy relaxations. The three-copy level has exact optimum $125/978 = 0.12781\ldots < 121/917$, and four copies are provably stronger (Section S9.1). The optimal three-copy pseudo-moment also rules out cubic sum-of-squares proofs of the sharp bound. Five copies are sharp: Section S11 gives the certificate and the equality case.

Proposition S43 (two-copy PPT formulation of the Rains LP)**.** *After local-unitary and site-permutation twirling, the Rains LP of Rains [4] (stated in Section S15 in the notation a_s, A_j, S_j used here) is exactly the two-copy positive-partial-transpose relaxation with support on the globally symmetric two-copy subspace: a twirled two-copy operator X and an LP point (a_k) correspond bijectively through $a_k = \mathrm{tr}(X G_{n,k})$ (with $G_{n,k}$ as in Theorem S44), and X is feasible exactly when the point is.*

Proof. Twirling is without loss of generality: the PPT constraint and the objective are invariant under $U \otimes U$ and site permutations, so averaging a feasible operator keeps it feasible with the same

value. Let F_i swap the two copies at qubit i, and $Q_i = 2F_i - I$. If R_r is the projector onto the sum of sectors with exactly r locally antisymmetric sites, then the local-unitary and site-twirled algebra is the commutative span of the R_r. The generating polynomial for the shadow coefficients gives $S_{n-r} = 2^n \operatorname{tr}(XR_r)$ for a twirled operator X, as an algebraic identity in which the enumerator of a general X is defined by $A_j = \sum_{|T|=j} \operatorname{tr}(XQ_T)$, $Q_T = \prod_{i\in T} Q_i$, so $S_j \geq 0$ is equivalent to $X \succeq 0$. Since $F_{\text{all}} = \prod_i F_i = 2^{-n} \prod_i (I + Q_i)$, the equality $a_n = a_0 = 1$ says $\operatorname{tr}(XF_{\text{all}}) = 1$, forcing globally symmetric support for positive normalised X. Let $\mathcal{T}$ be partial transpose on the second copy followed by the local spin flip $\epsilon^{\otimes n}$, acting on the second copy only. Since $\mathcal{T}(F_i) = I - F_i$, and $\mathcal{T}$ is its own Hilbert–Schmidt adjoint on the span of the F_S (so $\operatorname{tr}[\mathcal{T}(X)R_r] = \operatorname{tr}[X\mathcal{T}(R_r)]$),

$$F_{\text{all}} \sum_{r=0}^{n} \mathcal{T}(R_r) t^r = 2^{-n} \prod_{i=1}^{n} (tI + Q_i).$$

On the globally symmetric support this yields $\operatorname{tr}[\mathcal{T}(X)R_r] = 2^{-n} A_{n-r}$, so $A_j \geq 0$ is equivalent to $\mathcal{T}(X) \succeq 0$. Conversely, the shadow moments reconstruct X uniquely in the twirled algebra; global bosonic support gives $a_s = a_{n-s}$, and PPT gives $A_j \geq 0$. □

Theorem S44 (the spin-flip blocks are the bosonic PPT hierarchy)**.** *For general n, k, the two-copy operator $G_{n,k} = \binom{n}{k}^{-1} \sum_{|S|=k} \prod_{i\in S} F_i$ satisfies $\pi_k(z) = \operatorname{tr}[G_{n,k}(|z\rangle\langle z|)^{\otimes 2}]$. For $m \geq 2$, minimise the same expectation over normalised positive m-copy operators supported on* $\operatorname{Sym}^m((\mathbb{C}^2)^{\otimes n})$ *and positive under partial transposition on every collection of* global copies*. By Proposition S43, L_2 is exactly the Rains LP. The lower bounds L_m satisfy $L_m \leq L_{m+1} \leq \min_z \pi_k(z)$. Their limit equals $\min_z \pi_k(z)$ by compactness and the quantum de Finetti representation for infinitely extendible bosonic states. Local-unitary and site-permutation twirling reduce each finite SDP. For qubits, let $\mathcal{A}_m$ be the local copy commutant and, for $1 \leq s \leq \lfloor m/2 \rfloor$, define*

$$\mathcal{T}_s(X) = E_s X^{T_{[s]}} E_s^\dagger, \qquad E_s = (\epsilon^{\otimes n})^{\otimes s} \otimes I.$$

The operator X is feasible exactly when it is positive, normalised, globally bosonic, and every $\mathcal{T}_s(X)$ is positive. After twirling these are operators in $\mathcal{A}_m^{\otimes n}$. Thus the complete *Fourier ω- and flipped γ-blocks, including all site types, give precisely this hierarchy. In particular the three- and four-copy SDPs in §S9.1–S9.2 are its levels $m = 3, 4$ in conjugation-even coordinates. Restricted focused-contraction cones in §S13 are smaller objects and are not being identified with the complete hierarchy.*

Proof. Partial trace maps feasible level $m+1$ operators to feasible level m operators and preserves the objective, proving monotonicity. The product $(|z\rangle\langle z|)^{\otimes m}$ is feasible, so $L_m \leq \min_z \pi_k(z)$. For the limit, let X_m attain L_m (the feasible set is compact). For each fixed j, the j-copy marginals of X_m, $m \geq j$, lie in a compact set; a diagonal subsequence gives limits $X^{(j)}$, each the partial trace of $X^{(j+1)}$ and supported on $\operatorname{Sym}^j((\mathbb{C}^2)^{\otimes n})$. By the quantum de Finetti theorem for such an infinitely exchangeable family [14], $X^{(j)} = \int \sigma^{\otimes j} \, d\mu(\sigma)$ for a probability measure μ on states. Bosonic support gives $1 = \operatorname{tr}(FX^{(2)}) = \int \operatorname{tr}\sigma^2 \, d\mu(\sigma)$ for the swap F, so μ is carried by pure states $\sigma = |z\rangle\langle z|$. Hence $\lim_m L_m = \operatorname{tr}[G_{n,k}X^{(2)}] = \int \pi_k(z) \, d\mu \geq \min_z \pi_k(z)$; the PPT constraints are not needed for the limit.

Unitary conjugation by E_s preserves positivity, so $\mathcal{T}_s(X) \succeq 0$ is exactly the usual global-copy PPT constraint. Global copy permutations equate all subsets of the same size; full transposition equates a subset with its complement. These are all PPT constraints. The identity $\epsilon\overline{U}\epsilon^{-1} = U$ for $U \in SU(2)$ shows that $\mathcal{T}_s$ preserves the local commutant. Bosonic support of X implies left and right invariance of $\mathcal{T}_s(X)$ under $S_s \times S_{m-s}$, so its other blocks vanish identically. Positivity on every remaining Fourier block is equivalent to positivity of the full operator; invertible basis changes preserve it. For $m = 3$ there is just the $1|2$ flip; for $m = 4$ there are the $1|3$ and $2|2$ flips used in the construction. Finally $X \mapsto \overline{X}$ preserves all constraints and the real purity objective. Averaging X with $\overline{X}$ loses no objective value and justifies the conjugation-even coordinate reduction. □

Bosonic realignment at four copies. One natural convex strengthening of PPT is the realignment (or computable cross-norm) condition. The copy-2|2 version, however, is *already implied* by the bosonic PPT constraints of Theorem S44. For an operator X on $V \otimes V$ write its matrix elements $X_{ij,kl}$ and define its realignment by $\mathcal{R}(X)_{ik,jl} = X_{ij,kl}$. Let F swap the two V factors. If $XF = X$, then

$$\mathcal{R}(X)_{ik,jl} = X_{ij,lk} = (X^{T_2})_{ik,lj}, \qquad \text{i.e.} \quad \boxed{\mathcal{R}(X) = X^{T_2}F}.$$

Consequently for trace-one X with $X^{T_2} \succeq 0$, $\|\mathcal{R}(X)\|_1 = \|X^{T_2}\|_1 = \operatorname{tr} X = 1$. At four copies set $V = H^{\otimes 2}$ and group $(1,2)|(3,4)$. Bosonic support on $\mathrm{Sym}^4(H)$ implies $XF = X$ for the group swap $(13)(24)$; the Y_2 constraint of Section S9.2 is precisely PPT on copies $(3,4)$, up to its unitary spin-flip congruence. Thus a 2|2 CCNR inequality adds no restriction to the four-copy SDP.

For completeness, the *unequal* copy split 1|3 supplies the distinct, valid necessary condition

$$\|\mathcal{R}_{1|3}(X_4)\|_1 \le 1$$

for any actual (possibly twirled) fourth-power moment, or any convex mixture of fourth powers: each unaveraged pure tensor power is a product across that copy partition, its realignment has trace norm one, and convexity survives twirling. No claim is made that this 1|3 condition is independent of the existing bosonic PPT constraints or that it improves the numerical seven-qubit bound; both questions remain untested.

S9.1 The three-copy spin-flip relaxation

The sharp four-qubit inequality of Theorem 3 is proved by an identity among squared norms of local contractions of $z \otimes z \otimes \tilde{z}$, $\tilde{z} = \epsilon^{\otimes n}\bar{z}$, $\epsilon = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ (Appendix A). The same positivity construction gives a semidefinite relaxation for general numbers of qubits.

Moments. For a unit $z \in (\mathbb{C}^2)^{\otimes n}$ and m copies, the per-site twirl of $(zz^*)^{\otimes m}$ by $U^{\otimes m}$, $U \in U(2)$ (which has the same commutant as $SU(2)$), lies in $\mathcal{A}_m^{\otimes n}$, where $\mathcal{A}_m \subset \mathrm{End}((\mathbb{C}^2)^{\otimes m})$ is the image of $\mathbb{C}[S_m]$, and it is positive. For $m = 3$, $\mathcal{A}_3 \cong \mathbb{C} \oplus M_2$ (trivial and standard S_3 blocks; the sign block vanishes on qubits). The twirl is determined by the word values $\mu(w) = \langle z^{\otimes 3}| \bigotimes_i \pi_i |z^{\otimes 3}\rangle$, $w = (\pi_1, \ldots, \pi_n) \in S_3^n$. Since π_3 is invariant under site permutations, global copy permutations and $z \mapsto \bar{z}$, we may average μ over these; the averaged values $\bar{\mu}$ are constant on orbits of $S_n \times S_3 \times S_3$ and inversion. Two families of positive matrices are linear in $\bar{\mu}$:

- *ω-blocks*: the Fourier blocks $\sum_w \bar{\mu}(w) \bigotimes_i \rho(\pi_i^{-1})$ of the twirl of $(zz^*)^{\otimes 3}$, compressed to the copy-symmetric subspace;
- *γ-blocks*: the same for the twirl of $|w\rangle\langle w|$, $w = z \otimes z \otimes \tilde{z}$. Because $\epsilon \bar{U} \epsilon^{-1} = U$ on $SU(2)$, this twirl is the image of the first under the per-site spin-flip map $t(\pi) = (I \otimes I \otimes \epsilon)^T \pi^{T_3} (I \otimes I \otimes \epsilon)$, which is expanded exactly in permutations (TCB-0).

Every one-site antisymmetriser of three qubit copies vanishes, which gives linear relations among the $\bar{\mu}$. For $n = 7$ there are 37 orbit values, 23 relations, 7 ω-blocks and 8 γ-blocks, built exactly in integer bases (TCB-0).

Theorem S45 (exact certificate; TCB-0–TCB-4)**.** *Every pure seven-qubit state satisfies* $\pi_3(\psi) \ge 125/978 = 0.12781186\ldots$*, and the exact optimum of the* $n = 7$ *three-copy spin-flip relaxation is* $L_3 = 125/978$*.*

Proof. At the boundary rational pseudo-moment $m^*/70416$ (given by 37 four-digit integers with $m^*_{\mathrm{norm}} = 70416$ and $m^*_{\mathrm{obj}} = 9000$), all 23 antisymmetriser relations vanish, nine of the 15 integer block matrices $B_b(m^*)$ are strictly positive definite, and the remaining six active blocks $b \in \mathcal{A} = \{\omega_6, \omega_7, \gamma_4, \gamma_5, \gamma_6, \gamma_7\}$ have integer kernels $V_b = \ker B_b(m^*)$ of dimensions $k_b = 1, 14, 1, 5, 22, 56$ whose complementary principal pivot submatrices are strictly positive definite by fraction-free

Bareiss elimination (TCB-2, TCB-3), proving $B_b(m^*) \succeq 0$ and hence $L_3 \le 125/978$. On each active block $b \in \mathcal{A}$, let $G_{\mathrm{site},b} \succ 0$ be the integer site-invariant Gram matrix, $M_{b,0} = V_b^{\mathsf{T}} G_{\mathrm{site},b} V_b \succ 0$, and $K_{b,q} = V_b^{\mathsf{T}} B_b(R_q) V_b$ for the 14 integer RREF basis vectors R_q of $\ker(\mathrm{rel})$. The 25 canonically normalised S_m-invariant kernel matrices $\widehat{Y}_{b,p} = M_{b,0}^{-1} K_{b,p} M_{b,0}^{-1} / \operatorname{tr}(M_{b,0}^{-1} K_{b,p} M_{b,0}^{-1} K_{b,p})$ have basis-independent trace pairings $\operatorname{tr}(\widehat{Y}_{b,p} K_{b,q})$. Solving the 14 reduced dual equations at $\lambda = 125/978$ with 12 rational free parameters yields symmetric rational multipliers $Y_b = \sum_p w_{b,p} \widehat{Y}_{b,p}$ on the six active kernels that TCB-3 proves strictly positive definite by Bareiss elimination, together with rational relation multipliers $\nu \in \mathbb{Q}^{23}$ satisfying the exact 37-orbit identity (TCB-2)

$$\bar{\mu}_{\mathrm{obj}} - \frac{125}{978} \bar{\mu}_{\mathrm{norm}} = \sum_{b \in \mathcal{A}} \langle Y_b, V_b^{\mathsf{T}} B_b(\bar{\mu}) V_b \rangle + \nu \cdot \mathrm{rel}(\bar{\mu}).$$

For a genuine state $\bar{\mu}_{\mathrm{norm}} = 1$, $\mathrm{rel}(\bar{\mu}) = 0$, $B_b(\bar{\mu}) \succeq 0$ and $\bar{\mu}_{\mathrm{obj}} = \pi_3$, since each $B_b(\bar{\mu})$ is obtained from a multiplicity block of a positive twirled moment by compression to the copy-invariant subspace followed by congruence with a fixed invertible integer matrix. The identity uses only normalisation, the relations and block positivity, so every feasible pseudo-moment satisfies $\bar{\mu}_{\mathrm{obj}} \ge 125/978$, giving $L_3 = 125/978$. The four-copy bound of Theorem S47 supersedes this value; the three-copy certificate is kept for the strict separation of Corollary S46 and the cubic-SOS obstruction. □

Since $L_3 = 125/978 < 121/917$, the three-copy positivity and linear relations cannot certify the sharp seven-qubit bound. The boundary point $m^*/70416$ is a pseudo-moment, not a quantum state, so it is no physical counterexample. Through Theorem S44, which identifies block positivity with operator positivity on every local representation sector, it also excludes every homogeneous real cubic SOS proof of the sharp bound (Theorem S100).

Corollary S46 (quantitative strictness of the copy hierarchy)**.** *Let L_3 and L_4 denote the optima of the specific three- and four-copy spin-flip SDPs defined in this section. Then the exact certificates imply*

$$L_4 - L_3 \;\ge\; \frac{211}{1600} - \frac{125}{978} = \frac{3179}{782400} > 0.$$

Thus four copies yield a provably strictly stronger seven-qubit SDP bound, by more than 0.00406*. Neither level attains* 121/917*: Proposition S49 gives $L_3 < L_4 < 6597/50000 < 121/917$.*

Proof. Theorem S47 gives $L_4 \ge 211/1600$ and Theorem S45 gives $L_3 = 125/978$. □

Numerical (TCN-1): the relaxation reproduces the four-qubit value 1/3 at $n = 4$ and the AME values 1/4, 1/8 at $n = 5, 6$. At $n = 7$ the tight-tolerance solve (TCN-1b) gives $0.1278118609\ldots$, which agrees with $125/978$ to 3×10^{-13}, so the three-copy bound stays below 121/917. Rico et al. [11] report 0.127(8), without a certificate, by strengthening the linear programme with the quantum-code SDP of Anglès Munné, Nemec and Huber [12].

S9.2 The four-copy spin-flip relaxation

Construction. At four copies, the per-site algebra is $\mathcal{A}_4 \cong \mathbb{C} \oplus M_3 \oplus M_2$: the S_4 irreps $[4], [3,1], [2,2]$, with $U(2)$ multiplicity spaces of dimension $5, 3, 1$ (the others vanish on qubits). The twirled moment X of $(|\psi\rangle\langle\psi|)^{\otimes 4}$ lies in $\mathcal{A}_4^{\otimes n}$; after averaging over site permutations and $\psi \mapsto \overline{\psi}$ it is real, S_n-symmetric and fixed by global copy permutations on either side, so each Fourier block of a site-type pattern (a, b, c) (a sites in $[3,1]$, b in $[2,2]$, c in $[4]$) is supported on the S_4-invariant vectors of $[3,1]^{\otimes a} \otimes [2,2]^{\otimes b}$. Splitting these by the $S_a \times S_b$ isotypic components leaves blocks $\omega_{(a,b),\lambda\mu}$ whose entries are the free parameters: 167 at $n = 7$. Two flipped moments are linear in X:

$$Y_1 = \text{twirl of } |\psi\psi\psi\tilde{\psi}\rangle\langle\psi\psi\psi\tilde{\psi}| = T_1^{\otimes n} X,$$
$$Y_2 = \text{twirl of } |\psi\psi\tilde{\psi}\tilde{\psi}\rangle\langle\psi\psi\tilde{\psi}\tilde{\psi}| = T_2^{\otimes n} X,$$

where T_1, T_2 are the per-site maps $a \mapsto E\, a^{T_{\text{flip}}} E^T$ (partial transpose on the flipped copies, $E = \epsilon$ on those copies), exact rational 14×14 matrices with $12T$ integral. Y_1 is supported on S_3-invariant and Y_2 on $S_2 \times S_2$-invariant vectors of the copies, and both are positive. The relaxation minimises the mean purity $\operatorname{tr}(X\, \mathrm{SWAP}_{12}^S)$ (averaged over $|S| = k$) subject to $\operatorname{tr} X = 1$, $\omega \succeq 0$ and the γ-blocks of $Y_1, Y_2 \succeq 0$. Tracing out a copy maps these constraints onto the corresponding three-copy ones, so the four-copy bound is at least the three-copy bound. The numerical construction uses orthonormal irreps; the certificate uses integer irreps and integer symmetry-adapted bases.
Numerical (FS-0–FS-2): the values at $n = 4, 5, 6$ are 1/3, 1/4, 1/8 (to 10^{-9}), the true minima. At $n = 7$:

constraints	ω	$\omega + Y_1$	$\omega + Y_2$	$\omega + Y_1 + Y_2$
bound on π_3	$-1/7$	0.130888	0.131821	0.131877

The combined numerical value is about 7.5×10^{-5} below 121/917; Y_2 alone nearly reaches the combined value. To improve the conditioning of the numerical solve, every block is first congruence-centred at the Haar average of the moment, proportional to $\sum_{\pi \in S_4} \pi^{\otimes n}$, on the retained nonzero blocks. The two largest $n = 7$ solves are flagged inaccurate by the solver, so these values are numerical evidence. The certified bounds are Theorem S47 and, from above, Proposition S49: $211/1600 \le L_4 < 6597/50000 < 121/917$.

Theorem S47 (exact certificate; FC-0–FC-4)**.** *Every pure seven-qubit state satisfies* $\pi_3(\psi) \ge 211/1600 = 0.131875$.

Proof. Write x for the 167 real coordinates of the symmetrised moment, with $\mathrm{on}(x) = \operatorname{tr} X = 1$ and $\mathrm{oo}(x) = \pi_3(\psi)$. The trace weights each site type (a, b, c) by $\frac{7!}{a!b!c!} 3^a 5^c$. In the certificate's convention each multiplicity matrix has $\omega_{kk} = x_{kk}$ and $\omega_{kl} = x_{kl}/2$ for $k \ne l$. These matrices are positive semidefinite for genuine states. Indeed each nontrivial Fourier block is $2R(\omega \otimes \operatorname{adj} K)R^T H$, with H the positive Gram form, R an integer basis of the allowed isotypic support and K a positive definite Gram matrix. Conjugation by $H^{1/2}$ makes this a congruence of $\omega \otimes \operatorname{adj} K$; the scalar trivial block is treated separately. For a Fourier block B_j of Y_j, $j = 1, 2$, the flip blocks are $\gamma_g(x) = Q_g^T H B_j Q_g \succeq 0$, where Q_g spans the corresponding row-invariant isotypic component. These are Gram compressions of the positive Y_j blocks.

Put $\lambda = 211/1600$. The certificate supplies a positive integer Q_z and integer matrices $Z_g \succ 0$ such that

$$d_p = \mathrm{oo}_p - \lambda\, \mathrm{on}_p - \frac{1}{Q_z} \sum_g \langle Z_g, \gamma_g(e_p) \rangle$$

forms a positive definite matrix $D_b = [d_{kl}]$ on every ω-block b. Here $\langle A, B \rangle = \operatorname{tr}(A^T B)$. FC-3 checks positive definiteness by exact fraction-free elimination. It computes the integer numerator $12^7 \sum_g \langle Z_g, \gamma_g(e_p) \rangle$ using $12T_1, 12T_2$, by adjoint contraction modulo 35 primes and Chinese remaindering under an integer triangle-inequality bound. Thus, with the off-diagonal convention above,

$$\pi_3(\psi) - \lambda = \sum_b \langle D_b, \omega_b(x) \rangle + \frac{1}{Q_z} \sum_g \langle Z_g, \gamma_g(x) \rangle \ge 0.$$

The rational basis construction uses modular pivots only to prove independence. For the copy-group invariant space, the exact trace of its averaging projector gives the matching upper bound on dimension. In the subsequent Young-isotypic decomposition the row subgroup has a one-dimensional fixed line in the matching Specht module, so each selected row-invariant vector accounts for one multiplicity; the sum of its irrep dimensions equals that same exact rank. Thus no component is omitted. A floating rank estimate is not used for completeness. □

Remark S48 (regressions; numerical). Two numerical regressions accompany the exact proof and are not used in it. FC-1 checks the parametrisation and the identities $\mathrm{on} = \operatorname{tr} X$, $\mathrm{oo} =$ mean purity, in the same exact coordinates, on random three-qubit states; FC-2 compares the CRT-recovered adjoint sum with a direct forward computation at $n = 3, 4$ for random integer Z.

Proposition S49 (the four-copy relaxation is not sharp; exact certificate, NS-0–NS-4)**.** *Let L_4 be the optimum of the $n = 7$ four-copy relaxation above ($k = 3$; constraints $\operatorname{tr} X = 1$, all 78 ω-blocks and all 271 flip blocks of Y_1, Y_2 positive semidefinite; these are all of its constraints, with no marginal or further flip equalities). Then*

$$\frac{211}{1600} \;\le\; L_4 \;<\; \frac{6597}{50000} = 0.13194 \;<\; \frac{121}{917} = 0.131952\ldots, \qquad \frac{121}{917} - \frac{6597}{50000} = \frac{551}{45850000}.$$

The feasible point is a pseudo-moment, not a seven-qubit state; it excludes only proofs that use this relaxation unchanged.

Proof. The lower bound is Theorem S47. For the upper bound, the data file supplies an integer vector x of the 167 coordinates of the proof of Theorem S47. Apart from $\operatorname{tr} X = 1$ every constraint is homogeneous, so $x/\mathrm{on}(x)$ is feasible once every block of x is positive semidefinite, and its objective is $\mathrm{oo}(x)/\mathrm{on}(x)$. NS-1 computes $\mathrm{on}(x) > 0$ and this ratio exactly; it is $0.1319308\ldots < 6597/50000$. NS-2 checks that all 78 matrices $2\omega_b(x)$ (integer) are positive definite by exact fraction-free elimination. NS-3 forms the moment $X(x)$ and $(12T_j)^{\otimes 7}X(x)$ modulo each of 34 primes, recovers every $12^7\gamma_g(x) = Q_g^T H\, B_j\, Q_g$ as an integer matrix by the Chinese remainder theorem under an integer triangle-inequality bound on its entries (the bound covers the block sums of the moment, the seven flip contractions and the compression; the modulus exceeds twice it), and checks that all 271 are exactly symmetric and positive definite by exact fraction-free elimination. Hence $L_4 \le \mathrm{oo}(x)/\mathrm{on}(x) < 6597/50000$. □

Remark S50 (provenance of the pseudo-moment; numerical). The point x is a floating-point optimum of the Haar-centred relaxation (0.1318766470), mixed with the Haar point at weight 10^{-3}, moved to the exact coordinates by a per-type least-squares fit and rounded to a dyadic grid; its least centred block eigenvalue is about 10^{-3}. None of this is used in the proof, which checks the stored integers only. NS-0 is a regression of the modular forward map against a direct floating-point computation at $n = 3, 4$.

S10 The complex conference family

The preceding replica hierarchy shows how the lower-copy relaxations fall short of the sharp seven-qubit purity bound. An explicit candidate is provided by the complex conference family, whose states are parametrised by a complex symmetric 6×6 matrix. We prove a sum-of-squares purity identity *within this family*, classify its equality cases, and derive its spin-flip equation. This explains the icosahedral geometry, but does not establish a bound for arbitrary states: the five-copy certificate in Section S11 supplies globality, and the invariant reconstruction there supplies equality rigidity. Figure S1 separates these steps from the later entropy proof.

S10.1 A sharp complex conference family

The icosahedral family has the two-branch form $\Omega_{u,v}$ of Theorem S51, with branch supports far apart in Hamming distance. This separation makes the low-weight odd Pauli correlations vanish, as the first result shows. We then apply that cancellation to the family with complex pair amplitudes, which contains the icosahedral state. In Corollary S52 through Theorem S53 below, the distinguished qubit is labelled 0 and the ordinary qubits $1, \ldots, 6$; in Proposition S54 matrix indices are the ordinary labels $0, \ldots, 5$ of (14).

Theorem S51 (spin-flip-doubled odd-weight selection)**.** *Let u, v be vectors on m qubits whose computational-basis supports have minimum Hamming distance $D > 0$. Let $J_m = \epsilon^{\otimes m} K$, where K is complex conjugation, and set*

$$|\Omega_{u,v}\rangle = \frac{|0\rangle(u+v) + |1\rangle J_m(v-u)}{\sqrt{2(\|u\|^2 + \|v\|^2)}}.$$

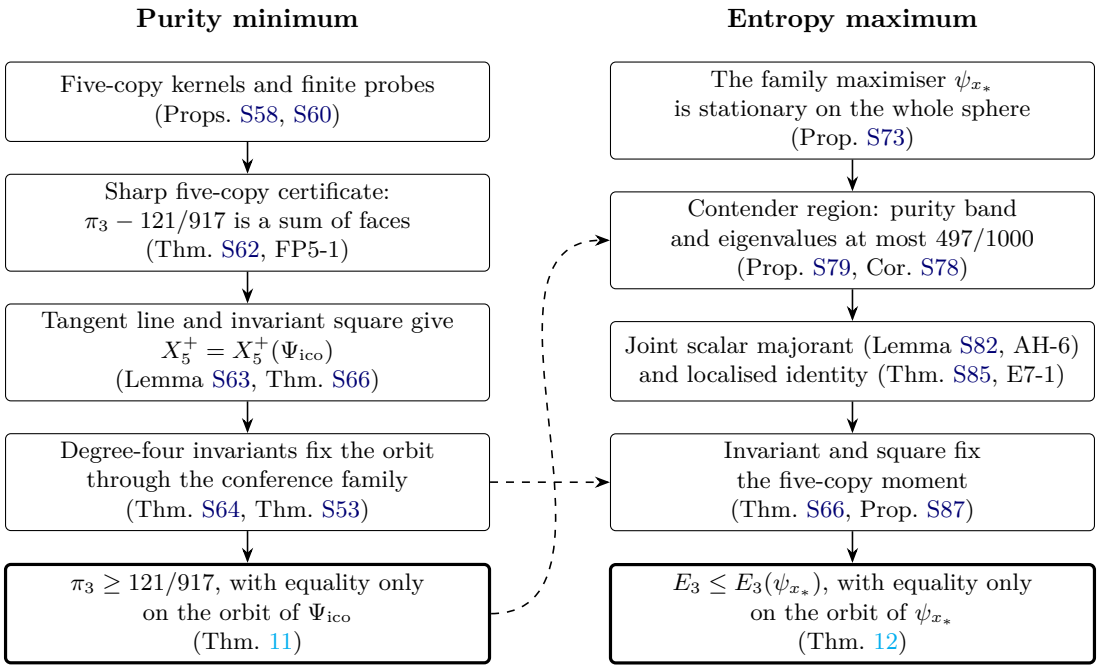


Figure S1: How the seven-qubit optima are proved. Solid arrows are the steps of each proof; dashed arrows mark what the entropy proof takes from the purity proof: the purity bound, which confines the contenders, and the orbit step. Section S10 treats the conference family, Section S11 the purity minimum and Section S12 the entropy maximum.

Every Pauli expectation supported entirely on the last m qubits of odd weight $q < D$ vanishes. If u is supported at Hamming weight w and $v = b|1^m\rangle$ with $0 < w \leq m-2$ and $2w \neq m$, then all $m+1$ *one-qubit marginals of $\Omega_{u,v}$ are maximally mixed.*

Proof. A Pauli word of weight $q < D$ cannot connect the support of u to that of v. Thus its expectations in $u+v$ and in $v-u$ are equal. Spin flip reverses the sign of every odd-weight Hermitian Pauli expectation: $\langle J_m a|P|J_m a\rangle = (-1)^{\mathrm{wt}\,P}\langle a|P|a\rangle$. The two branches therefore cancel. For the last assertion, $D = m - w \geq 2$ gives maximal mixing on each of the ordinary sites. The two branches have equal norm, while their supports are disjoint when $w \notin \{0, m/2, m\}$, so the distinguished qubit also has marginal $I/2$. □

Corollary S52 (ordinary triple reductions in the complex conference ansatz; SDS-1–SDS-3)**.** *The seven-qubit pair/complement family of Theorem S53, which contains the icosahedral family of Theorem 41, is $\Omega_{u,v}$ with $m = 6$, $w = 2$, $u = \sum_{i<j} t_{ij}|ij\rangle$ and $v = b|1^6\rangle$. For every three ordinary sites $I = \{i,j,k\} \subseteq \{1,\ldots,6\}$:*

$$\langle P_I\rangle = 0 \quad \textit{for every Pauli word of exact weight } 3, \qquad \Phi_{I,0} = 0,$$

and

$$\operatorname{tr}\rho_{ijk}^2 = \frac{1}{2}\left(\operatorname{tr}\rho_{ij}^2 + \operatorname{tr}\rho_{ik}^2 + \operatorname{tr}\rho_{jk}^2\right) - \frac{1}{4}. \tag{S10.1}$$

This holds for all complex pair amplitudes, not only for conference matrices or the optimal parameter.

Proof. The Hamming separation is $D = 4$, so all ordinary-site Pauli weights one and three vanish by Theorem S51. Lemma S96, where the focused covariants $\Phi_{I,r}$ are defined (Section S13), then proves the focused-covariant zero. The Pauli expansion of the normalised triple purity is $8\operatorname{tr}\rho_{ijk}^2 = 1 + \sum_{\{a,b\}\subseteq I}(4\operatorname{tr}\rho_{ab}^2 - 1)$, giving (S10.1). SDS-1–SDS-3 check Pauli cancellation and all 20 triple-purity relations using Gaussian integer amplitudes. □

Theorem S53 (proved; CC-1)**.** *Let $T = (t_{ij})$ be any complex symmetric 6×6 matrix with zero diagonal, and let $b \geq 0$ be real. Define the normalised seven-qubit state*

$$z_{T,b} = |0\rangle \left(\sum_{i<j} t_{ij} |ij\rangle + b|1^6\rangle \right) + |1\rangle \left(b|0^6\rangle - \sum_{i<j} \overline{t_{ij}} |\overline{ij}\rangle \right), \quad 2\left(b^2 + \sum_{i<j} |t_{ij}|^2 \right) = 1.$$

Here $|ij\rangle$ is the six-bit weight-two vector supported on i, j, and $|\overline{ij}\rangle$ its complement. Put $x = b^2$, $\sigma = \sum_{i<j} |t_{ij}|^2$, $p_i = \sum_{j \neq i} |t_{ij}|^2$. Then

$$\begin{aligned} \pi_3(z_{T,b}) - \frac{121}{917} = \frac{(524x - 37)^2}{137550} + \frac{4}{35} \sum_i (p_i - \sigma/3)^2 \\ + \frac{48}{35} \sum_{i<j} (|t_{ij}|^2 - \sigma/15)^2 + \frac{24}{35} \sum_{i<j} |(TT^\dagger)_{ij}|^2. \end{aligned} \tag{S10.2}$$

Consequently the minimum on this entire complex-amplitude family is $121/917$. Equality requires $x = 37/524$, uniform magnitudes $|t_{ij}|$ and $TT^\dagger = (\sigma/3)I$: T is a scaled complex symmetric conference matrix. The icosahedral state is one equality case.

Proof. Fix an ordinary pair $\{i, j\} \subset \{1, \ldots, 6\}$ and let $K = \{1, \ldots, 6\} \setminus \{i, j\} = \{k, l, m, n\}$. When K is traced out to form ρ_{0ij}, the five Hamming-weight sectors on K map to four orthogonal two-dimensional subspaces of $\mathcal{H}_{0ij}$:

- weights 0 and 4 on K give orthogonal kets $t_{ij}|011\rangle + b|100\rangle$ and $b|011\rangle - \overline{t_{ij}}|100\rangle$, contributing $(|t_{ij}|^2 + b^2) I_2$ on $\mathrm{span}\{|011\rangle, |100\rangle\}$;
- weight 1 on K (at $r \in K$) gives $|0\rangle(t_{ir}|10\rangle + t_{jr}|01\rangle)$, contributing $M_{ij} = \left(\begin{smallmatrix} p_i - |t_{ij}|^2 & (TT^\dagger)_{ij} \\ \overline{(TT^\dagger)_{ij}} & p_j - |t_{ij}|^2 \end{smallmatrix}\right)$ on $\mathrm{span}\{|010\rangle, |001\rangle\}$, and weight 3 on K gives $\overline{M_{ij}}$ in the ordered basis $(|101\rangle, |110\rangle)$;
- weight 2 on K gives $\left(\begin{smallmatrix} Q_K & -2\,\mathrm{Hf}(T_K) \\ -2\overline{\mathrm{Hf}(T_K)} & Q_K \end{smallmatrix}\right)$ on $\mathrm{span}\{|000\rangle, |111\rangle\}$, where $Q_K = \sum_{\{u,v\} \subset K} |t_{uv}|^2 = \sigma - p_i - p_j + |t_{ij}|^2$ and $\mathrm{Hf}(T_K) = t_{kl}t_{mn} + t_{km}t_{ln} + t_{kn}t_{lm}$.

Hence

$$\mathrm{tr}\, \rho_{0ij}^2 = 2(|t_{ij}|^2 + b^2)^2 + 2(p_i - |t_{ij}|^2)^2 + 2(p_j - |t_{ij}|^2)^2 + 4|(TT^\dagger)_{ij}|^2 + 2Q_K^2 + 8|\,\mathrm{Hf}(T_K)|^2.$$

Tracing out qubit 0 from ρ_{0ij} leaves ρ_{ij} diagonal on $\{|00\rangle, |11\rangle\}$ with entry $\frac{1}{2} - d_{ij}$ ($d_{ij} = p_i + p_j - 2|t_{ij}|^2$, using $\sigma + b^2 = \frac{1}{2}$) and equal to $\left(\begin{smallmatrix} d_{ij} & 2(TT^\dagger)_{ij} \\ 2\overline{(TT^\dagger)_{ij}} & d_{ij} \end{smallmatrix}\right)$ on $\mathrm{span}\{|10\rangle, |01\rangle\}$, so $\mathrm{tr}\, \rho_{ij}^2 = \frac{1}{2} - 2d_{ij} + 4d_{ij}^2 + 8|(TT^\dagger)_{ij}|^2$. Summing (S10.1) over the 20 ordinary triples gives $\sum_{i<j<k} \mathrm{tr}\, \rho_{ijk}^2 = 2\sum_{i<j} \mathrm{tr}\, \rho_{ij}^2 - 5$.

To combine the marginal purities, expand $|\,\mathrm{Hf}(T_K)|^2$ and $|(TT^\dagger)_{ij}|^2$. Both contain the same 4-cycle cross terms $2\,\mathrm{Re}(t_{kl}\overline{t_{km}}t_{mn}\overline{t_{ln}})$, while the squared moduli partition edge pairs of K_6 into disjoint and adjacent pairs:

$$2 \sum_{|K|=4} |\,\mathrm{Hf}(T_K)|^2 - \sum_{i<j} |(TT^\dagger)_{ij}|^2 = \sigma^2 - \frac{3}{2} \sum_{i=1}^{6} p_i^2 + 2 \sum_{i<j} |t_{ij}|^4.$$

Adding the 20 ordinary triples and the 15 cuts $\{0, i, j\}$, with $\sum_i p_i = 2\sigma$ and $\sigma = \frac{1}{2} - x$, gives

$$35\, \pi_3(z_{T,b}) = \frac{7}{2} - 4x + 64x^2 + 4 \sum_i p_i^2 + 48 \sum_{i<j} |t_{ij}|^4 + 24 \sum_{i<j} |(TT^\dagger)_{ij}|^2.$$

Since $\sum_i (p_i - \sigma/3)^2 = \sum_i p_i^2 - 2\sigma^2/3$ and $\sum_{i<j} (|t_{ij}|^2 - \sigma/15)^2 = \sum_{i<j} |t_{ij}|^4 - \sigma^2/15$, (S10.2) reduces to the scalar identity $\frac{7}{2} - 4x + 64x^2 = \frac{35 \cdot 121}{917} + \frac{7(524x - 37)^2}{27510} - \frac{88}{15}\left(\frac{1}{2} - x\right)^2$, checked by expansion. CC-1 independently checks all 226 formal quartic coefficients. Each term on the right of (S10.2) is nonnegative, and equality holds iff $x = 37/524$, $|t_{ij}|^2 = \sigma/15$, $p_i = \sigma/3$, and $(TT^\dagger)_{ij} = 0$ $(i \neq j)$. □

The classification of order-six complex symmetric conference matrices is due to Et-Taoui and Makhlouf [13], who classify the complex symmetric, skew-symmetric and Hermitian conference matrices of order six. We give the diagonal-congruence form needed here, with a self-contained proof, and then identify its action on the seven-qubit state. The application is an equality classification within the pair/complement ansatz.

Proposition S54 (order-six symmetric conference matrices and restricted-ansatz equality; proved, CP-4)**.** *Let C be the real icosahedral 6×6 conference matrix ($C_{ij}=s_{ij}$ for $i\neq j$) of Theorem 41. Every complex symmetric matrix $\widetilde{C}$ of order six with $\widetilde{C}_{ii}=0$, $|\widetilde{C}_{ij}|=1$ for $i\neq j$, and $\widetilde{C}\widetilde{C}^{\dagger}=5I$ has the form*

$$\boxed{\widetilde{C}=DPCP^{\mathsf{T}}D},$$

where D is diagonal unitary and P is a permutation matrix. Consequently, every *state attaining $\pi_3=121/917$ within the complete complex pair/complement ansatz of Theorem S53 is related to Ψ_{ico} by single-qubit diagonal unitaries and a permutation of the six ordinary qubits. This is a complete* restricted-ansatz *equality classification; the unrestricted equality statement is Corollary S67, proved independently by moment-orbit rigidity.*

Proof. A diagonal unitary congruence first makes the five off-diagonal entries in row and column 0 all equal to 1. Write the remaining 5×5 symmetric core as B, with zero diagonal. Because $B^{\mathsf{T}}=B$, the conference equation $\widetilde{C}\widetilde{C}^{\dagger}=5I_6$ on the core reads

$$B\mathbf{1}=0,\qquad B\overline{B}=5I_5-J_5,$$

where $J_5=\mathbf{1}\mathbf{1}^{\mathsf{T}}$. Four unit complex numbers summing to zero form two opposite pairs: for any three unit complex numbers z_1,z_2,z_3, the identity $|z_1+z_2+z_3|^2-1=(z_1+z_2)(z_2+z_3)(z_3+z_1)/(z_1z_2z_3)$ shows that $|z_1+z_2+z_3|=1$ iff two of them sum to zero and the third equals the sum.

After removing the common unit phase B_{12} from B and relabelling vertices $\{2,3,4,5\}$, row 1 of B is $(0,1,-1,t,-t)$ with $|t|=1$. The off-diagonal equation $(B\overline{B})_{12}=-1$ reads

$$-\overline{B_{23}}+t\overline{B_{24}}-t\overline{B_{25}}=-1.$$

The three unit summands on the left sum to -1, so one equals -1 and the other two sum to zero. The relabelling $(2\,3)(4\,5)$ followed by $B\mapsto -B$ fixes row 1 and sends B_{23} to $-B_{23}$, so every configuration with $B_{23}=-1$ is carried to one with $B_{23}=1$. Hence the excluded subcases $B_{23}=-1$ of the first two cases below reduce to the third case, $B_{23}=1$, and need no separate argument.

- If $t\overline{B_{24}}=-1$ and $-\overline{B_{23}}-t\overline{B_{25}}=0$, then $B_{24}=-t$ and $B_{25}=-B_{23}t$, so the sum of row 2 is $1+B_{23}-t-B_{23}t=(1+B_{23})(1-t)=0$, forcing $t=1$ unless $B_{23}=-1$.
- If $-t\overline{B_{25}}=-1$ and $-\overline{B_{23}}+t\overline{B_{24}}=0$, then $B_{25}=t$ and $B_{24}=B_{23}t$, so the sum of row 2 is $1+B_{23}+B_{23}t+t=(1+B_{23})(1+t)=0$, forcing $t=-1$ unless $B_{23}=-1$.
- Otherwise $-\overline{B_{23}}=-1$, so $B_{23}=1$. Then $(B\overline{B})_{12}=-1$ gives $B_{24}=B_{25}$, and zero row-2 sum gives $B_{24}=B_{25}=-1$; meanwhile $(B\overline{B})_{13}=1+t\overline{B_{34}}-t\overline{B_{35}}=-1$ gives $t\overline{B_{34}}-t\overline{B_{35}}=-2$, forcing $B_{34}=-t$ and $B_{35}=t$. Zero row-4 sum ($t-1-t+B_{45}=0$) then gives $B_{45}=1$, and $(B\overline{B})_{14}=-1+\overline{t}-t=-1$ forces $t=\overline{t}=\pm 1$.

Hence $t=\pm 1$ in all cases. Applying the same argument to each row i shows that row i is a unit phase ω_i times a real sign vector; since $B_{ij}=\pm\omega_i=\pm\omega_j$, all ω_i agree up to sign, so B is a single unit phase times a real $\{\pm 1\}$ matrix (also checked by CP-4 across all 3^5 row-matching patterns).

A diagonal congruence removes this common core phase while preserving row 0. The now-real core has exactly two negative and two positive entries per row. Therefore its negative-edge graph is a simple 2-regular graph on five vertices, necessarily a pentagon. All such pentagons differ by a permutation, so the real conference matrix is PCP^{T}. Restoring the removed diagonal phases proves the boxed classification.

Finally, if $D=\mathrm{diag}(e^{i\phi_i})$ and $\Sigma=\sum_i\phi_i$, applying $\mathrm{diag}(e^{i\Sigma/2},1)$ to the distinguished qubit, and $\mathrm{diag}(1,e^{i(\phi_i-\Sigma/4)})$ to ordinary qubit i, realises $C\mapsto DCD$ on the pair/complement state without changing either endpoint amplitude. Ordinary-qubit permutations implement P. □

S10.2 A spin-flip equation for the conference family

Proposition S55 (an exact spin-flip equation forces the conference shell)**.** *Let $J = J_7$. Choose a distinguished qubit s and local Z axes, and define*

$$H_s = -\frac{1}{2}Z_s - \frac{1}{4}\sum_{i\neq s} Z_i, \qquad \mathcal{A}_s = H_s J.$$

A normalised pure seven-qubit state satisfying $z = \mathcal{A}_s z$ lies in the complex pair/complement family *of Theorem S53, up to a special-qubit diagonal $SU(2)$ phase. Conversely all members of that family satisfy this equation in that phase frame. Thus every such z has $\pi_3(z) \geq 121/917$, with equality only on the icosahedral LU/permutation orbit. In its displayed conference frame the icosahedral state satisfies $\Psi_{\text{ico}} = H_s J \Psi_{\text{ico}}$.*

Proof. Since $J^2 = -I$ and $JH_sJ^{-1} = -H_s$, the real-linear operator $\mathcal{A}_s = H_sJ$ obeys $\mathcal{A}_s^2 = H_s^2$. For a computational string with special-site bit x_s and ordinary Hamming weight w, the eigenvalue of H_s is

$$h(x) = \begin{cases} -2 + w/2, & x_s = 0, \\ -1 + w/2, & x_s = 1. \end{cases}$$

Hence $z = \mathcal{A}_s z$ implies $H_s^2 z = z$, restricting support to $h(x)^2 = 1$: exactly $15 + 1 + 1 + 15 = 32$ strings, those of ordinary weights $2, 6$ at $x_s = 0$ and $0, 4$ at $x_s = 1$.

On this shell the equation identifies the weight-four amplitudes with the *negative complex conjugates* of the complementary weight-two amplitudes, and makes the two endpoint amplitudes complex conjugates. A diagonal $SU(2)$ phase at the special site makes both endpoint amplitudes real and nonnegative. This is exactly the complex pair/complement ansatz, and the converse follows by the same spin-flip sign calculation. Theorem S53 and Proposition S54 give the purity assertions. The icosahedral relation follows coefficientwise from (14). □

S11 The five-copy level at seven qubits

The five-copy certificate proves that Ψ_{ico} minimises π_3 over all pure seven-qubit states, uniquely up to local unitaries and qubit permutations (Corollary S67). The proof has three parts.

1. *Value.* A rational five-copy dual supported on twelve exact physical icosahedral kernels, with every Gram positive definite (Theorem S62). The identity is determined by 3424 direct probes (Section S11.2) and a verified right-inverse repair of an identity-Gram baseline, using 3754 independent symmetric coordinates and at most seven columns per frame.

2. *Equality moment.* The repair directions span the whole critical space, so at equality the five-copy moment lies on the tangent line of the icosahedral family (Lemma S63); an invariant and its square fix the tangent parameter at zero (Theorem S66).

3. *Orbit.* Thirteen invariants of bidegree at most $(4, 4)$ already determine the icosahedral orbit (Theorem S64).

Steps 2 and 3 hold at every family point with $1/32 < x < 1/12$ and combine into Theorem S66, which the entropy proof of Section S12.8 uses as well. Levels two, three and four of the hierarchy are provably not sharp (§§S9–S9.2; for level four, Proposition S49), so five copies is the first sharp level.

S11.1 Site algebra and the icosahedral kernels

Lemma S56 (Exact five-copy site algebra; S5-1)**.** *On one qubit's five-copy space, the commutant of collective* SU(2) *is $\mathcal{A}_5 \simeq \mathbb{C} \oplus M_4(\mathbb{C}) \oplus M_5(\mathbb{C})$. Choose rational bases of the highest-weight*

spaces in Hamming weights $m = 0, 1, 2$ by taking the kernels of the qubit raising operator. In the resulting matrix-unit coordinates, the spin-flip partial-transpose maps on copy sets $\{5\}$ and $\{4, 5\}$ are rational 42×42 involutions; every entry has denominator dividing 30.

Proof. The highest-weight kernels have dimensions $1, 4, 5$; the corresponding SU(2) irreducibles have dimensions $6, 4, 2$, giving $1{\cdot}6{+}4{\cdot}4{+}5{\cdot}2 = 32$. Schur–Weyl duality gives the stated commutant. For a weight-m integer representation ρ_m of S_5, its matrix unit is $E_{ij}^{(m)} = \frac{d_m}{120} \sum_{\sigma \in S_5} \rho_m(\sigma^{-1})_{ji} U_\sigma$. Apply $M \mapsto (\epsilon_J^{\mathsf{T}} M \epsilon_J)^{T_J}$ to these units, then read the transformed highest-weight blocks. The exact integer calculation S5-1 checks all 42 matrix units against their physical 32×32 operators, reconstructs each transformed operator, and checks both involutions and the denominator claim. □

Lemma S57 (Rational coordinates for the leading two-flip blocks; Y5-1)**.** *Let $F \cong \mathbb{Q}^4$ and $V \cong \mathbb{Q}^5$ carry the S_5 representations $[4, 1]$ and $[3, 2]$ from Lemma S56. For the seven-site type with one F sector and six V sectors, the $\mathrm{GL}(V)$ modules of shapes $(5, 1)$ and $(4, 2)$ both have dimension 420. Under the two-flip copy subgroup $H = S_3 \times S_2$, their F-tensored invariant spaces have dimensions 137 and 148, respectively. Both invariant spaces have explicit rational bases.*

Proof. Pieri decomposition identifies the two modules as

$$\ker\big(\mathrm{Sym}^5 V \otimes V \to \mathrm{Sym}^6 V\big), \qquad \ker\big(\mathrm{Sym}^4 V \otimes \mathrm{Sym}^2 V \xrightarrow{\delta} \mathrm{Sym}^5 V \otimes V\big),$$

where the first map is multiplication and δ polarises one factor of the second symmetric power. Both are surjective, so their kernel dimensions are $5\binom{9}{4} - \binom{10}{4} = 420$ and $\binom{8}{4}\binom{6}{2} - 5\binom{9}{4} = 420$. Y5-1 gives sparse integer generators annihilated by each map, checks rank 420 over a prime field, and hence obtains rational bases of the full kernels. Averaging their tensor products with F over the twelve elements of H gives invariant bases. The character averages are 137 and 148, and Y5-1 checks that many independent averaged columns and their invariance under every element of H exactly. □

Proposition S58 (Exact leading icosahedral five-copy kernels; IK5-1–IK5-4, PK5-1–PK5-3)**.** *At Ψ_{ico}, twelve five-copy moment blocks have the following ranks and kernel dimensions in their H-invariant multiplicity spaces, where j is the number of flipped copies:*

j	*site type*	GL(5) *shape*	rank	dim ker
0	$(0, 0, 7)$	(7)	2	2
0	$(0, 0, 7)$	$(6, 1)$	3	3
0	$(0, 0, 7)$	$(5, 2)$	3	6
1	$(0, 0, 7)$	(7)	2	14
1	$(0, 0, 7)$	$(6, 1)$	3	27
1	$(0, 0, 7)$	$(5, 2)$	3	36
2	$(0, 0, 7)$	(7)	2	31
2	$(0, 0, 7)$	$(6, 1)$	3	57
2	$(0, 0, 7)$	$(5, 2)$	3	72
2	$(0, 1, 6)$	(6)	8	69
2	$(0, 1, 6)$	$(5, 1)$	12	125
2	$(0, 1, 6)$	$(4, 2)$	20	128

For the four original blocks (the $j = 2$ rows of shapes (7), (6), $(5, 1)$ and $(4, 2)$), each physical kernel is $F^{-\mathsf{T}} K$ for the stored rational feature-annihilator basis K and the rational Gram transport F of the proof. PK5-1 constructs compact integer physical bases for all four blocks. PK5-2 directly constructs physical integer bases for the two additional all-V blocks, with maximum coefficient 3 in both cases. PK5-3 constructs exact physical kernel bases for the six $j = 0, 1$ blocks. The stored feature-annihilator bases are sparse; the two one-F Young blocks have at most 13 and 21 nonzero coordinates per vector, respectively.

Proof. In the explicit gauge of Theorem 41, write $\Psi_{\mathrm{ico}} = (\sqrt{15}\,u + \sqrt{37}\,v)/\sqrt{524}$, where u has thirty signed unit entries and v has two unit entries. The flipped rank-one moment is generated by $\Psi_{\mathrm{ico}}^{\otimes 3} \otimes (J_7\Psi_{\mathrm{ico}})^{\otimes 2}$. The pentagon-site cycle preserves u and v. For the three blocks with one $[4,1]$ site it permutes the six ordinary-site tensor slots, acting only on Young and spin-multiplicity *columns*. The GL(5) and H-invariant row coordinates stay fixed. Hence the seven site placements have the same joint row span as three representatives. The fully symmetric block has one placement. The scripts IK5-1–IK5-4 check the appropriate pentagon-cycle covariance entrywise in integer tensors.

Using the exact local maps of Lemma S56, the four scripts contract those representatives at six integer amplitude ratios and interpolate the resulting degree-five polynomial matrix. Every binary64 tensor contraction is integer-exact: an absolute-sum bound is $32^5 5^5 \cdot 24 \cdot 3^6 < 2^{53}$. The subsequent binary64 matrix products are chunked below 2^{52}, and integer accumulations are checked against their type bounds. The four feature matrices have sizes 33×128, 77×768, 137×3840 and 148×11520, respectively. For the last, the integral projector numerator $40I - 14FE + (FE)^2$ selects the kernel of the polarisation map of Lemma S57.

At the physical ratio each matrix is, up to a nonzero scalar, $15A + \sqrt{555}\,B$ with integral A, B collecting the even and the odd powers of t in the degree-five polynomial matrix $C(t) = \sum_{d=0}^{5} t^d C_d$. The key simplification is that these two parts have disjoint column supports. To see this, note that the diagonal local unitary

$$D = \mathrm{diag}(-i,-1)_s \otimes \mathrm{diag}(1, e^{i\pi/4})^{\otimes 6}$$

multiplies $|s,\mathbf{a}\rangle$ by $e^{i\pi(6-2s+|\mathbf{a}|)/4}$, which is 1 on the pair and complement strings and -1 on the two endpoints, so $Du = u$ and $Dv = -v$. For the spin flip $J_7 = \epsilon^{\otimes 7}K$, with $\epsilon = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and K complex conjugation, every diagonal unitary G satisfies $\epsilon \bar{G} \epsilon^{-1} = \overline{\det G}\, G$, so $J_7 D = cDJ_7$ with $|c| = 1$. Hence D maps $(u+tv)^{\otimes(5-j)} \otimes (J_7(u+tv))^{\otimes j}$ to a constant multiple of the same tensor at $-t$. On each site, $D^{\otimes 5}$ commutes with the copy permutations and multiplies every weight vector of the highest-weight bases of Lemma S56 by a scalar, so it fixes the row coordinates and acts on the columns by a diagonal unimodular matrix Δ. Thus $C(-t) = C(t)\Delta$. Since $C(t)$ is real, each column that is not identically zero is either even or odd in t. In particular

$$C_{\mathrm{even}}(t)\, C_{\mathrm{odd}}(s)^{\mathsf{T}} = 0 \quad (\forall t, s), \qquad AB^{\mathsf{T}} = 0,$$

for the Euclidean column pairing and for any diagonal column weighting. For any $k = x + \sqrt{555}\,y \in \mathbb{Q}(\sqrt{555})^n$, the relation $k^{\mathsf{T}}(15A + \sqrt{555}\,B) = (15x^{\mathsf{T}}A + 555y^{\mathsf{T}}B) + \sqrt{555}(x^{\mathsf{T}}B + 15y^{\mathsf{T}}A) = 0$ forces both summands, which are supported on disjoint columns, to vanish in each component, so $x^{\mathsf{T}}A = x^{\mathsf{T}}B = y^{\mathsf{T}}A = y^{\mathsf{T}}B = 0$. Every left kernel over $\mathbb{Q}(\sqrt{555})$ is therefore the scalar extension of the rational intersection $\ker A^{\mathsf{T}} \cap \ker B^{\mathsf{T}}$ (explaining why all $\sqrt{555}$ coefficients vanish across all twelve blocks), and $\mathrm{rank}_{\mathbb{Q}(\sqrt{555})}(15A + \sqrt{555}\,B) = \mathrm{rank}_{\mathbb{Q}}[A\ B]$. IK5-1 checks that the integer realification for shape $(5,1)$ has rational rank 24. For shape $(4,2)$, IK5-2 checks a nonzero 20×20 minor over $\mathbb{Q}(\sqrt{555})$ and 128 independent annihilators of all 11520 columns. IK5-3 and IK5-4 check rational realification ranks 4 and 16 for shapes (7) and (6). All four scripts verify their sparse annihilators against every column in both field components. To identify the physical kernels, let U be the integer invariant basis in the appropriate Young module, H its rational positive-definite physical Gram, and Y a column of the contracted moment. The scripts store $A_{\mathrm{feat}} = U^{\mathsf{T}}Y$. Since $Y \in \mathrm{im}\, U$, its physical covariant column is

$$U^{\mathsf{T}}HY = FA_{\mathrm{feat}}, \qquad F = (U^{\mathsf{T}}HU)(U^{\mathsf{T}}U)^{-1} \in \mathrm{GL}_d(\mathbb{Q}).$$

Thus, if K is the stored rational basis satisfying $K^{\mathsf{T}}A_{\mathrm{feat}} = 0$, then $F^{-\mathsf{T}}K$ is a rational basis of the physical moment kernel. The invertible change preserves all four ranks and kernel dimensions. This distinction matters: the stored K itself need not be a physical kernel.

PK5-1 avoids forming the dense rational transport. Select r independent columns Q from the common feature image, whose exact dimension r is already certified above. Reconstruct their ambient columns as $Y_Q = U(U^{\mathsf{T}}U)^{-1}Q$, then form the physical column basis $C_Q = U^{\mathsf{T}}HY_Q$. The

reconstruction is integral and $\ker C_Q^{\mathsf{T}}$ is the physical moment kernel. Integer Hermite normal form and lattice reduction give bases whose largest absolute coefficients are respectively $3, 8, 14, 28$ in the four original blocks' order. The compact artifact stores the selected columns, their exact physical images and the four kernel bases. PK5-1 reconstructs those images and checks both ranks and exact annihilation. These data do not assert any dual equality or positive multiplier.

For the two additional all-V shapes, use the complete Young models

$$\mathcal{Y}_{61} = \ker(\mathrm{Sym}^6 V \otimes V \to \mathrm{Sym}^7 V), \qquad \mathcal{Y}_{52} = \ker(\mathrm{Sym}^5 V \otimes \mathrm{Sym}^2 V \xrightarrow{\delta} \mathrm{Sym}^6 V \otimes V).$$

The maps are surjective and the model dimensions are $720, 840$. Their characters are $h_6h_1 - h_7$ and $h_5h_2 - h_6h_1$; averaging over H gives invariant dimensions $60, 75$. PK5-2 constructs integral invariant bases U, checks these characters exactly, and verifies the spanning ranks by nonzero modular minors and matching dimension upper bounds.

Contract the all-V physical tensor and row-symmetrise its six-plus-one or five-plus-two multiplicity indices. The seven choices of singleton have three pentagon-cycle orbits, and the twenty-one choices of pair have five. Taking one representative of each therefore captures every Young and spin column. Projection against U with the *physical* Gram gives covariant features $U^{\mathsf{T}}HY$ directly; other Pieri summands are orthogonal and need no explicit projector. The six integer amplitude ratios and interpolation above give physical feature pairs $15A + \sqrt{555}B$ of sizes 60×384 and 75×640, with $AB^{\mathsf{T}} = 0$ by the same column-parity argument. In each case PK5-2 checks the exact rational rank $\mathrm{rank}[A\ B] = 3$ (which already equals the rank over $\mathbb{Q}(\sqrt{555})$ by row orthogonality $AB^{\mathsf{T}} = 0$, and is also checked by a nonzero three-dimensional minor after mapping $\sqrt{555}$ into a finite prime field). Integer HNF and lattice reduction then give 57 and 72 independent rational physical annihilators, checked against every column of both A and B.

For $j = 0, 1$, use the same three all-V Young modules with the invariant subgroups S_5 and $S_4 \times S_1$. Projection of their two-flip invariant spaces onto these subgroups gives complete rational bases: their exact character dimensions are respectively $(4, 6, 9)$ and $(16, 30, 39)$. PK5-3 repeats the physical contraction with $\psi^{\otimes(5-j)} \otimes (J_7\psi)^{\otimes j}$. The common feature ranks are $(2, 3, 3)$ in both cases, determined by $\mathrm{rank}[A\ B]$ via $AB^{\mathsf{T}} = 0$ and checked against matching finite-field lower bounds and full rational annihilators.

□

A compact rational baseline uses all twelve kernels. Its Gram matrices are identities in the supplied dyadic bases; PK5-4 checks their exact support and nonsingularity. Thus its multipliers are strictly positive on every listed kernel. The baseline is repaired to an exact sharp dual by the verified bounds of Section S11.4.

S11.2 Direct probes for a sharp five-copy certificate

Let $W = (\mathbb{C}^2)^{\otimes 5}$, let U_σ permute its five copies, and let G be any element of the rational local commutant $\mathcal{A}_5$. On seven sites put

$$P = \frac{1}{120} \sum_{\sigma \in S_5} U_\sigma^{\otimes 7}, \qquad X(G) = PG^{\otimes 7}P.$$

These linear probes determine identities on the moment space; they need not themselves be positive moments. For $j = 0, 1, 2$, let $H_j = S_{5-j} \times S_j$, let ϕ_j be the local j-copy spin-flip transpose, and put $\Phi_j = \phi_j^{\otimes 7}$.

Lemma S59 (direct contracted dual probes; DP5-1)**.** *Suppose Z is a site-permutation-invariant commutant operator supported on the global H_j-invariant subspace. Choose left-coset representatives $a \in H_j\backslash S_5$ and right-coset representatives $b \in S_5/H_j$. Then*

$$\mathrm{tr}[Z\Phi_j(X(G))] = \binom{5}{j}^{-2} \sum_{a,b} \mathrm{tr}\left[Z\, \phi_j(U_aGU_b)^{\otimes 7}\right].$$

There are $1, 25, 100$ *terms for* $j = 0, 1, 2$*, respectively. The traces on the right are evaluated directly in the chosen Schur models; no full gamma map is needed. Write* $a_\sigma = \mathrm{tr}(U_\sigma G)$*, and let* $\mathcal{T}$ *be the ten transpositions in* S_5*. The sharp target on this probe is*

$$q(G) = \frac{1}{1200} \sum_{\sigma \in S_5,\, t \in \mathcal{T}} a_{t\sigma}^3 a_\sigma^4 - \frac{121}{917 \cdot 120} \sum_{\sigma \in S_5} a_\sigma^7.$$

Equality of this polynomial with the sum of the contracted dual probes for every $G \in \mathcal{A}_5$ *is equivalent to the sharp dual identity on the real, site-averaged bosonic moment space.*

Proof. The spin-flip transpose is self-adjoint for the physical trace pairing. Expanding both copies of P gives 120^2 terms. Both Z and $\Phi_j(Z)$ have left and right H_j-invariant support: partial transpose exchanges left and right permutations of the flipped copies and preserves the unflipped ones. The terms therefore group into $\binom{5}{j}^2$ equal-weight coset pairs. Moving Φ_j back to the probe gives the first formula. The same proof applies with any numbers of copies and sites in place of 5 and 7.

On the bosonic subspace, the purity operator may be averaged over all ten choices of the swapped global-copy pair. This average commutes with P, so cyclicity removes one of the two projectors in its trace against $X(G)$. All 35 choices of the three swapped sites have the same trace against an identical-site product $G^{\otimes 7}$, yielding the formula for q. Finally the span of the powers $G^{\otimes 7}$ is the symmetric seventh tensor power of $\mathcal{A}_5$, by polarisation. The trace pairing on this site-symmetric algebra is nondegenerate. This proves the last claim. Probes over $\mathbb{Q}(i)$ are also allowed: real and imaginary parts of $G = A + iB$ evaluations are rational combinations of the eight real probes $A + tB$, $t = 0, \ldots, 7$, by degree-seven interpolation. □

Proposition S60 (the critical space and finite probes; DP5-1)**.** *Let* $\mathcal{M}_5$ *be the real span of the normalised five-copy moments of pure seven-qubit states after local-unitary twirling, site averaging and averaging with the complex conjugate state (the conjugation-even part). Then* $\dim \mathcal{M}_5 = 3285$*. The* critical space $\mathcal{C} \subset \mathcal{M}_5^*$ *of functionals vanishing at the icosahedral moment* X_0 *and at its derivative* X_0' *along the normalised family* ψ_x *of* (14) *at* $x_0 = 37/524$ *has dimension* 3283*. If functionals in* $\mathcal{C}$ *have an evaluation matrix of rank* 3283 *on some finite set of probes, they span* $\mathcal{C}$*, and evaluation on those probes is injective on* $\mathcal{C}$*.*

Proof. By the last part of Lemma S59, $\mathcal{M}_5$ is dual to the site-symmetric probe polynomials, that is, to the permutation-symmetric local-unitary invariants of bidegree $(5, 5)$ for $N = 7$ sites and local dimension $r = 2$. Their Schur–Weyl character sum ((ID2) and (ID3) of Section S2) gives dimension 5844, and the trace of the conjugation swap is 726 (DP5-1), so the conjugation-even part has dimension $(5844 + 726)/2 = 3285$. This is the dimension of the linear span $\mathcal{M}_5$; the normalised moments lie in its affine hyperplane $\nu = 1$. The normalisation functional ν has $\nu(X_0) = 1$ and $\nu(X_0') = 0$, and DP5-1 checks that the weight-two Pauli enumerator A_2 has derivative $96/131 \neq 0$ there, so $X_0' \neq 0$ and X_0, X_0' are independent. Hence $\dim \mathcal{C} = 3283$. A rank-3283 evaluation matrix of elements of $\mathcal{C}$ gives both remaining assertions. □

The criterion is applied to the twelve-face baseline in Section S11.4, where rank and repair bounds are verified for the rational data.

S11.3 A positive repair lemma

The seven-qubit purity and entropy certificates (Theorems S62 and S85) both correct a strictly positive baseline, which fails the target identity, into an exact positive identity. The correction is found from finitely many evaluations, and its existence and positivity follow from two verified norm bounds. The common step is the following finite-dimensional real linear algebra; each application supplies its own spaces and verifies the bounds.

Lemma S61 (positive repair)**.** *Let $\mathcal{V}$ be a finite-dimensional real vector space, $\mathcal{C} \subseteq \mathcal{V}$ a subspace of dimension d, and $\mathcal{E} : \mathcal{V} \to \mathbb{R}^m$ a linear map. Let $\Phi : \mathbb{R}^N \to \mathcal{V}$ be linear with $\Phi(\mathbb{R}^N) \subseteq \mathcal{C}$, let $\rho \in \mathcal{C}$, and put $A = \mathcal{E}\Phi \in \mathbb{R}^{m\times N}$ and $r = \mathcal{E}\rho \in \mathbb{R}^m$. Suppose $L \in \mathbb{R}^{d\times m}$ and $R \in \mathbb{R}^{N\times d}$ satisfy*

$$\|I_d - LAR\|_2 \le \delta < 1. \tag{S11.1}$$

(a) A has rank d; Φ maps onto $\mathcal{C}$, and $\mathcal{E}$ is injective on $\mathcal{C}$.

(b) $z = R(LAR)^{-1}Lr$ satisfies $\Phi(z) = \rho$ and $\|z\|_2 \le \|R\|_2\|Lr\|_2/(1-\delta)$. If the entries of A, r, L, R lie in a subfield $K \subseteq \mathbb{R}$, then $z \in K^N$.

(c) Let $\Delta_b : \mathbb{R}^N \to \mathrm{Sym}_{k_b}(\mathbb{R})$, $b = 1, \dots, B$, be linear maps with $\|\Delta_b(y)\|_2 \le c\|y\|_2$ for every y. If

$$\frac{c\,\|R\|_2\|Lr\|_2}{1-\delta} < 1, \tag{S11.2}$$

then $I + \Delta_b(z) \succ 0$ for every b.

Proof. By (S11.1) the Neumann series for $(LAR)^{-1}$ converges, so LAR is invertible with $\|(LAR)^{-1}\|_2 \le 1/(1-\delta)$. Hence $\operatorname{rank} A \ge d$. Since $A = \mathcal{E}\Phi$ factors through the d-dimensional space $\mathcal{C}$, also $\operatorname{rank} A \le d$. Equality forces $\Phi(\mathbb{R}^N) = \mathcal{C}$ and $\ker \mathcal{E} \cap \mathcal{C} = 0$, proving (a).

Since AR has rank d, its image is $A(\mathbb{R}^N) = \mathcal{E}(\mathcal{C})$, which contains r. Write $r = ARy$; applying L gives $y = (LAR)^{-1}Lr$, so $Az = r$. Thus $\mathcal{E}(\Phi(z) - \rho) = 0$ with $\Phi(z) - \rho \in \mathcal{C}$, and (a) gives $\Phi(z) = \rho$. The norm bound follows from the bound on $\|(LAR)^{-1}\|_2$, and $z \in K^N$ because $(LAR)^{-1}$ has entries in K. For (c), $\|\Delta_b(z)\|_2 \le c\|z\|_2 < 1$ by (b) and (S11.2), so every eigenvalue of $I + \Delta_b(z)$ is positive. □

No singular value enters the conclusion: δ and the margin bound the exact matrices, and L, R are arbitrary candidates (in both applications, binary64 singular-vector data read as exact dyadic rationals).

S11.4 A positive twelve-face certificate

We construct the certificate by correcting a positive baseline on twelve exact physical kernels. In each supplied rational kernel basis, the identity Gram is strictly positive. A sufficiently small correction can therefore enforce the polynomial identity while preserving positivity. The same rank computation also gives the equality case (Corollary S67).

Theorem S62 (sharp five-copy certificate; verified-arithmetic certificate, FP5-1)**.** *There is a rational sharp five-copy dual supported on the twelve physical kernels of Proposition S58: rational $C_b \succ 0$ with*

$$q(G) = \sum_b \operatorname{tr}\left[C_b\, W_b^{\mathsf{T}} \Gamma_b(G) W_b\right] \qquad \text{for every } G \in \mathcal{A}_5.$$

Consequently the five-copy relaxation has value 121/917*. Moreover the functionals $X \mapsto \operatorname{tr}[W_b S W_b^{\mathsf{T}} \Gamma_b(X)]$, S symmetric, span the* 3283*-dimensional critical space of Proposition S60.*

The C_b solve a rational linear system whose solvability is certified with rigorous floating-point bounds; they are not computed in exact arithmetic, which is why the theorem is a verified-arithmetic certificate. The rational coefficients therefore exist but are not printed: their existence and the strict positivity of every C_b follow from the verified inverse test (S11.3) in the proof.

Proof. Write $W_b = U_b K_b B_b$ for the supplied rational basis of the bth physical kernel (PK5-4 checks that it annihilates the icosahedral moment block), and let $\Gamma_b(G)$ be the contracted probe matrix, including its positive multiplicity weight. The physical inner product is used on the ambient Young model, so an identity Gram in these coordinates is a positive physical multiplier even though the columns of W_b need not be orthonormal.

Let Q_b be the rational orthogonal matrix given by the supplied Householder reflections, and put $r_b = \min(7, k_b)$, where $k_b = \operatorname{cols} W_b$. Restrict repairs to

$$C_b = I + Q_b \begin{pmatrix} D_b & E_b \\ E_b^{\mathsf{T}} & 0 \end{pmatrix} Q_b^{\mathsf{T}}, \qquad D_b = D_b^{\mathsf{T}} \in \mathbb{R}^{r_b \times r_b}.$$

Each reflection is $I - 2vv^{\mathsf{T}}/(v^{\mathsf{T}}v)$; orthogonality is checked with integers. Let z_b list the upper-triangular entries of D_b and all entries of E_b. Then $\|C_b - I\|_2 \le \sqrt{2}\|z_b\|_2 \le \frac{3}{2}\|z_b\|_2$. There are $\sum_b [r_b k_b - r_b(r_b-1)/2] = 3754$ independent coordinates. Width six would give only 3249, too few to span the critical space (FP5-4), so seven is the smallest possible uniform width for this argument.

For every vector v, the smooth function $v^{\mathsf{T}} W_b^{\mathsf{T}} \Gamma_b(X_5^+(\psi_x)) W_b v$ is nonnegative and vanishes at $x = 37/524$, so its derivative vanishes there. Thus every repair functional is critical. The target q is critical by Theorem 41. We apply Lemma S61 with $\mathcal{V} = \mathcal{M}_5^*$, $\mathcal{C}$ the critical space ($d = 3283$, Proposition S60), $N = 3754$, $\Phi(z) = \sum_b \operatorname{tr}[(C_b - I) W_b^{\mathsf{T}} \Gamma_b W_b]$, $\Delta_b(z) = C_b - I$ with $c = \frac{3}{2}$, and

$$\rho = q - \sum_b \operatorname{tr}[W_b^{\mathsf{T}} \Gamma_b W_b].$$

By the preceding paragraph $\Phi(\mathbb{R}^N) \subseteq \mathcal{C}$ and $\rho \in \mathcal{C}$. Let $\mathcal{E}$ evaluate at the 3424 supplied rational probes, with the supplied nonzero dyadic row weights, so that A is the rational evaluation matrix of the repair directions and $r = \mathcal{E}\rho$. The verifier takes the leading 3283 singular triples of a floating evaluation of A, reads their binary64 entries as exact dyadic rationals, and forms from them rational matrices L and R of sizes 3283×3424 and 3754×3283. With the error bounds below it establishes

$$\|I - LAR\|_2 \le \delta < 1, \qquad \frac{3\|R\|_2 \|Lr\|_2}{2(1-\delta)} < 1; \tag{S11.3}$$

the logged upper bounds are $\delta < 0.070$ and 0.324. These are (S11.1) and (S11.2) for the exact rational matrices. FP5-3 reimplements the probe evaluator independently and reproduces all face traces and repair coefficients on non-scalar probes.

Lemma S61(a) gives rank 3283, so the repair functionals span the critical space (the rank criterion of Proposition S60). Part (b), with $K = \mathbb{Q}$, gives rational z with $\Phi(z) = \rho$, which is the asserted identity, and part (c), using $\|C_b - I\|_2 \le \frac{3}{2}\|z_b\|_2 \le \frac{3}{2}\|z\|_2$, gives every $C_b \succ 0$. Lemma S59 turns the identity into an exact rational feasible sharp dual, and the icosahedral moment is feasible with value 121/917. □

Lemma S63 (equality moments lie on the tangent line)**.** *Let $\mathcal{M}$ be a finite-dimensional real moment space with normalisation functional ν. Fix $X_0, X_0' \in \mathcal{M}$ with $\nu(X_0) = 1$, $\nu(X_0') = 0$ and $X_0' \neq 0$, and let $\mathcal{C} = \{f \in \mathcal{M}^* : f(X_0) = f(X_0') = 0\}$. Let Γ_b be linear maps to symmetric matrices, and W_b matrices such that the functionals $X \mapsto \operatorname{tr}[W_b S W_b^{\mathsf{T}} \Gamma_b(X)]$ span $\mathcal{C}$. If $\nu(X) = 1$ and $W_b^{\mathsf{T}} \Gamma_b(X) W_b = 0$ for every b, then $X = X_0 + tX_0'$ for some $t \in \mathbb{R}$.*

Proof. Every functional of the spanning set, hence every $f \in \mathcal{C}$, vanishes at X. Since $\nu(X_0) = 1$ and $\nu(X_0') = 0$, the vectors X_0, X_0' are independent, $\mathcal{C}$ has codimension two, and its annihilator is their span. Normalisation fixes the coefficient of X_0 at one. The lemma uses no property of the number of copies, the number of qubits or the target. □

Rational models and exact target values. The verification uses only the exact certificate data. Each baseline factor B_b is lower triangular with positive dyadic diagonal; its invertibility is therefore checked directly over the rationals. There are 24502 potentially nonzero factor entries, rather than 48434 entries in dense factors. Each is stored as an odd signed integer of at most 53 bits times an integral power of two, and integer shifts reconstruct it exactly. The twelve frames are regenerated from fixed PCG64 seeds; FP5-4 fingerprints all 36,480 original Householder integers before checking their exact orthogonality. The probe archive uses lossless compression, with signed 16-bit probe numerators in column order. The baseline, frames and probes occupy 441612 bytes.

PK5-4 fingerprints the triangular factors and checks their full physical kernel support; the probe fingerprint preserves every probe value and row weight. In the rational one-site model write its physical Gram as $G_d = L_d D_d L_d^{\mathsf{T}}$, with rational L_d and positive diagonal rational D_d. Changing coordinates by L_d^{T} makes the physical Gram diagonal without square roots. In a symmetric power the monomial of exponent m has squared norm

$$\frac{\prod_i m_i!}{(\sum_i m_i)!} \prod_i (D_d)_{ii}^{m_i}.$$

Thus both the kernel columns and their physical dual columns are rational. They, and their products with the rational repair frames, are formed exactly before conversion to binary64. The conversion is correctly rounded; exact zero counts and exponent guards exclude underflow. The local spin-flip and copy-permutation maps are formed exactly as rational 42×42 matrices.

Each source probe has integer coordinates divided by 8192. Consequently all a_σ in Lemma S59 are rational with this denominator. If their integer numerators are a'_σ, the target is evaluated first as the exact fraction

$$\frac{917 \sum_{\sigma,t} (a'_{t\sigma})^3 (a'_\sigma)^4 - 1210 \sum_\sigma (a'_\sigma)^7}{917 \cdot 1200 \cdot 8192^7}.$$

For probes commuting with the copy transposition (12), the paired cosets are also identified under $(a, b) \mapsto (a(12), (12)b)$. The commutation is checked exactly before using this reduction. An independent exact scalar control (FP5-2) takes G to be the identity on the local spin-5/2 block and zero elsewhere. Then $q(G) = (796/917)6^7$, and only the two-flip all-V shape (7) and one-F shape (6) survive. Their forms are respectively $6^7 d_7 d_7^{\mathsf{T}}$ and $(7/3)6^7 d_6 d_6^{\mathsf{T}}$, where the right functionals on the ambient polynomial models are e_5^7 and $(e_3 + e_4) \otimes e_5^6$. This follows because, for this G, every local two-flip block has rank one, so each contracted gamma form is a symmetric power of a rank-one local form. FP5-2 checks every thin coefficient and the baseline trace against these exact rational forms, including the ten zero faces.

Forward-error bounds for the probes. These bounds, and those of the entropy certificate E7-1, hold under one arithmetic model: IEEE-754 binary64 with round-to-nearest, so that every floating-point addition, multiplication, division or square root (including any fused multiply–add inside the single-threaded BLAS calls) returns the exact result times $1+\delta$ with $|\delta| \le u$, away from underflow and overflow, which are treated separately below. Put $u = 2^{-53}$ and $\gamma_s = su/(1-su)$. For a large face the baseline trace is evaluated as $\operatorname{tr}(TC)$, where $C = V_R V_L^{\mathsf{T}}$ is formed once. Its tensor indices are contracted in decreasing factor dimension, with inner products blocked at 128 terms. The first repair term is evaluated as $(T^{\mathsf{T}} V_{LU})^{\mathsf{T}} V_R$, so both repair actions use only $r_b \le 7$ columns. The smaller faces use the factored baseline trace. These are cyclic trace and transpose identities; no additional certificate data are retained. Forming C costs error depth $k+2$, including its two column conversions. For every signed contraction the verifier evaluates a positive companion. For a column matrix V, write $n(V)_i = \|V_{i,:}\|_2$. If T_+ is the nonnegative tensor action obtained by replacing the local matrices by their absolute values, Minkowski and Cauchy give

$$n(TV) \le T_+ n(V), \qquad \|A^{\mathsf{T}} T V\|_F \le \big\| |A|^{\mathsf{T}} T_+ n(V) \big\|_2, \qquad |\operatorname{tr}(A^{\mathsf{T}} T V)| \le n(A)^{\mathsf{T}} T_+ n(V).$$

Thus single-column tensor actions on $n(V_R)$ and $n(V_{RU})$ bound the two coefficient terms and the trace. The positive coefficient vectors are summed over coset terms before taking their norms; the two term bounds are added, and the face bounds are combined in Euclidean norm. The resulting scalar radius bounds the Frobenius norm of the entire coefficient error. Row norms are rounded upwards with $1 + 2^{-20}$; their squared inputs are normal by the conversion guards below. The polynomial functionals being evaluated are unchanged. A degree-n symmetric-power recurrence has at most five terms per entry at each step. Including conversion of the local matrix gives error depth at most $6n$. A matrix product with inner dimension d, split into chunks of length c and summed pairwise, has depth at most

$$\min(d, c) + \lceil \log_2 \lceil d/c \rceil \rceil .$$

If the ambient model has factor dimensions d_i, total dimension D, kernel dimension k, and t coset terms, define

$$\begin{aligned} a &= \sum_i \left(\min(d_i, 128) + \lceil \log_2 \lceil d_i/128 \rceil \rceil\right), \\ f &= \min(D, 256) + \lceil \log_2 \lceil D/256 \rceil \rceil, \\ h &= \min(D, 128) + \lceil \log_2 \lceil D/128 \rceil \rceil + \lceil \log_2 k \rceil + 1, \\ s &= 8 + \lceil t/8 \rceil + 16. \end{aligned}$$

With $g \le 7$ the local degree of the model, the coefficient and trace error depths are respectively $a + f + 6g + s$ and $a + h + 6g + s$ for a factored trace. For a multiplier trace its depth is

$$k + 2 + 6g + s + \sum_i \left[\min(d_i^2, 128) + \lceil \log_2 \lceil d_i^2/128 \rceil \rceil\right].$$

The final 16 covers the two rational column conversions, the addition of the two thin repair terms, multiplicity scaling, and summation over all twelve models. A common error budget for the positive companion is

$$b = Dk + \sum_i d_i + 6g + 2\binom{5}{j}^2 + 64.$$

If its computed value is p, the corresponding error radius is bounded by $\gamma_{a+f+6g+s} p/(1-\gamma_b)$, or with the corresponding trace depth for the trace. An additional factor $1 + 2^{-20}$ bounds rounding in the positive bounds themselves. Every scalar accumulation and Frobenius norm used in these bounds has fewer than 2^{29} terms. The input guards give $|h_{ij}| < 2^{10}$ for every exact local matrix, $D < 2^{13}$, $k < 2^8$, multiplicity below 2^{15} and model-column entries below 2^{20}. Each degree-n symmetric-power entry is bounded by $(5 \cdot 2^{10})^n < 2^{13n}$. Since the total degree is seven, signed tensor actions are bounded by $2^{13+91+20} = 2^{124}$ and norm companions by 2^{128}; subsequent column products, traces, multiplicities and at most 100 coset terms keep all mathematical intermediates below 2^{200}. These bounds also hold with a factor of two for the rounded computations, so overflow is excluded without repeatedly scanning the large arrays. An underflow error is at most 2^{-1074} per operation. Even allowing 2^{100} operations and amplification by 2^{200} gives less than 2^{-774}; the verifier adds 2^{-500} to each coefficient and trace radius. The initial row-norm squares are normal. Norms of the accumulated positive coefficient vectors are taken after the multiplicity and coset scales: loss of subnormal squares changes such a norm by less than 2^{-520}, and its contribution to a radius is multiplied by $\gamma_{a+f+6g+s} < 2^{-30}$. The same 2^{-500} allowance covers it. FP5-2 checks row scales between 1 and 2^{32}, and model conversion guards keep every nonzero input column above 2^{-400}. In the inverse validation, possible dot-product underflows, even after multiplication by a guarded d_i and taking a Frobenius norm, contribute less than 2^{-600}. The separate norm allowances and these terms sum to less than 2^{-500}, far below the coarse rational margins.

Validation of the rational inverse. The probe evaluator first gives rectangular coefficients in the first r_b frame columns. To obtain the independent coordinates, multiply each coefficient array M_b by the full Q_b, halve the diagonal, average the two entries on each off-diagonal of the leading square, and retain its upper triangle and the remaining rectangular block. These selections and averages contract the Euclidean error norm. If the initial row radius is ϵ, a radius after this change is

$$\epsilon + \sum_b \left[\left(\gamma_{k_b} + \frac{u}{1-u}\right) \|M_b\|_F \|\widetilde{Q}_b\|_F + u \| \operatorname{fl}(M_b \widetilde{Q}_b)\|_F\right],$$

with the same upward norm allowance. Here $\widetilde{Q}_b$ is the correctly rounded rational orthogonal matrix. FP5-4 checks this packing against independent exact matrix contractions on every supplied frame.

Let $\widetilde{A}, \widetilde{r}$ be the computed centres, and let e_j bound the Euclidean norm of row j of $A - \widetilde{A}$ and τ_j bound $|r_j - \widetilde{r}_j|$. The preceding radii provide these bounds. Take the first 3283 left and right singular vectors of $\widetilde{A}$ merely as candidate data, and regard their binary64 entries, and the computed reciprocal singular values d_i, as exact dyadic rationals. Write these matrices as U, V, d and put

$$L = \operatorname{diag}(d)U^{\mathsf{T}}, \qquad R = V^{\mathsf{T}}.$$

An independent Gram-product error bound verifies $\|R\|_2 \le v = 1 + 2^{-12}$. For the computed centres $H = \mathrm{fl}(\operatorname{diag}(d)\,\mathrm{fl}(U^{\mathsf{T}}\widetilde{A}))$ and $M = \mathrm{fl}(HV^{\mathsf{T}})$, the inverse defect is at most

$$\begin{aligned}\|I - M\|_F + (\gamma_m + u + u\gamma_m)d_{\max}\|U\|_F\|\widetilde{A}\|_F v + \gamma_n\|H\|_F\|V\|_F \\ + v\|\operatorname{diag}(d)|U|^{\mathsf{T}}e\|_2, \qquad m = 3424, \quad n = 3754.\end{aligned}$$

Likewise, if $p = \mathrm{fl}(\operatorname{diag}(d)\,\mathrm{fl}(U^{\mathsf{T}}\widetilde{r}))$, then

$$\|Lr\|_2 \le \left\| |p| + \operatorname{diag}(d)|U|^{\mathsf{T}}(\tau + \gamma_m|\widetilde{r}|) + \frac{u}{1-u}|p| \right\|_2.$$

All norms and positive bounds are rounded upwards using the same $1+2^{-20}$ allowance. Substituting these bounds into $3v\|Lr\|_2/[2(1-\delta)]$ verifies (S11.3). Thus neither a numerical rank assumption nor a dense exact inverse is needed.

S11.5 Degree-four invariants fix the orbit

The moment certificate fixes invariant values; we must now recover the state. The invariants needed for this step have degree at most four and are linear in the four-copy marginal of the five-copy moment. Combined with the tangent-line lemma this gives one equality theorem for every family point with $1/32 < x < 1/12$ (Theorem S66), applied below to the purity and in Section S12.8 to the entropy.

Theorem S64 (Degree-four invariants identify the icosahedral family; proved, FO-1)**.** *Fix $y \in (1/32, 1/12)$. Let z be a unit pure seven-qubit state whose thirteen product-*SU(2)*, site-permutation and complex-conjugation invariant polynomials*

$$A_1,\ C,\ D,\ \mathcal{E}_2,\ \mathcal{E}_3,\ \mathcal{E}_4,\ B_1,\ S_1,\ Q,\ R,\ F,\ L,\ \pi_3,$$

all of bidegree at most $(4,4)$ and defined below, take their values at ψ_y. Then z is local-unitarily and site-permutation equivalent to ψ_y. In particular, relabelling the distinguished site as 0 and the ordinary sites as $1,\dots,6$, in a suitable local frame its one-body marginals are $I/2$ and

$$T_{0j} = ve_z e_z^{\mathsf{T}}, \qquad T_{ij} = we_z e_z^{\mathsf{T}} \quad (i,j>0), \qquad v = \frac{1-8y}{3}, \quad w = \frac{32y-1}{15}.$$

The interval contains both $37/524$ and the entropy parameter x_*.

Proof. Write $r_i^a = \langle \sigma_a^{(i)} \rangle$ and $T_{ij}^{ab} = \langle \sigma_a^{(i)}\sigma_b^{(j)} \rangle$. At site i, concatenate T_{ij} for $j > i$ and T_{ji}^{T} for $j < i$ into the real 3×18 matrix M_i. The incident-axis sum of squares

$$C = \sum_i \sum_{\substack{a<b \\ c<d}} (M_i^{ac}M_i^{bd} - M_i^{ad}M_i^{bc})^2 = \frac{1}{2}\sum_i [(\operatorname{tr} M_i M_i^{\mathsf{T}})^2 - \operatorname{tr}(M_i M_i^{\mathsf{T}})^2]$$

has bidegree $(4,4)$ in $z, \bar{z}$. Also sum, over ordered quadruples of distinct sites, the nonnegative tetrad squares

$$D_{ijkl} = \sum_{a,b,c,d} (T_{ij}^{ab}T_{kl}^{cd} - T_{ik}^{ac}T_{jl}^{bd})^2.$$

The site-sum D likewise has bidegree $(4,4)$. Local $SU(2)$ actions rotate each Pauli index orthogonally; each displayed squared tensor norm is unchanged. Site permutations relabel summands. Complex conjugation reflects the Y Pauli axis, another orthogonal action. Hence C, D and $A_1 = \sum_{i,a}(r_i^a)^2$ are conjugation-even invariant polynomials.

By Theorem 41 and its proof, at ψ_y every one-body marginal is $I/2$ and the two-body computational-basis marginals are $\rho_{si} = \mathrm{diag}(10u, 5u+y, 5u+y, 10u)$ and $\rho_{ij} = \mathrm{diag}(\frac{1}{2}-8u, 8u, 8u, \frac{1}{2}-8u)$ with $u = (1-2y)/30$. In this displayed frame the axial pair tensors are $T_{0j} = ve_z e_z^{\mathsf{T}}$ and $T_{ij} = we_z e_z^{\mathsf{T}}$ with $v = 2(10u) - 2(5u+y) = (1-8y)/3$ and $w = 2(\frac{1}{2}-8u) - 2(8u) = (32y-1)/15$, so $A_1 = C = D = 0$ at ψ_y (checked independently by FO-1). On the stated interval $v > w > 0$. The hypothesis forces $A_1(z) = C(z) = D(z) = 0$. Thus all one-body Bloch vectors vanish and $\operatorname{rank} M_i \le 1$ for every i. At each site choose a unit vector n_i spanning the column space of M_i, or any unit vector if $M_i = 0$. Every T_{ij} then has the form $c_{ij} n_i n_j^{\mathsf{T}}$. Product local rotations send all n_i to e_z, leaving a real symmetric scalar edge array (c_{ij}). The tetrad equalities become

$$c_{ij}c_{kl} = c_{ik}c_{jl} = c_{il}c_{jk} \qquad (i,j,k,l \text{ distinct}). \tag{S11.4}$$

The following four scalar graph invariants are again conjugation-even, site-symmetric and of bidegree at most $(4,4)$:

$$\mathcal{E}_2 = \sum_{i<j} \|T_{ij}\|_F^2, \qquad \mathcal{E}_3 = \sum_{i<j<k} \mathrm{tr}(T_{ij}T_{jk}T_{ki}),$$

$$\mathcal{E}_4 = \frac{1}{3} \sum_{i<j<k<l} \sum_{\text{three cycles on } \{i,j,k,l\}} \mathrm{tr}(T_{ij}T_{jk}T_{kl}T_{li}), \qquad B_1 = \sum_{i<j} \|T_{ij}\|_F^4.$$

In the $\mathcal{E}_4$ expression the indices inside the trace follow each listed cycle. Their family values are

$$\mathcal{E}_2 = 6v^2 + 15w^2, \quad \mathcal{E}_3 = 15v^2 w + 20w^3, \quad \mathcal{E}_4 = 20v^2w^2 + 15w^4, \quad B_1 = 6v^4 + 15w^4.$$

In particular $\mathcal{E}_3 > 0$, so the edge array has a nonzero triangle. By (S11.4), a vertex outside this triangle is either connected to all its vertices or to none; in the latter case it has no edge to any other vertex. Two vertices connected to the triangle have a nonzero mutual edge. Thus the nonzero edges form a complete graph on one subset, with the other vertices isolated. For the scalar edge reconstruction, temporarily index the seven sites $1, \dots, 7$, with a nonzero triangle on $1, 2, 3$, and put $\tau = c_{12}c_{13}/c_{23} \neq 0$ and $\varepsilon = \operatorname{sgn}\tau$. Set $t_1 = \sqrt{|\tau|}$ and $t_j = c_{1j}/(\varepsilon t_1)$ on the active subset, and $t_j = 0$ elsewhere. The tetrads give $c_{ij} = \varepsilon t_i t_j$ for every pair. Every nonzero triangle product therefore has sign ε; since $\mathcal{E}_3 > 0$, $\varepsilon = 1$.

Put $x_i = t_i^2 \ge 0$, $e_k = e_k(x_1, \dots, x_7)$, and $p_k = \sum_i x_i^k$. The graph invariants above become $\mathcal{E}_2 = e_2$, $\mathcal{E}_3 = e_3$, $\mathcal{E}_4 = e_4$, $B_1 = e_2(x_1^2, \dots, x_7^2)$. Newton's identities give

$$B_1 = e_2^2 - 2e_1e_3 + 2e_4, \qquad e_1 = \frac{\mathcal{E}_2^2 + 2\mathcal{E}_4 - B_1}{2\mathcal{E}_3} = r + 6\ell, \qquad r = \frac{v^2}{w}, \quad \ell = w.$$

They also determine $p_2 = e_1^2 - 2e_2$ and $p_3 = e_1^3 - 3e_1e_2 + 3e_3$. For $\mu = e_1/7$ and $S_k = \sum_i (x_i - \mu)^k$, evaluating at the family profile $(r, \ell, \dots, \ell)$ with deviations $\frac{6}{7}(r-\ell)$ and $-\frac{1}{7}(r-\ell)$ (six times) gives (FO-1)

$$S_2 = \frac{6}{7}(r-\ell)^2 > 0, \qquad S_3 = \frac{30}{49}(r-\ell)^3 > 0, \qquad S_3^2 = \frac{25}{42} S_2^3.$$

The sharp seven-variable skewness inequality is $S_3 \le (5/\sqrt{42}) S_2^{3/2}$, with equality only for one high value and six equal low values. Indeed, maximising S_3 at fixed zero mean and S_2 by Lagrange multipliers yields at most two values; if the higher value occurs m times, the ratio is $(7-2m)/\sqrt{7m(7-m)}$, maximal only at $m = 1$. The recovered mean and variance give one value $r = v^2/w$ and six values $\ell = w$. All are positive. Flipping the local axis directions makes all t_i

positive, so there is one distinguished vertex, with incident weight $\sqrt{r\ell} = v$, and every ordinary edge has weight w.

The pair correlations have now fixed the local axes and edge weights. Return to labels $0, \dots, 6$, with 0 distinguished. We next use the remaining invariants to recover the state in that frame. The symbols Q, R, F, L below denote seven-qubit invariants local to this proof. Let $J_7 = \epsilon^{\otimes 7} K$ be the spin flip, with K complex conjugation, and form the 21 one-body spin-flip vectors $V_{ia} = \sigma_a^{(i)} J_7 z$ ($0 \le i \le 6$, $a \in \{x, y, z\}$), their Gram matrix $G = V^\dagger V$, the overlaps $h = V^\dagger z$ and $q_i = \sum_a |h_{ia}|^2$. Put $d_i = \sum_{j \ne i} \|T_{ij}\|_F^2$ and define

$$S_1 = \sum_i q_i, \qquad Q = \sum_i q_i^2, \qquad R = \sum_i d_i q_i,$$

$$F = \sum_{i \ne j} (\|T_{ij}\|_F^2 q_i - \|T_{ij}^{\mathsf{T}} h_i\|^2), \qquad L = \sum_{i<j} \mathrm{Re}(h_i^\dagger T_{ij} h_j),$$

where $T_{ij} = T_{ji}^{\mathsf{T}}$ for $i > j$. Each summand of F is nonnegative because the operator norm of a matrix is at most its Frobenius norm. The respective bidegrees in $(z, \bar{z})$ are $(2,2)$ for S_1, $(4,4)$ for Q, R, F, and $(3,3)$ for L. All five quantities are invariant under product $SU(2)$ and site permutations: $J_7 U = U J_7$, so the Pauli indices of h_i and T_{ij} rotate together. Under complex conjugation, with $\Delta = \mathrm{diag}(1, -1, 1)$, they transform as $h_i(\bar{z}) = \Delta \overline{h_i(z)}$ and $T_{ij}(\bar{z}) = \Delta T_{ij}(z) \Delta$; the real part makes L invariant. A global phase multiplies all h_i by one phase. Their values equal those of ψ_y by hypothesis.

In the displayed frame of (14), $J_7 |b\rangle = (-1)^{7-|b|} |\bar{b}\rangle$. The pair and complement strings have total weights 2 and 5, and the endpoint strings have weights 6 and 1, so their spin-flip signs must be retained separately. Comparing complementary amplitudes, the antilinear map $\sigma_z^{(s)} J_7$ acts on each string of the pair/complement pattern with sign -1 and on the endpoint pattern with sign $+1$. At an ordinary site i the map is $\sigma_z^{(i)} J_7 = (-1)^{b_s + b_i} \sigma_z^{(s)} J_7$, which reverses the sign on both endpoints and on the 10 pair/complement strings with $b_i \ne b_s$; these 10 strings, now with sign $+1$, and the other 20, with sign -1, contribute $-10u$, while the endpoints contribute $-2y$. The transverse components of every h_i vanish: $J_7 \psi_y$ has the same support as ψ_y ($s = 0$ with ordinary weights $\{2, 6\}$, $s = 1$ with $\{0, 4\}$), X_i or Y_i at an ordinary site changes the parity of the ordinary weight, and X_s or Y_s moves a string into the other branch, whose weight set is disjoint. This yields the family frame values (checked by FO-1)

$$h_s^* = -1 + 4y, \qquad h_o^* = -\frac{1 + 4y}{3}, \qquad q_0^* = (h_s^*)^2, \quad q_o^* = (h_o^*)^2.$$

Both h_s^* and h_o^* are negative. The pair profile gives $d_0 = 6v^2$ and $d_i = v^2 + 5w^2$ for $i > 0$, so $d_0 > d_i$. Since $F(\psi_y) = 0$, the hypothesis gives $F(z) = 0$; every pair weight is positive, so each $h_i = \zeta_i e_z$ is axial.

The equalities for S_1 and R, with $d_0 \ne d_i$, determine $q_0 = q_0^*$ and $\sum_{i=1}^6 q_i = 6 q_o^*$. The equality for Q then gives $\sum_{i=1}^6 q_i^2 = 6 (q_o^*)^2$. Equality in Cauchy–Schwarz forces $q_i = q_o^*$ for every $i > 0$. Thus all $|\zeta_i|$ equal their nonzero family values. Moreover

$$L(z) = \sum_{i<j} a_{ij} \,\mathrm{Re}(\bar{\zeta}_i \zeta_j) \le \sum_{i<j} a_{ij} |\zeta_i| |\zeta_j| = L(\psi_y),$$

where all a_{ij} are positive. The hypothesis on L makes this an equality, so every pair of phases agrees. A global phase of z can make $h_{0z} = h_s^*$ and $h_{iz} = h_o^*$ exactly.

The one-site Bloch vectors vanish; the Gram G therefore has identity diagonal 3×3 blocks and off-diagonal blocks T_{ij}. Let c have axial entries $c_{0z} = -1/2$, $c_{iz} = -1/4$ for $i > 0$, and zero transverse entries. Since $T_{0j} = v e_z e_z^{\mathsf{T}}$ and $T_{ij} = w e_z e_z^{\mathsf{T}}$, only axial entries of Gc survive, with $(Gc)_{0z} = -\frac{1}{2} - \frac{3}{2} v = h_s^*$ and $(Gc)_{iz} = -\frac{1}{4} - \frac{v}{2} - \frac{5}{4} w = h_o^*$; so $Gc = h$, and $c^\dagger h = -\frac{1}{2} h_s^* - \frac{3}{2} h_o^* = 1$. Consequently

$$\|z - Vc\|^2 = 1 - 2\,\mathrm{Re}(c^\dagger h) + c^\dagger G c = 0.$$

Thus $z = (-Z_0/2 - \sum_{i=1}^{6} Z_i/4)J_7 z$. Proposition S55 therefore puts z in the complex pair/complement shell $z_{T,b}$, where T is the shell's symmetric amplitude matrix (not the correlation blocks T_{ij} above), with entries t_{ij}. On that shell $\langle Z_0 Z_j\rangle = 1 - 4b^2 - 4p_j$ and $\langle Z_i Z_j\rangle = 1 - 4(p_i + p_j) + 8|t_{ij}|^2$, with σ and p_j as in (S10.2). Equating these with v and w, and using $\sum_j p_j = 2\sigma = 1 - 2b^2$, gives $b^2 = y$, $p_j = (1-2y)/6 = \sigma/3$ and $|t_{ij}|^2 = (1-2y)/30 = \sigma/15$, so the second and third squares of (S10.2) vanish and its first is $\pi_3(\psi_y) - 121/917$. Since z has the purity of ψ_y, (S10.2) gives $\sum_{i<j} |(TT^\dagger)_{ij}|^2 = 0$: T is a scaled symmetric conference matrix. Proposition S54 classifies it by diagonal congruence and ordinary-site permutations. Its displayed local unitaries preserve the endpoint amplitudes for every b, so z is equivalent to ψ_y. At $y = 37/524$ this recovers the purity orbit and the pair values $19/131$ and $11/131$ checked independently by MO-1–MO-2. □

Corollary S65 (Fourth-moment orbit rigidity)**.** *Let $X_r^+(z)$ denote the normalised r-copy moment of a pure seven-qubit state after product-local-unitary and site-permutation twirling and the complex-conjugation average. Fix $y \in (1/32, 1/12)$. If $X_r^+(z) = X_r^+(\psi_y)$ for some $r \geq 4$, then z is local-unitarily and site-permutation equivalent to ψ_y. In particular this applies to both the purity and entropy optimisers.*

Proof. A twirl-invariant polynomial of bidegree (k,k) with $k \leq r$ is a linear functional of X_r^+, after multiplying by $\mathrm{tr}(zz^*)^{r-k} = 1$. The thirteen invariants of Theorem S64 are of this kind with $k \leq 4$; the r-copy moment determines the four-copy moment by partial trace. □

Lemma S63 and Corollary S65 combine into the equality step shared by the purity and entropy theorems.

Theorem S66 (Equality from a sharp five-copy identity on the family; proved)**.** *Fix $y \in (1/32, 1/12)$, and put $X_y = X_5^+(\psi_y)$ and $X_y' = \frac{d}{dx} X_5^+(\psi_x)|_{x=y}$. Let Γ_b be linear maps to symmetric matrices, and W_b matrices such that the functionals $X \mapsto \mathrm{tr}[W_b S W_b^{\mathsf{T}} \Gamma_b(X)]$, with b varying and S ranging over symmetric matrices, span the critical space of linear functionals f on the span of the twirled five-copy moments with $f(X_y) = f(X_y') = 0$. If a normalised pure seven-qubit state z has $W_b^{\mathsf{T}} \Gamma_b(X_5^+(z)) W_b = 0$ for every b, then $X_5^+(z) = X_y$, and z is local-unitarily and site-permutation equivalent to ψ_y.*

Proof. The family values above give $H(\psi_x) = 155 + 96x$ for the invariant $H = 104S_1 - 25\mathcal{E}_2$, so $X_y' \neq 0$. Lemma S63 gives $X_5^+(z) = X_y + tX_y'$. Writing $h = H(\psi_y)$, both H and $(H-h)^2$ are linear functionals of the normalised five-copy moment, since their bidegrees are at most $(4,4)$. The latter is critical, hence vanishes at z. But $H(z) - h = 96t$, so $t = 0$. Thus $X_5^+(z) = X_y$, and Corollary S65 with $r = 5$ gives the orbit. □

Corollary S67 (seven-qubit purity minimum and rigidity; verified-arithmetic certificate, FP5-1)**.** *Every normalised pure seven-qubit state satisfies $\pi_3 \geq 121/917$. Equality holds precisely on the icosahedral local-unitary/permutation orbit, and there the whole normalised five-copy moment equals that of Ψ_{ico}. Quantitatively, with $c > 0$ the least eigenvalue of the C_b,*

$$\pi_3(z) - \frac{121}{917} \;\geq\; c \sum_b \mathrm{tr}\big[W_b^{\mathsf{T}} \Gamma_b(X_5^+(z)) W_b\big].$$

Proof. For a pure state the flipped five-copy moments are averages of rank-one projectors, so every $\Gamma_b(X_5^+(z)) \succeq 0$. The identity of Theorem S62 gives the bound and the displayed inequality. At equality every $W_b^{\mathsf{T}} \Gamma_b(X_5^+(z)) W_b$ vanishes, and Theorem S66 with $y = 37/524$ gives the moment and orbit, using the spanning assertion of Theorem S62. Conversely, Theorem 41 gives equality on that orbit. □

Corollary S68 (Labelled LU orbits in the family; proved, EO-1)**.** *For each fixed $y \in (1/32, 1/12)$, the states obtained by permuting the qubits of ψ_y form exactly 42 distinct orbits under local unitaries*

and global phase. They form one orbit when qubit permutations are also allowed. Representatives are indexed by a choice of distinguished qubit and a permutation of the last three of the six ordered ordinary labels, leaving the first three ordinary labels fixed.

Proof. The incident pair weights are $d_s = 6v^2$ at the distinguished site and $d_o = v^2+5w^2$ elsewhere, with $d_s > d_o$. These labelled quantities are LU invariants, so different choices of distinguished site give seven different LU classes of configurations.

Fix that site. Every canonical permuted state has the same nonzero positive axial pair profile. A local unitary mapping two such states must send each local Z axis to itself or its negative. Positivity of every pair coefficient forces the seven signs to agree. Thus all local unitaries are diagonal, or all are anti-diagonal. A simultaneous bit flip sends the real conference matrix C to $-C$, which is already realised by diagonal congruence $D = iI$ and the local unitaries of Proposition S54.

For diagonal unitaries, equality of the two nonzero endpoint amplitudes shows that the pair matrices are related by $C' = DCD$ with D diagonal unitary. Conversely every such congruence is implemented by the local unitaries displayed in that proposition, preserving the endpoints. When C, C' have real sign entries, $d_i d_j \in \{1, -1\}$ for every $i \neq j$. Using three distinct indices gives $D = \eta\,\mathrm{diag}(\epsilon_i)$, where $\eta \in \{1, i\}$ and $\epsilon_i \in \{1, -1\}$, up to an irrelevant common sign. The exact LU criterion for ordinary permutations is therefore

$$C' = \tau\,\mathrm{diag}(\epsilon_i) C\,\mathrm{diag}(\epsilon_i), \qquad \tau \in \{1, -1\}.$$

Normalise the first ordinary row to all positive signs. Orthogonality to that row says that every row of the five-vertex core has two negative entries. Its negative-edge graph is therefore a pentagon. There are $(5-1)!/2 = 12$ labelled pentagons, all attained by ordinary permutations of C. In the normalised frame the remaining equivalence above is identity or complementation: after choosing $\epsilon_0 = 1$, the first row forces $\epsilon_i = \tau$ for every $i > 0$. Complementation pairs the twelve pentagons without fixed points, leaving six classes.

The six permutations fixing ordinary labels $0, 1, 2$ and permuting $3, 4, 5$ represent all six classes. In the base pentagon $1-3-2-4-5$ the vertices $1, 2$ are not adjacent, and a permutation fixing them keeps them non-adjacent, whereas they are adjacent in every complement; and two such images coincide only if the permutations differ by a dihedral symmetry fixing two vertices, that is, the identity. EO-1 independently checks all 720 ordinary permutations: the switching and switching-complement automorphism counts are 60 and 60, and the six classes each contain 120 permutations. It also verifies the explicit diagonal local gauges on both amplitude-pattern basis vectors. Combining these with the seven distinguished-site choices gives $7 \cdot 6 = 42$ LU orbits. □

S11.6 Proof of Theorem 11

Theorem 11 (the seven-qubit purity minimum; verified-arithmetic certificate FP5-1, exact FO-1). *Every pure seven-qubit state satisfies $\pi_3(\psi) \geq 121/917$. Equality holds if and only if ψ is equivalent to Ψ_{ico} under local unitaries and permutations of the qubits.*

Proof. Corollary S67 combines the positive twelve-face identity of Theorem S62 with the equality-moment and orbit reconstruction of Theorem S66. Theorem 41 computes the attaining value at Ψ_{ico}. □

S11.7 Nondegeneracy of the purity minimiser

We now quantify the purity increase near the minimising orbit of Corollary S67. The Lagrange Hessian is positive transverse to the local-unitary orbit, giving an explicit quadratic basin.

Theorem S69 (exact full-space Hessian spectrum; LH-1–LH-4)**.** *The state Ψ_{ico} is a strict local minimiser of π_3 on $\mathbb{C}P^{127}$ modulo the action of local unitaries. The real 256×256 Lagrange-Hessian of the purity functional at Ψ_{ico} is positive semidefinite, with kernel exactly the 22-dimensional*

span of $i\Psi_{\mathrm{ico}}$ *and the* 21 *one-site Pauli rotations* $iX_j\Psi_{\mathrm{ico}}, iY_j\Psi_{\mathrm{ico}}, iZ_j\Psi_{\mathrm{ico}}$, $j \in \{s, 0, \dots, 5\}$ *(qubits labelled as in* (14)*). For the constrained quadratic form* Q *described below and every nonzero real vector* v *orthogonal to those rotations,*

$$Q(v) \geq \frac{350 - 6\sqrt{2005}}{4585}\|v\|^2 > \frac{78}{4585}\|v\|^2,$$

with equality in the first bound on the 18*-dimensional eigenspace inside the special-qubit spin-flip sector described below. Global minimality is Corollary S67; this theorem adds nondegeneracy.*

Proof. For each $|S| = 3$, flatten z into M_S and set $R_S = M_S M_S^\dagger$. Direct differentiation of the homogeneous quartic objective gives the stationary equation

$$\frac{1}{35}\sum_{|S|=3}(R_S \otimes I_{S^c})z = \pi_3(z)z,$$

where the operator notation includes the canonical reshuffling of tensor factors. The family quadratic in Theorem 41 is stationary at $x = 37/524$, so Lemma S72 (Section S12.1, which uses only the symmetry of the family) proves full-space stationarity. LH-1 independently checks the displayed equation for $z = \Psi_{\mathrm{ico}} = \sqrt{15/524}\,U + \sqrt{37/524}\,V$ (where U and V are the parts of (14) multiplying α and β), with the signed integer support tensors from (14).

Write v for a variation and N_S for its flattening. The Lagrange second-variation form (half the usual real Hessian) is

$$Q(v) = \frac{1}{35}\sum_{|S|=3}\Big(\mathrm{tr}\big[(N_S M_S^\dagger + M_S N_S^\dagger)^2\big] + 2\,\mathrm{tr}(R_S N_S N_S^\dagger)\Big) - 2\pi_3(\Psi_{\mathrm{ico}})\|v\|^2.$$

LH-2 produces a real symmetric integer-matrix representation $H = (A + \sqrt{555}\,B)/18340$ of this form on all 256 real amplitude directions, and LH-3 verifies that the 22 displayed symmetry tangent vectors are annihilated in $\mathbb{Q}(\sqrt{555})$.

To analyse H and the entropy Hessian along the family (14), we use a common symmetry decomposition defined over $\mathbb{Z}$. First observe that the local diagonal unitary $T = -Z_s \otimes \mathrm{diag}(1, i)^{\otimes 6}$ of Lemma S72 acts on $|s, b_0 \cdots b_5\rangle$ with eigenvalue $-(-1)^s i^w$, where $w = \sum_{j=0}^5 b_j$. Hence T has eigenvalue $+1$ on the 32-dimensional support subspace $\mathcal{H}_{\mathrm{supp}}$ ($(s, w) \in \{(0, 2), (1, 4), (0, 6), (1, 0)\}$), eigenvalue -1 on the 32-dimensional special-qubit flip subspace $\mathcal{H}_{\mathrm{flip}} = X_s\mathcal{H}_{\mathrm{supp}}$ ($(s, w) \in \{(1, 2), (0, 4), (1, 6), (0, 0)\}$), and eigenvalues $\pm i$ on the 64-dimensional odd-weight subspace $\mathcal{H}_{\mathrm{odd}}$ ($w \in \{1, 3, 5\}$). Both objectives are invariant under local unitaries and complex conjugation. Since $T\psi_x = \psi_x$ and ψ_x is real, their Hessians commute with these two symmetries, decoupling into the six computational-basis blocks

$$\mathbb{R}\mathcal{H}_{\mathrm{supp}} \oplus i\mathbb{R}\mathcal{H}_{\mathrm{supp}} \oplus \mathbb{R}\mathcal{H}_{\mathrm{flip}} \oplus i\mathbb{R}\mathcal{H}_{\mathrm{flip}} \oplus \mathbb{R}\mathcal{H}_{\mathrm{odd}} \oplus i\mathbb{R}\mathcal{H}_{\mathrm{odd}} \tag{S11.5}$$

of dimensions $32, 32, 32, 32, 64, 64$. T maps $\mathbb{R}\mathcal{H}_{\mathrm{odd}}$ isometrically onto $i\mathbb{R}\mathcal{H}_{\mathrm{odd}}$, and on $\mathcal{H}_{\mathrm{flip}}$ a signed bit-complement $DX^{\otimes 7}$ ($D = +1$ on $(0, 4), (1, 6)$, -1 on $(1, 2), (0, 0)$) conjugates the real block into the imaginary one.

Two signed incidence structures of the conference matrix finish the reduction (LH-4 checks every identity below). On the 15 ordinary pairs, the disjoint-pair matching matrix $(L_{\mathrm{disj}})_{p,q} = s_p s_q s_{(p\cup q)^c}$ ($p \cap q = \emptyset$) satisfies $(L_{\mathrm{disj}} - 2)(L_{\mathrm{disj}} - 3)(L_{\mathrm{disj}} + 1)(L_{\mathrm{disj}} + 3) = 0$, with integer eigenprojectors P_2, P_3, P_{-1}, P_{-3} of ranks $1, 5, 5, 4$. Pairing each pair string with its bit-complement and adding the two endpoints reduces $\mathcal{H}_{\mathrm{supp}}$ and $\mathcal{H}_{\mathrm{flip}}$ to a 4×4 block on P_2 and the endpoints and 2×2 blocks on P_3, P_{-1}, P_{-3} (superscripts $\pm$ below denote the bit-complement sign). On the 20 ordinary triples, the vertex-incidence matrices $(E_{\mathrm{in}})_{T,i} = s_{kl}$ ($T = \{i, k, l\}$) and $(E_{\mathrm{out}})_{T,i} = (E_{\mathrm{in}})_{T^c,i}$ satisfy $E_{\mathrm{in}}^{\mathsf{T}}E_{\mathrm{in}} = E_{\mathrm{out}}^{\mathsf{T}}E_{\mathrm{out}} = 10I_6$ and $E_{\mathrm{in}}^{\mathsf{T}}E_{\mathrm{out}} = 2C$ by $C^2 = 5I_6$. The orthogonal columns of $F_1 = E_{\mathrm{in}}$ and $F_2 = E_{\mathrm{out}}C - E_{\mathrm{in}}$ and their rank-8 complement $P_{8,40}$ reduce $\mathcal{H}_{\mathrm{odd}}$ to scalars on $P_{8,40}^{\pm}$ and two 4×4 blocks tensored with I_6.

In every block of $18340Q = A + \sqrt{555}\,B$, A preserves and B exchanges the U- and V-type subspaces. Changing the sign on one of them is orthogonal and sends $\sqrt{555} \mapsto -\sqrt{555}$, so each characteristic polynomial is Galois-invariant and lies in $\mathbb{Z}[t]$ (LH-4):

- on $\mathbb{R}\mathcal{H}_{\text{supp}}$ (dimension 32), $18340Q$ is strictly positive definite with eigenvalues 1440^4 (on P_{-3}^+), 1648^9 (on $P_3^- \oplus P_{-3}^-$), 3600^5 (on P_3^+), 4080^5 (on P_{-1}^+), 5488^5 (on P_{-1}^-), 4440^1 and 9680^1 (on $\operatorname{span}\{U, V\}$), and $6864 \pm 16\sqrt{10726}$ (on the complementary P_2/endpoint plane);
- on $i\mathbb{R}\mathcal{H}_{\text{supp}}$ (dimension 32), the eigenvalues are 0^8 (spanned by $i\Psi_{\text{ico}}$ and $iZ_s\Psi_{\text{ico}}, iZ_0\Psi_{\text{ico}}, \ldots, iZ_5\Psi_{\text{ico}}$: three in the P_2/endpoint block and five on P_{-1}^-), 688^5 (on P_{-1}^+), 1048^1 (in the P_2/endpoint block), 1440^5 (on P_3^-), 1648^9 (on $P_3^+ \oplus P_{-3}^+$), and 3600^4 (on P_{-3}^-);
- on $\mathbb{R}\mathcal{H}_{\text{flip}} \cong i\mathbb{R}\mathcal{H}_{\text{flip}}$ (32 dimensions each), they are $1400 \pm 24\sqrt{2005}$ on $P_3 \oplus P_{-3}$ (multiplicity 9 per sector, 18 total), $1400 \pm 24\sqrt{1195}$ on P_{-1} (multiplicity 5 per sector, 10 total), 0^1 ($iY_s\Psi_{\text{ico}}$ and $iX_s\Psi_{\text{ico}}$), and the three roots of $t^3 - 13600t^2 + 55582720t - 63383040000 = 0$ (all > 1900) on the P_2/endpoint 4×4 block;
- on $\mathbb{R}\mathcal{H}_{\text{odd}} \cong i\mathbb{R}\mathcal{H}_{\text{odd}}$ (64 dimensions each), they are the scalars 2248^8 and 840^8 on $P_{8,40}^{\pm}$ (multiplicity 16 each across real and imaginary parts), 0^6 ($iY_0\Psi_{\text{ico}}, \ldots, iY_5\Psi_{\text{ico}}$ and $iX_0\Psi_{\text{ico}}, \ldots, iX_5\Psi_{\text{ico}}$), 2248^6, and $2484 \pm 4\sqrt{286081}$ (multiplicity 6 each per sector) on the first 4×4 block, and the four roots of $t^4 - 10400t^3 + 32376000t^2 - 37446912000t + 14433914880000 = 0$ (multiplicity 6 per sector, 12 total, all > 900) on the second 4×4 block.

The kernel therefore has dimension $0 + 8 + 2 + 12 = 22$, spanned by $i\Psi_{\text{ico}}$ and the 21 one-site Pauli rotations. Since $44^2 = 1936 < 2005 < 2025 = 45^2$, the smallest positive eigenvalue of $18340Q$ is $1400 - 24\sqrt{2005} > 1400 - 24 \cdot 45 = 320 > 312 = 4 \cdot 78$, giving the exact transverse minimum $(1400 - 24\sqrt{2005})/18340 = (350 - 6\sqrt{2005})/4585 > 312/18340 = 78/4585$. The usual nondegenerate transverse-Hessian argument gives strict local minimality modulo the compact local-unitary group. □

Lemma S70 (quantitative orbit basin for quartic swap observables)**.** *Let $F(z) = \langle z^{\otimes 2}, Gz^{\otimes 2}\rangle$ for a Hermitian G with $\|G\|_{\text{op}} \le 1$. Suppose F is invariant under a compact group of unitary transformations, and z_0 is a unit stationary point of F on the unit sphere. Let $\mathcal{O}$ be the orbit of z_0 including global phases. If the Lagrange second-variation form satisfies $Q(v) \ge h\|v\|^2$ on the real orthogonal complement of the orbit tangent space at z_0, then every unit vector z satisfies, with $\delta = \min_{\phi \in \mathcal{O}} \|z - \phi\|$,*

$$F(z) - F(z_0) \ge (h - 4\delta - \delta^2)\delta^2.$$

Proof. The compact orbit contains a closest point ϕ to z. Put $v = z - \phi$. Differentiating the squared-distance function along the orbit shows that v is real orthogonal to every orbit tangent at ϕ. Because the group preserves F, the same transverse Hessian bound holds at ϕ. Write $p = F(\phi)$ and expand the quartic polynomial $F(\phi + v) - 2p\|\phi + v\|^2 + p$. Its constant and linear terms vanish by normalisation and stationarity, and its quadratic term is exactly $Q(v)$. The remaining terms have degrees three and four in v. In the two-copy matrix-element expression, each multilinear term is bounded by the product of its input vector norms because $\|G\|_{\text{op}} \le 1$. There are $\binom{4}{3} = 4$ cubic terms and one quartic term, so their total absolute value is at most $4\|v\|^3 + \|v\|^4$. Finally $\|\phi + v\| = \|z\| = 1$, so the polynomial on the left equals $F(z) - p$. Combining the bounds proves the claim. □

Theorem S71 (exact quantitative basin; QB-1–QB-2)**.** *Let $\mathcal{O}_{\text{ico}}$ be the full local-unitary orbit of Ψ_{ico} including global phase, and for any normalised pure seven-qubit state z put*

$$\delta(z) = \min_{\phi \in \mathcal{O}_{\text{ico}}} \|z - \phi\|_2.$$

If $\delta(z) \le 1/250$, then

$$\boxed{\pi_3(z) \ge \frac{121}{917} + \frac{\delta(z)^2}{1024}.}$$

Here $\delta(z)^2 = 2 - 2\max_{U\in\mathrm{LU}} |\langle z|U\Psi_{\mathrm{ico}}\rangle|$.

Proof. Take $G = \binom{7}{3}^{-1} \sum_{|S|=3} F_S$ where F_S is the unitary two-copy subsystem swap. Thus $\|G\|_{\mathrm{op}} \le 1$, and its quartic expectation is the unnormalised mean three-body purity. The exact full-space Hessian certificate Theorem S69 gives $h = 78/4585$ in Lemma S70. QB-1 checks the rational strict inequality

$$\frac{78}{4585} - \frac{4}{250} - \frac{1}{250^2} - \frac{1}{1024} = \frac{285123}{14672000000} > 0.$$

Since $4\delta + \delta^2$ increases with $\delta \ge 0$, the displayed quadratic basin follows. For unit vectors, minimising the distance over global phase gives the overlap form of δ (QB-2). □

S12 The sharp mean entropy at seven qubits

The family entropy maximiser ψ_{x_*} has, over all its triple reductions, four distinct positive marginal eigenvalues. A one-variable polynomial majorant of the entropy touching all four would require degree at least seven (Proposition S81). The proof instead pairs each marginal with its spin flip to obtain a degree-five joint majorant. An entropy–purity slope and an eigenvalue cap first restrict the states that could improve on ψ_{x_*}; a localised five-copy identity then proves the bound on that region. The equality-moment reconstruction used for purity applies again at x_*, but selects a different orbit. The final subsections establish nondegeneracy, exclude mixed-state maximisers, and show that every local entropy maximum has full-rank marginals.

S12.1 The family maximiser

Write $E_3(z) = 35^{-1} \sum_{|S|=3} S(\rho_S)$ for a unit seven-qubit vector, and retain the two real amplitude patterns

$$U = \sum_{i<j} s_{ij}\big(|0, e_i + e_j\rangle - |1, \mathbf{1} - e_i - e_j\rangle\big), \qquad V = |0, \mathbf{1}\rangle + |1, \mathbf{0}\rangle,$$

so $\psi_x = \alpha U + \beta V$, $\alpha = \sqrt{(1-2x)/30}$, $\beta = \sqrt{x}$, $\langle U, V\rangle = 0$, $\|U\|^2 = 30$, and $\|V\|^2 = 2$.

Lemma S72 (Symmetric criticality of the conference family; proved, ES-1)**.** *Let $\mathcal{F}$ be a real function on the unit sphere, invariant under local unitaries, qubit permutations and complex conjugation, and differentiable at ψ_x for some $0 < x < 1/2$. If x is a stationary point of $x \mapsto \mathcal{F}(\psi_x)$, then ψ_x is stationary on the whole unit sphere. This applies to π_3 (a polynomial) and to E_3, whose marginals have full rank for $0 < x < 1/2$ by Theorem 41.*

Proof. We exhibit a common symmetry group of U, V whose real fixed space is exactly $\mathrm{span}_{\mathbb{R}}\{U, V\}$. Ordinary labels are $0, \dots, 5$, with the distinguished qubit denoted by s. The diagonal unitary

$$T = -Z_s \otimes \mathrm{diag}(1, i)^{\otimes 6}$$

fixes U, V. Its eigenvalue on $|s, b\rangle$ is $-(-1)^s i^{|b|}$, so its fixed space consists of the pair and complement sectors $(s, |b|) = (0, 2), (1, 4)$ and the two endpoints $(0, 6), (1, 0)$.

Put $g = (1\,3\,2\,4\,5)$, $p = (0\,1)(2\,4)$ and $\epsilon = (1, 1, 1, -1, 1, -1)$. The conference signs satisfy $s_{g(i)g(j)} = s_{ij}$ and $\epsilon_i \epsilon_j s_{p(i)p(j)} = s_{ij}$. If P_π permutes the ordinary qubits by π, then P_g and

$$W = \left(-Z_s \otimes \bigotimes_{i=0}^{5} \mathrm{diag}(1, -i\epsilon_i)\right) P_p$$

fix U, V and both endpoints. On either pair sector, dividing the coefficients by s_{ij} turns their actions into ordinary permutations of the fifteen pairs. These permutations act transitively: g has

three pair orbits (pairs through 0, pentagon edges, and pentagon diagonals), while p sends $\{0,3\}$ to $\{1,3\}$ and $\{0,2\}$ to $\{1,4\}$, connecting all three. Thus a real vector fixed by T, P_g, W has one coefficient on each signed pair sector and one on each endpoint.

Finally the local unitary

$$Q = \left(\mathrm{diag}(i,1)_s \otimes \mathrm{diag}(1, e^{i\pi/4})^{\otimes 6}\right) X^{\otimes 7}$$

fixes U, V. On the displayed support it exchanges complementary pair strings with a minus sign and exchanges the endpoints with a plus sign. Its fixed space within the preceding four-dimensional space is therefore exactly $\mathrm{span}_{\mathbb{R}}\{U, V\}$.

At a real ψ_x, conjugation invariance makes the spherical gradient real. The four displayed symmetries act orthogonally on $\mathbb{C}^{128} \cong \mathbb{R}^{256}$, preserve $\mathcal{F}$ and fix ψ_x, so they fix that gradient, which therefore lies in this plane. Orthogonality to ψ_x leaves only the one-dimensional family tangent. Stationarity within the family makes that last component vanish, proving full-space stationarity. □

Proposition S73 (Full-space stationarity; proved, ES-1)**.** *The unique maximiser ψ_{x_*} of E_3 within the icosahedral family is a stationary point on the whole projective space $\mathbb{C}P^{127}$. Its parameter is the unique root in $(0, 1/2)$ of*

$$2^{32}(1-2x_*)^{15} = 3^8(1+8x_*)^8(1+28x_*)^7, \qquad x_* = 0.069828\ldots. \tag{S12.1}$$

In particular $E_3(\psi_{x_*}) = 2.96535602947767\ldots$ bits, and $1/32 < x_* < 37/524$, so $x_*/u_* \in (1, 37/15)$ with $u_* = (1-2x_*)/30$.

Proof. By Lemma S72, it suffices to find the family stationary point. The marginal spectra of Theorem 41 give

$$E_3'(x) = \frac{4}{35\ln 2} \ln \frac{(4u)^{13}(8u)^2}{(3u+x)^8(u+x)^7}, \qquad u = (1-2x)/30.$$

Strict concavity and the endpoint derivative signs in Corollary 42 give a unique interior zero, equivalent to (S12.1). At this zero the symmetry lemma gives full-projective stationarity. At $x = 1/32$ one has $u = x$ and $E_3'(1/32) = 4\ln 512/(35\ln 2) > 0$, so $1/32 < x_* < 37/524$ by concavity and Corollary 42; since x/u increases with x and equals $37/15$ at $x = 37/524$, $x_*/u_* \in (1, 37/15)$. □

Throughout the entropy sections, $\phi = \psi_{x_*}$, $u_* = (1-2x_*)/30$ and $E_* = E_3(\phi)$; a *competitor* is a pure state z with $E_3(z) \geq E_*$.

S12.2 The fixed-purity entropy envelope

For $1/8 \leq p \leq 1$, let

$$t(p) = \frac{1+\sqrt{7(8p-1)}}{8}, \qquad g_8(p) = -t(p)\log_2 t(p) - (1-t(p))\log_2 \frac{1-t(p)}{7}. \tag{S12.2}$$

To bound entropy at fixed purity, we use a quadratic majorant of $f(x) = -x\ln x$, in the spirit of the Hermite majorants of the four-qubit sections.

Lemma S74 (Quadratic majorant of $-x\ln x$; proved)**.** *Let $0 < b < a$, let $T(x) = f(b) + f'(b)(x-b)$ be the tangent of f at b, and put $c = (f(a) - T(a))/(a-b)^2$ and $Q(x) = T(x) + c(x-b)^2$. Then $f(x) \leq Q(x)$ for $0 \leq x \leq a$, with equality only at $x \in \{b, a\}$.*

Proof. Put $R = f - T$, so $R(b) = R'(b) = 0$ and $R'(y) = -\ln(y/b)$. The function $u(y) = R(y) + (y-b)^2/(2b)$ has $u(b) = 0$ and $u'(y) = y/b - 1 - \ln(y/b) \geq 0$, with equality only at $y = b$. Hence $R(x) < -(x-b)^2/(2b)$ for $0 \leq x < b$, and $R(a) > -(a-b)^2/(2b)$, that is $c > -1/(2b)$; together, $R(x) < c(x-b)^2$ on $[0, b)$. On (b, ∞) the divided difference $\varrho(y) = R(y)/(y-b)^2$ has $\varrho'(y) = N(y)/(y-b)^3$ with $N(y) = (y+b)\ln(y/b) - 2(y-b)$. Since $N(b) = 0$ and $N'(y) = \ln(y/b) + b/y - 1 > 0$ for $y \neq b$, ϱ is strictly increasing, so $R(x) < \varrho(a)(x-b)^2 = c(x-b)^2$ for $b < x < a$. □

Lemma S75 (Entropy envelope at fixed purity; all dimensions)**.** *For every $d \geq 2$ and every d-dimensional density matrix ρ,*

$$S(\rho) \leq g_d(\operatorname{tr}\rho^2), \qquad t_d(p) = \frac{1+\sqrt{(d-1)(dp-1)}}{d},$$

where $g_d(p)$ is the entropy of $(t_d(p), ((1-t_d(p))/(d-1))^{d-1})$. Equality holds exactly at that spectrum, including the uniform and pure endpoints. For $d = 8$ this is (S12.2)*.*

Proof. The cases $p = 1/d$ and $p = 1$ are immediate. Otherwise put $a = t_d(p)$, $b = (1-a)/(d-1)$, so $a^2 + (d-1)b^2 = p$. Cauchy–Schwarz on all eigenvalues except λ_i gives $p \geq \lambda_i^2 + (1-\lambda_i)^2/(d-1)$, hence $\lambda_i \leq a$. The preceding dimension-independent quadratic majorant gives

$$S(\rho)\ln 2 \leq \sum_i Q(\lambda_i) = g_d(p)\ln 2,$$

since the sum depends only on the trace and purity and can be evaluated on (a, b^{d-1}). Equality requires every $\lambda_i \in \{a, b\}$; if j eigenvalues equal a, the trace forces $(j-1)(da-1) = 0$, hence $j = 1$. □

S12.3 Spin-flip spectral pairing and the pure-state obstruction

We first bound opposite-pair sums in the marginal spectrum, then combine these bounds across overlapping cuts. For a normalised three-qubit state ρ, write its eigenvalues as $\lambda_1 \geq \cdots \geq \lambda_8$. Put

$$e(\rho) = 8\operatorname{tr}\rho^2 - 1, \quad b(\rho) = \sum_{1\leq i<j\leq 3}\sum_{a,b=1}^{3} \left[\operatorname{tr}(\rho\,\sigma_a^{(i)}\sigma_b^{(j)})\right]^2, \quad v(\rho) = \sum_{i=1}^{4}\left(\lambda_i + \lambda_{9-i} - \frac{1}{4}\right)^2.$$

The convention for b sums precisely the weight-two Pauli squares on these three sites; one- and three-body terms do not enter.

Proposition S76 (Spin-flip spectral pairing; SP7-1)**.** *Every three-qubit density matrix satisfies the* sharp spectral inequality

$$\boxed{b(\rho) \geq 4v(\rho).} \tag{S12.3}$$

Equality holds, in particular, for every

$$\rho_r = (1-r)\frac{I_8}{8} + r\,|g\rangle\langle g|, \qquad 0 \leq r \leq 1, \quad |g\rangle \in (\mathbb{C}^2)^{\otimes 3} \text{ pure},$$

where $e(\rho_r) = 7r^2$, $b(\rho_r) = 3r^2$, $v(\rho_r) = 3r^2/4$. For a pure seven-qubit state, letting $e_S = e(\rho_S)$ and $R = \sum_{|T|=3} e_T$, there is the stronger joint spectral compatibility *inequality*

$$\boxed{e_S + 16v(\rho_S) \leq R \qquad \text{for all } |S| = 3.} \tag{S12.4}$$

The latter also holds for mixed global states.

Proof. Let $\widetilde{\rho} = Y^{\otimes 3}\rho^T Y^{\otimes 3}$ be the spin-flipped marginal. This is positive, has the same eigenvalues as ρ, and flips the sign of each odd Pauli weight. Pauli orthogonality gives

$$\operatorname{tr}\rho^2 = \frac{1+e}{8}, \qquad \operatorname{tr}(\rho\widetilde{\rho}) = \frac{1-e+2b}{8},$$

hence $\operatorname{tr}\rho^2 + \operatorname{tr}(\rho\widetilde{\rho}) = (1+b)/4$. The von Neumann trace rearrangement inequality implies

$$\operatorname{tr}(\rho\widetilde{\rho}) \geq \sum_{i=1}^{8}\lambda_i\lambda_{9-i}.$$

Consequently

$$\frac{1+b}{4} \geq \sum_{i=1}^{4} (\lambda_i + \lambda_{9-i})^2 = \frac{1}{4} + v(\rho),$$

proving (S12.3).

For every pure three-qubit vector g, complementarity of one- and two-qubit marginal purities gives $\sum_{i<j} \operatorname{tr} \rho_{ij}(g)^2 = \sum_i \operatorname{tr} \rho_i(g)^2$. Its Pauli expansion shows that the total weight-two Pauli sum is exactly 3, independently of g. Depolarisation by r multiplies all nontrivial Pauli expectations by r, proving $b = 3r^2$ and $e = 7r^2$. The spectrum is $((1+7r)/8, ((1-r)/8)^7)$, whose four opposite pair sums give $v = 3r^2/4$; equality follows.

For a seven-qubit global state, let b_{ij} denote the sum of weight-two Pauli squares on the physical pair ij. Then $b(\rho_S) = \sum_{\{i,j\} \subset S} b_{ij}$ and each such pair occurs in *four other* three-site subsets. Since all weights in every e_T are nonnegative,

$$R - e_S \geq 4b(\rho_S) \geq 16v(\rho_S).$$

This proves (S12.4). □

Lemma S77 (Sharp entropy–purity slope; proved, SP7-5)**.** *Every d-dimensional density matrix, $d \geq 3$, satisfies*

$$\log_2 d - S(\rho) \geq c_d(d \operatorname{tr} \rho^2 - 1), \qquad c_d = \frac{(d-1)\log_2(d-1)}{d(d-2)}. \tag{S12.5}$$

The constant is sharp: equality holds exactly at the uniform spectrum and at $((d-1)/d, (1/[d(d-1)])^{d-1})$. In particular, for $d = 8$,

$$3 - S(\rho) \geq \frac{7\log_2 7}{48}(8 \operatorname{tr} \rho^2 - 1) \geq \frac{49}{120}(8 \operatorname{tr} \rho^2 - 1) \geq \frac{2}{5}(8 \operatorname{tr} \rho^2 - 1), \tag{S12.6}$$

and the last two comparisons are strict unless $\rho = I_8/8$. These bounds include rank-deficient matrices.

Proof. Put $m = d - 1 \geq 2$. By Lemma S75 it suffices to treat the one-spike spectrum $((1 + mr)/d, ((1-r)/d)^m)$, $0 \leq r \leq 1$. Its normalised purity excess is $d \operatorname{tr} \rho^2 - 1 = mr^2$ and its entropy loss is $\log_2 d - S = F_m(r)/(d \ln 2)$, where

$$F_m(r) = (1+mr)\ln(1+mr) + m(1-r)\ln(1-r).$$

The derivative of $F_m(r)/r^2$ has numerator $N = rF_m' - 2F_m$, with $N(0) = N'(0) = 0$ and

$$N'' = rF_m''' = \frac{m(m+1)r\big(2mr - (m-1)\big)}{(1-r)^2(1+mr)^2}.$$

So N'' changes sign once, from negative to positive: N' is negative and then increasing, and N has at most one interior zero. At $r_* = (m-1)/m$ one has $1 + mr_* = m$ and $1 - r_* = 1/m$, so

$$F_m(r_*) = (m-1)\ln m, \qquad F_m'(r_*) = 2m \ln m, \qquad N(r_*) = 0.$$

Hence F_m/r^2 strictly decreases on $(0, r_*]$ and strictly increases on $[r_*, 1]$, with minimum $m^2 \ln m/(m-1)$. Dividing by $dm \ln 2$ gives (S12.5), with equality only at $r \in \{0, r_*\}$. For $d = 8$, $c_8 = 7\log_2 7/48$; since $7^5 = 16807 > 16384 = 2^{14}$, $\log_2 7 > 14/5$ and $7\log_2 7/48 > 49/120 > 2/5$. SP7-5 checks these identities and constants. □

Corollary S78 (Sharpened universal high-entropy spectral restriction; proved, SC-2, SP7-6)**.** *For any (pure or mixed) seven-qubit state ω,*

$$E_3(\omega) \geq \frac{59307}{20000} \quad \Longrightarrow \quad \boxed{\lambda_{\max}(\omega_S) < \frac{497}{1000} = 0.497 \quad \textit{for every } |S| = 3.}$$

Every competitor satisfies the hypothesis, since $E_* \geq E_3(\psi_{7/100}) > 59307/20000$ (SC-2).

Proof. Fix a cut S with eigenvalues in descending order. Let $t = \lambda_1$, $a = \lambda_8$ and $b = (1-t-a)/6$, so $0 \le a \le (1-t)/7$. By concavity of entropy,

$$S(\omega_S) \le -t\log_2 t - a\log_2 a - 6b\log_2 b.$$

For the four opposite-pair sums of eigenvalues in $v(\omega_S)$, one sum is $t+a$, and the other three sum to $1-t-a$. Cauchy–Schwarz thus yields

$$v(\omega_S) \ge \frac{4}{3}\left(t + a - \frac{1}{4}\right)^2.$$

The spin-flip compatibility inequality (S12.4) and Lemma S77 (with the rational slope 49/120) applied to the other 34 marginals yield

$$\begin{aligned} 35(3 - E_3(\omega)) &\ge 3 - S(\omega_S) + \frac{49}{120}\sum_{T\neq S} e_T \\ &\ge 3 - S(\omega_S) + \frac{98}{15}v(\omega_S) \ge J(t,a), \end{aligned}$$

where, continuously at $a = 0$,

$$J(t,a) = 3 + t\log_2 t + a\log_2 a + 6b\log_2 b + \frac{392}{45}\left(t + a - \frac{1}{4}\right)^2.$$

Apply the tangent inequality $p\ln(p/q) \ge p - q$ $(p \ge 0, q > 0)$ to the three probability weights $(t, 6b, a)$ with comparison weights $q_1 = 1/2$, $6q_2 = 1/2$ $(q_2 = 1/12,\ 8q_2 = 2/3)$, and $q_3 = 1/192$ $(8q_3 = 1/24)$, whose sum is $q_1 + 6q_2 + q_3 = 1 + 1/192$. Since $3 + t\log_2 t + a\log_2 a + 6b\log_2 b = \big[t\ln(8t) + 6b\ln(8b) + a\ln(8a)\big]/\ln 2$ and $\ln(8q_1) = \ln 4 = 2\ln 2$, $\ln(8q_2) = \ln(2/3) = \ln 2 - \ln 3$, $\ln(8q_3) = \ln(1/24) = -3\ln 2 - \ln 3$, this gives

$$\begin{aligned} 3 + t\log_2 t + a\log_2 a + 6b\log_2 b &\ge 2t + (1-t-a)(1-\log_2 3) - a(3+\log_2 3) - \frac{1}{192\ln 2} \\ &= 1 + t - (1-t)\log_2 3 - 4a - \frac{1}{192\ln 2}. \end{aligned}$$

Completing the square in a gives

$$-4a + \frac{392}{45}\left(t + a - \frac{1}{4}\right)^2 = 4t - 1 - \frac{45}{98} + \frac{392}{45}\left(t + a - \frac{47}{98}\right)^2 \ge 4t - 1 - \frac{45}{98}.$$

The resulting lower bound $5t - \frac{45}{98} - (1-t)\log_2 3 - \frac{1}{192\ln 2}$ increases in t (slope $5 + \log_2 3 > 0$). For $t \ge t_0 = 497/1000$, using only $3^5 = 243 < 256 = 2^8$ $(\log_2 3 < 8/5)$ and $\ln 2 = 2\,\mathrm{arctanh}(1/3) > 2/3$ $(1/(192\ln 2) < 1/128)$, we obtain

$$J(t,a) > 5t_0 - \frac{45}{98} - \frac{8}{5}(1-t_0) - \frac{1}{128} = \frac{4755759}{3920000} > \frac{4851}{4000} = 35\left(3 - \frac{59307}{20000}\right),$$

with exact rational margin $1779/3920000 > 0$ (SP7-6), contradicting $E_3(\omega) \ge 59307/20000$. Therefore $t < 497/1000$ on all 35 cuts. □

Proposition S79 (Competitor purity band; AH-1)**.** *Every pure seven-qubit state with $E_3(z) \ge E_*$ obeys*

$$\frac{121}{917} \le \pi_3(z) < p_c := \frac{8693}{64000}. \tag{S12.7}$$

Proof. The lower bound is Corollary S67. Lemma S77 gives $3 - E_3 \ge (2/5)(8\pi_3 - 1)$. Since $E_* > 59307/20000$ (SC-2), rearrangement gives $\pi_3 < (1 + (5/2)(3 - 59307/20000))/8 = 8693/64000$. □

Lemma S80 (Quadratic joint spin-flip relations; proved, AH-5)**.** *For any triple marginal R of the real family ψ_x, put $\widetilde{R} = \epsilon^{\otimes 3} R^{\mathsf{T}} (\epsilon^{\otimes 3})^{\mathsf{T}}$ (with $\epsilon = i\sigma_y$ this is the spin flip $Y^{\otimes 3} R^{\mathsf{T}} Y^{\otimes 3}$ of Section S12.3), $E = (R + \widetilde{R})/2$, $O = (R - \widetilde{R})/2$, and $u = (1 - 2x)/30$. If the triple avoids the distinguished site, then*

$$O = 0, \qquad (E - 4uI)(E - (3u + x)I) = 0.$$

If it contains that site, then

$$[E, O] = 0, \qquad O(E - 6uI) = 0, \qquad 2u(E - 4uI)(E - (u + x)I) - (5u - x)O^2 = 0. \tag{S12.8}$$

Proof. In the computational basis $\{|b_1 b_2 b_3\rangle\}$, $\epsilon^{\otimes 3}$ maps $|b_1 b_2 b_3\rangle$ to $(-1)^{3-(b_1+b_2+b_3)}|\bar{b}_1 \bar{b}_2 \bar{b}_3\rangle$. By the block diagonalisation in the proof of Theorem 41, on any cut $I = \{i, j, k\}$ avoiding s the marginal $R = \mathrm{diag}(3u + x, (4u)^6, 3u + x)$ is diagonal with bit-complement-symmetric entries, so $\widetilde{R} = R$, $O = 0$, and $(E - 4uI)(E - (3u + x)I) = 0$. On any cut $\{s, i, j\}$ containing s, R is diagonal with bit-complement-symmetric entries on the six basis states outside $\mathrm{span}\{|000\rangle, |111\rangle\}$ (two eigenvalues $u + x$ and four eigenvalues $4u$, where $O = 0$), while on $\mathrm{span}\{|000\rangle, |111\rangle\}$ one has $R = \left(\begin{smallmatrix} 6u & \mp 2u \\ \mp 2u & 6u \end{smallmatrix}\right)$ and $\epsilon^{\otimes 3}$ maps $|000\rangle \mapsto -|111\rangle$ and $|111\rangle \mapsto |000\rangle$, flipping the off-diagonal sign: $\widetilde{R} = \left(\begin{smallmatrix} 6u & \pm 2u \\ \pm 2u & 6u \end{smallmatrix}\right)$. Thus $E = 6u\, I_2$ and $O = \left(\begin{smallmatrix} 0 & \mp 2u \\ \mp 2u & 0 \end{smallmatrix}\right)$ on $\mathrm{span}\{|000\rangle, |111\rangle\}$, where $O^2 = 4u^2 I_2$ and $2u(6u - 4u)(6u - u - x) - (5u - x)(4u^2) = 0$. This proves (S12.8) across all eight basis directions; AH-5 checks all 35 cuts directly. □

S12.4 The degree of a sharp scalar entropy majorant

Proposition S81 (Minimum replica degree for a sharp scalar entropy majorant)**.** *Let $f(t) = -t \log_2 t$ on $[0, 1]$, with $f(0) = 0$. For $k \geq 2$ distinct interior nodes $0 < a_1 < \cdots < a_k < 1$, any polynomial q satisfying $q(t) \geq f(t)$ on $[0, 1]$ and $q(a_j) = f(a_j)$ for all j has degree at least $2k - 1$. This threshold is attained: the degree-$(2k-1)$ Hermite interpolant matching both f and f' at every node majorises f on $[0, 1]$.*

At the seven-qubit entropy maximiser ψ_{x_*}, a single *scalar entropy majorant sharp on every triple reduction therefore requires degree at least 7, since the combined spectra have four distinct positive eigenvalues. A joint majorant in a marginal and its spin flip avoids this: Lemma S82 has degree five.*

Proof. If $h = q - f \geq 0$ and $h(a_j) = 0$ at an interior point, then $h'(a_j) = 0$. Thus h has at least $2k$ zeros counted with multiplicity, so iterated Rolle's theorem forces $h^{(2k-1)}$ to vanish somewhere in the interval spanned by the nodes. If $\deg q \leq 2k - 2$, however,

$$h^{(2k-1)}(t) = -f^{(2k-1)}(t) = -\frac{(2k-3)!}{t^{2k-2} \ln 2} < 0 \qquad (t > 0),$$

a contradiction. For the converse, let H denote the Hermite polynomial matching f, f' at all k nodes. For $0 < t < 1$, the Hermite remainder formula gives

$$f(t) - H(t) = \frac{f^{(2k)}(\xi)}{(2k)!} \prod_{j=1}^{k} (t - a_j)^2 \leq 0, \qquad f^{(2k)}(\xi) = -\frac{(2k-2)!}{\xi^{2k-1} \ln 2} < 0.$$

Continuity extends the inequality to $t = 0$. By Theorem 41 the cut eigenvalues of ψ_x are $4u, 3u + x, u + x, 8u$ with $u = (1 - 2x)/30$; two of them coincide only when $x/u \in \{1, 3, 5, 7\}$. By Proposition S73, $x_*/u_* \in (1, 37/15)$, so the cuts of ψ_{x_*} have four distinct nodes. The asserted degree bound follows. □

S12.5 A joint degree-five entropy majorant

The joint spin-flip relations let us build a majorant with the same contacts for all seven choices of distinguished site. Its moment functional is therefore invariant under qubit permutations.

Throughout this subsection $f(t) = -t\log_2 t$, with $f(0) = 0$, and $u = u_*$, $c = 4u$, $a = 3u + x_*$, $d = u + x_*$, $b = 8u$. Let P be the degree-five Hermite interpolant to $X \mapsto f(X/8)$ at $8c, 8a, 8d$, including the first derivatives. Put

$$e_0 = 48u, \quad v_0 = 256u^2, \quad h = \frac{f(b) + f(c)}{2}, \quad h_e = \frac{f'(b) + f'(c)}{16}, \quad h_v = \frac{f'(b) - f'(c)}{512u}.$$

Choose the following exact rational coefficients:

$$\alpha_2 = -\frac{115001089107}{10^{12}}, \qquad \alpha_3 = \frac{18166193766}{10^{12}}, \qquad \beta_1 = \frac{14438812336}{10^{12}},$$

and define the remaining coefficients by

$$\begin{aligned}
\beta_0 &= \frac{v_0 h_v - h + P(e_0)}{v_0^2} - \beta_1 e_0, \\
\alpha_1 &= \frac{h_e - P'(e_0) - v_0^2\beta_1}{v_0} - 2\alpha_2 e_0 - 3\alpha_3 e_0^2, \\
\alpha_0 &= h_v - 2v_0(\beta_0 + \beta_1 e_0) - \alpha_1 e_0 - \alpha_2 e_0^2 - \alpha_3 e_0^3.
\end{aligned}$$

For $e = 4(\lambda + \mu)$ and $v = 16(\lambda - \mu)^2$, set

$$H(\lambda, \mu) = P(e) + v\sum_{j=0}^{3} \alpha_j e^j + v^2(\beta_0 + \beta_1 e) = \sum_{k+l\le 5} h_{kl}\lambda^k \mu^l. \tag{S12.9}$$

It is a symmetric polynomial of total degree five.

Lemma S82 (Joint scalar majorant; exact-rational coefficient certificate AH-6)**.** *For all $\lambda, \mu \ge 0$,*

$$H(\lambda, \mu) \ge \frac{f(\lambda) + f(\mu)}{2}.$$

Equality holds exactly at $(c, c), (a, a), (d, d), (b, c), (c, b)$, with first-order contact at each pair.

Proof. The diagonal contacts follow from the definition of P. Substitution in the three formulas for $\beta_0, \alpha_1, \alpha_0$ gives the value and both first derivatives at (b, c); symmetry gives (c, b).

Section S12.6 proves the inequality by a positive Stieltjes-kernel bound and a cubic Hermite minorant for the binary entropy correction. A single univariate discriminant polynomial remains; its fixed positive-coefficient certificate is checked in exact rational arithmetic (AH-6). That proof also excludes any further equality points. □

Corollary S83 (Degree five is minimal)**.** *Let $G(\lambda, \mu)$ be a polynomial of total degree m with $G(\lambda, \mu) \ge \frac{1}{2}[f(\lambda) + f(\mu)]$ on $[0, 497/1000]^2$ and $G(t, t) = f(t)$ for $t \in \{c, a, d\}$. Then $m \ge 5$. Every joint majorant with equality at the joint eigenvalue pairs of the marginals of ψ_{x_*} satisfies these hypotheses, so H has the least possible degree.*

Proof. The points c, a, d are distinct (Proposition S81) and lie in $(0, 497/1000)$. The restriction $g(t) = G(t, t)$ has degree at most m, satisfies $g \ge f$ on $[0, 497/1000]$ and touches f at three interior points. The proof of Proposition S81 uses $g \ge f$ only on an interval containing the nodes in its interior, so it gives $m \ge \deg g \ge 5$. For the second statement, Lemma S80 shows that a marginal avoiding the distinguished site equals its spin flip, with eigenvalues c and a, and that on the eigenspace of E for the eigenvalue d of a marginal containing it the odd part O vanishes, because $O(E - 6uI) = 0$ and $d \ne 6u$ (as $x_*/u_* < 37/15$); so $(c, c), (a, a), (d, d)$ are joint eigenvalue pairs. This concerns majorants in a marginal and its spin flip only; it is not a lower bound on the copy hierarchy. □

Proposition S84 (The five-copy entropy target)**.** *For a positive semidefinite* 8×8 *matrix* R*, let* $\widetilde{R}$ *denote its three-qubit spin flip. Then*

$$\operatorname{tr} f(R) \le \mathcal{P}(R) := \sum_{k+l\le 5} h_{kl} \operatorname{tr}(R^k \widetilde{R}^l). \tag{S12.10}$$

Commutation of R *and* $\widetilde{R}$ *is not required. For normalised seven-qubit states put*

$$\mathcal{P}_5(z) = \frac{1}{35} \sum_{|S|=3} \mathcal{P}(\rho_S(z)).$$

Every normalised pure seven-qubit state satisfies $E_3(z) \le \mathcal{P}_5(z)$*, and* $\mathcal{P}_5(\phi) = E_*$ *for every choice of the distinguished qubit of* ϕ*. After homogenisation,* $\mathcal{P}_5$ *is a permutation-invariant linear functional on five-copy moments.*

Proof. Take orthonormal eigenbases u_i, v_j of $R, \widetilde{R}$, with eigenvalues λ_i, μ_j. The matrix $w_{ij} = |\langle u_i, v_j\rangle|^2$ is doubly stochastic, so

$$\mathcal{P}(R) = \sum_{ij} w_{ij} H(\lambda_i, \mu_j) \ge \frac{\operatorname{tr} f(R) + \operatorname{tr} f(\widetilde{R})}{2} = \operatorname{tr} f(R).$$

Lemma S80 shows that every marginal of ϕ commutes with its spin flip and has only the five joint pairs listed in Lemma S82. This proves equality. Multiplication of the term of degree $k + l$ in unnormalised marginals by ν^{5-k-l}, where $\nu = \|z\|^2$, gives the homogeneous five-copy functional. □

Theorem S85 proves the matching quantum inequality for every competitor, completing the entropy bound.

S12.6 A positive-kernel proof of the joint entropy majorant

We now prove the scalar-majorant inequality of Lemma S82 for the polynomial H defined there. The proof separates the diagonal Hermite remainder from a binary entropy correction, then reduces nonnegativity of the combined gap to the sign of one univariate polynomial. Only the fixed coefficient bounds in Table S5 require arithmetic verification (AH-6, exact-rational interval enclosures); no subdivision of the entropy gap is used.

Retain $A(e) = \sum_{j=0}^{3} \alpha_j e^j$, $B(e) = \beta_0 + \beta_1 e$, $e_0 = 48u$ and $v_0 = e_0^2/9$ from the definition of H before (S12.9), and put

$$q(e) = (e - 8a)(e - 8c)(e - 8d), \qquad K(e) = \frac{P(e) - f(e/8)}{q(e)^2}.$$

At the three apparent singularities take the continuous extensions. The family bound $x_*/u_* < 37/15$ places e_0 strictly above these three nodes. The Stieltjes remainder formula for quintic Hermite interpolation gives

$$K(e) = \frac{1}{8 \ln 2} \int_0^\infty \frac{s\, ds}{(s + e)(s + 8a)^2(s + 8c)^2(s + 8d)^2} > 0. \tag{S12.11}$$

Indeed, $-t \ln t = \int_0^\infty [t/(t+s) - t/(1+s)]\, ds$, and only the term $-s/(t+s)$ is not affine in t; the affine difference between $f(e/8)$ and $f(e)/8$ contributes no interpolation error. The Hermite remainder of $e \mapsto 1/(s + e)$ at the double nodes $8a, 8c, 8d$ is $q(e)^2/[(s + e)(s + 8a)^2(s + 8c)^2(s + 8d)^2]$, which gives the identity.

Set

$$k_0 = K(e_0), \qquad \gamma = -K'(e_0)/k_0, \qquad L(e) = 1 + \gamma(e - e_0).$$

The kernel has the rational lower bound

$$K(e) \geq \frac{k_0}{L(e)}, \qquad e \geq 0, \qquad k_0 > 0, \quad 0 < \gamma < 1/e_0. \tag{S12.12}$$

To see this, normalise the positive integrand of $K(e_0)$ to a probability measure. Then $K(e)/k_0$ is the mean of $\phi(y) = [1+(e-e_0)y]^{-1}$ at $y = 1/(s+e_0)$. Since $1+(e-e_0)y = (s+e)/(s+e_0) > 0$ for $e \geq 0$, ϕ is convex on the support, and Jensen's inequality gives (S12.12), because the mean of y is γ. Its support also gives $0 < \gamma < 1/e_0$, so $L(e) > 0$ for all $e \geq 0$. Both value and first derivative agree at e_0.

For the binary correction define

$$\Phi(z) = \frac{(1+\sqrt{z})\ln(1+\sqrt{z}) + (1-\sqrt{z})\ln(1-\sqrt{z})}{2}, \qquad 0 \leq z \leq 1.$$

Its cubic Hermite interpolant, with double contacts at 0 and 1/9, is

$$\Phi(z) \geq \frac{z}{2} + \eta_2 z^2 + \eta_3 z^3, \quad \eta_2 = \frac{1593}{4}\ln 2 - 243\ln 3 - 9, \quad \eta_3 = \frac{81}{2} - \frac{9477}{4}\ln 2 + 1458\ln 3 > 0. \tag{S12.13}$$

The interpolation data at 1/9 are $\Phi(1/9) = \frac{5}{3}\ln 2 - \ln 3$ and $\Phi'(1/9) = \frac{3}{4}\ln 2$. The power series $\Phi(z) = \sum_{j\geq 1} z^j/[2j(2j-1)]$ gives $\Phi^{(4)} > 0$ on $[0,1)$, so the Hermite error has the sign of $z^2(z-1/9)^2$. Continuity includes $z = 1$. The positive third divided difference also gives $\eta_3 > 0$.

For $\lambda, \mu \geq 0$, put $e = 4(\lambda+\mu)$, $v = 16(\lambda-\mu)^2$. If $e > 0$ then $0 \leq v/e^2 \leq 1$ and

$$\frac{f(\lambda)+f(\mu)}{2} = f(e/8) - \frac{e}{8\ln 2}\Phi(v/e^2).$$

Consequently, for $e > 0$, the entropy gap is bounded below by a cubic in v:

$$H(\lambda,\mu) - \frac{f(\lambda)+f(\mu)}{2} \geq \frac{F(e,v)}{8\ln 2\, e^5 L(e)}, \tag{S12.14}$$

where

$$\begin{aligned} U(e) &= 8\ln 2\, k_0 q(e)^2, \\ V(e) &= \tfrac{1}{2} + 8\ln 2\, eA(e), \\ W(e) &= \eta_2 + 8\ln 2\, e^3 B(e), \\ F(e,v) &= e^5 U(e) + e^4 L(e)V(e)v + e^2 L(e)W(e)v^2 + \eta_3 L(e)v^3. \end{aligned}$$

This follows by substituting the two lower bounds above into $q(e)^2 K(e) + vA(e) + v^2 B(e) + e\Phi(v/e^2)/(8\ln 2)$.

Reduction to one polynomial sign. The discriminant of $F(e,\cdot)$ is

$$\operatorname{disc}_v F(e,v) = -e^{10} L(e)^2 (e-e_0)^2 \Pi(e), \tag{S12.15}$$

where Π is the degree-eighteen polynomial obtained by dividing

$$\begin{aligned} R(e) = {} & 27\eta_3^2 U^2 + 4eLUW^3 + 4\eta_3 e^2 L^2 V^3 \\ & - e^2 L^2 V^2 W^2 - 18\eta_3 eLUVW \end{aligned} \tag{S12.16}$$

by $(e-e_0)^2$; U, V, W, L in this expression are evaluated at e. The factor is exact. The contact $(\lambda,\mu) = (b,c)$ of the Lemma maps to (e_0, v_0), and since $b \neq c$ (so $v_0 > 0$) the map $(\lambda,\mu) \mapsto (e,v)$ is a local diffeomorphism there. The entropy gap has value and gradient zero at this contact, and both minorant defects vanish to second order there: $q^2(K - k_0/L)$ at e_0, and the binary defect at $v/e^2 = 1/9$. Hence $F = F_v = F_e = 0$ at (e_0, v_0). The cubic $F(e_0,\cdot)$ has a double root at v_0, so

its discriminant vanishes; and the differential of the discriminant at a cubic with double root r is a multiple of $\delta p \mapsto \delta p(r)$, so its e-derivative is a multiple of $F_e(e_0, v_0) = 0$. Dividing out $e^{10}L(e)^2$ preserves this double zero (AH-6b checks the cubic identity symbolically).

The sign needed is $\Pi(e) > 0$ for every $e \geq 0$. On each finite interval $[l, h]$ of Table S5, expand $\Pi(l + (h - l)t)$ in the degree-eighteen Bernstein basis; on the final half-line expand $\Pi(5/4 + t)$ in powers of t. Every coefficient is strictly positive.

e range	basis	lower bound on every coefficient
$[0, 3/4]$	Bernstein	$2 \cdot 10^{-7}$
$[3/4, 1]$	Bernstein	$5 \cdot 10^{-10}$
$[1, 5/4]$	Bernstein	$9 \cdot 10^{-11}$
$[5/4, \infty)$	powers of $e - 5/4$	$2 \cdot 10^{-10}$

Table S5: Exact rational coefficient certificate AH-6c for $\Pi > 0$. Each row checks nineteen coefficients.

AH-6 computes outward dyadic rational enclosures of the coefficients of Π from the exact root equation for x_* and atanh series with rational tails, converts them to the bases of the table, and verifies the bounds by integer comparisons. These are coefficient signs of a fixed polynomial, not sampled values of the entropy inequality.

Completion and equality. For $e > 0$, $e \neq e_0$, the discriminant is strictly negative. Thus the cubic $F(e, \cdot)$ has exactly one real root. Its leading coefficient is positive, and $F(e, 0) = e^5 U(e) \geq 0$, so that root is nonpositive and $F(e, v) \geq 0$ for $v \geq 0$. Equality requires $v = 0$ and $e \in \{8a, 8c, 8d\}$. At $e = e_0$, the double contact gives

$$F(e_0, v) = (v - v_0)^2(\eta_3 v + m), \qquad m = e_0^5 U(e_0)/v_0^2 > 0,$$

so equality requires $v = v_0$. Equation (S12.14) therefore proves the majorant, with exactly the five asserted contacts. For $e = 0$, $\lambda = \mu = 0$ and the gap is $P(0) = q(0)^2 K(0) > 0$ by (S12.11). This completes the proof on the full nonnegative quadrant.

S12.7 The localised five-copy inequality

The scalar majorant bounds entropy by a five-copy functional. We now prove the matching bound on that functional using a quantum compatibility identity. Write $\nu = \|z\|^2$, $R_S = \mathrm{tr}_{S^c} |z\rangle\langle z|$, and $\Pi(z) = 35^{-1} \sum_{|S|=3} \mathrm{tr}\, R_S^2$. Let $\mathcal{H}(z)$ be the homogeneous lift of $\mathcal{P}_5$: its term $h_{kl}\,\mathrm{tr}(R_S^k \widetilde{R}_S^l)$ is multiplied by ν^{5-k-l}. Thus $\mathcal{H}(z) = \mathcal{P}_5(z)$ when $\nu = 1$. Put $p_0 = 121/917$, $p_c = 8693/64000$, and $\kappa = 497/1000$. For a vector that need not be normalised, $X_5^+(z)$ denotes the unnormalised twirled moment $\nu^5 X_5^+(z/\|z\|)$, with $X_5^+(0) = 0$; it is a real polynomial of degree ten in $z, \bar{z}$, so both sides of the identity below are homogeneous of degree ten.

Theorem S85 (Localised entropy identity; verified-arithmetic certificate E7-1)**.** *There are seventeen positive definite real Gram matrices C_b, two nonnegative cubic-moment polynomials K_+, K_-, and quadratic marginal matrix covariants A_j, B_j, such that the following polynomial identity holds for every complex seven-qubit vector z:*

$$\begin{aligned} E_*\nu^5 - \mathcal{H}(z) = {} & \sum_{b=1}^{17} \mathrm{tr}[C_b G_b(z)] + (p_c\nu^2 - \Pi(z))K_+(z) + (\Pi(z) - p_0\nu^2)K_-(z) \\ & + \frac{1}{35} \sum_{|S|=3} \left[\nu \sum_j \mathrm{tr}\, A_j(R_S)^\dagger A_j(R_S) + \sum_j \mathrm{tr}\, B_j(R_S)^\dagger (\kappa\nu I - R_S) B_j(R_S) \right]. \end{aligned} \tag{S12.17}$$

Here $G_b(z) = W_b^{\mathsf{T}}\Gamma_b(X_5^+(z))W_b \succeq 0$ are Gram matrices of five-copy features: Γ_b is a block of the twirled five-copy moment with zero, one or two spin-flipped copies, as in Section S11, and the columns of W_b lie in its kernel at $\phi = \psi_{x_*}$. Every term on the right vanishes at ϕ, and the functionals $X \mapsto \operatorname{tr}[W_b S W_b^{\mathsf{T}}\Gamma_b(X)]$ ($b \le 12$, S symmetric) span the 3283-dimensional space of linear functionals vanishing on $X_5^+(\phi)$ and on its family derivative. In particular,

$$p_0 \le \pi_3(z) \le p_c, \qquad R_S \preceq \kappa I \quad (|S| = 3), \qquad \nu = 1 \implies \mathcal{P}_5(z) \le E_*. \tag{S12.18}$$

The positive sources. The construction uses the one-site multiplicity modules $F = \mathbb{R}^4$ and $V = \mathbb{R}^5$ of the five-copy qubit Schur algebra from Section S11.1. In the following table $[a, b]V$ denotes the two-row Young module inside $\operatorname{Sym}^a V \otimes \operatorname{Sym}^b V$. The middle column is the number of spin-flipped copies; the last column is the number of kernel features.

module	flips	features	module	flips	features
$\operatorname{Sym}^7 V$	2	31	$F \otimes \operatorname{Sym}^6 V$	2	69
$F \otimes [5,1]V$	2	125	$F \otimes [4,2]V$	2	128
$[6,1]V$	2	57	$[5,2]V$	2	72
$\operatorname{Sym}^7 V$	0	2	$[6,1]V$	0	3
$[5,2]V$	0	6	$\operatorname{Sym}^7 V$	1	14
$[6,1]V$	1	27	$[5,2]V$	1	36
$\operatorname{Sym}^2 F \otimes \operatorname{Sym}^5 V$	2	94	$\bigwedge^2 F \otimes \operatorname{Sym}^5 V$	2	46
$[4,3]V$	0	2	$[4,3]V$	1	20
$[4,3]V$	2	44			

Each feature is projected onto the exact physical kernel at ϕ; G_b includes the positive multiplicity weight and the average over local unitaries and site permutations. Positivity holds for every complex pure state. The last five modules supply directions absent from the twelve-face purity certificate.

For three copies the corresponding one-site multiplicity module has dimension two. The modules $\operatorname{Sym}^7 \mathbb{R}^2$, $[6,1]\mathbb{R}^2$, and the one-spin-3/2 $\operatorname{Sym}^6 \mathbb{R}^2$ module have respectively two, one and one exact kernel features at ϕ. Positive sums of their squared norms give K_+ and K_-. In the invariant bases U_b of E7-K3, the kernel features in $[6,1]\mathbb{R}^2$ and one-spin-3/2 $\operatorname{Sym}^6 \mathbb{R}^2$ are constant rational vectors annihilating the entire family ψ_x, while the two kernel features in $\operatorname{Sym}^7 \mathbb{R}^2$ depend on x and are taken at $x = x_*$. E7-K3 checks their physical coordinate forms directly. Section S13 gives a structural interpretation of these features: the spin-3/2 feature is the one-site focused covariant $\Phi_{i,0}$ of Lemma S96, and the $\operatorname{Sym}^7 \mathbb{R}^2$ pair are the one-body and purity-gradient Wronskian relations (S13.5) of Proposition S97. That proposition also proves their completeness analytically: the entire symmetric scalar cubic Gram has rank two on $0 < x < 1/2$, with the positive two-source determinant $72x(1-2x)$ in (S13.6). Thus its kernel is exactly this pair.

To define the marginal sources put, for normalised R,

$$E = (R + \widetilde{R})/2, \quad O = (R - \widetilde{R})/2, \quad p = \operatorname{tr} R^2, \quad q = \operatorname{tr} R\widetilde{R},$$

$$p_o = 6c^2 + 2a^2, \quad p_i = b^2 + 5c^2 + 2d^2, \quad \alpha = (p_i - p_o)/c^2,$$

and set

$$Q_0 = ((p - p_o) - \alpha(p - q))I, \quad Q_1 = O(E - 6u_*I), \quad Q_2 = [E, O].$$

The A_j and B_j are fixed real linear combinations of these three covariants. Homogenisation replaces p_o by $p_o\nu^2$ and $6u_*$ by $6u_*\nu$. Lemma S80 gives $Q_0 = Q_1 = Q_2 = 0$ for every marginal of ϕ: in the out-class $p = q = p_o$, while in the in-class $p = p_i$ and $p - q = c^2$. Consequently both the ordinary and cap-weighted Gram matrices of the Q_j are tight at ϕ.

Proof. We describe the finite verification and the exact identity it establishes. All choices of bases and fixed coefficients are supplied as integers, rationals or binary rationals. The only other constants are x_*, its positive square root ratios, and the logarithms in H.

Let U_b be an integer basis fixed by the copy subgroup, D_b its physical Gram matrix, and $Y_b(t)$ the degree-five physical image of $U + tV$. With $S_b = U_b^{\mathsf{T}} D_b U_b$ and $X_b(t) = U_b^{\mathsf{T}} D_b Y_b(t)$, select independent columns X_b^0. The physical kernel projector in invariant coordinates is

$$P_b = I - S_b^{-1} X_b^0 ((X_b^0)^{\mathsf{T}} S_b^{-1} X_b^0)^{-1} (X_b^0)^{\mathsf{T}}, \qquad t = \sqrt{x_*/u_*}. \tag{S12.19}$$

The fixed target factors B_b are lower-trapezoidal dyadic matrices in invariant coordinates; the exact features are $U_b P_b B_b$, with the rational diagonal-Gram transport used in the purity certificate. These baselines need no justification of their own: the complete repair check below establishes the identity for them. Repeated image columns are stored once and expanded exactly, and no least squares or QR is performed during verification. E7-K0 checks the added two-row Young modules. E7-K1 verifies the image polynomials by integer contractions at six determining homogeneous amplitude pairs, and checks subgroup fixedness. E7-K2 proves the asserted generic image rank r_b: all $(r_b + 1)$-minors have degree at most $5(r_b + 1)$, and exact rank factorisations at $5(r_b + 1) + 1$ integer points make those minors identically zero. Modular elimination only selects pivots; the factorisations themselves use integers. Arb verifies invertibility of the displayed r_b-column Gram matrix at x_*. The symmetric cubic kernel instead follows from the closed Gram and its two analytic relations in Proposition S97. E7-K3 checks their physical coordinates, the other cubic kernels and the exact generating polynomials. These steps prove kernel membership exactly, rather than inferring it from small singular values.

Let $\mathcal{C}_*$ be the real, conjugation-even, permutation-invariant five-copy polynomial space annihilating the moment of ϕ and its family tangent. The invariant space has dimension 3285, by Proposition S60. Evaluation and family differentiation are independent: normalisation has value one and zero derivative, while $\pi_3'(x_*) \neq 0$ by the explicit family quadratic. Hence $\dim \mathcal{C}_* = 3283$. Every source above belongs to $\mathcal{C}_*$, since its kernel feature vanishes at ϕ. The target belongs to $\mathcal{C}_*$ by first-order scalar contact and the entropy stationarity equation for x_*.

Start with the identity Gram in each projected five-copy basis, and with the fixed positive cubic and marginal sources. Repair only the first twelve Grams, using the same rational orthogonal frames Q_b as in Theorem S62:

$$C_b = I + Q_b \begin{pmatrix} J_b & E_b \\ E_b^{\mathsf{T}} & 0 \end{pmatrix} Q_b^{\mathsf{T}} \quad (b \le 12), \qquad C_b = I \quad (b > 12),$$

where $J_b = J_b^{\mathsf{T}}$ has size $r_b = \min(7, k_b)$. Let z_b list the upper triangle of J_b and all entries of E_b. Then $\|C_b - I\|_2 \le \sqrt{2}\|z_b\|_2 \le \frac{3}{2}\|z\|_2$. There are 3754 independent real repair coordinates $z = (z_b)_{b \le 12}$, with the same exact frame and coordinate audit as FP5-4. The columns of the evaluation matrix are the repair functionals $X \mapsto \operatorname{tr}[W_b S W_b^{\mathsf{T}} \Gamma_b(X)]$, $b \le 12$, which lie in $\mathcal{C}_*$: each face $W_b^{\mathsf{T}} \Gamma_b(X_5^+(\psi_x)) W_b$ is positive semidefinite along the family and vanishes at ϕ, so its derivative vanishes there too. We apply Lemma S61 over $K = \mathbb{R}$ with $\mathcal{C} = \mathcal{C}_*$ ($d = 3283$) inside the 3285-dimensional invariant space, $N = 3754$, $\Phi(z) = \sum_{b \le 12} \operatorname{tr}[(C_b - I) G_b]$, $\Delta_b(z) = C_b - I$ with $c = 3/2$, and ρ the left side of (S12.17) minus the baseline right side (identity Grams and the fixed cubic and marginal sources). Then $\rho \in \mathcal{C}_*$, since the target, every source and each identity Gram $\operatorname{tr} G_b$ (the $S = I$ repair functionals above) lie in $\mathcal{C}_*$. Let $\mathcal{E}$ evaluate, with fixed dyadic row weights, at the 3424 rational direct probes of Proposition S60. E7-1 encloses every entry of the exact evaluation matrix A and residual r by explicit forward rounding bounds and Arb input enclosures. Its proposed binary matrices L, R satisfy

$$\|I - LAR\|_2 \le \delta < 1, \qquad \frac{3\|R\|_2 \|Lr\|_2}{2(1 - \delta)} < 1,$$

which are (S11.1) and (S11.2). The complete cold log gives verified upper bounds $\delta < 0.0311$ and a Gram correction below 0.1354. Lemma S61 shows that these repair functionals span $\mathcal{C}_*$, that evaluation is injective there, and that $z = R(LAR)^{-1} Lr$ yields the exact identity with $C_b \succ 0$ for every repaired block. The unmodified Grams are identities. This proves (S12.17); every term is nonnegative under the hypotheses of (S12.18). □

Arithmetic bounds in E7-1. E7-1 uses the IEEE-754 arithmetic model and the row-norm companions of the purity certificate (the paragraphs following Theorem S62). Large-face traces use the multiplier $V_R V_L^{\mathsf{T}}$ formed once, while both repair actions use at most seven columns. The symmetric and exterior F^2 actions are quadratic products of the local 4×4 matrix; replacing their signs by plus gives their positive companions, and their depth is bounded by the same $6g$ degree allowance. In particular the exterior action retains its alternating sign in the signed contraction. The coefficient and factored-trace depths are $a + f + 6g + s$ and $a + h + 6g + s$, with chunks $(128, 256, 128)$; multiplier traces use the corresponding squared-factor dimensions, as in FP5-1. Here $g = 7$, and the positive-companion budget is $b = Dk + \sum_i d_i + 42 + 2\binom{5}{j}^2 + 64$. The extra summation allowance covers all seventeen faces. Every signed path has depth at least 64, asserted by the verifier.

The entropy certificate also has Arb input radii E_W on the feature bases. Replace each $|W|$ by $|W| + E_W/(64u)$, rounded upwards, *before* forming its row norms or left companions. Since $\gamma_m \geq 64u$, the bilinear terms containing E_W bound input errors, and the $|W|$ term bounds rounding errors. Minkowski and Cauchy therefore give the same coefficient Frobenius and trace bounds for the exact Arb-enclosed features. The input guards and explicit 2^{-500} allowances for propagated subnormal errors are those of FP5-1. The rectangular coefficients are then rotated and packed using the exact rational frames and the rigorous transformation radius in that certificate; this yields one row radius for the 3754 independent coordinates. The validated inverse uses these row radii, the source radii and the unchanged 3283-dimensional critical space. The joint trace products have local degree seven and at most 120^2 copy-coset terms, with integer dot products below 2^{53} before rational division. The two cubic localisers are combined into one degree-seven polynomial in ten local variables (5560 monomials), built from rational local maps that E7-K3 checks. These bound the exact functions in the identity.

Theorem S86 (Sharp seven-qubit mean entropy; verified-arithmetic certificate E7-1)**.** *Every pure seven-qubit state satisfies*

$$\boxed{E_3(z) \leq E_3(\psi_{x_*}) = \frac{39f(c) + 8f(a) + 6f(d) + 3f(b)}{7} = 2.965356029477675899\ldots} \tag{S12.20}$$

where x_* is the root specified in Proposition S73. Equality holds if and only if z is local-unitarily and site-permutation equivalent to ψ_{x_*}. With the qubit labels fixed, this equality set consists of exactly 42 LU orbits.

Proof. It suffices to consider $E_3(z) \geq E_*$. Proposition S79 and Corollary S78 put every such state in the region of (S12.18). Therefore Proposition S84 and Theorem S85 give $E_3(z) \leq \mathcal{P}_5(z) \leq E_*$, with equality at ϕ. Proposition S87 and Corollary S68 prove the equality assertions. □

S12.8 The entropy equality orbit

The positive identity and the invariant-square equality theorem shared with purity fix the five-copy moment, after which the degree-four invariants recover the orbit.

Proposition S87 (Entropy equality fixes the five-copy moment)**.** *If a normalised pure seven-qubit state z satisfies $E_3(z) = E_*$, then*

$$X_5^+(z) = X_5^+(\psi_{x_*}).$$

Consequently it is local-unitarily and site-permutation equivalent to ψ_{x_*}.

Proof. Proposition S79 and Corollary S78 apply to z, so Proposition S84 and Theorem S85 give $E_* \leq \mathcal{P}_5(z) \leq E_*$. Every term of the positive identity therefore vanishes; since each C_b is positive definite, so does every face $W_b^{\mathsf{T}} \Gamma_b(X_5^+(z)) W_b$. Apply Theorem S66 with $y = x_*$, which lies in its interval by Proposition S73: the spanning hypothesis is the last assertion of Theorem S85. The converse follows from local-unitary and permutation invariance and $E_3(\phi) = E_*$. □

Theorem 12 (seven-qubit mean-entropy maximum and equality; verified-arithmetic E7-1; exact AH-6, FO-1, EO-1; uses Theorem 11). *Every pure seven-qubit state satisfies*

$$E_3(\psi) \le E_3(\psi_{x_*}) = 2.965356029477675899\ldots \text{ bits.}$$

Equality holds if and only if ψ is local-unitarily and qubit-permutation equivalent to ψ_{x_*}. With fixed qubit labels, the equality set consists of exactly 42 LU orbits.

Proof. Theorem S86 combines the competitor restrictions (Proposition S79 and Corollary S78), the joint majorant (Proposition S84), and the localised five-copy identity (Theorem S85). Proposition S87 fixes the equality moment; Corollary S65 identifies the orbit, and Corollary S68 counts its 42 labelled LU orbits. □

S12.9 Nondegeneracy of the maximiser

Having identified the global maximum and its equality orbit, we now quantify the entropy loss near that orbit. The constrained Hessian at ϕ is negative definite transverse to the symmetry orbit, giving an explicit quadratic basin.

Theorem S88 (Strict full-space local entropy maximum; exact certificate SEH-1–SEH-3 with an analytic perturbation bound)**.** *The state ψ_{x_*} is a* strict local maximiser *of E_3 modulo local unitaries and global phase. More precisely, for every nonzero real unit-sphere tangent vector v orthogonal to the local-unitary and phase tangent space at ψ_{x_*},*

$$D^2(E_3|_{\|z\|=1})_{\psi_{x_*}}[v,v] < -\frac{239}{1000}\,\|v\|^2. \tag{S12.21}$$

The 22 phase and local-unitary symmetry infinitesimals have zero constrained Hessian.

Proof. Extend $E_3(z)$ off the unit sphere by the same formula $-35^{-1}\sum_{|S|=3}\operatorname{tr}(R_S\log_2 R_S)$, with R_S not renormalised: for unit z and $r>0$, $E_3(rz) = r^2E_3(z) - r^2\log_2 r^2$. Hence E_3 is the mean entropy only on the unit sphere, and its radial derivative $\frac{1}{2}\frac{d}{dr}E_3(rz)|_{r=1} = E_3(z) - 1/\ln 2$ is the multiplier λ below. For a three-qubit cut flatten z to an 8×16 matrix M and put $R = MM^\dagger$. For a variation v with flattening $\dot M$, write $\dot R = \dot M M^\dagger + M\dot M^\dagger$, and denote the Fréchet derivative of the matrix logarithm by

$$\mathcal{L}_R(H) = \int_0^\infty (R+tI)^{-1}H(R+tI)^{-1}\,dt.$$

The unnormalised second variation of the mean entropy is

$$D^2E_3(z)[v,v] = -\frac{1}{35\ln 2}\sum_{|S|=3}\left[2\operatorname{tr}\dot M^\dagger(\ln R + I)\dot M + \operatorname{tr}\dot R\,\mathcal{L}_R(\dot R)\right]. \tag{S12.22}$$

At the stationary unit vector put $\lambda = \langle z, \partial_{\bar z}E_3(z)\rangle = E_3(z) - 1/\ln 2$. The constrained Hessian is $H = D^2E_3 - 2\lambda I_{256}$; it splits into a real and imaginary 128×128 block because z is real. Its phase/LU symmetry generators, given by iz and $i\sigma_a^{(j)}z$, span G.

For the certificate take the nearby *rational* vector $z_0 = a_0U + b_0V$, with $a_0 = 16934604/10^8$, $b_0 = 26425139/10^8$. Exact integer signs in (S12.1), evaluated at $698287976009/10^{13}$ and $698287976010/10^{13}$, bracket x_*. Squaring rational endpoints proves $|a-a_0|, |b-b_0| < 5\cdot 10^{-9}$ and hence $\|\psi_{x_*} - z_0\| < 3\cdot 10^{-8}$ (SEH-1). Every reduced eigenvalue along this short segment is at least $1/12$, and each difference between two *distinct* reduced eigenvalues at z_0 exceeds $1/64$; both are exact rational checks.

At z_0 the spectral projectors are rational: if ℓ ranges over the two or three distinct eigenvalues of R, then

$$P_\ell = \prod_{h\ne\ell}\frac{R-hI}{\ell-h}, \qquad \ln R = \sum_\ell (\ln\ell)P_\ell, \qquad \mathcal{L}_R(H) = \sum_{\ell,h}\frac{\ln\ell - \ln h}{\ell-h}P_\ell H P_h,$$

with diagonal coefficient $1/\ell$. SEH-2 uses rational proxies for the four logarithms and $1/\ln 2$, each with error less than 10^{-13}, bounded by exact rational arctanh series. It forms a rational fixed-point matrix $\widehat{B}$ approximating $B = -H + 4GG^T$, including the symmetry penalty, with entrywise rounding checked by integer arithmetic.

The exact Hessian at $z_0 = a_0U + b_0V$ respects the symmetry-adapted orthogonal integer decomposition of Theorem S69. Rounding could disrupt this structure in $\widehat{B}$, so SEH-2 checks the decomposition exactly for $\widehat{B}$. The decomposition consists of the six computational-basis sectors (S11.5), the disjoint-pair projectors P_2, P_3, P_{-1}, P_{-3} on $\mathcal{H}_{\text{supp}}$ and $\mathcal{H}_{\text{flip}}$, and the triple-incidence projectors $F_1, F_2, P_{8,40}$ on $\mathcal{H}_{\text{odd}}$. In both 128×128 blocks, SEH-2 verifies by exact integer multiplication that the 16 isotypic blocks (sizes 1×1, 2×2, and 4×4, with diagonal integer Gram matrices D_{blk}) span $\mathbb{R}^{128}$ and are exact invariant subspaces of $\widehat{B}$, and that every 1×1, 2×2, and 4×4 symmetric integer matrix $V_{\text{blk}}^{\mathsf{T}}(\widehat{B} - 0.239395\, I_{128})V_{\text{blk}}$ has strictly positive leading principal minors. Hence $\lambda_{\min}(\widehat{B}) > 0.239395$ with no floating-point basis or Gram conditioning step.

It remains to bound $B(\psi_{x_*}) - \widehat{B}$ analytically. Write $\varepsilon = \|z_0 - \psi_{x_*}\| < 3 \cdot 10^{-8}$. Using $R \succeq I/12$ along the segment,

$$\|R(z_0) - R(z_*)\|_{\text{op}} \le 3\varepsilon, \quad \|\ln R(z_0) - \ln R(z_*)\|_{\text{op}} \le 36\varepsilon, \quad \|\mathcal{L}_{R(z_0)} - \mathcal{L}_{R(z_*)}\|_{F\to F} \le 432\varepsilon.$$

Also $\|\dot{R}(v)\|_F \le 3\|v\|$ and $\|\dot{R}_{z_0}(v) - \dot{R}_{z_*}(v)\|_F \le 2\varepsilon\|v\|$. Inserting these estimates in (S12.22) with $\|\dot{M}\|_F = \|v\| = 1$ and $\|\mathcal{L}_R\| \le 12$, each cut changes by at most $2 \cdot 36\varepsilon + 2 \cdot (2\varepsilon) \cdot 12 \cdot 3 + 3 \cdot 432\varepsilon \cdot 3 = 4104\varepsilon$, hence the mean by at most $4104\varepsilon/\ln 2 < 5.93 \cdot 10^3\varepsilon$. Parametrise the segment from z_* to z_0 over $[0,1]$; then $\|\dot{R}\|_F \le 3\varepsilon$ along it. Hence $|\operatorname{tr} \dot{R}(\ln R + I)| \le \|\dot{R}\|_F \|\ln R + I\|_F < 3\sqrt{8} \cdot \frac{3}{2}\,\varepsilon$ per cut and $|\|z_0\|^2 - 1| < 3\varepsilon$, so $\lambda(z) = E_3(z) - \|z\|^2/\ln 2$ obeys $2|\lambda(z_0) - \lambda(z_*)| < 50\varepsilon$; the 22 columns $iz, i\sigma_a^{(j)}z$ of G have norm $\|z\|$, so $4\|G(z_0)G(z_0)^T - G(z_*)G(z_*)^T\|_{\text{op}} \le 8 \cdot 22(1+\varepsilon)\varepsilon < 180\varepsilon$. These add less than $4 \cdot 10^2\varepsilon$. The operator-norm change is therefore below $10^4\varepsilon$. This budget is analytic; SEH-3 checks only the resulting margin. The four rational logarithm enclosures, their divided differences, and the explicit 10^{-10}-grid rounding used to assemble $\widehat{B}$ add less than 10^{-5} in operator norm. Consequently

$$\|B(\psi_{x_*}) - \widehat{B}\|_{\text{op}} < 10^4(3 \cdot 10^{-8}) + 10^{-5} = 0.00031.$$

The exact block positivity bound now implies $B(\psi_{x_*}) \succ 0$, with $\lambda_{\min}(B) > 0.239395 - 0.00031 = 0.239085 > 0.239$ (SEH-3). By Proposition S73 and LU/phase invariance, H annihilates every vector in G. On $G^{\perp}$ the penalty vanishes, so $-H = B \succ 0$ and (S12.21) follows. A local slice transverse to the compact unitary-group orbit gives the claimed strict local maximum modulo symmetry. □

Theorem S89 (quadratic stability near the maximiser; proved from SEH-1–SEH-3, rational constants EB-0–EB-3)**.** *Let $\mathcal{O}_E$ be the local-unitary, global-phase and qubit-permutation orbit of $\phi = \psi_{x_*}$, the equality set of Theorem S86. For any normalised pure seven-qubit state z write $\delta(z) = \min_{\phi \in \mathcal{O}_E} \|z - \phi\|_2$. If $\delta(z) \le 1/20000$, then*

$$\boxed{E_3(z) \le E_3(\psi_{x_*}) - \frac{\delta(z)^2}{20}.} \tag{S12.23}$$

Proof. Let $\phi' \in \mathcal{O}_E$ be a closest point to z; by symmetry we may take $\phi' = \phi$. After the allowed global phase its overlap with z is real and nonnegative. Thus

$$z = \cos\theta\, \phi + \sin\theta\, w, \qquad \|w\| = 1, \quad \langle \phi, w\rangle = 0, \quad \delta(z) = 2\sin(\theta/2).$$

Here $\theta = 0$ is trivial. The nearest-point condition implies that w is real orthogonal to every local-unitary and phase tangent vector at ϕ. The full-space constrained Hessian certificate of Theorem S88 then gives, for $e(t) = E_3(\cos t\, \phi + \sin t\, w)$,

$$e'(0) = 0, \qquad e''(0) < -\frac{239}{1000}.$$

We obtain a uniform third-derivative bound with no numerical spectral estimates. For each $|S| = 3$, let $M(t)$ be the 8×16 flattening of the unit vector $\gamma(t) = \cos t\, \phi + \sin t\, w$, and let $N = M'$, $R = MM^\dagger$. Along this unit-sphere geodesic, $M'' = -M$, $\|M\|_F = \|N\|_F = 1$, so

$$\|R'\|_F \le 2, \quad \|R''\|_F \le 4, \quad R''' = -4R', \quad \|R'''\|_F \le 8.$$

At $t = 0$ all triple marginals have smallest eigenvalue at least $1/12$ by the exact eigenvalue enclosures in SEH-1. As $\|R'(t)\|_{\mathrm{op}} \le 2$, Weyl's inequality gives, for $0 \le t \le 1/10000$,

$$\lambda_{\min}(R(t)) \ge \frac{1}{12} - 2t \ge \frac{1}{12} - \frac{1}{5000} > \frac{1}{16} =: m.$$

The eigenvalues are at most 1 because $\gamma(t)$ is normalised. Hence $\|\ln R + I\|_F < 6$.

Write $\mathcal{L}_R(K) = \int_0^\infty (R + sI)^{-1} K (R + sI)^{-1}\, ds$ for the Fréchet derivative of $\ln R$. The resolvent integral bounds imply

$$\|\mathcal{L}_R(K)\|_F \le m^{-1} \|K\|_F, \qquad \|D\mathcal{L}_R[H](K)\|_F \le m^{-2} \|H\|_F \|K\|_F.$$

Differentiating the entropy of one cut three times (with natural logarithms internally) yields

$$\frac{d^3}{dt^3}[-\operatorname{tr} R \log_2 R] = -\frac{1}{\ln 2} \left[\operatorname{tr} R'''(\ln R + I) + 3 \operatorname{tr} R'' \mathcal{L}_R(R') + \operatorname{tr} R'\, D\mathcal{L}_R[R'](R')\right].$$

Therefore its absolute value is bounded by

$$\frac{8 \cdot 6 + 3 \cdot 4 \cdot 2 \cdot 16 + 2^3 \cdot 16^2}{\ln 2} = \frac{2480}{\ln 2} < 3720,$$

using $\ln 2 > 2/3$. Averaging the 35 cuts gives the same bound $|e'''(t)| < 3720$. Every step holds uniformly along the geodesic.

For $\delta \le 1/20000$, $\theta = 2 \arcsin(\delta/2) \le 2\delta \le 1/10000$. Taylor's theorem and the negative Hessian give

$$e(\theta) - e(0) \le -\frac{239}{2000}\theta^2 + \frac{3720}{6}\theta^3 \le -\left(\frac{239}{2000} - \frac{620}{10000}\right)\theta^2 = -\frac{23}{400}\theta^2 \le -\frac{1}{20}\delta^2,$$

as $\theta \ge \delta$. EB-0–EB-3 check all needed rational constant inequalities independently; the spectral and Hessian inputs are exactly SEH-1–SEH-3. This proves (S12.23). □

Corollary S90 (Global quadratic stability at seven qubits; proved from the certified theorems cited in the proof)**.** *Let $\delta_p(z)$ be the distance from a normalised pure seven-qubit state z to the orbit of Ψ_{ico} under local unitaries, global phase and qubit permutations, and $\delta(z)$ the distance to $\mathcal{O}_E$ as in Theorem S89. There are constants $c_p, c_e > 0$ such that every such z satisfies*

$$\pi_3(z) - \frac{121}{917} \ge c_p\, \delta_p(z)^2, \qquad E_3(\psi_{x_*}) - E_3(z) \ge c_e\, \delta(z)^2.$$

Proof. Both distances are at most $\sqrt{2}$, since $\min_\theta \|z - e^{i\theta}\phi\|^2 = 2 - 2|\langle z, \phi\rangle|$. If $\delta_p(z) \le 1/250$, a qubit permutation σ carries a nearest point of the orbit into the local-unitary and phase orbit of Ψ_{ico}; π_3 is permutation invariant, so Theorem S71 applied to σz gives the bound with $1/1024$. The set $\{\delta_p \ge 1/250\}$ of unit vectors is compact and $\pi_3 - 121/917$ is continuous on it. It is positive there, because equality holds only on the orbit (Corollary S67), so it has a positive minimum m_p. Hence $c_p = \min\{1/1024, m_p/2\}$ works. The entropy bound follows in the same way from Theorem S89 for $\delta \le 1/20000$, continuity of E_3, and the equality case of Theorem S86. The constants come from compactness and are not explicit. □

S12.10 Mixed states attain neither optimum

For a seven-qubit density matrix ρ and $|S| = 3$ let

$$E_S(\rho) = \inf_{\rho=\sum_\ell p_\ell|\psi_\ell\rangle\langle\psi_\ell|} \sum_\ell p_\ell\, S(\rho_S^{(\ell)}), \qquad L_S(\rho) = \inf_{\rho=\sum_\ell p_\ell|\psi_\ell\rangle\langle\psi_\ell|} \sum_\ell p_\ell(1-\operatorname{tr}(\rho_S^{(\ell)})^2)$$

be the convex roofs of the entropy (in bits) and of the linear entropy of the cut $S|S^c$, where $\rho_S^{(\ell)}$ is the marginal of ψ_ℓ.

Lemma S91 (no flat extremal planes)**.** *Let f be a real function on the unit sphere of $(\mathbb{C}^2)^{\otimes 7}$, invariant under phases and local unitaries, and let ϕ be a global extremum of f such that f is C^2 near ϕ, every one-qubit marginal of ϕ is $I/2$, and the constrained Hessian H of f at ϕ satisfies $H[w,w] \neq 0$ for every nonzero real tangent vector w ($\operatorname{Re}\langle\phi, w\rangle = 0$) orthogonal to*

$$G = \operatorname{span}_{\mathbb{R}}\{i\phi,\ i\sigma_a^{(j)}\phi\}.$$

Then no complex subspace of dimension at least two containing ϕ has all its unit vectors global extrema of the same kind.

Proof. Suppose $V \ni \phi$ is such a subspace with $\dim V \geq 2$. Pick a unit vector $\psi \in V$ with $\langle\phi,\psi\rangle = 0$, and let w be ψ or $i\psi$. The great circle $t \mapsto \cos t\,\phi + \sin t\, w$ lies in the unit sphere of V, so f is constant on it, and at the critical point ϕ its second derivative along this geodesic is $H[w,w]$; hence $H[w,w] = 0$. The vectors spanning G are tangent, as $\langle\phi, \sigma_a^{(j)}\phi\rangle$ is real, and the orbit of ϕ consists of critical points, so differentiating the vanishing constrained gradient along it gives $Hg = 0$ for $g \in G$. Writing $w = g + w'$ with $g \in G$ and w' tangent and orthogonal to G, $0 = H[w,w] = H[w',w']$, so $w' = 0$: both ψ and $i\psi$ lie in G.

But $G \cap iG = \{0\}$. Let $\mathfrak{k}$ be the real span of iI and the $i\sigma_a^{(j)}$, and suppose $g = A\phi = iA'\phi$ lies in $G \cap iG$, with $A, A' \in \mathfrak{k}$. Put $B = -A'$, so $A\phi = -iB\phi$. Since A, B are anti-Hermitian,

$$\langle\phi, [A,B]\phi\rangle = \langle B\phi, A\phi\rangle - \langle A\phi, B\phi\rangle = -2i\|B\phi\|^2.$$

The commutator $[A,B]$ is a sum of traceless one-site operators, and every one-qubit marginal of ϕ is $I/2$, so the left side vanishes. Hence $B\phi = 0$ and $A\phi = 0$. Thus $\psi = 0$, a contradiction. □

Proposition S92 (mixed-state roofs; proved from the certified theorems cited in the proof)**.** *Every seven-qubit density matrix ρ satisfies*

$$\frac{1}{35}\sum_{|S|=3} E_S(\rho) \leq E_3(\psi_{x_*}), \qquad \frac{1}{35}\sum_{|S|=3} L_S(\rho) \leq 1 - \frac{121}{917}.$$

Equality holds in the first bound only for pure states local-unitarily and permutation equivalent to ψ_{x_*}, and in the second only for pure states equivalent to Ψ_{ico}.

Proof. Using one ensemble of ρ for all 35 infima bounds the left sides by $\sum_\ell p_\ell E_3(\psi_\ell)$ and $\sum_\ell p_\ell(1-\pi_3(\psi_\ell))$, so Theorem S86 and Corollary S67 give the bounds and the pure equality cases.

Suppose the entropy bound is attained. The left side is at most the ensemble mean of $E_3(\psi_\ell)$, and each term is at most $E_3(\psi_{x_*})$, so every member of every ensemble of ρ is a pure maximiser. Every unit vector χ in the support of ρ belongs to some ensemble, since $\rho - p|\chi\rangle\langle\chi| \succeq 0$ for $p = \langle\chi, \rho^+\chi\rangle^{-1}$, with ρ^+ the inverse of ρ on its support; so every unit vector of the support is a maximiser. Pick a unit vector ϕ in the support; after a local unitary, a qubit permutation and a phase, applied to ρ, which change neither side, $\phi = \psi_{x_*}$. Lemma S91 applies to $f = E_3$ at ϕ: near ϕ all triple marginals have full rank, so E_3 is smooth there, the one-qubit marginals are $I/2$ (Theorem 41), and Theorem S88 gives $H[w,w] < -\frac{239}{1000}\|w\|^2$ for tangent w orthogonal to G. So the support has dimension one and ρ is pure.

The linear-entropy case is the same with $f = \pi_3$, a global minimum and Ψ_{ico} in place of ψ_{x_*}: the Lagrange form Q of Theorem S69 is the constrained Hessian, positive semidefinite with kernel exactly G. □

Thus mixed-state exclusion follows from two properties of the pure optimiser, maximally mixed one-qubit marginals and a Hessian nondegenerate transverse to the local-unitary orbit (Lemma S91); the four-qubit proof of Corollary 10 instead uses the square flattenings, which the 3|4 cuts do not have.

S12.11 Full-rank necessity at every local entropy maximum

At a rank-deficient cut, a perturbation can create a small positive eigenvalue. The resulting entropy gain dominates the smooth losses on the other cuts, ruling out a local maximum.

Proposition S93 (No rank-deficient local maxima of mean balanced entropy)**.** *Let z be a normalised pure state of seven qubits. If z is a local maximiser of the mean three-body von Neumann entropy $E_3(z)$ on the full pure-state space, then*

$$\rho_S(z) \succ 0 \qquad \text{for every } |S| = 3.$$

More generally, for any finite multipartite pure-state system, a positive weighted sum of subsystem entropies cannot have a local maximum at a state with a rank-deficient included marginal, provided the included subsystem has dimension no greater than its complement. This holds without assuming permutation or local-unitary symmetry.

Proof. Suppose the reduction at a cut $A|B$, with dimensions $d_A \le d_B$, has rank $r < d_A$. Flatten the normalised state z into a $d_A \times d_B$ matrix M. Choose unit vectors $u \in \ker M^\dagger$ and $w \in \ker M$, and let v have flattening $uw^\dagger$. Then $\langle z, v\rangle = 0$ and $\|v\| = 1$. The normalised perturbation

$$z_t = \frac{z + tv}{\sqrt{1+t^2}}, \qquad p = \frac{t^2}{1+t^2}$$

has, at the selected cut, the exact reduced state

$$\rho_A(z_t) = (1-p)\rho_A(z) + p\, uu^\dagger,$$

with orthogonal supports. Hence

$$S(\rho_A(z_t)) - S(\rho_A(z)) = h_2(p) - pS(\rho_A(z)) = \frac{2t^2|\ln|t||}{\ln 2} + O(t^2) \qquad (t \longrightarrow 0),$$

where h_2 is the binary entropy.

For every other included subsystem T, its reduced state $\rho_T(z_t)$ is a real-analytic positive semidefinite Hermitian matrix in t. Its eigenvalues separated away from zero at $t = 0$ contribute

$$S_T^{\text{positive}}(t) - S_T^{\text{positive}}(0) = a_T t + O(t^2)$$

by analytic functional calculus on the positive spectral cluster. The eigenvalues emanating from zero contribute $-\sum_{\lambda_i(t)\to 0} \lambda_i(t)\log_2\lambda_i(t) \ge 0$ for sufficiently small t, since all such eigenvalues lie in $[0,1]$. Thus

$$S(\rho_T(z_t)) - S(\rho_T(z)) \ge a_T t - C_T t^2$$

for small positive and negative t; this also holds for the selected cut with its strictly positive $t^2|\ln|t||$ improvement kept separately.

Summing with strictly positive weights, including the selected cut, gives

$$E_w(z_t) - E_w(z) \ge at + c\,t^2|\ln|t|| - Ct^2, \qquad c > 0,$$

for both signs of sufficiently small t. Choose the sign of t so that $at \ge 0$. Then the right side is positive for small nonzero t, contradicting local maximality. For seven qubits all 35 balanced cuts have dimensions $8 \le 16$ and positive equal weights, proving the claim. □

Corollary S94 (Stabilizer exclusion at every non-AME size)**.** *For $n \geq 2$ and $k = \lfloor n/2 \rfloor$, a pure stabilizer state is a local maximiser of mean k-body von Neumann entropy if and only if it is AME$(n, 2)$. Consequently no stabilizer state is an unrestricted entropy maximiser at $n = 4$ or at any $n \geq 7$; in particular, neither are the graph-state minimisers of π_k in Table S6.*

Proof. An AME state attains the cut-dimension upper bound k bits on every cut. Conversely, every marginal of a stabilizer state is a normalised projector, $\rho_S = 2^{-r_S}\Pi_S$ with $\operatorname{rank}\Pi_S = 2^{r_S}$ (Hein–Eisert–Briegel [5]). If every k-qubit marginal had full rank 2^k, each would be maximally mixed and the state would be AME$(n, 2)$. So a non-AME stabilizer state has a rank-deficient k-qubit marginal, and $k \leq n - k$, so Proposition S93 applies with equal weights. The AME qubit sizes are $2, 3, 5, 6$ (Section S15). □

The remaining sections. The four- and seven-qubit proofs are now complete. The remaining sections collect independent consequences and generalisations, with one exception: Lemma S96 and Proposition S97 of Section S13 supply the focused-covariant zero of the conference family and the symmetric cubic kernel of the seven-qubit entropy certificate. Apart from these two statements, none of the remaining sections is needed for the preceding global optima.

S13 Focused covariants of the four-qubit sextic

Throughout this section n is the number of qubits and $N = \|z\|^2$ (in the four-qubit sections the norm squared is denoted by N, and in the seven-qubit entropy certificate by ν), and $W_1 = \sum_i \|\mathbf{r}_i\|^2$ with $\mathbf{r}_i^a = \langle z|\sigma_a^{(i)}|z\rangle$. The four-qubit moment identity (S) is not an ordinary positive three-copy replica operator: the latter has a negative sector on the copy-symmetric cube (Appendix C). The successful proof uses the spin-flipped mixed tensor $w = z_x \otimes z_y \otimes \widetilde{z}_t$, $\widetilde{z} = \epsilon^{\otimes n}\bar{z}$, with $\epsilon = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. A complementary Pauli-coordinate description of third moments is derived in (S13.7) below. At one site write

$$L|xyt\rangle = \epsilon_{yt}|x\rangle, \qquad M|xyt\rangle = \epsilon_{xt}|y\rangle,$$

and let S_i project the local three-copy tensor at site i onto $\mathrm{Sym}^3(\mathbb{C}^2)$.

Proposition S95 (focused-site sextic covariants)**.** *For a subset $I \subseteq [n]$ and $0 \leq r \leq n - |I|$, define*

$$\Phi_{I,r}(z) = \left(\prod_{i\in I} S_i\right) \sum_{\substack{A\subseteq I^c \\ |A|=r}} \left(\prod_{j\in A} M_j\right)\left(\prod_{j\in I^c\setminus A} L_j\right)(z_x \otimes z_y \otimes \widetilde{z}_t).$$

Each $\Phi_{I,r}$ is a local-$SU(2)$ covariant of bidegree $(2, 1)$ (quadratic in z, linear in $\bar{z}$), and

$$\Phi_{I,r} = \Phi_{I,n-|I|-r}. \tag{S13.1}$$

Its Gram matrix $G_I(z) = (\langle\Phi_{I,r}, \Phi_{I,s}\rangle)_{r,s}$ is positive semidefinite. After (S13.1), at most $\lfloor (n - |I|)/2 \rfloor + 1$ distinct vectors remain.

For $n = 4$ and $I = \{i\}$ put $\mathcal{R}_i = 4\Phi_{i,1} - 15\Phi_{i,0}$, a residual different from the $V_i - 8U_i$ of (S). The homogeneous sextic identity then has the reduced form

$$32(17NP - 12T - 12N^3) = 3\sum_{i=1}^{4}\left(15\|\Phi_{i,0}\|^2 + \|4\Phi_{i,1} - 15\Phi_{i,0}\|^2\right), \tag{S13.2}$$

where $N = \|z\|^2$, $P = \sum_{j=2}^4 \operatorname{tr} R_{1j}^2$ and $T = \sum_{j=2}^4 \operatorname{tr} R_{1j}^3$.

Proof. Interchanging the two identical factors z_x, z_y swaps L with M at every contracted site, while the symmetric projectors at the focused sites are unchanged. This gives (S13.1). Gram positivity follows from the Hilbert-space inner product. In the notation of Appendix A, $U_i = \Phi_{i,0}$

and $V_i = \sum_{r=0}^{3} \Phi_{i,r} = 2(\Phi_{i,0} + \Phi_{i,1})$. Thus $\mathcal{R}_i = 2V_i - 19U_i = 2Y_i - 7U_i$ with $Y_i = V_i - 6U_i$. Writing $F = 17NP - 12T - 12N^3$ for the left side of (S13.2) divided by 32, the contractions (CY) give

$$\sum_i \|\mathcal{R}_i\|^2 = 4\sum_i \|Y_i\|^2 - 28\,\mathrm{Re}\sum_i \langle U_i, Y_i\rangle + 49\sum_i \|U_i\|^2 = \tfrac{32}{3}F - 5NW_1.$$

Hence $32F = 15NW_1 + 3\sum_i \|\mathcal{R}_i\|^2$, and $\sum_i \|U_i\|^2 = NW_1/3$ turns this into (S13.2). SF-1 checks the identity on one integer four-qubit tensor by exact rational arithmetic. □

Lemma S96 (the exact-support Pauli selection rule)**.** *Let R_I be the unnormalised reduction of an arbitrary pure n-qubit state to the focused sites I. Then, up to the fixed sign $(-1)^{n-|I|}$,*

$$\Phi_{I,0} = \left(\prod_{i\in I} S_i\right)\left(z_x \otimes (R_I \epsilon_I^T)_{y_I,t_I}\right), \qquad \epsilon_I = \epsilon^{\otimes |I|},$$

with the x-slots on I^c left intact. In particular, if $\langle z|P|z\rangle = 0$ for every Pauli word P of exact support I, then $\Phi_{I,0}(z) = 0$.

Proof. At a contracted site, $\sum_t \epsilon_{yt}\epsilon_{tu} = -\delta_{yu}$, so the two remaining replicas form R_I. Expand $R_I = 2^{-|I|}\sum_{P\in\{I,X,Y,Z\}^I}\langle z|P|z\rangle P$. If P has an identity factor at some $i \in I$, then $\mathrm{vec}(I\epsilon^T)$ is a local singlet in the y,t slots. Its tensor product with any x-qubit is killed by S_i. Only terms supported on every site survive. □

Proposition S97 (complete scalar cubic gradients; analytic, SF-5 controls)**.** *For any n, let $Q_r(z) = \sum_{|S|=r} \mathrm{tr}\, R_S^2$, $N = \|z\|^2$, and $V_r(z) = \sum_{|S|=r}(R_S \otimes I_{S^c})z$. Then*

$$\Phi_{\varnothing,r}(z) = (-1)^n V_r(z), \qquad \frac{\partial Q_r}{\partial \bar z} = 2V_r(z), \qquad \langle z, V_r(z)\rangle = Q_r(z).$$

More generally, let v be a local-unitary scalar covariant of bidegree $(2,1)$ with real coefficients in the L, M tensor basis. Its closure $p_v = \langle z, v(z)\rangle$ is real and $v = \frac{1}{2}\partial_{\bar z} p_v$. The vectors $V_S = (R_S \otimes I)z$, with one S from each complementary pair $\{S, S^c\}$, form a basis of this real space, of dimension 2^{n-1}. Its permutation-invariant subspace has basis $V_0, \dots, V_{\lfloor n/2\rfloor}$ and dimension $\lfloor n/2\rfloor + 1$. Thus the zero-focus weight sums exhaust the symmetric scalar cubic source. Consequently, for any fixed real λ,

$$\mathcal{G}_{r,\lambda}(z) := \|\Phi_{\varnothing,r}(z) - \lambda\Phi_{\varnothing,0}(z)\|^2 = \|V_r(z) - \lambda N z\|^2 \geq 0 \tag{S13.3}$$

is a universal homogeneous sextic. It vanishes exactly when z is a projective stationary point of Q_r/N^2 with $Q_r/N^2 = \lambda$, for nonzero z. Moreover,

$$\begin{aligned} N\mathcal{G}_{r,\lambda}(z) &= (N\|V_r\|^2 - Q_r^2) + (Q_r - \lambda N^2)^2, \\ N\|V_r\|^2 - Q_r^2 &= \sum_{a<b} |z_a (V_r)_b - z_b (V_r)_a|^2 \geq 0. \end{aligned} \tag{S13.4}$$

Along the seven-qubit icosahedral family ψ_x $(0 \leq x \leq 1/2)$,

$$\begin{aligned} &\Phi_{\varnothing,1}(\psi_x) = \tfrac{7}{2}\Phi_{\varnothing,0}(\psi_x), \\ &4(104x-7)\Phi_{\varnothing,3}(\psi_x) - 2(524x-37)\Phi_{\varnothing,2}(\psi_x) + (96x^2 + 3628x - 263)\Phi_{\varnothing,0}(\psi_x) = 0. \end{aligned} \tag{S13.5}$$

At $\Psi_{\mathrm{ico}} = \psi_{37/524}$ the second relation simplifies to $\Phi_{\varnothing,3} = \frac{605}{131}\Phi_{\varnothing,0}$. (Both relations are proved below; SF-5 checks them exactly.) In fact the entire zero-focus Gram on this family is

$$G(x) = q(x)q(x)^T + \frac{x(1-2x)}{8} q'(x)q'(x)^T, \qquad q = (1, \tfrac{7}{2}, 21\pi_2, 35\pi_3)^T. \tag{S13.6}$$

It has rank two for $0 < x < 1/2$ and rank one at both endpoints. For the two source combinations V_0 and $131V_2 - 52V_3$, its 2×2 Gram determinant is exactly $72x(1-2x)$. Thus $\mathcal{G}_{3,605/131}$ is a nontrivial sextic SOS vanishing at Ψ_{ico}, while at $x = x_*$ the two relations in (S13.5) give the $\mathrm{Sym}^7\mathbb{R}^2$ three-copy kernel features of Theorem S85. Neither gives a lower bound on π_3; Theorem S100 excludes a sharp fixed-coefficient cubic SOS certificate even after including this form.

Proof. At an L site, the first replica index survives and the other two are identified after spin flip; at an M site the second replica index survives and the first is identified with the conjugate. For a set S of M sites, the surviving tensor is $(-1)^n(R_S \otimes I_{S^c})z$. Differentiating $\operatorname{tr} R_S^2$ by the product rule yields $2(R_S \otimes I)z$. Euler's homogeneous identity gives the inner product.

To prove that these scalar cubics exhaust the possibilities, contract the output of a general scalar cubic with $\bar{z}$. Locally its two-copy invariant operator is a real combination of identity and swap. Their tensor products are Hermitian and commute with the simultaneous copy exchange. Since the two holomorphic inputs are already symmetric, the two antiholomorphic inputs are symmetric as well; hence p_v is real and $\partial_{\bar{z}} p_v = 2v$. All quartic local-unitary invariants of bidegree $(2,2)$ are linear combinations of the swap expectations $\operatorname{tr} R_S^2$. On the symmetric square, complementary swap words agree. These are their only relations: the local antisymmetric-site sets A have even cardinality, and the words act on them by the characters $(-1)^{|S\cap A|}$. Characters of this even-subset group are indexed exactly by S modulo complementation and are linearly independent. Each such sector occurs on qubits. Expectations on $z^{\otimes 2}$ determine a Hermitian operator on the symmetric square, by its polynomial coefficients. This proves the full basis statement. After site averaging, only $|S|$ remains. The weight-r swap sum has value $[t^r](1+t)^{n-|A|}(1-t)^{|A|}$, a polynomial of degree r in $|A|$ with leading coefficient $(-2)^r/r!$. For $0 \le r \le \lfloor n/2 \rfloor$ these polynomials are independent on the equally many distinct even values of $|A|$, proving the symmetric basis statement.

Projective stationarity of Q_r/N^2 is $V_r = (Q_r/N)z$, proving the equality criterion. The octic equation is the orthogonal decomposition of V_r into its component along z and its perpendicular component; its final equality is the usual complex Lagrange identity. Along the icosahedral family ψ_x, one-body maximal mixing gives the first relation in (S13.5). For $r \in \{2,3\}$, the fixed-space argument of Lemma S72 places the real gradient $4V_r(\psi_x)$ in the plane $\operatorname{span}\{U,V\} = \operatorname{span}\{\psi_x, \psi_x'\}$, where $\psi_x' \perp \psi_x$, $\langle \psi_x, V_r(\psi_x)\rangle = Q_r(\psi_x)$ and, by the gradient formula, $\langle \psi_x', V_r(\psi_x)\rangle = \frac{1}{4}Q_r'(\psi_x)$. Hence $V_r = Q_r\psi_x + \frac{1}{4}Q_r'\psi_x'/\|\psi_x'\|^2$. Since $30(a')^2 + 2(b')^2 = 1/[2x(1-2x)]$, taking inner products gives (S13.6). The explicit purity polynomials give $131q_2' - 52q_3' = 24$. The displayed Gram determinant is therefore $24^2 x(1-2x)/8 = 72x(1-2x)$, proving rank two throughout the open interval; continuity gives rank one at its endpoints. Eliminating ψ_x' between $r=2$ and $r=3$ gives $Q_2'\Phi_{\varnothing,3} - Q_3'\Phi_{\varnothing,2} - (Q_2'Q_3 - Q_3'Q_2)\Phi_{\varnothing,0} = 0$; substituting $Q_2 = 21\pi_2$ and $Q_3 = 35\pi_3$ from Theorem 41 yields the second relation in (S13.5). Thus the two displayed relations span its kernel for every interior x: their source rows are independent, since $104x - 7$ and $524x - 37$ cannot vanish together. SF-5 compares the closed Gram formula with every coefficient of an independent exact tensor contraction, and checks its specialisation at $x = 37/524$ with null vectors $(-7,2,0,0)$ and $(-605,0,0,131)$ in the $\Phi_{\varnothing,0},\dots,\Phi_{\varnothing,3}$ basis. □

Remark S98 (scope of cubic-contraction searches). Theorem S100 rules out every fixed-coefficient cubic SOS certificate of the sharp seven-qubit purity bound, including all nonempty-focus forms, zero-focus gradients and their cross-sector couplings. The covariant identities above remain useful algebraic descriptions; they do not evade the complete three-copy obstruction.

Remark S99 (joint quadratic/cubic Pauli enumerators). For a trace-one density matrix ρ define $r_P = \operatorname{tr}(\rho P)$ for Hermitian Pauli strings P. The ordinary weight enumerator $A_j = \sum_{\mathrm{wt}\,P=j} r_P^2$ controls mean subsystem purity. There is a natural signed *triple* enumerator:

$$C_j = \sum_{\substack{P,Q:[P,Q]=0 \\ |\operatorname{supp} P \cup \operatorname{supp} Q| = j}} \eta(P,Q)\, r_P r_Q r_{R(P,Q)}, \qquad PQ = \eta(P,Q)R(P,Q),\ \eta(P,Q) \in \{\pm 1\},$$

The sum is over ordered pairs of Pauli words, including the identity; $R(P,Q)$ is the canonical Hermitian Pauli word. For $b_k = \binom{n}{k}^{-1} \sum_{|S|=k} \operatorname{tr} \rho_S^3$,

$$b_k = \frac{1}{4^k \binom{n}{k}} \sum_{j=0}^{k} \binom{n-j}{k-j} C_j. \tag{S13.7}$$

Expand ρ_S^3 in the Pauli basis: products of anticommuting P, Q cancel between orders and every commuting product is $\pm R(P,Q)$. Group by the union of supports to obtain the binomial coefficient. Thus A_j, C_j are natural joint second/third-moment coordinates. For pure states the third moments are sextic in the amplitudes; unlike A_j, the individual C_j need not be positive. This coordinate identity is independent of the covariant certificates and is not used in the cubic-SOS obstruction below.

S14 The complete cubic-SOS obstruction

For seven qubits let $N(z) = \|z\|^2$, let $\mathcal{F}_3(z)$ be the homogeneous quartic mean three-body purity ($\mathcal{F}_3 = N^2\pi_3$), and put

$$F_\lambda(z) = N(z)\big(\mathcal{F}_3(z) - \lambda N(z)^2\big), \qquad \lambda_{\mathrm{SOS},6} = \max\{\lambda : F_\lambda \text{ is a sum of squares of real cubics}\}.$$

The polynomials are in all 256 real amplitude coordinates. No restriction to the conference family or to covariant subsets is imposed.

Theorem S100 (Complete real cubic-SOS obstruction; TCB)**.** *The maximal homogeneous degree-six real SOS lower bound satisfies*

$$\boxed{\lambda_{\mathrm{SOS},6} = L_3 = \frac{125}{978} < \frac{121}{917}.}$$

In particular, for every $\lambda > 125/978$ there is no identity

$$F_\lambda(z) = \sum_a |p_a(z)|^2 + \sum_b |q_b(z,\bar{z})|^2$$

with fixed-coefficient holomorphic cubics p_a and bidegree-$(2,1)$ cubics q_b, nor a real homogeneous cubic SOS. This includes arbitrary local $[2,1]$ sectors and cross-focused couplings. It excludes neither higher-degree SOS nor non-SOS inequalities or state-dependent reductions.

Proof. Every squared holomorphic cubic is a positive linear functional of $|z\rangle\langle z|^{\otimes 3}$. Every squared mixed cubic is a positive linear functional of $|zz\tilde{z}\rangle\langle zz\tilde{z}|$, since spin flip is invertible on the conjugate factor. Average any proposed real cubic SOS over global phase: only opposite phase weights pair, so its terms become squared moduli of their $(3,0)$ and $(2,1)$ components, since $(\mathrm{Re}\, p)^2 + (\mathrm{Im}\, p)^2 = |p|^2$. Twirling their positive Gram matrices over the local unitaries, sites and conjugation preserves positivity and F_λ; an average of sums of squares is again a sum of squares. By Theorem S44, the complete three-copy Fourier blocks contain both kinds of Gram forms, with no omitted local representation sector.

Polynomial equality on every z implies equality of the compressed Hermitian coefficient operators on $\mathrm{Sym}^3 H$: the expectations in all $z^{\otimes 3}$ determine such an operator by its polynomial coefficients. Thus the twirled identity holds on any feasible averaged moment, not only on physical pure powers. The argument uses three properties of the moment: its twirled operator X is positive semidefinite on every ω-block of the $\mathrm{Sym}^3 H$ sector, the flipped image $\mathcal{T}(X)$ is positive semidefinite on every γ-block of the $\mathrm{Sym}^2 H \otimes H$ sector, and its word values satisfy the one-site antisymmetriser relations, so that the pairing is well defined. In the isotypic decomposition of the local three-copy algebra the compressed blocks $\omega_1, \ldots, \omega_7$ and $\gamma_0, \ldots, \gamma_7$ have sizes $(1,1,1,3,5,11,21)$ and $(1,1,2,4,8,16,32,64)$. The boundary pseudo-moment $m^*/70416$ of Theorem S45 has positive semidefinite blocks, satisfies the relations and has objective value $125/978$, so it evaluates the left side as

$$F_\lambda(m^*/70416) = \frac{125}{978} - \lambda,$$

while every positive Gram form pairs nonnegatively with it. An SOS representation therefore forces $\lambda \le 125/978$. Conversely, the TCB dual certificate gives an SOS at $125/978$: its positive ω and γ Gram forms factor into cubic squared moduli, and its antisymmetriser relations vanish identically. □

Remark. The relations have exact rank 23 (TCB); the $37 - 23 = 14$ free coordinates match the independent invariant count from (ID2) (HI). This count is not used in the proof.

S15 Exact minima of the balanced purity

For a pure state ψ of n qubits and $k = \lfloor n/2 \rfloor$ write $\pi_k(\psi) = \binom{n}{k}^{-1} \sum_{|S|=k} \operatorname{tr} \rho_S^2$ for the mean balanced-cut purity. Since $\operatorname{tr} \rho_S^2 \geq 2^{-k}$, $\pi_k \geq 2^{-k}$, with equality iff ψ is absolutely maximally entangled (AME$(n, 2)$), which happens only for $n = 2, 3, 5, 6$ (Rains [4] and Scott [6] for $n \geq 8$; Huber–Gühne–Siewert [2] for $n = 7$). This section shows that for $n \leq 18$, $n \neq 7, 13$, the minimum of π_k is the value of the Rains linear programme and is attained by a graph state (Table S6; $n = 18$ is Theorem S105).

n	2^{-k}	$\min \pi_k$ (the Rains LP value)	attained by
4	1/4	1/3	cluster (P_4) or Higuchi–Sudbery state
5	1/4	1/4	AME(5,2), e.g. $C_5(1)$
6	1/8	1/8	AME(6,2), e.g. $C_6(2, 3)$
8	1/16	3/35	graph state (Table S7)
9	1/16	1/14	$C_9(3, 4)$
10	1/32	1/21	$C_{10}(1, 2, 4)$ (Remark S102)
11	1/32	1/28	graph state (Table S7)
12	1/64	5/224	$C_{12}(3, 5, 6)$
14	1/128	331/27456	graph state (Theorem S103)
15	1/128	1/104	graph state (Theorem S103)
16	1/256	1/156	induced subgraph of $C_{17}(1, 3, 4, 5)$
17	1/256	1/208	$C_{17}(1, 3, 4, 5)$
18	1/512	7/2288	graph state (Theorem S105)

Table S6: Minimum of π_k for $n \leq 18$ qubits, $n \neq 7, 13$ (for $n = 7$ the minimum 121/917 is Corollary S67). $C_n(D)$ is the circulant graph joining i and $i \pm d$ for $d \in D$.

Write $A_j = \sum_{\mathrm{wt}(P)=j} \langle\psi|P|\psi\rangle^2$ for the weight distribution over the 4^n Pauli strings P ($A_0 = 1$, $\sum_j A_j = 2^n$). Expanding ρ_S in Pauli strings gives

$$a_s := \binom{n}{s}^{-1} \sum_{|S|=s} \operatorname{tr} \rho_S^2 = \frac{1}{2^s \binom{n}{s}} \sum_{j \leq s} \binom{n-j}{s-j} A_j, \tag{S15.1}$$

and for pure states $a_s = a_{n-s}$. Rains' shadow inequalities [4] state that the coefficients S_j of $S(x, y) = \sum_j A_j \left(\frac{x+3y}{2}\right)^{n-j} \left(\frac{y-x}{2}\right)^j$ are non-negative. The *Rains LP* minimises a_k over $A \in \mathbb{R}^{n+1}_{\geq 0}$ with $A_0 = 1$, $a_s = a_{n-s}$ and $S_j \geq 0$.

Theorem S101 (exact certificate; LP-1, GS-1)**.** *1. For $n \in \{4, 5, 6, 8, 9, 10, 11, 12\}$, $\pi_k(\psi)$ is at least the value in Table S6 for every pure n-qubit state.*

2. For $n \in \{4, 5, 6, 8, 9, 10, 11, 12\}$ the graph state of Table S7 has π_k equal to the LP bound. Hence $\min \pi_k$ equals the LP bound for these n.

Proof. (1) For each n, LP-1 exhibits rational multipliers $e_s \in \mathbb{Q}$, $y_i \geq 0$ such that $a_k - LA_0 - \sum_s e_s(a_s - a_{n-s}) - \sum_i y_i S_i$, as a linear form in A, has non-negative coefficients. On any feasible A the right-hand terms vanish or are non-negative, so $a_k \geq L$. (Part (2) shows that L is the LP value.) (2) For the graph state $|G\rangle$ and a cut S, $\operatorname{tr} \rho_S^2 = 2^{-\operatorname{rk}_2 \Gamma[S, S^c]}$, with $\Gamma[S, S^c]$ the biadjacency matrix over $\mathbb{F}_2$ (Hein–Eisert–Briegel [5]). GS-1 computes all ranks exactly. □

n	graph	ranks over all k-sets S
4	path P_4 (linear cluster)	$1^2\,2^4$
5	$C_5(1)$	2^{10}
6	$C_6(2,3)$	3^{20}
8	edges 20 21 31 32 41 43 50 51 52 54 60 61 63 64 70 71 76	$2^6\,3^8\,4^{56}$
9	$C_9(3,4)$	$3^{18}\,4^{108}$
10	$C_{10}(1,2,4)$	$4^{132}\,5^{120}$
11	edges 20 21 31 40 41 42 51 53 54 60 61 63 64 70 71 76 81 83 84 85 86 87 92 93 96 98 {10}0–6 {10}9	$4^{66}\,5^{396}$
12	$C_{12}(3,5,6)$	$4^6\,5^{378}\,6^{540}$

Table S7: Graph states attaining the LP bound (r^m: m cuts of rank r). For $n = 11$, "{10}0–6" means vertex 10 is joined to $0, \ldots, 6$.

Remark S102 (ten qubits). The $C_{10}(1,2,4)$ graph state is 3-uniform, with weight distribution $A = (1,0,0,0,10,72,160,240,285,200,56)$ (GS-2): a $[[10,0,4]]$ stabilizer state. By (S15.1), π_5 depends on A_4 and A_5 as well as the distance, so not every $[[10,0,4]]$ state attains 1/21: the Petersen graph state has $\pi_5 = 3/56$ (GS-3). Scott's Table I [6] lists only the Petersen weight distribution for $n = 10$ (value 3/56, with a caption noting that other distributions exist); Zha et al. [10] give 13/336 as a heuristic "theoretical limitation". The distribution of $C_{10}(1,2,4)$ is the extremal Type I enumerator $W_{10,1}$ of Danielsen–Parker [7], so the value 1/21 is implicit in published data. We have found no earlier statement that $\min \pi_5 = 1/21$.

Theorem S103 (exact finite certificates; LP-14, LP-15, GS-14, GS-15)**.** *The minimum mean balanced-cut purity over all pure qubit states is*

$$\min_{\psi\in(\mathbb{C}^2)^{\otimes 14}} \pi_7(\psi) = \frac{331}{27456}, \qquad \min_{\psi\in(\mathbb{C}^2)^{\otimes 15}} \pi_7(\psi) = \frac{1}{104}.$$

Both minima are attained by graph states.

Proof. For each $n = 14, 15$, the checks LP-14 and LP-15 verify, coefficient by coefficient in all $n+1$ Pauli-weight coordinates, the exact rational Rains-LP dual identity

$$a_{\lfloor n/2\rfloor}(A) - LA_0 = \sum_{s<n/2} e_s(a_s - a_{n-s}) + \sum_{i=0}^{n} y_i S_i(A) + \sum_{j=0}^{n} w_j A_j, \quad y_i, w_j \geq 0,$$

where the finite lists of rational coefficients are supplied literally in the script. This proves the universal lower bounds.

For the graph witnesses with symmetric binary adjacency matrices specified by the hexadecimal row strings in that script, exact enumeration of the graph stabilizer gives, respectively,

n	$(A_0, \ldots, A_7)$	balanced-cut binary rank histogram
14	$(1,0,0,0,0,0,161,576)$	$5^{60}\,6^{1684}\,7^{1688}$
15	$(1,0,0,0,0,0,105,540)$	$5^{35}\,6^{1380}\,7^{5020}$.

The 14- and 15-vertex graphs have 47 and 56 edges. An independent enumeration of every $7|(n-7)$ binary cut matrix computes its rank r_S, yielding $\operatorname{tr}\rho_S^2 = 2^{-r_S}$. Averaging the rank histograms gives exactly 331/27456 and 1/104, matching the certified lower bounds. □

Theorem S104 (exact finite certificates; GL-1, GL-2)**.** *The minimum mean balanced purity is* 1/156 *for* $n = 16$ *and* 1/208 *for* $n = 17$*. It is attained by the graph state* $C_{17}(1,3,4,5)$ *at* $n = 17$*, and by its induced graph on vertices* $0, \ldots, 15$ *at* $n = 16$*.*

Proof. GL-1 verifies exact Rains-LP dual identities with nonnegative rational shadow and Pauli-weight coefficients, proving the stated universal lower bounds. GL-2 computes every balanced cut rank over $\mathbb{F}_2$. For $n = 16$ the ranks are $5^{10}6^{500}7^{6680}8^{5680}$; for $n = 17$ they are $6^{510}7^{4080}8^{19720}$. Inserting 2^{-r} for each cut of rank r yields $1/156$ and $1/208$ respectively, matching the lower bounds. □

S16 An exact eighteen-qubit balanced-purity minimum

Theorem S105 (exact Rains dual and graph certificate; E18-1–E18-4)**.** *For every normalised pure* 18*-qubit state,*

$$\boxed{\pi_9(\psi) \geq \frac{7}{2288}, \qquad \min_{\psi} \pi_9(\psi) = \frac{7}{2288}.}$$

Equality is attained by the graph state whose eighteen adjacency rows (vertex 0 *is the lowest bit, in hexadecimal) are*

`36e72 0fb7d 1b80a 0cad6 25e0b 30583`
`1c10b 2b228 1a262 3999b 27031 3221f`
`2a696 1dd87 2a45b 273ce 22b65 1deb1.`

Its entire graph-state Pauli weight distribution is

$$A_0 = 1, \quad (A_8, A_{10}, A_{12}, A_{14}, A_{16}, A_{18}) = (2754, 18360, 77112, 110160, 50949, 2808)$$

and all other weights vanish. Its $9|9$ *cut-rank distribution is* $7^{9180}9^{39440}$*. The graph is of Type II (all vertex degrees odd) and has global spin-flip overlap* $\tau_{18} = 1$*.*

Proof. E18-1 verifies the rational dual identity for the 18-qubit Rains LP, with constant $7/2288$. Specifically, using $a_s = a_s(A)$ and shadow coefficients $S_i(A)$, it verifies coefficientwise

$$a_9 - \tfrac{7}{2288}A_0 = \sum_{s=0}^{8} e_s(a_s - a_{18-s}) + \tfrac{1}{1166880}S_4 + \tfrac{1}{1633632}S_6 + \tfrac{1}{2498496}S_{11} + \sum_{j=0}^{18} w_j A_j$$

with $w_j \geq 0$ and the rational multipliers

$$(e_0, \ldots, e_8) = \left(\tfrac{5}{4004}, -\tfrac{291}{2002}, \tfrac{81}{364}, \tfrac{278}{1001}, -\tfrac{333}{1001}, 0, -\tfrac{101}{143}, -\tfrac{54}{143}, \tfrac{51}{26}\right).$$

For a pure state the complementarity terms vanish and every remaining term is nonnegative.

E18-2 checks the displayed binary graph is simple, symmetric and has odd degrees; exhaustive enumeration of its 2^{18} stabilizers gives the displayed weight distribution, corresponding to an extremal Type II self-dual additive $GF(4)$ code of minimum weight eight. E18-3 computes every one of the $\binom{18}{9} = 48620$ binary cut ranks, obtaining 9180 cuts of rank 7 and 39440 of rank 9. Thus

$$\pi_9 = \frac{9180/128 + 39440/512}{48620} = \frac{7}{2288}.$$

E18-4 independently checks the same purity from the Pauli weight formula. The matching lower and upper bounds prove the theorem. □

Remark S106 (relationship to the known quaternary S_{18} code). An extremal Hermitian self-dual quaternary $[18, 9, 8]$ code is classical and well documented (see Cheng–Sloane and Harada–Munemasa's code database). The graph above is a local-Clifford graph form of a generator in that database. E18-1–E18-4 make the purity/cut-rank attainment claim entirely reproducible without downloading the external generator. We do not claim novelty for the underlying code or the existence of an extremal $[18, 9, 8]$ code. The equality between its purity and the Rains-LP lower bound is the conclusion here.

Corollary S107 (further sharp time-reversal-symmetric minima; TY-1–TY-2)**.** *Inside the full spin-flip-invariant marginal class,*

$$\boxed{\min_{\tau_{16}=1} \pi_8 = \frac{1}{143}, \quad \min_{\tau_{18}=1} \pi_9 = \frac{7}{2288}, \quad \min_{\tau_{20}=1} \pi_{10} = \frac{131}{77792}.}$$

For $n = 16$ and 20 the respective attaining graph states are the circulants $C_{16}(2,3,4,8)$ and $C_{20}(3,4,5,10)$. Their balanced-cut rank distributions are

n	*ranks*	$\pi_{n/2}$
16	$6^{1096}7^{6882}8^{4892}$	1/143
20	$8^{12312}9^{96900}10^{75544}$	131/77792.

In particular the positive spin-flip coefficients $c_{16} = 1/1716$ and $c_{20} = 103/1478048$ in Theorem S108 are sharp with their stated constant terms. *The $n = 20$ constant* 1193/739024 *remains only a global Rains lower bound, not a proven global minimum.*

Proof. Theorem S108 at $\tau_n = 1$ gives the lower values 1/143 and 131/77792 for $n = 16, 20$. TY-1–TY-2 check all 2^n graph stabilizers and all balanced-cut ranks, with exact fraction arithmetic. Each circulant has odd degree at every vertex, hence $\tau_n = 1$. Theorem S105 handles $n = 18$ and also exhibits a $\tau_{18} = 1$ minimiser. A stronger coefficient at 16 or 20 would fail on the respective circulant witness, proving its sharpness. □

S17 Sharp purity–time-reversal tradeoffs

At four qubits, $\pi_2 = \frac{1}{3} + \frac{1}{6}(W_1 + \tau_4)$ (Lemma 2): a large time-reversal overlap τ_4 costs purity. This section proves the corresponding inequality for larger even n, using the exact minima of Section S15. The paper's Discussion explains the motivation. For odd n, τ_n vanishes identically; Section S17.1 gives the odd-n tradeoff.

For a normalised pure even-n qubit vector ψ let

$$\tau_n(\psi) = |\langle \psi, J_n \psi \rangle|^2 = |\psi^T \sigma_y^{\otimes n} \psi|^2, \qquad J_n = \epsilon^{\otimes n} K,$$

so that τ_4 is the four-tangle τ of Lemma 2, and let $\pi_{n/2}$ be the mean purity of the $\binom{n}{n/2}$ balanced marginals. The Pauli weight distribution A_j, the mean s-site purities a_s (so $\pi_{n/2} = a_{n/2}$, and $a_s = a_{n-s}$ for pure states) and Rains' non-negative shadow coefficients S_i are as in Section S15. The new ingredient is the identity $\tau_n = 2^{-n} \sum_j (-1)^j A_j$, since $\sum_j (-1)^j A_j = 2^n \operatorname{tr}(\rho\, J_n \rho J_n^{-1})$. The inequalities below hold on the *unrestricted* pure state space: no spin-flip symmetry is imposed.

Theorem S108 (exact certificates; ST-1–ST-6)**.** *For each n in the table below, every normalised pure n-qubit state satisfies*

$$\boxed{\pi_{n/2}(\psi) \geq L_n + c_n \tau_n(\psi),}$$

with

n	4	8	10	14	16	20
L_n	1/3	3/35	1/21	331/27456	1/156	1193/739024
c_n	1/6	1/70	1/168	5/6864	1/1716	103/1478048

For each n, c_n is the largest coefficient compatible with the constant L_n: a graph state with every vertex of odd degree has $\tau_n = 1$ and $\pi_{n/2} = L_n + c_n$.

Proof. The checks ST-1–ST-6 supply rational multipliers e_s, $y_i \geq 0$ and residual coefficients $w_j \geq 0$ satisfying the exact coefficientwise identity

$$a_{n/2}(A) - L_n A_0 - c_n\, 2^{-n} \sum_{j=0}^{n} (-1)^j A_j = \sum_{s<n/2} e_s (a_s - a_{n-s}) + \sum_{i=0}^{n} y_i S_i + \sum_{j=0}^{n} w_j A_j \qquad \text{(S17.1)}$$

of linear forms in A (the $s=0$ term encodes the normalisation $a_0 = A_0 = 1 = 2^{-n}\sum_j A_j = a_n$). For a pure state $A_0 = 1$, $a_s = a_{n-s}$ and $A_j, S_i \geq 0$, and the left side is $\pi_{n/2} - L_n - c_n\tau_n$; this gives the inequalities. ST-1–ST-6 check every coefficient and sign in rational arithmetic, with no numerical solver.

For sharpness, let G be a graph whose vertices all have odd degree. The product of the stabilizer generators $X_v Z_{N(v)}$ of $|G\rangle$ is, up to a phase, $X^{\otimes n}Z^{\otimes n}$, a phase times $\sigma_y^{\otimes n}$, so $|G\rangle$ is an eigenvector of $\sigma_y^{\otimes n}$; as $|G\rangle$ is real, $\tau_n = |\langle G|\sigma_y^{\otimes n}|G\rangle|^2 = 1$. ST-1–ST-6 compute $\pi_{n/2}$ exactly for such a graph at each n and find $L_n + c_n$: at $n = 4, 8, 10, 14$ the spin-flip extension of a graph on $n-1$ vertices, at $n = 16, 20$ the circulants $C_{16}(2,3,4,8)$ and $C_{20}(3,4,5,10)$. □

For $n \leq 16$ the constant L_n is the minimum of $\pi_{n/2}$ over pure states, attained by a graph state (Theorems S101, S103 and S104); at $n = 20$ it is an exact lower bound whose attainment is open.

Corollary S109 (time-reversal breaking and stability)**.** *For $n \in \{4, 8, 10, 14, 16\}$ every minimiser of $\pi_{n/2}$ has $\langle\psi, J_n\psi\rangle = 0$, and for $n \in \{4, 8, 10, 14, 16, 20\}$ every pure state with $\pi_{n/2}(\psi) \leq L_n + \delta$ has $\tau_n(\psi) \leq \delta/c_n$.*

Proof. Each $c_n > 0$; subtract L_n in Theorem S108. □

Remark S110 (four qubits and the parity-preserving sizes). At $n = 4$ the inequality is the identity $\pi_2 = \frac{1}{3} + \frac{1}{6}(\sum_i \|\mathbf{r}_i\|^2 + \tau_4)$ of Lemma 2, with the Bloch-vector term dropped. For larger n the certificates come from Rains' shadow inequalities, not from an analogue of the four-qubit sextic remainder, and they exclude time-reversal-symmetric minimisers without any graph or conference-matrix ansatz. At $n = 6$ and $n = 12$ the known balanced-purity minima are attained by states with $\tau_n = 1$, so no positive coefficient can accompany the sharp constant there.

S17.1 Odd n: the time-reversal overlap on single-qubit complements

For odd n every pure state has $\tau_n = 0$, but the time-reversal overlap survives on reduced states. For a k-qubit state ρ put $\tilde\rho = \sigma_y^{\otimes k}\bar\rho\,\sigma_y^{\otimes k}$ and $R_\rho = \mathrm{tr}(\rho\tilde\rho)$, the spin-flip overlap of Serrano-Ensástiga, Giraud and Martin [15] (Lemma S111 below is the odd-sector companion of their even-sector identity $\mathrm{tr}\,\rho^2 + R_\rho = 2^{1-k}\sum_{\mathrm{wt\ even}}\langle P\rangle^2$). For a pure state ψ of n qubits write $\rho_{\bar s}$ for the $(n-1)$-qubit reduction without qubit s, and

$$O(\psi) = \frac{1}{n}\sum_{s=1}^{n} \tfrac{1}{2}(\mathrm{tr}\,\rho_{\bar s}^2 - R_{\rho_{\bar s}}),$$

the mean *T-odd purity* of the single-qubit complements. As before π_m is the mean purity of the $\binom{n}{m}$ marginals on m qubits, and A_j and a_s are as above.

Lemma S111 (sector formulas; proved)**.** *For a state ρ_S on a set S of k qubits with Pauli expectations $\langle P\rangle$, $\mathrm{tr}\,\rho_S^2 = 2^{-k}\sum_{P\subseteq S}\langle P\rangle^2$ and $\frac{1}{2}(\mathrm{tr}\,\rho_S^2 - R_{\rho_S}) = 2^{-k}\sum_{P\subseteq S,\ \mathrm{wt}P\ \mathrm{odd}}\langle P\rangle^2$. Hence $a_s = \sum_j A_j\binom{n-j}{s-j}/(2^s\binom{n}{s})$ and, for odd n, $O = \sum_{j\ \mathrm{odd}}\dfrac{n-j}{n\,2^{n-1}}A_j$.*

Proof. The first formula is the Pauli expansion $\rho_S = 2^{-k}\sum_P\langle P\rangle P$. For each of $\sigma = X, Y, Z$ one has $\sigma_y\bar\sigma\sigma_y = -\sigma$, so $\sigma_y^{\otimes k}\bar P\sigma_y^{\otimes k} = (-1)^{\mathrm{wt}\,P}P$ and $\tilde\rho_S = 2^{-k}\sum_P(-1)^{\mathrm{wt}\,P}\langle P\rangle P$; therefore $R_{\rho_S} = 2^{-k}\sum_P(-1)^{\mathrm{wt}\,P}\langle P\rangle^2$. A weight-$j$ Pauli word lies in $n-j$ of the n complements $\bar s$ and in $\binom{n-j}{s-j}$ of the s-subsets. □

Theorem S112 (odd-n tradeoff; proved, exact checks ST-7, ST-8)**.** *Let $n = 2m+1 \geq 5$ be odd, and put $c_5 = 1$ and $c_n = (n+1)/4$ for $n \geq 7$. Every normalised pure n-qubit state satisfies the identity*

$$\pi_m(\psi) + c_n\,O(\psi) = \frac{1}{2} + \sum_{j\ \mathrm{odd},\, j<n} w_j^{(n)} A_j(\psi), \tag{S17.2}$$

where, with $\binom{a}{b} = 0$ *for* $b < 0$ *or* $b > a$,

$$w_j^{(n)} = \frac{2^m\left[\binom{m}{j-1} + 3\binom{m}{j}\right] + (-1)^{(n-j)/2}\binom{m}{(n-j)/2} + 2c_n\binom{n-1}{j} - \binom{n}{j}}{2^n\binom{n}{j}}. \tag{S17.3}$$

For example $\pi_2 + O = \frac{1}{2} + \frac{1}{5}A_1$ *at* $n = 5$, $\pi_3 + 2O = \frac{1}{2} + \frac{3}{28}A_1 + \frac{3}{140}A_3$ *at* $n = 7$, *and* $\pi_4 + \frac{5}{2}O = \frac{1}{2} + \frac{5}{96}A_1 + \frac{5}{448}A_3 + \frac{11}{4032}A_5$ *at* $n = 9$. *The weights are positive on the odd* $j \le n-4$ *and zero for* $j = n-2, n$. *Consequently*

$$\pi_m(\psi) + c_n\, O(\psi) \;\ge\; \frac{1}{2},$$

with equality if and only if $A_j(\psi) = 0$ *for every odd* $j \le n-4$. *The bound is attained by the GHZ state for every odd* n. *In particular, at* $n = 5$ *equality holds exactly when all one-qubit marginals are maximally mixed.*

Proof. Put $B_j = A_j/\binom{n}{j}$. By Lemma S111, $a_s = 2^{-s}\sum_j \binom{s}{j}B_j$ and $O = 2^{1-n}\sum_{j\ \mathrm{odd}}\binom{n-1}{j}B_j$. For a pure state $A_0 = 1$ and $a_s = a_{n-s}$, so it suffices to find e_s with

$$a_m + c\,O - \tfrac{1}{2}A_0 - \sum_{s=2}^{m} e_s(a_s - a_{n-s}) = \sum_{j\ \mathrm{odd}} w_j A_j$$

as an identity of linear forms in B, for $c = c_n$. Multiply the coefficient of B_j by y^j, sum over j, and put $x = (1+y)/2$. Since $\sum_j 2^{-s}\binom{s}{j}y^j = x^s$ and $2^{1-n}\sum_{j\ \mathrm{odd}}\binom{n-1}{j}y^j = \frac{1}{2}(x^{n-1} - (1-x)^{n-1})$, the left side becomes $x^m + \frac{c}{2}(x^{n-1} - (1-x)^{n-1}) - \frac{1}{2} - \sum_s e_s(x^s - x^{n-s})$. On the right, the four parts of the numerator of (S17.3) sum over odd j to the odd parts in y of $2^m(3+y)(1+y)^m$, $y(y^2-1)^m$, $2c(1+y)^{n-1}$ and $-(1+y)^n$. In x, after division by 2^n, these are $\frac{1}{2}[(1+x)x^m - (2-x)(1-x)^m]$, $\frac{1}{2}(-1)^m(2x-1)x^m(1-x)^m$, $\frac{c}{2}(x^{n-1} - (1-x)^{n-1})$ and $-\frac{1}{2}(x^n - (1-x)^n)$. The terms in c cancel, for every c, and the identity becomes $2\sum_{s=2}^m e_s(x^s - x^{n-s}) = Q(x)$, where

$$Q(x) = x^m - x^{m+1} - 1 + x^n + (2-x)(1-x)^m - (1-x)^n - (-1)^m(2x-1)x^m(1-x)^m.$$

Each of the pairs $\{x^m, -x^{m+1}\}$ and $\{-1, x^n\}$ and the remaining three terms together change sign under $Q(x) \mapsto x^nQ(1/x)$ (for the last group use $(x-1)^n = -(1-x)^n$). Hence $Q = \sum_{s\le m} q_s(x^s - x^{n-s})$ with $q_s = [x^s]Q$, and reading off coefficients for $s \le m$,

$$q_s = (-1)^s\left[2\binom{m}{s} + \binom{m}{s-1} - \binom{n}{s}\right] - \delta_{s0} + (1 + (-1)^m)\delta_{sm},$$

so $q_0 = q_1 = 0$ (as $n = 2m+1$) and $e_s = q_s/2$ proves (S17.2).

Let $N_j = 2^n\binom{n}{j}w_j$ and $k = (n-j)/2$. At $j = n$, $N_n = 0 + 1 + 0 - 1 = 0$. At $j = n-2$, $N_3 = 4 - 2 + 8 - 10 = 0$ for $n = 5$, and for $m \ge 3$ both binomials in the first term vanish and $2c_n = m + 1$, so $N_{n-2} = -m + (m+1)2m - m(2m+1) = 0$. For $j \le n-4$ ($k \ge 2$, so $n \ge 7$), $(m+1)\binom{n-1}{j} - \binom{n}{j} = \frac{2k(m+1)-n}{n}\binom{n}{j}$, which exceeds $\binom{n}{j}$ because $2k(m+1) - n \ge 4(m+1) - n = 2m + 3 > n$; and $\binom{n}{j} = \binom{n}{2k} \ge \binom{m}{k}$ (send a k-subset T of $\{1,\dots,m\}$ to $T \cup (T+m)$). So $N_j > 0$, and for $n = 5$, $j = 1$ directly $N_1 = 28 + 1 + 8 - 5 = 32$. The equality statement follows because $A_j \ge 0$. For $\mathrm{GHZ}_n = (|0\rangle^{\otimes n} + |1\rangle^{\otimes n})/\sqrt{2}$, a Pauli word with an X or Y on some but not all sites has zero expectation, and a Z-word of weight $j < n$ has $\langle P\rangle \ne 0$ only if j is even, so $A_j = 0$ for odd $j < n$. ST-7 checks (S17.3) and the multipliers e_s against an independent exact elimination for odd $n \le 41$; ST-8 evaluates both sides of (S17.2) exactly on Gaussian-integer states at $n = 5, 7, 9$ (three random states, GHZ, the ring graph state and W) from the reduced density matrices and the Pauli expectations, without Lemma S111. □

Remark S113 (interpretation). The inequality is the odd-n counterpart of Theorem S108: a small balanced purity forces a large mean T-odd purity, $O \ge (\frac{1}{2} - \pi_m)/c_n$, with $1/c_n = 4/(n+1)$ for $n \ge 7$. The coefficient c_n is the least for which (S17.2), with the same multipliers e_s, has nonnegative

weights: O carries the weight $(n-j)/(n2^{n-1}) > 0$ on A_j, so lowering c_n makes $w^{(n)}_{n-2} = 0$ negative. It is also optimal over all pure states. For $\Psi_n = \mathrm{GHZ}_{n-2} \otimes \Phi^+$ the only odd Pauli weights are $A_{n-2} = 2^{n-3}$ (the full-support X/Y words of the GHZ factor) and $A_n = 3 \cdot 2^{n-3}$ (those words times the three Bell correlations), so the residual of (S17.2) vanishes, while $O = 1/(2n) > 0$ by Lemma S111. Any smaller coefficient therefore fails at Ψ_n (ST-9). For example, at $n = 7$ the hypothesis $\pi_3 < 19/140$, the purity of the best stabilizer states (Proposition 43), gives $O > 51/280$.

S18 Thirteen qubits: purity interval and an entropy witness

At thirteen qubits the Rains linear programme (level two of the hierarchy of Section S9) has value $25/1408$, but a state attaining it would be a pure $((13, 1, 6))_2$ code, which does not exist. The three-copy level, in the Schur–Weyl coordinates of Section S18.1, gives a lower bound strictly above the LP value; measuring one qubit of the fourteen-qubit graph state of Theorem S103 gives an upper bound (Proposition S116). The exact minimum remains open. A graph-code superposition built on the same witness gives $\max E_6 > 5.8336$ (Corollary S119). The paper uses only these thirteen-qubit statements and Theorem S114(a); the other sizes are recorded with the same certificates.

S18.1 The three-copy level for all n: Schur–Weyl form

The orbit coordinates of §S9.1 grow exponentially in n. Schur–Weyl duality for the site symmetry reduces level $m = 3$ of Theorem S44 to $O(n^3)$ real parameters and blocks of size $O(n)$, with every step exact over $\mathbb{Q}$ (`three_copy_schur_weyl.py`).

Reduction. Per site, $\mathcal{A}_3 = \mathrm{End}(S^{[3]}) \oplus \mathrm{End}(S^{[21]}) \cong \mathbb{C} \oplus M_2$, with $S^{[3]} = \langle |000\rangle\rangle$ and $S^{[21]} = \langle b_1 = |100\rangle - |010\rangle,\ b_2 = |010\rangle - |001\rangle\rangle$ (Gram matrix $G_0 = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$), on which global copy permutations act by the standard representation ρ of S_3. Sort the sites by type: a sites in the [21] sector. The site-symmetrised twirl Ω of $(zz^*)^{\otimes 3}$ lies in $\bigoplus_a (M_2^{\otimes a})^{S_a}$, and by Schur–Weyl duality for GL_2, $(M_2^{\otimes a})^{S_a} = \bigoplus_\nu \mathrm{End}(U_\nu)$ over two-row partitions $\nu = (a-q, q)$, with $U_\nu = \mathrm{Sym}^{a-2q} \otimes \det{}^q$ and multiplicity $f_\nu = \binom{a}{q} - \binom{a}{q-1}$. Global copy symmetry confines the block on U_ν to the $\rho(S_3)$-invariant vectors $w_1, \dots, w_r$ of U_ν, so Ω is the family $X_{a,\nu} = \sum_{k,l} M_{kl} |w_k\rangle\langle w_l|$ with real symmetric $M = M_{a,\nu} \succeq 0$ (the ω-blocks; their entries are the parameters). For a test element $g = (g_0, g_1) \in \mathbb{C} \oplus M_2$ of $\mathcal{A}_3$,

$$P(g) = \mathrm{tr}[\Omega\, g^{\otimes n}] = \sum_{a=0}^{n} \binom{n}{a} g_0^{n-a} \sum_\nu f_\nu\ \mathrm{tr}[X_{a,\nu}\, \rho_\nu(g_1)],$$

and the normalisation and every mean purity π_k are fixed linear functionals of the parameters: $\sum_k \binom{n}{k} \pi_k\, t^k$ is P evaluated on the identity plus t times a copy transposition. The flipped moment of $z \otimes z \otimes \tilde{z}$ has generating polynomial $P(T^* g)$, where T^* is the per-site spin-flip map on test elements (a fixed rational 5×5 matrix), and the same block form with S_2-invariant vectors (copies $1, 2$): the γ-blocks $\Gamma_{a,\nu}$. Matching the two polynomial forms determines each $\Gamma_{a,\nu}$ as an exact rational linear function of the parameters (the *γ-map*); it is computed by isolating each type a on rational points of the unit circle and solving one square rational system per type. Level $m = 3$ is then

$$L_3(n, k) = \min\Big\{ \pi_k(M) :\ \text{normalised},\ M_{a,\nu} \succeq 0,\ \Gamma_{a,\nu}(M) \succeq 0 \Big\}.$$

The sizes are small: $(14, 12, 18)$ parameters, ω- and γ-blocks at $n = 7$, $(68, 41, 53)$ at $n = 13$, $(209, 87, 105)$ at $n = 19$ and $(502, 152, 176)$ at $n = 25$, with blocks of size at most 13. SW-1 checks the construction against genuine states: for a random integer state at $n = 4, 5$ the exact generating polynomial lies in the parameter family, its flipped values equal those of the γ-map, and its γ-blocks are positive semidefinite to floating tolerance (a regression, not a proof input); the Haar moment gives the Haar purity $(2^k + 2^{n-k})/(2^n + 1)$ exactly.

Theorem S114 (three-copy bounds for all n; exact certificates SW-0–SW-4)**.** *For every pure state of n qubits:*

(a) $n = 13$*:* $\pi_6 \geq 18171249/10^9 = 0.018171249 > 25/1408$*;* $n = 19$*:* $\pi_9 \geq 23947609/10^{10} = 0.0023947609 > 93/38896$*;*

(b) $n = 13$*:* $\pi_5 \geq 15699039/(5 \cdot 10^8) > 1/32$*;* $n = 19$*:* $\pi_7 \geq 78133177/10^{10} > 1/128$*;*

(c) $n = 25$*:* $\pi_9 \geq 4882879/(25 \cdot 10^8) = 0.0019531516 > 1/512$*.*

Consequently no pure 13*-qubit state is* 5*-uniform, no pure* 19*-qubit state is* 7*-uniform and no pure* 25*-qubit state is* 9*-uniform, that is, there is no pure* $((13, 1, 6))_2$*,* $((19, 1, 8))_2$ *or* $((25, 1, 10))_2$ *quantum code.*

Proof. For each case `data/three_copy_sw_certificates.json` gives λ and rational matrices $Z_{a,\nu}$, which SW-2 proves positive definite. SW-3 forms the exact residual $d = \pi_k - \lambda \cdot \mathrm{norm} - \sum\langle Z_{a,\nu}, \Gamma_{a,\nu}(\cdot)\rangle$, a linear functional of the parameters, and proves each of its ω-blocks positive definite. For a genuine state the parameters are normalised with $M_{a,\nu} \succeq 0$ and $\Gamma_{a,\nu} \succeq 0$, so $\pi_k - \lambda = \sum\langle D_{a,\nu}, M_{a,\nu}\rangle + \sum\langle Z_{a,\nu}, \Gamma_{a,\nu}\rangle \geq 0$. Since $\operatorname{tr}\rho_S^2 \geq 2^{-k}$ for $|S| = k$, with equality only for a maximally mixed marginal, a k-uniform state has $\pi_k = 2^{-k}$; and a pure $((n, 1, d))_2$ code is exactly a $(d-1)$-uniform state. □

Code-theoretic priority. The nonexistence of pure $((13, 1, 6))_2$ and $((25, 1, 10))_2$ codes is *not new.* Huber's 2017 dissertation [8, §5.11.4] explicitly excludes those code parameters, and $((31, 1, 12))_2$, by applying Rains's shortening theorem for pure quantum codes and a linear programming nonexistence bound for the shortened codes. The later Terwilliger-algebra SDP certificates of Anglès Munné and Huber [9] independently exclude $((13, 1, 6))_2$ and $((19, 1, 8))_2$. Theorem S114(b,c) supplies a *different certificate family* and, importantly, the explicit *quantitative* inequalities $\pi_5(13) > 1/32$, $\pi_7(19) > 1/128$, and $\pi_9(25) \geq 4882879/(25 \cdot 10^8) = 1/512 + 133/(5 \cdot 10^9)$. These quantitative lower bounds concern *all* pure states, not just their exact uniformity status. The three-copy SDP is tighter than the Rains LP for the 25-qubit 9-body problem, but that fact does not imply a new code-nonexistence theorem. The seven-qubit three-copy numerical value is $125/978 > 1/8$ (TCN-1, TA-1), compatible with the known nonexistence of AME$(7, 2)$. The quantitative bounds in part (a) do not require the previous code-nonexistence results.

Where three copies beat the Rains LP (TA-1; numerical). The Rains LP is level L_2 (Proposition S43). For $4 \leq n \leq 20$ and $k = \lfloor n/2 \rfloor$, L_3 agrees with L_2 to 10^{-12} except at $n = 7$, where $L_3 = 0.1278118609407\ldots$ (agreeing with $125/978$ to 10^{-16}), at $n = 13$, where $L_3 = 0.0181712542184\ldots$, and at $n = 19$, where $L_3 = 0.0023947628089\ldots$. For $n \leq 18$ every other L_2 is attained by a graph state (Table S6 and Theorem S105), so $L_3 = L_2$ is forced there. The three exceptions are exactly the cases $n \leq 20$ in which the LP optimum would require a nonexistent code: $((7, 1, 4))_2$, $((13, 1, 6))_2$ and $((19, 1, 8))_2$ ([2] and Theorem S117). At $n = 20$ level three gives no obstruction to the LP equality profile, consistent with its open status; nor (numerically) at $n = 23, 24$ (at the uniformity levels $k = 8, 9$ of the putative codes, not $\lfloor n/2 \rfloor$), while at $n = 25$, $k = 9$ it excludes the code (Theorem S114(c)): so far it obstructs exactly at $n \equiv 1 \pmod 6$. The double-precision solver loses accuracy from $n = 15$; TA-1 uses an exactly re-centred multiprecision solve (`search_three_copy_sw_certificate.py`).

S18.2 The thirteen-qubit interval

Proposition S115 (Product Pauli postselection convexity for stabilizer states)**.** *Let* $|\Omega\rangle$ *be a pure n-qubit stabilizer state, and let M be a set of t sites with $\rho_M = I/2^t$. On each $i \in M$ choose an arbitrary pure vector $|u_i\rangle$ with Bloch coordinates $r_i = (r_{i,x}, r_{i,y}, r_{i,z})$, and set* $|\psi_u\rangle =$

$2^{t/2}(\bigotimes_{i\in M}\langle u_i|)\Omega\rangle$. *For each* $a\in\{x,y,z\}^M$, *let* ψ_a *be the normalized state obtained by projecting every* $i\in M$ *onto either eigenstate of* σ_{a_i}. *Then, for every subset* T *of the unmeasured sites,*

$$\operatorname{tr}\rho_T(\psi_u)^2=\sum_{a\in\{x,y,z\}^M}\left(\prod_{i\in M}r_{i,a_i}^2\right)\operatorname{tr}\rho_T(\psi_a)^2.$$

Thus the entire subsystem-purity profile after arbitrary product postselection on M *is in the convex hull of the* 3^t *product-Pauli profiles. No such product measurement can improve the minimum of any fixed linear combination of these purities beyond its best product-Pauli measurement. The one-site statement is the case* $t=1$.

Proof. Expand the stabilizer density operator as $|\Omega\rangle\langle\Omega|=2^{-n}\sum_{g\in\mathcal{S}}g$. The normalized postselected density is

$$\rho_u=2^{-(n-t)}\sum_{g\in\mathcal{S}}\left(\prod_{i\in M}\langle u_i|g_i|u_i\rangle\right)g_{M^c}.$$

The condition $\rho_M=I/2^t$ says that no nonidentity stabilizer is supported inside M. Hence distinct stabilizers have distinct Pauli restrictions to M^c, up to their fixed signs. In the Pauli expansion of ρ_T, each nonzero coefficient is therefore a single product $\pm\prod_{i\in M}r_{i,b_i}$, with $r_{i,I}=1$ and $b_i\in\{I,x,y,z\}$. Purity is the sum of the squares of these coefficients divided by $2^{|T|}$. Since $\sum_{a_i=x,y,z}r_{i,a_i}^2=1$ for each measured site, expanding the identity factors gives the displayed convex combination. Changing a Pauli outcome only changes coefficient signs, so every ψ_a has the same purity profile for its 2^t possible outcomes. □

Proposition S116 (exact thirteen-qubit interval; LP-13, QP-13, SW-2–SW-4)**.** *For all pure thirteen-qubit states,*

$$\frac{18171249}{10^9}\ \le\ \min_{\psi}\pi_6(\psi)\ \le\ \frac{85}{4576}.$$

The upper bound is attained by a stabilizer state obtained by measuring X *on vertex* 4 *(indexed from* 0*) of the 14-vertex graph in Theorem S103. The Rains LP gives only* 25/1408 *(Theorem S117).*

Proof. The lower bound is Theorem S114(a). LP-13 checks an exact rational Rains dual for 25/1408 as in Theorem S103. For the X-projection witness, QP-13 counts the restricted graph stabilizers exactly. Its Pauli-weight distribution has $A_0=1$, $A_1=\cdots=A_4=0$, $A_5=15$, $A_6=204$. Formula (S15.1) gives

$$\pi_6=\frac{\binom{13}{6}+8A_5+A_6}{64\binom{13}{6}}=\frac{85}{4576}.$$

QP-13 also checks all 14×3 choices of distinguished qubit and Pauli measurement and, by Proposition S115, establishes that this is the best projection from this particular 14-qubit stabilizer witness in *any* one-qubit pure basis. No global thirteen-qubit optimality is asserted. □

Theorem S117 (quantitative Rains-LP gaps at 13 and 19 qubits; LP-13, LP-19, SW-2–SW-4)**.** *The unrestricted balanced-purity minima satisfy*

$$\boxed{\frac{25}{1408}<\frac{18171249}{10^9}\le\min_{\|\psi\|=1}\pi_6(\psi)\le\frac{85}{4576}\quad(n=13),}$$

$$\boxed{\frac{93}{38896}<\frac{23947609}{10^{10}}\le\min_{\|\psi\|=1}\pi_9(\psi)\le\frac{15201}{5912192}\quad(n=19).}$$

The values 25/1408 *and* 93/38896 *are the exact optima of the Rains* linear programmes. *The improved nineteen-qubit upper bound is attained by the explicit graph in* `code/verify_nineteen_graph_improvement.py` *(NG-1–NG-3).*

Proof. The middle inequalities are Theorem S114(a); the upper bounds are Proposition S116 and the nineteen-qubit graph of NG-1–NG-3. The exact LP-13 dual of Proposition S116 has strictly positive Pauli-weight slack coefficients for $j = 1, \dots, 5$. Thus equality in the LP would force $A_1 = \cdots = A_5 = 0$, and hence every five-body reduced state would be maximally mixed. Such a vector would be a pure one-dimensional quantum code with parameters $((13, 1, 6))_2$. A second, qualitative proof that the LP value is not attained is the nonexistence of $((13, 1, 6))_2$ codes, proved by Anglès Munné and Huber [9] and independently by Theorem S114(b), together with compactness.

For $n = 19$, the script LP-19 checks the exact rational dual identity

$$a_9 - \frac{93}{38896} A_0 = \sum_{s \in \{0,1,2,7\}} e_s (a_s - a_{19-s}) + \sum_{i \in \{0,1,2,3,4,8,12,16\}} y_i S_i + \sum_{j=1}^{7} w_j A_j, \quad y_i > 0, \ w_j > 0.$$

All rational multipliers are literal in the script, and every coefficient of $A_0, \dots, A_{19}$ is checked exactly. Starting from the graph witness specified by nineteen hexadecimal adjacency row masks in GS-19, toggle precisely the three edges $(2, 6)$, $(2, 16)$, and $(2, 18)$ (vertices indexed from zero). NG-1–NG-3 check the resulting symmetric binary adjacency matrix, enumerate all 2^{19} stabilizer elements, and obtain the improved purity numerator 121608, giving $\pi_9 = 15201/5912192$. An independent enumeration of all $\binom{19}{9} = 92378$ cut ranks yields $6^2\, 7^{840}\, 8^{26696}\, 9^{64840}$ and the same purity. The improvement over the earlier GS-19 upper bound $449/173888$ is exactly $5/454784$. Neither graph is claimed globally optimal.

Any state attaining the LP bound at $n = 19$ would have $A_1 = \cdots = A_7 = 0$, hence would define a pure $((19, 1, 8))_2$ code, which [9] and Theorem S114(b) exclude: the same qualitative second proof. □

S18.3 A thirteen-qubit entropy witness

Let G_{13} be the graph obtained from the fourteen-vertex graph of Theorem S103 by pivoting on edge $(4, 1)$ and deleting vertex 4, with the remaining vertices relabelled $0, \dots, 12$ in order. This is the graph rule for the X measurement of vertex 4, so up to local Clifford gates G_{13} is the witness of Proposition S116; as $\pi_6(G_{13}) = 85/4576 > 2^{-6}$, some of its balanced cuts are rank deficient. A superposition of 256 graph-basis vectors of G_{13}, all fixed by five graph stabilizers, removes every rank deficiency and raises the attained mean entropy above 5.8336 bits. The five stabilizers make each balanced marginal unitarily block diagonal with blocks of size at most eight, so the entropy is enclosed from small exact blocks.

Lemma S118 (Pauli-code compression of a marginal; proved)**.** *Let $\psi \in (\mathbb{C}^2)^{\otimes k} \otimes (\mathbb{C}^2)^{\otimes m}$ be a joint eigenvector of commuting Pauli operators $P^{(j)} = P_A^{(j)} \otimes P_B^{(j)}$, $j = 1, \dots, r$, and let $\rho_A = \mathrm{tr}_B\, |\psi\rangle\langle\psi|$. Then ρ_A commutes with every $P_A^{(j)}$. Suppose the group generated by the $P_A^{(j)}$ modulo phases is also generated by Paulis $\bar{X}_1, \bar{Z}_1, \dots, \bar{X}_e, \bar{Z}_e, \bar{Z}_{e+1}, \dots, \bar{Z}_{e+c}$, which commute except that $\bar{X}_i$ and $\bar{Z}_i$ anticommute for $i \le e$. Extend this set to a full canonical set $\bar{X}_i, \bar{Z}_i$, $1 \le i \le k$, and let V be a matrix with $V^\dagger V = dI$, $d > 0$, such that $V^\dagger \bar{Z}_i V/d$ and $V^\dagger \bar{X}_i V/d$ are the standard Z_i and X_i. Then*

$$\frac{V^\dagger \rho_A V}{d} = I_{2^e} \otimes \bigoplus_{u \in \mathbb{F}_2^c} D_u, \qquad \dim D_u = 2^{k-e-c},$$

where the first factor is qubits $1, \dots, e$ and u labels the computational values of qubits $e+1, \dots, e+c$.

Proof. If $P^{(j)}\psi = \lambda\psi$, then $|\lambda| = 1$ and $(P_A^{(j)} \otimes I)\psi = \lambda (I \otimes P_B^{(j)\dagger})\psi$, so

$$P_A^{(j)} \rho_A P_A^{(j)\dagger} = \mathrm{tr}_B\big[(I \otimes P_B^{(j)\dagger})|\psi\rangle\langle\psi|(I \otimes P_B^{(j)})\big] = \rho_A.$$

Commuting with a Pauli does not depend on its phase, so ρ_A commutes with the generated group. After conjugation by $V/\sqrt{d}$ it commutes with X_i, Z_i for $i \le e$, hence is the identity on those qubits, and with Z_i for $e < i \le e + c$, hence is block diagonal in their computational values. □

Corollary S119 (Thirteen-qubit entropy witness; verified arithmetic E13L-1–E13L-2)**.** *Let g be the graph state of G_{13} and $g_v = Z^v g$, $v \in \mathbb{F}_2^{13}$, its orthonormal graph basis, with qubit i as bit i of v. Let a_v, b_v be the integer coefficients in* `data/thirteen_entropy_local.json`*, ordered by increasing labels v in the binary annihilator of*

$$3987, \quad 3302, \quad 3039, \quad 2939, \quad 968.$$

The normalized state

$$\zeta = \frac{\sum_v (a_v + ib_v) g_v}{\sqrt{\sum_v (a_v^2 + b_v^2)}}$$

has every balanced marginal positive definite, with all eigenvalues above $1/2100^2$, and

$$5.83360583836270199 < E_6(\zeta) < 5.83360583836270200.$$

Hence the thirteen-qubit balanced-entropy maximum satisfies $\max_{\|z\|=1} E_6(z) > 5.8336$.

Proof. Write $K_j = X_j Z^{N(j)}$ for the graph stabilizers and $K^w = \prod_j K_j^{w_j}$. Since $K_j Z^v = (-1)^{v_j} Z^v K_j$, one has $K^w g_v = (-1)^{w \cdot v} g_v$. Every occupied label annihilates the five displayed words, so ζ is fixed by their five graph stabilizers, which are $\pm X^w Z^{\sum_j w_j N(j)}$. Orthonormality of the graph basis makes the denominator exact. The computational amplitudes are $\langle x|\zeta\rangle = (-1)^{q(x)} \sum_v (a_v + ib_v)(-1)^{v \cdot x} / \sqrt{2^{13} \sum_v (a_v^2 + b_v^2)}$, with q the edge form of G_{13}; E13L forms the integer numerators by a Walsh transform and checks the exact norm identity.

For each of the 1716 balanced cuts $A|\bar{A}$, E13L restricts the five stabilizers to A, checks that the restrictions are independent, and computes a canonical set by symplectic elimination whose span equals theirs. It builds a Gaussian-integer V_A with $V_A^\dagger V_A = d_A I$, d_A a power of two, and checks exactly that V_A conjugates every canonical Pauli to its standard form. Lemma S118 then gives

$$\rho_A \simeq I_{2^e} \otimes \bigoplus_{u \in \mathbb{F}_2^c} D_u, \qquad 2^e \sum_u \operatorname{tr} D_u = 1,$$

with $(e, c) = (1, 3)$ at 312 cuts and $(e, c) = (2, 1)$ at 1404 cuts. The blocks D_u are therefore 4×4 or 8×8, and $S(\rho_A) = 2^e \sum_u S(D_u)$. E13L encloses the entries of each D_u in Arb from the exact normalized amplitudes, isolates its eigenvalues rigorously and checks that each exceeds $1/2100^2$. This proves positive definiteness. Summing the enclosed $-\lambda \log_2 \lambda$ over all blocks and cuts gives $E_6(\zeta) \in [5.8336058383627019939735 \pm 10^{-40}]$, which lies strictly inside the stated interval. □

Remark S120 (Scope)*.* The state ζ is not claimed stationary, and no global thirteen-qubit entropy bound is asserted here; the global maximum remains open. The script E13L also checks Hessian and roundoff data (E13L-0, E13L-3–E13L-5) and prints a conclusion about a nearby local maximum; that statement is not part of this supplement, and only E13L-1–E13L-2 are used here.

S19 Lean formalisation

The formalisation is the `lean/` folder of the companion repository `cmbant/qubit-entropy`, a standalone Lake project (`lake build`) using Lean 4 and Mathlib, whose theorem docstrings give the TeX labels of the paper and this supplement. Table S8 lists what it proves, with the main Lean declarations; the companion README gives the build commands. The default `lake build` omits the final overlap certificate and the unconditional quadratic-variance certificate; these have separate targets described below. The formalised results use only Lean's standard three axioms (`propext`, `Classical.choice`, `Quot.sound`), except for the finite compiled checks behind Theorems 18 and 27 and any results that depend on them. For the first, three finite checks of the certificate data (the identity (S7.7) with the Gram test (S7.8), the target polynomial, and the constraint matrices) are evaluated by compiled code (`native_decide`, which adds one axiom per

check); their soundness and the whole reduction to the normal slice are kernel-checked. The 32-variable polynomial identities behind (S) and (FGI) are split into 7 and 22 pieces, each checked by kernel evaluation in its own declaration.

For Theorem 27 the formalisation proves the identity (S6.1) for all z without the dimension count of Section S6.1. The difference of its two sides is written as a sesquilinear form $\overline{z^{\otimes 6}}^{\mathsf{T}} K z^{\otimes 6}$ with an integer matrix K assembled from the covariants Φ_{ki} and from copy permutations; the form vanishes once $PKP = 0$ for the symmetriser P of the six copies. The integer spin intertwiner of $(\mathbb{C}^2)^{\otimes 6}$ block-diagonalises each qubit factor into sectors of twice-spin $0, 2, 4, 6$, and $PKP = 0$ becomes one integer matrix identity in each of the 164 spin sectors that carry symmetrised vectors, with bases of their images certified by a character average and pivot certificates. These finite checks run as one compiled checker (`native_decide`); a second compiled check identifies its intertwiner array with the mathematical definition. The remaining table facts are kernel-evaluated. The soundness of the checker, the block calculus, and the deduction of (QV), (QK) and the Rényi envelope above one are kernel-checked.

To include these results, build `FourQubitEntropy.QVEqualityMain` separately (it imports `FourQubitEntropy.QVCheck.Unconditional` and adds the equality and optimality statements of (QV) and (QK)), and, after generating the overlap data as directed in the companion README, `FourQubitEntropy.GOGlue.Final`. Their consequences inherit the same compiled checks: in particular, the unconditional Rényi envelope inherits the quadratic-variance checks, and the unconditional global orbit-stability bounds inherit the overlap checks.

The analytic arguments and certificates of the paper do not depend on the Lean development.

Table S8: Scope of the Lean formalisation.

Topic	Formalised statements	Main declarations
Four-qubit maximum	Theorem 1: the bound, attainment by M_4 and the equality case; one-cut rigidity on every cut (Proposition 4); the critical normal form (Theorem 16).	`meanEntropy_le`, `meanEntropy_eq_max_iff`, `one_cut_rigidity_12`, `critical_normal_form`, `critical_iff_LU_Gv`
Moment inequality	Theorem 3: the sharp $F \geq \frac{3}{4} N W_1$ from (S), optimality of $\frac{3}{4}$ at product states, the zero set of F (Proposition 14) and the sharp third-moment lemma.	`sexticF_ge`, `sexticF_ge_sharp`, `moment_zeros`, `third_moment`
Envelopes	The spin-flip derivative identity (FGI) (Lemma S1); the general envelope of Theorem 15 in the homogeneous form (GE), with optimality of its coefficient $\frac{7}{8}$ and its complete equality classification; the critical envelope (Proposition 13) in full: the closed form, its equality classification and the critical moment boundary.	`flipGradient`, `general_envelope_homog`, `general_envelope_coeff_iff`, `envelope_equality_iff`, `critical_closed_form`, `critical_envelope_eq_iff`
Orbit distance	The distance d to the M_4 orbits as a minimum, with $d^2 = 2 - 2\sqrt{\Phi}$; the fidelity floor (Lemma S27) with its attainment at GHZ; hence the global estimates of Theorem 19 and Corollary 22 (the local-radius clauses and the limiting coefficient are not formalised).	`orbitDist_sq`, `fidelity_floor`, `fidelity_floor_ghz_attained`, `entropy_gap_orbit_stability_dist_all`
HS overlap stability	Theorem 18, $Q \geq 1 - \Phi$ for every unit state: a closest representative on the 19-coordinate normal slice with all its constraints, Q on the slice, and the certificate (S7.7) on data generated from the certificate file.	`globalOverlap_all`, `globalOverlap_of_frame_cert`, `go_certificate_sound`
Critical orbit stability	Theorem 20 with its sharp constants (the case $\Phi \geq \frac{1}{2}$ from Theorem 18) and its null-critical refinement (8), attained at the cluster state.	`critical_orbit_sharp_dist_all`, `Phi_eq_max`, `bell_product_sharp`, `null_critical_sharp`

Topic	Formalised statements	Main declarations
Spectral extensions	The cubic-majorant criterion with its equality clause (Proposition 6); Tsallis $0 < q < 2$ and Rényi $0 < q < 1$ with equality cases; failure of M_4 for Tsallis $q > 2$; the quadratic deficit bounds in W_1 and τ; the rank-two and $q \geq 2$ Rényi bounds; the HS–cluster crossing $1587/1000 < q_* < 1589/1000$; Lemma 31 and Proposition 32 with their equality cases, the necessity of q_b and $\frac{3}{2} < q_b < \frac{313}{200}$; the single crossing (Lemma 25) with its equality case and the order transfer (Corollary 26), also for the three cut spectra of a state.	`majorant_le`, `majorant_eq_iff_of_pos`, `tsallis_le`, `renyi_le`, `tsallis_cluster4_gt`, `entropy_stability`, `renyi_crossing`, `renyi_grouping_eq_iff`, `renyi_cubic_eq_iff`, `renyi_cubic_iff`, `single_crossing_statement`, `meanRenyi_order_transfer`
Rényi above one	The envelope $\mathcal{E}_q \leq \max\{h_q, 5/3\}$ of Corollary 9 for every $q > 1$, with its equality cases (HS equality as the two M_4 orbits, cluster equality spectrally), from (QK); within it: the band-L bound (Lemma S7, from the bounded-head bound of Lemma S9 and the large-head Lemmas S10 and S11), the capped cubic bound (Lemma S12), the binary quartic lemma at q_s (Lemma S8, by a real-variable route that differs from the Cauchy-formula proof in the text), the HS supporting inequality (Lemma S6, with the low-tilt reduction), the cluster sliver (Lemma S17), the dual bound (Lemma 34), determinant monotonicity (Lemma 29), the near-uniform dual bound (Lemma S16), the large-moment-slack regime (Corollary 33) and the case analysis of Theorem 36.	`renyi_envelope_above_one_unconditional`, `renyi_envelope_above_one_QK`, `largeHeadDual`, `smallHeadDual`, `det_reduction`, `nearUniformDual`, `meanRenyi_high_F`
Quadratic variance	The sharp bound (QV) of Theorem 27 through the identity (S6.1) for every z, with the certificate data committed as Lean literals generated from the certificate file and the identity checked by compiled code, its equality along the cluster-to-Bell family and optimality of the coefficient 3; (QK) of Corollary 28 with its equality classification and both coefficients sharp; the critical-state bound of Proposition S21 with its equality case and attainment.	`qvCertIdentity`, `sharpQuadraticVarianceBound`, `quadraticVarianceBound`, `checkAll_true`, `qv_coefficient_iff`, `qk_equality_classification_product`, `qk_coefficients_sharp`, `critical_rf`
Convex roofs	The von Neumann, Tsallis ($0 < q < 2$) and Rényi ($0 < q < 1$) roof ceilings, attained only by pure states in the two M_4 orbits (Corollary 10).	`convexRoof_vn_eq_iff` and its Tsallis and Rényi analogues
Spin-flip bounds	Both inequalities of (TF), their sharpness, the equality case on the critical locus and the Schmidt-rank-two consequence and the identification of states with two maximally mixed cuts with the cluster-to-Bell family, with their Rényi and von Neumann values (Proposition 40); the spectral spin-flip bound (Proposition 38) and the overlap inequality behind it (Lemma 39, with attainment); Lemma 17.	`two_flat_TF_12`, `two_flat_eq_iff_12`, `two_flat_QF`, `two_flat_renyi`, `spectral_spinflip_12`, `isospectral_overlap_rect`, `real_gram_rect`
Icosahedral family	Normalisation, maximally mixed one-qubit marginals, and the purity and spectrum of every pair and three-qubit marginal; hence $\min_x \pi_3(\psi_x) = 121/917$ at $x = 37/524$, and $b_3'(37/524) = -432/120127$. The family entropy $E_3(x)$ is strictly concave with unique maximiser x_*, the root in $(0, \frac{1}{2})$ of $2^{32}(1-2x)^{15} = 3^8(1+8x)^8(1+28x)^7$, with $0.0698287 < x_* < 0.0698288$ and $2.9653560294776 < E_3(x_*) < 2.9653560294777$ bits.	`icosahedralState_eq_ico`, `pairCut_spectrum`, `star_value`

Topic	Formalised statements	Main declarations
Seven-qubit reductions	The global joint scalar majorant with its equality set (Lemma S82, AH-6); the spectral pair (Proposition S76); the joint spin-flip identity on the family (Lemma S80); the minimal majorant degree (Proposition S81); the eigenvalue cap of Corollary S78 for pure states. Assuming $\pi_3 \geq 121/917$: the competitor band (Proposition S79), so every pure state with E_3 above the SC-2 threshold has purity in the band and eigenvalues at most 497/1000, and E_3 is bounded by the joint functional; with the localised bound E7-1 as a further hypothesis this gives $E_3 \leq E_3(\psi_{x_*})$ for every state.	`joint_scalar_majorant`, `joint_scalar_majorant_eq_iff`, `seven_entropy_max_reduction_joint`, `spinflip_spectral_pair`, `tripleCut_spinflip_family`, `majorant_degree_isLeast`, `eigenCap_497_pure`, `competitor_purity_band`
Spin-flip doubling and five-copy steps	Theorem S51 and Corollary S52; the tangent-line lemma (Lemma S63) in abstract form, for any copy number and number of qubits; the moment-level step of Theorem S66; the family relations of Proposition S97.	`spinflip_double_odd`, `ordinary_triple_purity`, `tangent_line`, `equality_moment_centred`, `zeroFocus_wronskian_unconditional`
Balanced-purity minima	For $4 \leq n \leq 18$, $n \neq 7, 13$: $\min \pi_{\lfloor n/2 \rfloor}$ equals the value of Table S6 (Theorems S101, S103, S104 and S105), with Rains' shadow inequalities and the graph-state purity $2^{-\mathrm{rk}_2}$ proved rather than quoted.	`minPurity`n, `shadow_nonneg`, `purity_gsn`
Entropy–purity slope	For every $d \geq 3$ the sharp slope (S12.5) of Lemma S77 and, for every $d \geq 2$, the fixed-purity envelope of Lemma S75, both with their equality cases; hence the entropy ceilings of Table 5 for $8 \leq n \leq 18$, $n \neq 13$, from the formalised purity minima.	`entropy_purity_slope_bits`, `entropy_le_gEnv`, `meanEntropy_le`, `entropyCeiling`n
Odd-n tradeoff	The identity of Theorem S112 for every odd $n \geq 5$, with the coefficients c_n and its equality case.	`odd_tradeoff_sharp`, `odd_tradeoff_sharp_eq_iff`
Even-n tradeoff	The bound of Theorem S108 for $n \in \{8, 10, 14, 16, 20\}$, with its sharpness, the optimality of c_n for L_n and Corollary S109.	`purity_spinflip_tradeoff_even`, `purity_spinflip_tradeoff_even_sharp`
Not formalised: four qubits	The local-radius and limiting-coefficient clauses of Theorem 19 and Corollary 22 (Lemma S28); Theorem 21 and its covering proof (Proposition S40 and its lemmas); the critical fidelity profile (S8.8) (Lemmas S30 and S31); Theorem S32 for $\theta > 0$; Proposition S29; the near-one proof of Theorem S18 (Lemma S19, the defect estimate (RE) and the Rényi clause of Corollary 22) and Corollary S24, although the bound and equality case of Theorem S18 follow from the formalised envelope; Proposition S3; the equality and defect clauses of Proposition S22 (its inequality is (QV)) and the equality clause of Theorem S25; Lemma S26; the Rényi roofs for $1 < q < q_*$ in Corollary 10.	

Topic	Formalised statements	Main declarations
Not formalised: seven qubits	Theorems 11 and 12 (FP5-1, E7-1, attainment and the longer decimal; only the family values and the reductions under explicit hypotheses above are formalised), with the five-copy chain (Lemmas S56, S57, S59, S61, Propositions S58, S60, Theorem S62, Corollary S67), orbit rigidity (Theorem S64, Corollaries S65 and S68; of Theorem S66 only the moment-level step in abstract form: the tangent line and the elimination of t by the centred square, given $H(X'_y) \neq 0$; the family value $H(\psi_y) = 155 + 96y$ (FO-1) and the passage to the orbit are not formalised) and the entropy certificates (Theorems S85, S86, Proposition S87, the five-copy part of Proposition S84, Corollary S83, the mixed-state clause of Corollary S78); full-space stationarity (Lemma S72); the Hessian, basin and stability results (Theorems S69, S71, S88, S89, Lemmas S70 and S91, Corollary S90, Proposition S92); Propositions 43 and S93, Corollary S94; Theorem S53 and Propositions S54 and S55.	
Not formalised: replica hierarchy and covariants	Proposition S43 and Theorem S44; Theorems S45 and S47, Corollary S46, Proposition S49; Proposition S95, Lemma S96, Proposition S97 beyond its family relations, Theorem S100; the Pauli weight distribution and Type II claim of Theorem S105 and the $n = 18$ entry of Corollary S107.	

S20 Verification checks and certificate status

The labels below identify the finite checks used in the paper and this supplement. They have different roles: *exact symbolic or rational checks* establish algebraic identities, ranks and coefficient signs; *verified-arithmetic checks* prove bounds using explicit rounding-error control; and *exploratory numerical checks* are diagnostics, not proof inputs. In particular, the five-copy identities for purity (FP5-1, rational) and for entropy (E7-1, whose coefficients involve the algebraic parameter x_* and logarithms) are verified-arithmetic certificates. The surrounding reductions and equality arguments are given in the proofs, not delegated to a table of scripts.

Table 9 of the paper lists which results use which checks. The companion's `verifiers/MAP.md` identifies the script, supporting data and committed output for each label. The groups NV, TCN, FS and TA are numerical evidence only, and no proof uses them.

S20.1 Checks of the paper's appendices

Table 9 of the paper lists which results use the checks below. The four-qubit von Neumann, Tsallis and single-crossing proofs do not assume a computational certificate; their checks confirm algebra. Finite decimals used as bounds specify exact rationals.

ME-1–ME-6 Exact polynomial arithmetic in the sixteen amplitudes and their conjugates: two independent sum-of-squares forms of F, coefficient by coefficient (ME-1, ME-2); the purity identity of Lemma 2 (ME-3); the Hermite coefficients a, b and the entropy-gap identity of Section 2.3 (ME-4); the values of (P, T, F) at M_4 and at a product of two Bell pairs (ME-5); and, on the critical slice, the cut spectra, the spin-flip amplitude $\sum_k v_k^2$, Clifford realisations of the coordinate symmetries and the equality solutions (ME-6).

SP-1–SP-5 The elementary swap derivation of Appendix A: the ϵ-maps, spin-flip transpose and antisymmetry relations behind (K) (SP-1); the generic two-qubit swap, cycle and partial-transpose contractions with the marginal trace identities and the pair norms (Pr) (SP-2); the eleven type coefficients of the table and the two class reductions (SP-3); the three contractions (C) and the identity (S) (SP-4); and the local polarisation norm (U), including degenerate Schmidt values (SP-5).

SI-1–SI-3 An independent check of (S): the tensors U_i, V_i and the pair vectors built from their definitions (SI-1), the identity compared coefficient by coefficient in the sixteen amplitudes and their conjugates (SI-2), and the product-state value $F = \frac{3}{4}NW_1$ (SI-3).

CE-1–CE-5 The critical envelope (CE): on $G(v)$, with $v, \bar{v}$ independent symbols, $F = \frac{1}{2}N^3 + \frac{7}{4}N\tau + \frac{9}{4}\operatorname{perm}\Theta$, $P = N^2 + \tau/2$, $\det\Theta = -N\tau$ and the eigenvalues N^2, τ, τ of $\Theta^T\Theta$ (CE-1); the sign-triple and cyclic-diagonal sums (CE-2); the expansion of $\operatorname{perm}(\kappa I + (1-\kappa)uu^T)$ and its Maclaurin value (CE-3); the twelve equality matrices O among the signed permutations, and the bound attained on all of them (CE-4). CE-5 is a numerical sanity check, not used.

PZ-1–PZ-2 Bloch-operator invertibility, the positive-Bloch support exclusion, and the generic GHZ identity $H = (9/2)N\tau$ of Lemma S2.

FI-1–FI-5 The spin-flip gradient identity (FGI) of Lemma S1: the dimension 19 of the $SU(2)^4$-invariants of bidegree $(4,2)$ (and 8 for $(2,2)$) as an exact character constant term (FI-1); 19 site-wise ϵ/δ contraction invariants with an exact rank-19 evaluation matrix at 24 integer points, with $\bar{z}$ replaced by an independent vector (FI-2); (S) in the normalisation used, $\|A\|^2 = F - \frac{3}{4}NW_1$ (FI-3); the identity at all 24 points in exact integer arithmetic (FI-4); negative regressions with the coefficient $\frac{9}{8}$ replaced by 1 or the multiple of s dropped (FI-5).

FG-6–FG-8 The integration proving Theorem 15: the passage from (FGI) to (FG), the pencil derivative, the scaling of W_1, the integrating factor and the final substitution (FG-7); the scalar rational identities giving the exact transport law (SC) and its positive-norm integral (SC-SOS) (FG-8). FG-6 is a finite-difference sanity check of (FG) and of the envelope at random states, not used in the proof. FG-1–FG-5 and FG-9, in the same script, check the direct proof of (FGI) in Section S4.

ESH-1–ESH-3 Optimality of the coefficient 7/8 in Theorem 15: on the GHZ family $c|0000\rangle + s|1111\rangle$, τ, W_1 and F computed exactly from the tensor definitions, and the slack with 7/8 replaced by C equals $(7/2 - 4C)\tau + (1 + 4C)\tau^{3/2}$; exact symbolic.

ER-1–ER-5 The equality case: the one-cut spectrum and spin-flip amplitude on $G(v)$ with the Pauli actions that move its coordinates (ER-1); the weighted triangle-deficit identity (ER-2); the entropy-gap identity and the fact that $(\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ is the only spectrum in $\{\frac{1}{6}, \frac{1}{2}\}$ with trace one (ER-3); the critical-slice factorisation (Z)'s second proof of Proposition 14 against direct partial traces (ER-4); and the vanishing of F on every listed zero representative together with the determinant ratio that separates M_4 from $\overline{M_4}$ (ER-5).

CN-1–CN-4 The normal form of Appendix B: the magic-basis generators and moment maps (CN-1); the amplitude formula $G(v) = G_{abcd}$ (CN-2); the block-elimination identities of the real singular-value lemma (CN-3); and rank-deficient and repeated-block rectangular examples (CN-4).

EF-1–EF-7 The spectral corollaries of Section 3: the general Hermite coefficient identity (T) (EF-1); the cluster-state cut spectra (EF-2); the tie at $q = 2$, the Tsallis advantage 1/144 of the cluster state at $q = 3$ and the all-q moment identity (EF-3); the von Neumann and $q = 2$ limits and the stability constant 4.57862 (EF-4); the derivative and logarithmic-derivative identities, endpoint values and rational logarithm bounds of the Tsallis sign proof (EF-5); the strengthened polarisation and spin-flip coefficients, including the small-τ values 0.454 and

0.602 with and without the envelope term (EF-6); and the strengthened defect identity with the leading coefficient of the determinant pencil used in the mixed-state exclusion (EF-7).

RA-1–RA-5 The fourth-derivative identity, three endpoint derivative numerators, fifteen printed coarse Taylor-coefficient bounds and their tails in Lemma S20 (RA-1), logarithm absorption, moment-defect algebra, the HS–cluster crossing brackets of Proposition 24: the integer comparisons giving $3/2 < q_* < 8/5$ and the exact enclosures giving $1587/1000 < q_* < 1589/1000$ (RA-4), and the strict rational constants in the cap-free purity bound (RA-5). The concavity, derivative-shape and Hermite-remainder proof is analytic; no parameter grid is used.

SCB-1–SCB-5 Generic complex-matrix calibration algebra in dimensions two through six, the trace-rearrangement gap identity, exact unitary regressions, the generic two-qubit Pauli spin-flip identity, the physical sharpness family, and a classical counterexample to deriving QK from the separate-cut caps. Positivity and the arbitrary-dimensional overlap bound in Section F.9 are analytic.

UR-4 The quantitative crossed-purity algebra, including the unrestricted $W_1/4$ term and the critical rank-two equality case, for Proposition 40; the rigidity argument itself is analytic.

OT-1–OT-4 Exact symbolic algebra in the analytic single-crossing proof (Lemma 25, every $q > 1$): the family formula, its derivative and the gradient profile (OT-1); the Euler identity and the second variation splitting equal coordinates (OT-2); the hyperbolic identity (SR), its contact value, the derivative of $\sinh(sz)/z$ and the integrating factor (OT-3); and a negative control where increasing κ_q by $1/1000$ destroys HS stationarity (OT-4). Full support, exclusion of repeated larger entries and all sign and equality arguments are proved analytically; these checks are not proof inputs.

SHD-1–SHD-4 The interval enclosures of the eight node values of the binary quartic correction and $H_q(1/3, 2/3)$; the grouping constants and dual identities; the exact Bernstein coefficients of the eight one-variable quartics (SHD-3, with a negative control); and the large-tail coefficient bounds and the SHD column of Table S4 (Lemma S15) in Lemma S9. The monotonicity of A_q (Lemma S8, by Cauchy's formula), the grouping and global derivative arguments are analytic.

LHD-1–LHD-4 Lemmas S10 and S11. LHD-1: the identities for the biased binary fourth derivative $J(x, y)$, the range of K, the three envelopes of y, and the Bernstein expansions of degree 12, 16, 16 on $[0, 3/5]$ with all coefficients positive (with a negative control). LHD-2: the polynomial with positive coefficients at $a = 1/2$. LHD-3: the monotonicity constants in K, the cubic minorant κ, the logarithm constants and error budgets. LHD-4: the two Bernstein expansions of degree 13 and 26 behind the completed square, each with all coefficients at least the stated bound. The Hermite remainder signs and the quasi-convexity of J are proved in the text; no subdivision or external coefficient file is used.

BC-1–BC-3 Exact grouping derivative cancellation, geometric-convexity and cubic grouping identities, equality contacts, lower-order negative control, and the high-F transfer. Log-sum, the single-crossing inference and the global entropy bound are analytic in Section F.7. This covers the analytic high-F case $20\varepsilon + 2F \geq 1$ of Theorem 36 without interval arithmetic.

DA-1–DA-5 Exact-symbolic HS tangency, cluster contacts, evenness, dual and sliver algebra; exact comparison of every printed multiplier and band with the executable data; closed-band coverage and finite crossing brackets; exact wrong-multiplier and shifted-contact negative controls.

CV-1–CV-5 An independent quaternion parametrisation of the cut spectra on $P = 1$ (CV-1, CV-2), cyclic purity and moment identities, the cubic discriminant and nonnegative geometric-mean transfer remainder, and the full equality-family data for Proposition S21. The cut-spectrum formula and equality inference are analytic.

CJ-1–CJ-5 Directional permanent and geometric-concavity identities, the Fourier variance factor and shared-unitary block expansion, the scalar factorisation and quadratic discriminant, the sharp cluster-to-Bell family, and a scalar-only negative control for Proposition S22. The Cauchy–Schwarz bounds, global sign and equality arguments are analytic; these checks use no coefficient certificate.

QVA-1–QVA-4 The compact universal certificate for Theorem 27: exact six-copy SU(2) intertwiners and vanishing at HS and cluster (QVA-1); positivity of the denominator and 95 integer norm weights (QVA-2); and the exact identity (S6.1) at the 83 determining Gaussian-integer states of QG (QVA-3). There are 95 squared norms in 14 spin sectors, with denominator $D = 10235456181520$ and positive integer weights of at most 14 decimal digits. Their sparse multiplicity vectors have 716 nonzero integer entries, all of absolute value at most 22. There are no Gram matrices or matrix positivity checks. QVA-4 retains the invariance regression for every squared norm under conjugation, qubit permutations and four Gaussian-integer local maps per qubit. The replay uses integer arithmetic throughout. The data encodes each highest-weight row by its negative and positive supports in the zero-based six-bit computational basis. For $j = 0, 2, 4, 6$ these give $5, 9, 5, 1$ rows H_j, respectively; each is supported at Hamming weight $(6 - j)/2$ and killed by the raising matrix E. Concatenate $H_j E^k/k!$ in increasing j, then $k = 0, \ldots, j$, to obtain C_6. For each sector, `a` and `spins` specify $W_{a,p}$; each entry of `vectors` is a flat list $[j_1, c_1, \ldots, j_s, c_s]$ specifying $\sum_t c_t e_{j_t}$ with zero-based multiplicity indices, and the corresponding entry of `weights` is its positive coefficient q_{ki}. The proof needs no equality-curve jets, auxiliary rank computation or coefficient solve.

EU-1, EU-2 The sharp entropy–purity slope of Lemma S77 for every dimension: the one-spike identities for N'', $F_m(r_*)$, $F_m'(r_*)$ and c_d (EU-1); and the entropy ceilings of Table 5, evaluated at 60 digits and rounded up (EU-2).

QG-1–QG-5 Mixed-sextic recoupling, every permuted HS/cluster zero, and the nonsingular 83-point evaluation matrix of 83 raw Gram entries built from five-copy blocks, multiplied by N or coupled to one further copy (QG-1, QG-3): these make the 83 states determining for the invariant space of Section S6.1. The same script checks an independent direct proof of (QK): 80 squared covariant norms in 16 modules with strictly positive rational coefficients from an exact 80×80 solve, and a signed norm identity forcing $P \in \{1, 3/2\}$ at equality (QG-2, QG-4). QG-5 checks invariance of every raw Gram entry under complex conjugation, qubit transpositions and five Gaussian-integer multiples of local unitaries on full-support states (a regression of the construction). The ambient dimension 85 follows from (ID3) and SD-4.

RG-1–RG-3 The correlation geometry of Appendix C: the Pauli trace formulas for $\operatorname{tr} R^2$ and $\operatorname{tr} R^3$ of an arbitrary two-qubit matrix (RG-1); the determinant identity (CV) (RG-2); and the generalised eigenvalues $4, -47, -47$ of the replica witness K_{rep} on the three pairings (RG-3).

IC-1–IC-6 The icosahedral family of Appendix H: one-qubit marginals, the quadratic $\pi_3(\psi_x)$ and its exact minimiser $x = 37/524$, the marginal and pair purities, the computational-basis block diagonalisation of the 35 triple and 21 pair marginals (proved analytically from $C^2 = 5I_6$ in Theorem 41), and the exact signs of both entropy derivatives in Corollary 42.

EG-1 Interval-arithmetic enclosures of the gap $E_3(\psi_{x_*}) - E_3(\Psi_{\mathrm{ico}})$ and of $\pi_3(\psi_{x_*}) - 121/917$ on the icosahedral family, from its explicit spectra.

GS-4 The MacWilliams and shadow-integrality identity for seven-qubit stabilizer states and an explicit seven-vertex graph witness attaining $\pi_3 = 19/140$; the lower bound is proved analytically in Proposition 43.

SD-1, SD-2, SD-4 Dimensions of the spaces of local-unitary invariant polynomials of four qubits for copy numbers up to 8, by a character average and independently by a multiplicity count

(SD-1, SD-2), and the qubit-symmetric conjugation-even and odd counts of (ID3) (SD-4). The degree-six even count 85 is a proof input for QVA and QG.

Orbit-stability certificates

The entropy-maximum proof is analytic; the quantitative stability comparisons use exact or outward-rounded interval certificates. The global overlap identity is checked by GO-2 in exact integer arithmetic. The local quadratic forms (OS, EH), the Bell-frame witness (BF) and the closest-point normal slice (NSL) check proofs in this supplement and are listed below. The two sharp critical-locus theorems are proved in this supplement, and their checks, listed below, are the analytic critical-moment comparison supported by the exact symbolic checks CO-1–CO-3 and the covering profile FP-2 over $\mathbb{Q}$, the analytic mixed null-critical proof supported by the exact symbolic checks MS-1–MS-4, and for the sharp entropy coefficient the interval certificates PS, TC, LCC and CCV on the critical locus, with the exact checks PM (the slice $P = 1$) and EX. Normal-form coverage, constraint positivity, chart reductions and entropy-transfer arguments are analytic. Interval certificates use outward-rounded ball arithmetic, and every sign decision is a certain comparison of balls; no floating tolerance decides anything. Certificate data files give every Gram matrix as integers over a stated denominator, with an integer Cholesky approximation whose residual is checked to be strictly diagonally dominant with positive diagonal, using integer arithmetic; no search solver is a verification input.

GO-1–GO-4 The sharp overlap certificate for Theorem 18: independent verification of all 50 used unit HS rays (46 baseline plus four rational two-factor skew-rotation rays) by $n = 1$ and $Q = W_1 = \tau = 0$ (GO-1); exact rounded low-rank reconstruction and the integer dominance test (S7.8) for all 277 remaining rational Gram blocks in $\mathcal{C}_Q$; invariance of the two explicit analytic remainders; and exact equality of (S7.7) for $(Q - n^2 e)/36$ across all 3,277 degree-≤ 6 signed lower-permutation monomial orbits on the 19-variable closest-point normal slice (GO-2); the ordered-singular-value remainder of Lemma S27 and the rational chain giving $Q \geq 1 - \Phi \geq \frac{6+\sqrt{6}}{12} d^2 > \frac{7}{10} d^2$ (GO-3); rational series bounds giving $k > 16/225$, and the margin $k(6+\sqrt{6})/12 > 1/20$ of Theorem 19 (GO-4). Normal-slice coverage and the positivity of the closest-point constraints are analytic (Section S7.2).

S20.2 Further four-qubit and stability checks

FG-1–FG-5, FG-9 The direct proof of the spin-flip gradient identity (Section S4): Schouten symmetry of $V - 6U$ (FG-1); the copy-label Casimir selection rules, the moment operators (CM), and the polarisation defect (CP) on the full invariant multiplicities (FG-2); the cubic intertwiner's normalisation 54 and the Klein-four rule excluding the quartic spin-one covariant (FG-3); the mixed-norm algebra (CY) and its polarisation giving (FGI) (FG-4); nonzero coefficients for the wrong factor or omitted correction (FG-5). FG-9 evaluates every intermediate identity of the proof from the tensor definitions, in exact arithmetic at Gaussian-integer states, as a regression check. The checks FG-6–FG-8 of the same script are listed in Section S20.1.

CO-1–CO-3 The critical variance and three-term moment identity (S8.3), the 32-ray tight frame and its permutation up to sixth-root-of-unity phases under H_1, H_2 (CO-1); the exact polynomial identities and rational concavity bounds P_1, P_2 for $Q \geq \frac{3}{2} - 2\Phi$ on $\Phi \leq \frac{1}{2}$ (CO-2); and the sharp distance constant $(2 + \sqrt{2})/4$ and Bell equality (CO-3). The signed-Birkhoff reduction and case reductions are analytic (Section S8.2).

OS-1–OS-3 The full local quadratic form from direct flattenings, its exact spectrum and orbit kernel, the local sextic radius, and HS, Bell and cluster regressions (Section S7.3).

BF-1–BF-4 All coefficients of the quartic determinant and cubic gradient identities behind (S7.1), the magic Pauli normalisation of the W_1 formula, and the Bell-pair Gram algebra, compatible-plane geometry and HS/Bell representatives of Proposition S29, whose constructive plane argument is analytic.

EH-1–EH-3 The full entropy quadratic forms from all cut flattenings, explicit orthonormal normal blocks, exact coefficient multiplicities, the rational radius bound and a least-coefficient normal; the resolvent estimate and the sharp limiting argument are analytic (Section S7.4).

NSL-1–NSL-3 The closest-point normal slice: the thirteen first-order conditions have rank ten and are exactly (S7.3); lower rotations conjugate the gauge matrix; the stationary and gauge conditions cut out exactly the span of the nineteen slice coordinates. That a closest point exists and maximises the height is analytic.

NC-1–NC-4 The two auxiliary Bloch spheres on the null-critical locus (Section S8.2): the auxiliary magic transform is unitary with determinant $\sum_j v_j^2/2$ (NC-1); the 32 HS rays are exactly the opposite-parity cube pairs (NC-2); the Bell projectors and the moment formula (S8.4) (NC-3); and the scalar remainder identities, sharp constants and cluster equality of the elementary proof of (8) of the paper (NC-4). The unit-sphere inequalities and the overlap maximisation are analytic. NC-5 and NC-6 of the same script are further checks not used here.

RP-1–RP-4 The three real mutually unbiased bases and the tight tomography identity, all sixteen HS planes, the paired overlap/Bures compression formulas, purity, the fourth frame moment, sharp GHZ covering equality and entropy-gradient matrices; the rank-two coverage and the relative-entropy identity are analytic (Lemma S30).

FP-1–FP-4 All 70 interleavings and their 23 symmetry classes (FP-1), every coefficient and positive Gram of (S8.11) (184 Grams, all 4830 coefficients through degree four compared exactly; FP-2), the homogeneous area-weighted fusion identity and the magic-basis spin flip (FP-3), and every overlap of the matching family (S8.14) (FP-4); ordering coverage, the Bloch bridge and the area-weighted Bures argument are analytic.

PM-1, PM-2, PM-4, PS-1 The slice $P = 1$ (Section S8.4): exact cut correlation matrices from the physical magic basis, symmetry generators, and the exact equality spectra, fidelities and full 144-pair symmetry orbit (PM); C_* to eighteen digits in interval arithmetic (PS-1); the symmetry arguments and the wall lemma are analytic.

NV-1–NV-2 Numerical evidence only, no certificate: the minimum of $\delta - C_* d^2$ over critical states at fixed τ (NV-1) and random positive local filters of the equality state and of nearby critical states (NV-2), quoted in the remarks after Theorem 21.

TC-1–TC-8 The correlation-matrix chart of the critical locus (Section S8.5): exact Bell correlation matrices of the twelve rays and $\lambda_{ji} = (\|v\|^2 + \langle C^{ji}, T\rangle)/4$ for a rational v (TC-1); the analytic range $\tau \geq 0.36$ in ball arithmetic (TC-6) and the sign symmetry at the equality point (TC-7). TC-2–TC-5 are numerical regressions of the analytic Lemmas S36 and S37 and of the ball evaluator against floating point and finite differences; TC-8 is a sampled audit of the cell bounds and of the HS rule, a regression and not a proof input.

EX Exact symbolic statements behind the local cone lemma at all eight chart equality points: the sign symmetry permutes the rays within each cut and the planes, preserving the trace and determinant of each plane matrix; exactly four planes tie, at $4\Phi_* = (1 + 1/\sqrt{3})(1 + \sqrt{2/3})$, with $G_\Pi = 0$ there by the definition of C_*; and the tied gradients sum to zero identically.

LCC The four tied planes, their gradients and interval Hessians at the eight chart equality points, and the cone cells of Lemma S39.

CCV The covering of Proposition S40; a floating-point twin only orders the attempts and certifies nothing.

UR-3 The fixed-trace Newton identities $e_2 = (1-p)/2$ and $e_3 = (1-3p+2t)/6$ used in the proof of Lemma 29 (Section S6.2); the rest of that proof is analytic. The same verifier checks the quantitative crossed-purity bound UR-4 in Section S20.1; its UR-1 and UR-2 are further checks not used here.

HD-1–HD-5 Algebra in the analytic HS supporting proof of Lemma S6 at $q_u = 8/5$: the fixed-purity path to the one-large spectrum (HD-1), the slope constant $105^5 < 3 \cdot 2^{32}$ (HD-2), the one-large moment formula and the displayed scalar curvature identity (HD-3), the rational uniform-contact gap and the outer dual (HD-4), and the lower-band target and wrong-slope/excessive-tilt controls (HD-5). The maximisation, divided-difference and curvature sign arguments and all equality inferences are proved in the text.

LT-1–LT-3 The generic identities in the low-tilt chart reduction (Lemma S4): the divided-difference identities behind the bound (DD) and the slope constant $64/35 > 9/5$ (LT-1), fixed-purity path and derivative ratio (LT-2), and negative controls for extending q to 2 or dropping the tilt bound (LT-3). The feasible monotone path is proved in the text.

CG-1–CG-5 The explicit arithmetic in the capped strengthened cubic bound (Lemma S12 and Corollary S13): the coefficient bounds and the CG column of Table S4 (CG-1), the divided-difference algebra and two binary endpoints, with the kernel algebra of the Cauchy-integral proof of monotonicity in Lemma S8, which is all this lemma uses of that lemma (CG-2), and the quartic endpoint bounds (CG-3), the pair identities, the two weight polynomials and the capped-cluster assembly (CG-4), and negative controls for an excessive correction or an omitted purity cap (CG-5). All maximisation and sign arguments are in the proof; CG adds no Bernstein sign test or parameter covering.

NU-1–NU-3 Exact-symbolic fifth-order Newton and projection identities, rational coefficient and logarithm bounds at the order q_s, and the NU column of Table S4 in Lemma S16. NU-3, SHD-4 and CG-1 check their columns of that table, the hypotheses of Lemma S15, through one shared function. This is algebra for a written analytic proof, with no subdivision, Hessian grid or interval sign test.

SL-1–SL-4 Exact finite power/logarithm constants for the head/tail entropy estimates, four polynomial lower bounds, rational domain and ε-removal inequalities, and comparison of all 11 printed Bernstein coefficients with exact rationals. The distinguished HT cut is proved analytically in Lemma S17; there is no subdivision or interval derivative grid. The verifier checks both printed tables against the symbolic formulas.

QV-1–QV-3 The residual identity and the minimum-purity reduction (QV-1), the purity-box bounds for (QV) and (QK) (QV-2), and the cluster-to-Bell calibration (QF) of the paper with both forced coefficients (QV-3), used in Section S6.1; the box extremum and the equality inferences are analytic.

JI-1–JI-3 Exact symbolic verification of the implication identity (JI), all three purity-box residual formulas (JB), and rational data at HS, cluster and Bell products. These check algebra only; the fixed-sum box maximum and equality inference are analytic.

MS-1–MS-4 The strengthened sphere norm identities and all sixteen multiaffine vertices for the opposite-parity reduction (MS-1), the exact same-parity parametrisation, monotonicity in $\tilde{w}$ and polynomial minorants (MS-2), the three univariate polynomial decompositions closing the same-parity triangle on $[0,1]$ (MS-3), and the sharp cluster interpolation and equality family (MS-4); the sphere, monotonicity and coverage arguments are analytic.

S20.3 Seven-qubit and higher-qubit checks

ES-1 Exact phase-support and conference-switching identities, and the common real fixed space $\mathrm{span}_{\mathbb{R}}\{U, V\}$. Lemma S72 gives the analytic stationarity argument for every differentiable LU/permutation/conjugation-invariant functional; Proposition S73 applies it to entropy.

SEH-1–SEH-3 The root bracket for x_* and the spectral floor $1/12$ (SEH-1); exact invariance and Sylvester positive definiteness of all 16 canonical symmetry-adapted integer blocks (sizes at most 4×4) of the two 128×128 Hessian blocks above 0.239395, with no floating-point basis (SEH-2); and the perturbation bound giving $-239/1000$ (SEH-3), for Theorem S88.

EB-0–EB-3 The rational constants of the third-derivative and Taylor bounds in Theorem S89.

SC-2 The rational logarithm lower witness $E_3(\psi_{7/100}) > 59307/20000$ used for the competitor band and the cap.

SP7-1, SP7-5, SP7-6 Spin-flip spectral pairing and overlapping-cut compatibility, the sharp entropy–purity slope at $d = 8$ (proved analytically in Lemma S77 from $N(r_*) = 0$ and $7^5 > 2^{14}$), and the competitor eigenvalue cap $497/1000$ (proved analytically by a relative-entropy tangent lower bound and square completion in Corollary S78).

AH-1, AH-5 The exact competitor purity band and candidate joint spin-flip identities (proved analytically in Lemma S80).

AH-6 Exact rational coefficient certificate for the global joint scalar majorant (Section S12.6): rational root/logarithm enclosures and the positive kernel constants (AH-6a), the symbolic cubic discriminant and double factor (AH-6b), and positive Bernstein and shifted-power coefficients (four rows) of the degree-eighteen quotient (AH-6c). The positive-kernel and binary Hermite reductions are analytic.

E7-K0–E7-K3 Exact integer family contractions for seventeen five-copy sources, exact generic rank bounds by determining-point minor identities, and exact three-copy kernel source polynomials.

E7-1 Verified-arithmetic certificate of Theorems S85 and S86: exact physical kernel projection with Arb, evaluation of all 3424 direct probes with forward error bounds, a validated rank-3283 inverse, and a strictly positive Gram repair in 3754 independent symmetric coordinates. The cubic purity multipliers and marginal spectral-cap sources are included in the identity.

TCB-0–TCB-4 The three-copy certificate of Theorem S45: integer block kernels and antisymmetriser relations, primal feasibility of the boundary pseudo-moment $m^*/70416$, the exact dual identity at $L_3 = 125/978$ in all 37 orbit coordinates from the six active-block kernels, and positive definiteness of the primal pivot submatrices and rational kernel multipliers by fraction-free Bareiss elimination.

TCN-1–TCN-3 Numerical only: floating-point solves of the three-copy relaxation for $n = 4, \ldots, 7$ (TCN-1), with a tight-tolerance solve at $n = 7$ (TCN-1b), a strictly feasible dual at $n = 7$ (TCN-2) and its rounding to integer data (TCN-3).

FC-0–FC-4 The four-copy certificate of Theorem S47: exact integer S_4 irreps, matrix units and the flip maps T_1, T_2 (FC-0); numerical regressions of the parametrisation and of the modular adjoint contraction (FC-1, FC-2, not used in the proof); positive definiteness of the 271 integer flip multipliers and of every residual ω-block, with the residual sum recovered exactly by the Chinese remainder theorem over 35 primes under an integer magnitude bound (FC-3); and the conclusion $\pi_3 \geq 211/1600$ (FC-4). The integer certificate is a data file found by a floating-point SDP search; only its exact check is used. The reported smallest eigenvalue is diagnostic; exact Sylvester tests decide positivity. Runtime about 12 minutes.

NS-0–NS-4 The exact four-copy pseudo-moment of Proposition S49: a regression of the modular forward map against a direct floating-point computation at $n = 3, 4$ (NS-0, not used in the proof); the exact objective $\mathrm{oo}(x)/\mathrm{on}(x) < 6597/50000$ (NS-1); positive definiteness of all 78 ω-blocks (NS-2) and of all 271 flip blocks, recovered exactly by the Chinese remainder theorem over 34 primes under an integer magnitude bound (NS-3), by fraction-free elimination; and the conclusion $L_4 < 6597/50000 < 121/917$ (NS-4). The integer point comes from a floating-point solve; only its exact check is used. Runtime about 13 minutes.

FS-0–FS-2 Numerical only: floating-point solves of the four-copy relaxation at $n = 4, 5, 6, 7$, the source of the table in Appendix G; the two largest $n = 7$ solves are flagged inaccurate by the solver and are not certified bounds.

S5-1, Y5-1 The exact S_5 matrix units and physical spin-flip reconstruction of Lemma S56, and the rational two-flip Young models of Lemma S57.

IK5-1–IK5-4, PK5-1–PK5-4 The exact icosahedral feature ranks and annihilators, the twelve exact physical kernels, and the support and invertibility of the supplied kernel bases (Proposition S58); PK5-4 does not check the sharp identity.

DP5-1 The character count of the five-copy moment space, the trace-adjoint flip, the coset count and the target calibration of the direct probes (Lemma S59, Proposition S60).

FP5-1 Verified-arithmetic certificate of Theorem S62: evaluation of the 3424 rational direct probes with single-column norm companions, the rank-3283 right inverse in 3754 independent symmetric coordinates with $\|I - LAR\|_2 < 0.070$, and the positive repair of the identity Grams, with correction below 0.324. The triangular dyadic baseline, frames and probes occupy 441612 bytes.

FP5-2 An independent exact check, on scalar probes, of the baseline trace and of all 8965 evaluator coefficients in the width-16 frames, whose first $r_b \le 7$ columns give the 3754 coordinates of FP5-1.

FP5-3 An independent reimplementation of the probe evaluator (physical local transport, Gram transport and symmetric powers rebuilt from the stored data, all coset pairs summed) reproduces all twelve face traces and the 8965 width-16 evaluator coefficients inside the evaluator's error radii on twelve non-scalar probes, and refutes a wrong-convention control.

FP5-4 Exact orthogonality and fingerprints of the seed-generated repair frames, action and norm bound of the independent symmetric coordinates, rigorous rotation radii and a wrong-frame control.

TT-1 Exact contractions of the three unflipped all-V faces over the 14,432 allowed support five-multisets check disjoint even/odd column supports, as do the four stored IK5 arrays. The stored physical kernels annihilate both field components, as predicted by the column-parity proof in Proposition S58.

FO-1 Integer one-body, pair and spin-flip frame contractions for the entire real icosahedral family; exact symbolic graph, skewness and frame reconstruction identities on $1/32 < y < 1/12$ (proved analytically from the computational-basis marginals and spin-flip parity in Theorem S64); the affine invariant $H(\psi_y) = 155 + 96y$ and its tangent-square defect $96^2 t^2$ in Theorem S66.

LH-1–LH-4 Exact stationarity of Ψ_{ico} (LH-1); the integer field decomposition $H = (A + \sqrt{555}B)/18340$ of the real 256×256 Lagrange Hessian (LH-2); the 22 local-unitary and phase tangents in its kernel (LH-3); and the exact symmetry-adapted integer block decomposition into 1×1, 2×2, and 4×4 blocks with closed-form characteristic polynomials and

exact minimum transverse curvature $(350 - 6\sqrt{2005})/4585 > 80/4585 > 78/4585$ (LH-4), for Theorem S69.

QB-1–QB-3 The rational remainder margin 285123/14672000000 (QB-1), the overlap form of the orbit distance (QB-2), and the diagonal phases of the 32 nonzero amplitudes (QB-3), for Theorem S71.

EO-1 Exhaustive integer switching/complement classification of the 720 ordinary-site permutations: six classes of size 120, giving 42 labelled LU orbits after the distinguished-site choice (Corollary S68).

MO-1, MO-2 Independent purity-parameter checks of the pair tensors, tetrads, skewness arithmetic, frame overlaps and reconstruction coefficients in Theorem S64.

SF-1, SF-5 The focused sextic covariants, complete scalar-gradient basis controls, and a coefficient check of the closed zero-focus Gram with determinant $72x(1-2x)$ in Section S13. Completeness and rank throughout the family are analytic.

HI-1–HI-4 Exact copy-invariant counts from (ID2), used for the free coordinates in Theorem S100.

LP-0–LP-2, GS-1–GS-3 Exact Rains-LP dual identities for $4 \le n \le 12$, the exact primal point attaining $L_2 = 1/8$ at $n = 7$ (the AME weight distribution), the graph-state cut ranks of Table S7, the weight distribution of $C_{10}(1, 2, 4)$ and the Petersen purity (Theorem S101, Remark S102).

LP-14, LP-15, GS-14, GS-15, LP-13, QP-13 The same at $n = 14, 15$ (Theorem S103); at $n = 13$, the exact Rains dual for 25/1408 (LP-13) and the exact stabilizer count of the X-measured fourteen-qubit witness, with all 14×3 one-qubit Pauli measurements compared (QP-13), for Proposition S116.

LP-19, GS-19, NG-1–NG-3 The exact Rains dual at $n = 19$ (LP-19), the starting graph witness (GS-19), and the improved graph with its stabilizer enumeration and independent cut-rank count (NG-1–NG-3), for Theorem S117.

SW-0–SW-4 The three-copy Schur–Weyl certificates of Theorem S114: the per-site algebra (SW-0); regressions against the Haar moment and random states at $n = 4, 5$, not used in the proof (SW-1); exact positive definiteness of the rational multipliers (SW-2) and of every ω-block of the exact residual (SW-3); and the resulting purity bounds and code exclusions (SW-4). Runtime about 7 minutes, dominated by $n = 25$.

TA-1 Numerical only: the three-copy level compared with the Rains linear programme for $4 \le n \le 20$, in the paragraph after Theorem S114.

E13L-1–E13L-2 The thirteen-qubit entropy witness of Corollary S119: exact Gaussian-integer amplitudes and norm by a Walsh transform, the exact Pauli-code block compression of Lemma S118 at all 1716 balanced cuts with every block eigenvalue enclosed in Arb above $1/2100^2$ (E13L-1), and the Arb enclosure of E_6 (E13L-2). The same script's Hessian checks E13L-0, E13L-3–E13L-5 are not used. Runtime about 15 minutes.

GL-1, GL-2 The same at $n = 16, 17$ (Theorem S104).

E18-1–E18-4 The exact Rains dual and the attaining graph state at $n = 18$ (Theorem S105).

TY-1–TY-2 The circulant graph states $C_{16}(2, 3, 4, 8)$ and $C_{20}(3, 4, 5, 10)$ of Corollary S107: odd degree at every vertex ($\tau_n = 1$), the exact $\mathbb{F}_2$ ranks of all balanced cuts, and the purities 1/143 and 131/77792.

ST-1–ST-6 The rational multipliers and residual signs of (S17.1) for each $n \in \{4, 8, 10, 14, 16, 20\}$, and the exact purity $L_n + c_n$ of a $\tau_n = 1$ graph state at each n (Theorem S108).

ST-7, ST-8 The closed-form multipliers and weights of (S17.2), checked against an exact elimination for odd $n = 5, \ldots, 41$, and the exact evaluation of both sides on Gaussian-integer states at $n = 5, 7, 9$ (Theorem S112).

ST-9 The witness $\mathrm{GHZ}_{n-2} \otimes \Phi^+$ for the optimality of c_n over all pure states: its Pauli weights, $O = 1/(2n)$, and π_m from the subsystem-purity generating polynomials, with $\pi_m + c_n O = \frac{1}{2}$ exactly for odd $n = 5, \ldots, 43$ and a negative control.

SDS-1–SDS-3 Exact Pauli selection in spin-flip-doubled states (Theorem S51, Corollary S52).

CC-1 The quartic identity (S10.2), all 226 coefficients, in formal variables (proved analytically by Hamming-weight block decomposition and the Hafnian–Gram identity in Theorem S53).

CP-4 The algebraic matching-pattern identity in Proposition S54 (proved analytically from $B\mathbf{1} = 0$ and $B\overline{B} = 5I_5 - J_5$).

References

[1] A. Lewis, The most entangled states of four and seven qubits (the paper this document supplements).

[2] F. Huber, O. Gühne and J. Siewert, Absolutely maximally entangled states of seven qubits do not exist, Phys. Rev. Lett. **118**, 200502 (2017), arXiv:1608.06228, doi:10.1103/PhysRevLett.118.200502.

[3] A. C. Doherty, P. A. Parrilo and F. M. Spedalieri, Complete family of separability criteria, Phys. Rev. A **69**, 022308 (2004), arXiv:quant-ph/0308032, doi:10.1103/PhysRevA.69.022308.

[4] E. M. Rains, Quantum shadow enumerators, IEEE Trans. Inf. Theory **45**, 2361 (1999), arXiv:quant-ph/9611001, doi:10.1109/18.796376.

[5] M. Hein, J. Eisert and H. J. Briegel, Multiparty entanglement in graph states, Phys. Rev. A **69**, 062311 (2004), arXiv:quant-ph/0307130, doi:10.1103/PhysRevA.69.062311.

[6] A. J. Scott, Multipartite entanglement, quantum-error-correcting codes, and entangling power of quantum evolutions, Phys. Rev. A **69**, 052330 (2004), arXiv:quant-ph/0310137, doi:10.1103/PhysRevA.69.052330.

[7] L. E. Danielsen and M. G. Parker, On the classification of all self-dual additive codes over GF(4) of length up to 12, J. Combin. Theory A **113**, 1351 (2006), arXiv:math/0504522, doi:10.1016/j.jcta.2005.12.004.

[8] F. M. Huber, Quantum states and their marginals: from multipartite entanglement to quantum error-correcting codes, Ph.D. thesis, Universität Siegen (2017), §5.11.4, d-nb.info/1154308545/34.

[9] G. Anglès Munné and F. Huber, SDP bounds on quantum codes: rational certificates, arXiv:2603.19901 (2026).

[10] X.-W. Zha, J.-X. Qi and Y.-G. Zhang, Exploring maximally multipartite entanglement of even n qubits with N-tangle, arXiv:1410.5049 (2014).

[11] A. Rico, J. Abellanet-Vidal, N. B. T. Kothakonda, A. Sanpera and G. Anglès Munné, Entangling a quantum system to disentangle its parts, arXiv:2604.13169 (2026).

[12] G. Anglès Munné, A. Nemec and F. Huber, SDP bounds on quantum codes, arXiv:2408.10323 (2024).

[13] B. Et-Taoui and A. Makhlouf, Complex skew-symmetric conference matrices, Linear Multilinear Algebra **70**, 6648–6663 (2022), doi:10.1080/03081087.2021.1967848.

[14] R. L. Hudson and G. R. Moody, Locally normal symmetric states and an analogue of de Finetti's theorem, Z. Wahrscheinlichkeitstheorie verw. Gebiete **33**, 343–351 (1976), doi:10.1007/BF00534784.

[15] E. Serrano-Ensástiga, O. Giraud and J. Martin, Multiqubit monogamy relations beyond shadow inequalities, Phys. Rev. A **113**, 012415 (2026), arXiv:2507.12680, doi:10.1103/9fkf-hm8l.